THE PEOPLE'S REPUBLIC OF CHINA

THE EIGHTH NATIONAL REPORT FOR THE CONVENTION ON NUCLEAR SAFETY

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Contents

Abbreviations ..... **I**
A. Introduction **1**
A.1 General Situation of the Peaceful Uses of Nuclear Energy in China 1
A.2 Policies and Objectives for Nuclear Power Development in China 1
A.3 Nuclear Safety Policy in China 2
A.4 Preparation and Structural Features of the National Report 3
B. Summary **5**
B.1 Summary on the Seventh Review Meeting and Follow-up Actions 5
B.2 Carried out and Planned Activities on Safety Improvement 14
B.3 International Peer Review 18
B.4 Experience Feedback of Typical Operational Events Since the Previous National Report 19
B.5 Response to VDNS 23
B.6 Main Safety Issues Identified Since the Previous National Report 28
B.7 Lessons Learned from the Emergency Drills and Exercises Since the Previous National Report 29
B.8 Good Practices and Challenges Since the Previous National Report 30
6. Existing Nuclear Power Plants **36**
6.1 List of Existing NPPs 36
6.2 General Situation of Existing NPPs 36
6.3 Overall Safety of NPPs in China 40
7. Legislation and Regulatory Framework **42**
7.1 Legislation and Regulatory Framework 42
7.2 Nuclear Safety Licensing System 45
7.3 Nuclear Safety Supervision System 51
7.4 Enforcement 52
8. Regulation Organizations **54**
8.1 Regulation System 54
8.2 MEE (NNSA) 54
8.3 China Atomic Energy Authority 65
8.4 National Energy Administration 65
8.5 National Health Commission 66
9. Responsibilities of the Licensees **67**
10. Priority to Safety **69**
10.1 Safety Policies and Arrangements for Safety Management 69
10.2 Safety Culture of NPPs 69
10.3 Peer Review and Self-assessment of NPPs 71
10.4 Regulatory Review and Control Activities 73
11. Financial and Human Resources **78**
11.1 Financial Resources 78
11.2 Human Resources 80
12. Human Factors **86**
12.1 Regulatory Requirements on Prevention and Correction of Human Errors 86
12.2 Measures to be Taken by Licensees 87

12.3 Regulatory Review and Control Activities 89
13. Quality Assurance 91
13.1 Overview....91
13.2 Basic Principles and Requirements of Quality Assurance 91
13.3 Establishment, Implementation, Assessment and Improvement on Quality Assurance System of NPPs 93
13.4 Regulatory Review and Control Activities 96
14. Safety Assessment and Verification 97
14.1 Regulatory Requirements on Safety Assessment and Verification for NPPs 97
14.2 Practices on Safety Assessment and Verification for NPPs 98
14.3 Regulatory Review and Control Activities 108
15. Radiation Protection 111
15.1 Basic Requirements of Radiation Protection 111
15.2 Application of ALARA Principle in NPPs 115
15.3 Exposure Control for Workers in NPPs 117
15.4 Environmental Radioactivity Monitoring 117
15.5 Regulatory Review and Control Activities 120
16. Emergency Preparedness 122
16.1 Basic Requirements for Emergency Preparedness 122
16.2 Emergency Organizational System and Responsibilities 123
16.3 Emergency Classification and Reporting 127
16.4 Emergency Plan 128
16.5 Trainings, Drills and Exercises 129
16.6 Information Publicity and Public Communication on Emergency 131
16.7 Capability Building of Emergency Response and Rescue Arrangement 133
16.8 Regulatory Review and Control Activities 136
16.9 International Arrangements for Nuclear Emergency 137
17. Siting 139
17.1 Evaluation on Site-related Factors 139
17.2 Impact of NPP on Individuals, Society and Environment 143
17.3 Re-evaluation of Site-related Factors 146
17.4 Consultation with Other Contracting Parties Likely to Be Affected by NPPs 148
18. Design and Construction 149
18.1 Implementation of Defence in Depth 149
18.2 Incorporation of proven technologies 172
18.3 Design for Reliable, Stable and Manageable Operation 178
19. Operation 182
19.1 Initial Authorization 182
19.2 Operational Limits and Conditions 185
19.3 Procedures for Operation, Maintenance, Inspection and Testing of NPP 187
19.4 Procedures for Responding to Operational Occurrences and Accidents 189
19.5 Engineering and Technical Support 190
19.6 Event Reporting System of Operating NPPs 191
19.7 Operational Experience Feedback 193

19.8 Management of Spent Fuel and Radioactive Wastes on the Site 196
19.9 Cyber Security of NPPs 200
Appendix 1: List of NPPs in China (By December 31, 2018) 201
Appendix 2: Operational Events in NPPs of China (From 2016 to 2018).....204
Appendix 3: WANO Performance Indicators of Operating Nuclear Power Units in China (from 2016 to 2018) 206
Appendix 4: Laws, Administrative Regulations Departmental Rules and Guides on Nuclear Safety for NPPs in China.....212
Appendix 5: List of Domestic and Overseas Peer Review Received by NPPs in China (From 2016 to 2018).....221
Appendix 6: List of Scheduled Domestic and Overseas Peer Review for NPPs in China (From 2019 to 2021).....226
Appendix 7: Licensed Operators and Senior Operators of Operating NPPs in China (by December 31, 2018).....229
Appendix 8: Occupational Exposure in NPPs of China (From 2016 to 2018) 230
Appendix 9: List of Drills of NPPs in China (From 2016 to 2018) 233
Annex: Contributors to the Eighth National Report of China 235

Abbreviations

CAEA	China Atomic Energy Authority
CGN	China General Nuclear Power Group Co., Ltd.
CNEA	China Nuclear Energy Association
CNNC	China National Nuclear Corporation
CNNP	China National Nuclear Power Co., Ltd.
CPR	Corporate Peer Review
DCS	Digital Control System
FSAR	Final Safety Analysis Report
IAEA	International Atomic Energy Agency
INES	International Nuclear Events Scale
IRRS	Integrated Regulatory Review Service
MDEP	Multinational Design Evaluation Program
MEE(NNSA)	Ministry of Ecology and Environment (National Nuclear Safety Administration)
NEA	National Energy Administration
OLE	Operational License Extension
OSART	Operational Safety Review Team
PSAR	Preliminary Safety Analysis Report
PSA	Probabilistic Safety Assessment
PSR	Periodical Safety Review
PSUR	Pre-startup Peer Review
SAMG	Severe Accident Management Guideline
SOER	Significant Operating Experience Report
VDNS	Vienna Declaration on Nuclear Safety
WANO	World Association of Nuclear Operators

A. Introduction

China has attached high importance to nuclear safety, adhered to a rational, coordinated and balanced nuclear safety and security outlook, earnestly performed all obligations committed to international community, undertaken the safety responsibilities for its nationwide NPPs, and made unremitting efforts to meet and keep a high-level nuclear safety accepted internationally.

A.1 General Situation of the Peaceful Uses of Nuclear Energy in China

By the end of December 2018, China had 45 nuclear power units in operation, with total capacity of 45.90GWe, and 11 nuclear power units under construction, with total capacity of 12.18GWe.

From 2016 to 2018, China had put a total of 15 nuclear power units into operation, with capacity of 17.30GWe, among them, Unit 1 of Sanmen NPP and Unit 1 of Taishan NPP is the first operating unit of AP1000 and EPR in the world respectively.

The operating nuclear power units have continually maintained safe and steady operation. The annual accumulative electricity generated by nuclear power units in commercial operation in China increased steadily, being respectively 210.52 billion kWh, 247.47 billion kWh and 286.5 billion kWh in 2016, 2017 and 2018, accounting for about 3.51%, 3.94% and 4.22% of the total electric energy generated in the whole country.

The nuclear power units under construction met the current requirements of national nuclear safety regulations and standards. Strict management was performed in the process of siting, design, manufacture, construction, installation and commissioning. The quality assurance system functioned well, the construction quality met the design requirements and the overall quality is under control.

A.2 Policies and Objectives for Nuclear Power Development in China

The development of nuclear energy plays an important role in optimizing Chinese energy mix, ensuring energy security, reducing environmental pollution and tackling climate change. The development of nuclear power constitutes an important component of China's nuclear energy sector. China follows the policy of developing nuclear power in a safe and efficient manner, and adheres to the approach of enhancing safety for the sake of development and promoting development by upholding safety in an unrelenting effort to bring the dual goals of development and safety in alignment with each other.

In 2016, the National Development and Reform Commission and NEA jointly issued the *13th Five-Year Plan for Development of Power Industry*, which stated that

China adheres to the principle of safe development of nuclear power and speed up the construction of nuclear power projects in coastal regions. During the 13th Five-Year Plan period, China will put nuclear power units about 30GWe into operation, start construction of nuclear power units over 30GWe, and the capacity will reach 58GWe by 2020. Demonstration projects of AP1000 at Sanmen and Haiyang will be completed, the HPR1000 demonstration projects at Fuqing of Fujian and Fangchenggang of Guangxi will be constructed, and construction will be started for a number of new nuclear power projects in coastal regions. China will further carry out the research, justification and initial preparatory activities for inland nuclear power projects; and work earnestly for the protection of NPPs site resources.

A.3 Nuclear Safety Policy in China

China has consistently given top priority to nuclear safety in its peaceful use of nuclear energy and takes nuclear safety as an important part of the national overall security system. In order to ensure nuclear safety, protect the safety and health of the public and employees, and protect the ecological environment. China has taken defence-in-depth measures on prevention, protection and mitigation for nuclear facilities, implemented independent nuclear safety regulation to prevent nuclear accidents caused by technical or human factors or natural hazards, and taken measures to minimize the radioactive consequences in the event of a nuclear accident.

On January 1, 2018, the *Nuclear Safety Act of the People's Republic of China* (hereinafter referred to as *Nuclear Safety Act*) came into force, to provide high level legal guarantee for nuclear safety. It is stipulated in the *Nuclear Safety Act* that: "Development of nuclear energy and technology must be conducted in conformity with the policy of ensuring safety. Nuclear safety work shall adhere to principles of safety first, prevention first, clear responsibilities, strict management, defense in depth, independent regulation, and comprehensive safeguard." The *Nuclear Safety Act* has clearly taken the nuclear safety outlook as the overall guiding ideology in the nuclear safety work, specified that the state shall formulate nuclear safety policy, and take strengthening safety culture development as a legal requirement, defined the responsibilities of the regulatory body and the nuclear safety responsibilities of all participating parties, and enhanced the nuclear safety inspection and law enforcement efforts to guarantee the rights and interests of the public.

In February 2017, China issued the *13th Five-Year Plan and Prospective Plan of 2025 on Nuclear Safety and Radioactive Pollution Prevention and Control* (hereinafter referred to as *Nuclear Safety Plan*), as a guideline for nuclear safety work. The *Nuclear*

Safety Plan set specific objectives and main tasks, major projects and guaranteed measures on the aspects of improving the safety level of nuclear facilities, escalating the level of radioactive contamination prevention, security, emergency response and safety regulation, and made top-level design and strategic arrangement for the nuclear safety and radioactive contamination prevention in the 13th Five-Year Plan period. In the *Nuclear Safety Plan*, the targets of 2020 are set as: to significantly improve the safety level of nuclear facilities in operation and under construction, to maintain at the international advanced level for the nuclear power safety, to obtain effective results on the decommissioning of early nuclear facilities and treatment of radioactive contamination, to prevent nuclear accident with radioactive contamination to environment, to maintain the good quality of radiation environment , to enhance the nuclear emergency capacity, to substantially escalate the nuclear safety regulation level, and to effectively guarantee nuclear safety, environment safety and public health. The prospective targets of 2025 are set as: to maintain the international advanced level of nuclear power plant safety, to reach international advanced level of other nuclear facilities safety, to maintain a low level of radioactive source radiation accident occurrence rate, to make significant progress in the decommissioning of early nuclear facilities, to dispose radioactive wastes in a safety and timely manner, to maintain good quality of radiation environment, to modernize nuclear and radiation safety regulation system and capacity, and to continue effectively ensuring nuclear safety, environment safety and public health.

The Chinese government will continue to deepen the exchanges and cooperation with all countries and international organizations in the world in nuclear energy field and in nuclear safety, properly fulfill the obligations under conventions already signed, put the multilateral and bilateral commitments on nuclear safety into practice, and control and reduce nuclear safety risks jointly with the international society to make active contribution to improving the global nuclear safety.

A.4 Preparation and Structural Features of the National Report

In terms of structure, the report consists of three parts: introduction, overview and detailed reports. “Introduction” gives a comprehensive and brief description of current situation of the peaceful uses of nuclear energy in China, describes the development status and objectives of nuclear power in China and illustrates the policies and opinions of China in respect of nuclear safety. “Overview” expounds the response to the seventh review results, the focused areas and improvement actions taken on nuclear safety, the response to the *VDNS*, and the good practice and challenges.

Please refer to the part “Detailed Reports for” more details of the above-mentioned contents. This part covers Chapters 6 to 19 of this report, and is prepared according to contents of Article 6 to Article 19 of the *Guidelines Regarding National Reports under the Convention on Nuclear Safety*. All chapters of this part begin with the original text of the *Convention on Nuclear Safety* and expound how to fulfill the obligations specified in *Convention on Nuclear Safety* by China through presentation of requirements of laws and regulations, important activities, practices and corresponding progresses.

In preparing of the report, the requirements in the *Guidelines Regarding National Reports under the Convention on Nuclear Safety* (INFCIRC/572/Rev.6) and in the summary report of the seventh review meeting were referenced, and the three principles set in the VDNS for the goal to prevent accident and mitigate accident consequences and the suggestions made by the chairman of the eighth review meeting requesting attention of the signing parties for the preparation of the report were considered to compile this report.

This report does not include information of Taiwan Province of the People’s Republic of China.

B. Summary

B.1 Summary on the Seventh Review Meeting and Follow-up Actions

With a view to strictly performing commitments made by the Chinese government when signing the *Convention on Nuclear Safety* and obligations of contracting party prescribed in the Convention, the Chinese government has set up a coordinating group for the implementing the *Convention on Nuclear Safety*, which is in charge of organizing and coordinating the implementation of the Convention in China and ensures that the requirements to the contracting parties made by the Convention and that all resolutions made in the previous review meetings related to the national reports under the *Convention on Nuclear Safety* will be fulfilled in China.

B.1.1 Summary on the Seventh Review Meeting

In August 2016, China submitted officially the *Seventh National Report of the PRC for the Convention on Nuclear Safety* to the IAEA. Furthermore, all written questions raised to China by other contracting parties were addressed.

In the seventh review meeting, contracting parties seriously scrutinized the implementation of the Convention of China in forms of report review, on site presentations and questions&answers. Practices and progresses in several aspects made by China since last review meeting were recognized positively, in which seven good performances were recommended, including the following:

(1) The extensive use of social software and networking by the MEP(NNSA) in daily business ;

(2) Timely amended and promulgated *Code on the Safety of Nuclear Power Plant Design* in response to VDNS;

(3) Issued the *Generic Technical Requirements for Improvements of NPPs after Fukushima Nuclear Accident*, and effectively implemented by NPPs;

(4) Building and perfecting rapid emergency response teams;

(5) Establishing and maintaining the nuclear safety regulations system fully in line with the IAEA safety standards;

(6) Continually improving domestic peer review system for NPPs under construction and in operation;

(7) The MEP (NNSA) developed a full-scale simulator for 1000MW NPPs with the function to simulate severe accidents.

Challenges and the areas for improvement for China were also identified in the review meeting, including:

(1) Training and development of regulatory body staff in light of multiple reactor technologies;

(2) Fostering of safety culture in contractors and suppliers of NPPs;

(3) Enhancing public communication on the safety of NPPs;

(4) Developing dedicated regulations on cyber security for NPPs and further strengthening defence in-depth cyber security system;

(5) Improving knowledge management, especially knowledge transfer, for both regulatory body and the nuclear industry;

(6) Ageing management of NPPs, including optimization ageing management program of NPPs and the long term operation review by the regulatory body for the first time.

B.1.2 Response Actions to the Challenges from the Seventh Review Meeting

China worked out and implemented follow-up actions to address all challenges from the seventh review meeting of the *Convention on Nuclear Safety*, specifically as:

(1) Training and development of regulatory body staff in light of multiple reactor technologies

The MEE (NNSA) has established formalized training hierarchy and formed a multi-level nuclear and radiation safety training system covering junior training, intermediate training and senior training in regular and irregular training ways.

To improve the understanding and mastering of the technical features of different reactors for regulatory staff, especially new types of reactor, the MEE (NNSA) has taken a series of actions. Firstly, kept on optimizing the training program and curriculum design, strengthened knowledge management and extended the training means by network, full-scale simulator and NPP site training. Secondly, incorporated knowledge about AP1000, EPR and HPR1000 into the current training materials. Thirdly, revised and updated relevant supervision program and inspection procedures based on regulation experience for AP1000, EPR and HPR1000 reactors construction and operation

China also dispatched personnel to participate in relevant technical meetings, technical working groups or training via international organizations or bilateral cooperation programs, to expand the knowledge structure of nuclear and radiation safety regulatory staff and improve their regulatory competence.

In addition, to further strengthen the capability of regulatory staff and promote the accumulation and the sharing of regulation experience, the MEE (NNSA) has carried out staff exchanges and job rotation between the headquarter, all regional offices and

technical support organizations. Experts with rich working experience in nuclear power have been hired from NPP operating organizations and relevant design and research organizations, to continually replenish the technical force for nuclear safety regulation.

(2) Fostering of safety culture in contractors and suppliers of NPPs

In the *Nuclear Safety Act*, it is required that organizations providing equipment, engineering and services for nuclear facility operating organizations shall vigorously foster and build safety culture, and integrate the safety culture into each part of production, operation, scientific research and management.

It is clearly stated in the *Nuclear Safety Plan* that: safety culture pilots shall be established for the nuclear equipment and professional qualification area etc. Good practices in safety culture development shall be compiled, and experience exchange shall be strengthened, to push all organizations to incorporate the safety culture concepts and requirements into rules and regulations. The safety culture assessment mechanism shall be established, and safety culture pilot assessment activities shall be carried out in NPPs and nuclear equipment areas.

In April 2017, the MEE (NNSA), based on the *Safety Culture Policy Statement*, compiled and issued the *Safety Culture Features*, further consolidating the policy guiding foundation for improving safety culture level of nuclear power contractors and suppliers. This document was compiled on the basis of the safety culture practices of NPPs, but was of the general guiding significance, and can be further converted and applied in areas such as equipment according to their own features and actual conditions.

Since the issuance of the *Safety Culture Policy Statement* in 2014, the MEE (NNSA) has organized nationwide special actions in nuclear equipment area to strengthen the safety culture communication and implementation in license holders of nuclear equipment and promote safety culture development. Since 2016, the MEE (NNSA) has carried out safety culture assessment in conjunction with daily inspection, to popularize good practices on human error prevention and further advance the safety culture development in the nuclear equipment area.

The MEE (NNSA) entrusted CNEA to carry out safety culture assessment on ten domestic nuclear safety equipment design and manufacturing organizations in 2017, and conducted field assessment on the safety culture development in a number of important nuclear power equipment suppliers in 2018. Subsequently, it will conduct assessment for thirteen organizations, to provide reference for the method study and good practices on safety culture development. In addition, CNEA called experience exchange conferences on safety culture development in 2017 and 2018, to exchange the practical

experience in safety culture development in various specialized areas and at all levels in the nuclear industry.

In the meantime, contractors and suppliers of NPPs are also attaching increasing importance to the safety culture development. They issued documents such as safety culture manuals and safety culture development plans, and established specialized working groups to promote the work, and urged the employees at all levels to pay more attention. They actively carried out self-assessment activities, to assess their status quo on safety culture and the attention of employees on nuclear safety.

In addition, the NPP operating organizations in China also actively promoted the safety culture development of contractors, including assisting in preparing safety culture development scheme, carrying out safety culture training and human error prevention training for contractors, conducting activities such as human error prevention skill competition in contractors, and encouraging contractors to formalize personnel behavior. The NPPs developed safety culture assessment guideline with respect to the contractors management expectation, and carried out safety culture assessment on one of the main service contractors in 2018, to comprehensively assess its safety culture development and propose improvement suggestions.

(3) Enhancing public communication on the safety of NPPs

In Chapter V of the *Nuclear Safety Act*, clear stipulations are specifically made on information disclosure and public communication. It is pointed out in Article 63 that the relevant departments of the State Council and departments designated by the government of provinces, autonomous regions and municipalities where NPPs are located shall disclose the information about nuclear safety in accordance with the law within their respective scope of responsibilities. The nuclear safety regulation administration of the State Council shall disclose the administrative licensing related to nuclear safety, and the information such as inspection reports on activities related to nuclear safety, overall safety status, radiation environment quality and nuclear events in accordance with the law. It is pointed out in Article 64 that nuclear facility operating organizations shall disclose the information such as the nuclear safety management policies and relevant documents, safety status of nuclear facilities, effluents and surrounding environment radiation monitoring data, and annual nuclear safety reports of their own. The Law also clearly specifies that the nuclear safety information that should be made public in accordance with the law shall be made public in society via government announcement and website and other ways facilitating the information reaching the public. Citizen, corporations and other organizations can also request from the MEE(NNSA) and departments designated by the government of provinces,

autonomous regions and municipalities where NPPs are located to obtain information about nuclear safety in accordance with the law. The nuclear facility operating organizations and the government of provinces, autonomous regions and municipalities where they are located shall hold hearings, justification meetings and symposiums or other forms on major nuclear safety issues with effect on public interests, to solicit opinions of stakeholders, and make feedback in appropriate forms.

The *Nuclear Safety Plan* also clearly stated that public communication on nuclear safety will further be promoted during the 13th Five-Year Plan period, meanwhile "openness and transparency" is taken as one of the basic principles for nuclear safety work in the period. It is stated in the *Nuclear Safety Plan* that, the basic knowledge of nuclear safety will be included into the education and training system during the 13th Five-Year Plan period. China will continue to spread nuclear and radiation knowledge into communities, secondary and primary schools and cadre training classes, and will build 10 national level nuclear and radiation safety popular science publicity and education bases with the support of enterprises, to enhance the publicity function of network platform and new media, and to strengthen the communication and exchange with media. China will complete the information disclosure scheme and guidelines, and strengthen the construction of information disclosure platform. Enterprises will release to the public project construction information in accordance with the law in different phases, and the government will take initiative to disclose the information of license review and approval, inspection and enforcement, environmental monitoring, nuclear events, and enhance the interpretation of public information.

In April 2018, the MEE revised and issued the *Measures for Public Participation in Environmental Impact Assessment* to standardize public participation in environmental impact assessment and guarantee the public's rights to know, to participate in, to express opinions and to supervise the environmental protection. It is required that owner entities should disclose relevant information to solicit opinions related to the environmental impact of the project. Owner entities should prepare reports on the public participation, and submit it together with the environmental impact assessment report.

The MEE (NNSA) has long attached importance to establish an open and transparent nuclear safety regulation system, formulated documents such as *Management Measures for Information Disclosure in Nuclear and Radiation Safety Regulation*, *Work Plan for Public Communication on Nuclear and Radiation Safety* and *Work Guideline for Public Communication on Nuclear Power Projects*, and issued the *Notice on Strengthening Nuclear and Radiation Safety Information Disclosure of*

Nuclear Power Plants, and promoted the information disclosure for environmental impact assessment reports of nuclear-related projects, national radiation environmental quality monitoring results, results of project review and approval and administrative licensing documents. In the meanwhile, popularization of science on nuclear safety has been carried out to popularize nuclear safety knowledge in campuses and communities.

The NEA entrusted CNEA to carry out research on the increasement of social acceptability for new nuclear power projects. The public communication pattern with Chinese characteristics has been formed by analyzing the status quo of public communication on nuclear power and learning the good practices at home and abroad, to provide references on solving the issue of social acceptability for nuclear power and other projects in China. All nuclear power corporations have also carried out research on "not in my back yard (NIMBY)" for NPPs.

The nuclear power corporations and NPP operating organizations have formulated programs, management procedures and work plans for information disclosure, public communication and nuclear power science popularization, to continually improve the transparency. Platforms for regular news release and media communication have been built to strength the communication and exchange with news media by combining the respective advantages of traditional and new media. Nuclear safety publicity materials such as *Science Popularization Manual for CNNP* and *Public Communication Guideline for Nuclear Power* were issued for further strengthening the nuclear power science popularization on the own initiative. In the meanwhile, communication methods were innovated to increase the interaction and mutual trust with the society and to enhance the understanding of public to nuclear energy and safety via multiple means, such as inviting media and public to visit nuclear power sites, opening WeChat account and microblog account for science popularization, developing online intelligent question and answer program, online interaction with netizens, and offline interaction with robots. On the basis of activities such as regular public opening days and weeks, the nuclear power corporations have continued to build science and technology museum and science popularization exhibition halls on nuclear power, which open to the public all the year round and display environmental monitoring information. At present, all nuclear power sites in China have built facilities for public visit and science popularization. The nuclear power corporations and NPP operating organizations in China regularly organize science popularization summer camps and tours for secondary and primary school students every year.

(4) Cyber security for NPPs

China officially issued the *Cyber Security Act of the People's Republic of China* in

2016, as a legal guarantee for the protection of the critical information infrastructures including NPPs.

For cyber security supervision of NPPs, the MEE (NNSA) established the cyber security working group in September 2017, and completed the draft of *Technical Policy for Cyber Security Supervision of Nuclear Power Plants* in January 2019. At present, the opinions for the draft are being solicited from related parties. The technical policy has set targeted specific requirements on cyber security of NPPs.

Cyber security assessment had carried out by NPPs according to relevant requirements, especially on prevention of cyber-attack to digital control and protection systems. According to the assessment results, comprehensive initiatives are adopted to enhance cyber security and prepared emergency plan to cope with cyber-attack. To continuously improve the level of cyber security, the nuclear power corporations have established cyber security laboratories or R&D centers, and carried out research on cyber-attack means and protection measures, so as to identify new cyber-attack means and develop targeted protection measures.

(5) Knowledge management and transfer for both regulator and the nuclear industry

The MEE (NNSA) has taken a number of initiatives to strengthen knowledge management and transfer, and a certain division called “Personnel Qualification Division” is in charge of the work of capacity building such as personnel training and knowledge management development, etc. On the basis of summarizing the practice and experience on nuclear and radiation safety regulation in China, the MEE (NNSA) has completed the nuclear and radiation safety management system documents with reference to the relevant IAEA safety standards. This management system has covered all relevant elements of the nuclear and radiation safety regulation undertaken by the MEE (NNSA). For specific contents, please refer to Section 8.2.4 of this report. Meanwhile, the nuclear and radiation regulation training system has been completed step by step, the training program and training materials for nuclear and radiation safety regulatory personnel have been drawn up and annual training plan has been published. All units of the nuclear and radiation safety regulation system send personnel to participate in various trainings according to their own requirements for human resource and profession development. The main contents of training include the basic knowledge on nuclear and radiation safety, laws and regulations and standards on nuclear and radiation safety, nuclear and radiation safety regulatory law enforcement, environmental radiation monitoring management and technologies, and special subject training focused on important position personnel and the key professional work such as simulator

training, non-destructive testing and civil nuclear safety equipment regulation.

The MEE (NNSA) attaches importance to summarize experience in routing regulation, and associated experience has been taken into account in the preparation or revision of inspection procedures and work guidelines. To guide the review for special types of reactors and some specific issues, on the basis of summarizing associated research, expertise and experience, specific review principles and technical viewpoints have been compiled. The MEE (NNSA) has also established online database for information sharing, including the knowledge management system, over 30 specialized databases and experience feedback platforms. The centralized knowledge management information platform and public information database are planned in the near future, which will integrate existing relevant systems and data resources.

All nuclear power corporations in China have prepared their human resources planning to meet the needs in business development and make proper talent reserves. The talent fostering system and training management system have been established by learning the internationally advanced experience in conjunction with their features and demands. The transfer of knowledge and experience is effectively carried out through the teams of experienced teachers with high quality, complete course system and scaled training facilities, which met the demand for talents in the rapid development of the nuclear industry. All nuclear power corporations have established the talent fostering pattern of apprenticeship to actively play the role of technical experts. Forms of technological consultancy and professional guidance have not only brought into full play the roles of experts, but also helped rapid growth of young technical personnel. The three levels skill training capability including general, professional and special skill has been established in NPPs in conjunction with the skill training system. The full-course skill training from new-comers to high-skilled talents have been realized through the alternative training forms of centralized skill training, on-job training, centralized re-training and on-job re-training to promote the knowledge transfer in all expertise areas. In the establishment and operation of experience feedback system, attention was paid to incorporating important events into the training course.

In the meantime, special work on knowledge management has also been carried out in some NPPs. Knowledge management organizations have been set up; knowledge management assessment and incentive system have been established to promote knowledge sharing and innovation among employees; the knowledge management platforms which include knowledge encyclopedia, knowledge map and knowledge community have been developed to allow access by authorized users with computers and mobile phones.

(6) Ageing management of NPPs

In the *Code on the Safety of Nuclear Power Plant Design*, it is required that the factor of ageing must be taken into consideration in the design of NPPs. *Code on the Safety of Nuclear Power Plant Operation* has set principle requirements on ageing management, and the relevant nuclear safety guidelines have provided specific guidance on ageing management methods.

Ageing management programs have been established for all NPPs in China and continuously improved according to relevant requirements. Also, ageing management is reviewed as a safety factor during PSR according to the requirements of nuclear safety regulations. For example, the ageing management program for reactor internals, control rod drive mechanism and nuclear class 1 piping was improved in 2017 for Units 1 and 2 of Qinshan Phase II NPP according to PSR result; the operating organization of LingAo NPP updated the implementation guideline for ageing and life management program, optimized the field implementation process, made clear the specifications and requirements of programme implementation and put the requirements of ageing management program into the corresponding equipment inspection and maintenance procedures, according to the PSR result.

The NPPs operating organizations in China carried out samples reservation for materials such as cables and structures, established the long-term assets management system, implemented assessment of ageing for critical equipment and full-life ageing management, and formulated life management plan according to the ageing assessment result. The ageing testing and evaluation for nuclear safety structures have been performed, including testing and concrete repair of containment, anti-corrosion and bolt replacement for gantry steelworks, and modification for seawater piping lining. The ageing management databases for nuclear power corporations have been established, including equipment basic information, ageing management, obsolete management, action management modules, which used as the working platforms for ageing management in NPPs.

The specific application of ageing management is also embodied in the OLE for NPPs. The *Nuclear Safety Act* stipulates that the validity period of NPP operation license first issued shall be the designed lifetime. Where operation needs to continue upon the expiry of the validity period of the operation license, the NPP operating organization shall submit an application for extension to the nuclear safety regulation department of the State Council five years before the expiry of the validity period, and demonstrate and verify its conformity with the nuclear safety standards, and may continue the operation only after being subject to review and obtaining approval

The MEE (NNSA) issued the *Technical Policy for Extension of Validity Period of*

Nuclear Power Plant Operation license (for Trial Implementation) on December 31, 2015, which specified the routing, scope, objects, methods and review basis for OLE projects, providing technical guidance for OLE review. In 2017, the MEE (NNSA) prepared the *Safety Review Guideline for Extension of Validity Period of Qinshan Nuclear Power Plant Operation license* to ensure proper review of first application of OLE in China. To further formalize the nuclear safety supervision and inspection on the OLE related activities of Qinshan NPP, the MEE (NNSA) prepared and issued the *Nuclear Safety Inspection Program for Activities Associated with Extension of Validity Period of Operation License of Qinshan NPP*. In the meantime, experience exchange with international peers was carried out. In February 2018, a China-US symposium on NPP operation licenses renewal was organized to conduct in-depth exchange with experts of NRC on experience in OLE supervision and review.

The OLE safety evaluation was conducted for Qinshan NPP in 2016, covering definition and screening on OLE safety evaluation scope, review on the ageing management program validity and time-limited ageing analysis, etc. The OLE application has been submitted to the MEE (NNSA). By the end of 2018, Qinshan NPP had completed three review dialogues with the MEE (NNSA), and the main work for safety review had been completed. In 2018, Qinshan NPP completed 9 modification items associated with OLE.

The operating organization of Daya Bay NPP completed the OLE feasibility analysis in 2015, and started in all aspects the OLE evaluation in 2016. By the end of 2018, Daya Bay NPP had completed the OLE evaluation guideline and basically completed the OLE safety evaluation scope definition and screening and ageing management consistency review. It is planned to initially complete the OLE evaluation in June 2019, and to submit the OLE application to the MEE (NNSA) at the end of 2019.

B.2 Carried out and Planned Activities on Safety Improvement

To improve safety of NPPs, China has carried out and planned the following activities:

(1) Continuously improving the nuclear safety law and regulation system

In September 2017, *Nuclear Safety Act* was officially issued and came into force on January 1, 2018. As the host law in the nuclear safety area, the *Nuclear Safety Act* which is a footstone for effectively guarantee nuclear safety, and further completed the nuclear safety law and regulation system of China. This law has specified the policy, principle, responsibility and scientific, technological and institutional guarantee to

ensure nuclear safety, specified the qualification, responsibilities and obligations of nuclear facility operating organizations, specified licensing system for nuclear facility and nuclear material and the radioactive wastes management system, defined the system of nuclear emergency coordination committee, the emergency plan system and nuclear accident information releasing system, defined the nuclear safety information disclosure and public participation system; defined the main bodies and scopes for nuclear safety information disclosure and public participation; clearly stipulated on the specific approaches of nuclear safety inspection, penalty clauses for behavior violating the law and the compensation for damages caused by nuclear accidents. At present, the MEE (NNSA) is preparing and revising lower level regulations according to the requirements in the *Nuclear Safety Act*.

With reference to the *Safety of Nuclear Power Plant: Design* (SSR-2/1, Rev.1) issued by the IAEA in February 2016, the MEE (NNSA) revised and issued the *Code on the Safety of Nuclear Power Plant Design* in October 2016. The revised regulation has taken into account the lessons learned from the Fukushima nuclear accident and some initiatives in VDNS.

Since 2016, China has issued a total of six nuclear safety regulations related to NPPs, including two department rules and four nuclear safety guidelines. A number of department rules are being revised, including the *Measures for the Management of Nuclear Facilities Safety Licensing*, *Code on the Safety of Nuclear Power Plant Siting*, *Code on the Safety of Nuclear Power Plant Operation* and *Code on the Safety of Nuclear Power Plant Quality Assurance*; over 20 guidelines are being revised, including the *Design of Emergency Power System in Nuclear Power Plant*, *Seismic Design and Qualification for Nuclear Power Plant* and *Emergency Drills of NPP Operating Organization*.

Furthermore, public opinion soliciting was completed for the *Atomic Energy Act* on October 19, 2018.

(2) Updating and completing the nuclear safety inspection document system

On the basis of experience in nuclear safety inspection, the MEE (NNSA) recently revised the documents such as the *Inspection Program for the Construction Phase of Nuclear Power Plant*, *Inspection Program for the Commissioning of Nuclear Power Plant* and *Inspection Program for the Operation Phase of Nuclear Power Plant*, and revised the nuclear safety inspection procedures based on the updated inspection programs. The standardization and consistency of the inspection has been strengthened to improve the effectiveness and rationality of inspection by the revision of the nuclear safety inspection document system.

(3) Advancing activities related to the extension of operation license validity period

For the activities carried out by the MEE (NNSA) and Qinshan NPP and Daya Bay NPP on OLE, please refer to the contents in B.1.2 (6) Ageing management of NPPs.

By summarizing the OLE practices at Qinshan NPP, China started the preparation and review of the energy industry series standards of *Nuclear Power Plant Operation License Extension*, completed the technical review before the end of 2018. The standards will be issued later. The standards which provide technical guidance on OLE for NPPs covers the screening of ageing management review object, time-limited ageing analysis and identification, ageing management review for mechanical, electrical and instrumentation equipment, ageing management review for structures and components, time-limited ageing analysis and assessment for reactor pressure vessel and steam generator, metal fatigue analysis and assessment, environmental qualification review for electrical and instrumentation equipment, time-limited ageing analysis and assessment for prestressed concrete containment, guideline for FSAR supplement, and guideline for preparing OLE application reports.

(4) Continually promoting the application of risk-informed mode of regulation

The MEE (NNSA) has continually promoted the pilot on the application of PSA for NPPs, studied the risk-informed mode of regulation, and approved the application of risk-informed in the operation specification optimization, modification of periodical test surveillance requirements, in-service inspection optimization, and online maintenance and so on. With the existing equipment reliability database of operating NPPs, the relevant data in domestic NPPs are constantly collected, and the *Report on Equipment Reliability Data of Nuclear Power Plants in China* has been issued regularly.

In the past three years, the PSA has continuously extended and improved for NPPs in China. For most NPPs, the development of level I and level II PSA of internal events, flooding and fire PSA as well as PSA for spent fuel pool have been completed, and earthquake PSA have been launched to develop. The operating organizations in China actively carried out daily application of PSA, and developed risk-informed tools such as risk monitor, significance determination process and mitigation system performance indicators. The operating organizations developed the risk-informed specification and in-service inspection and carried out optimization of periodical test periods. Risk monitor has been applied to evaluate risks in the maintenance plan for safety systems and to track and monitor risks in daily operation and maintenance. The mitigation system performance indicators are applied to track and analyze the performance of mitigation system.

In past three years, the CNEA entrusted by the MEE (NNSA) has organized and carried out peer review on PSA work for Hongyanhe NPP, and made follow-up to Qinshan Phase II NPP. The improved quality of PSA through these review activities has provided basis for the further NPP application of PSA.

(5) Building national technical R&D base on nuclear and radiation safety regulation

To further improve the nuclear and radiation regulation capacity, China officially started construction of the national technical R&D base on nuclear and radiation safety regulation in March 2016, and put it into service in May 2019. By constructing scientific research and experimental facilities and computing analysis center, the technical base with nuclear safety regulation capacity of testing verification, calculation and analysis, nuclear emergency and anti-terrorism and radiation monitoring has been formed.

(6) Improving effectiveness of maintenance in NPPs

To guide the NPP operating organizations to improve the effectiveness of equipment maintenance and reasonably reduce risks in maintenance activities, the MEE (NNSA) started the preparation of *Technical Policy for Improving Effectiveness in NPP Maintenance (for Trial Implementation)* in 2016 and issued the technical policy in 2017. It is expected to promote the NPP operating organizations to complete the maintenance assessment system, strengthen the monitoring of equipment reliability, optimize the maintenance activities, strengthen risk evaluation for maintenance activities and continually raise the reliability level of equipment through the implementation of the technical policy.

In June 2018, the MEE (NNSA) organized and established a NPP maintenance rule working group, and formulated the working plan of the working group, to regularly organize meetings of the working group every quarter, to study and discuss work progress and relevant technical issues on maintenance rules, to make efforts to formalize the implementation of work related to maintenance rules in China, and provide the technical foundation for the subsequent formulation and revision of nuclear safety codes, standards and guidelines.

(7) Carrying out certification for NPP safety analysis software

In 2016, the MEE (NNSA) performed the certification for safety analysis software, to strengthen the verification and validation of safety analysis software independently developed by domestic suppliers.

In 2017, the MEE (NNSA) issued the nuclear safety guideline *Development and Application of Computer Software for Safety Analysis in Nuclear Power Plants (for*

Trial Implementation) to formalize and promote the development and application of computer software for safety analysis in NPPs.

The MEE (NNSA) has set up special safety analysis software evaluation department in technical support organizations, and established software evaluation teams and software evaluation implementation methods.

B.3 International Peer Review

China periodically requests IRRS and the follow-up mission by IAEA to continuously improve nuclear and radiation safety regulation system. Meanwhile, China sends senior officers and technical experts to participate in the IAEA's IRRS missions for the regulatory bodies of other contracting parties.

In September 2016, IAEA dispatched a review team to conduct a follow-up IRRS mission for China. The team reviewed the measures undertaken following the recommendations and suggestions of the 2010 IRRS mission, and paid special attention to regulatory implications to the Chinese framework for safety in relation to the lessons learned from the Fukushima nuclear accident.

The review team recognized China's political commitment to nuclear safety. The Chinese government has approved and implemented a Nuclear Safety Plan, increased the human and financial resources and strengthened the capacity building of the MEE(NNSA). The review team considers that the MEE(NNSA), as the leading organization, in cooperation with other governmental agencies, has acted promptly and effectively after the Fukushima nuclear accident in the interest of nuclear safety and protection of the public and environment. The MEE(NNSA) is an effective and credible regulatory body.

The review team concluded that the recommendations and suggestions from the 2010 IRRS mission have been taken into account and that significant progress has been made in many areas and many improvements were carried out following the implementation of recommendations and suggestions. Most of the recommendations and suggestions have been effectively addressed and therefore could be considered closed, and the recommendations and suggestions remain open are mainly related to the promulgation of the *Nuclear Safety Act, which* was officially promulgated in September 2017.

The NPP operating organizations in China participated in peer reviews on nuclear power safety. From 2016 to 2018, the NPP operating organizations in China accepted 3 IAEA review missions and 29 WANO peer review activities, and also actively selected experts to participate in peer review activities in foreign NPPs performed by IAEA or

WANO. In addition, an OSART mission was conducted in Fangjiashan NPP and a Pre-OSART follow-up mission was conducted in Taishan NPP in January 2019.

Furthermore, NPP operating organizations in China also carried out domestic peer review activities at different levels while accepting international peer reviews, as detailed in section 10.3 of this report.

B.4 Experience Feedback of Typical Operational Events Since the Previous National Report

(1) Scram due to algae swarming into Circulation Water Filtration system in Unit 3 of Hongyanhe NPP

Event description:

On May 25, 2017, Unit 3 of Hongyanhe NPP was in full power operation. Since algae swarmed into the inlet of Circulation Water Filtration system, differential pressure of trains A/B drum screens increased in synchronism, in which differential pressure of train B drum screen increased relatively faster. Actuation of high drum screen differential pressure protection caused tripping of CRF002PO (Circulation Water pump). The shift on duty manually tripped the turbine according to the heat sink protecting scheme. Subsequently, actuation of high drum screen differential pressure protection caused tripping of CRF001PO (Circulation Water pump). Reactor trip protection signal actuated, and the reactor tripped automatically. Unit response was normal.

INES level: 0

Cause analysis:

In severe weather, large amount of algae fell off interception screens under the force of wave, clogging the drum screens, resulting in tripping of circulating water pump for protection. This indicates that measures against swarming of micro marine organism in severe weather are not sufficient.

Corrective actions

1) To promote feasibility study and analysis, design, and construction of large cofferdam project.

2) To complete the weather forecast mechanism for the water intake and confirm source of forecast information.

3) To implement installation of fine mesh nets and continuously monitor and record interception by these nets.

4) To install screens at once when installation conditions are ready each year.

5) To clarify criterion for cleaning of interception screens at water intake.

6) To follow with changes of surrounding environment and monitor marine

organism and alarm as early as possible.

7) To construct permanent high pressure back flushing system instead of temporary one.

8) To optimize the material used for interception screens.

(2) Self-starting of auxiliary feedwater system during shutdown of in unit 1 of Fangjiashan NPP

Event description:

On September 10, 2017, the power of Fangjiashan Unit 1 was reduced to disconnection for outage as scheduled. During the power reduction, the three level gauges of stage II drain tank 1GSS202BA in the Moisture Separator Reheater system (GSS) failed, and stage II reheaters were isolated. The turbine vibration started to rise, and the operator quickly reduced the power to 50MWe when they noticed the turbine rotor/bearing vibration relative high alarm. Subsequently, the water level signal of drain tank 1GSS103BA on the shell side of moisture separator-reheater increased to 1846mm in a short time, and the turbine automatic shutdown protection was triggered. During inertial rotation the turbine, vibration of 2# pad bearing increased quickly. The steam discharge of the Turbine Bypass system (GCT) to the condenser was switched to the atmosphere after vacuum of the condenser was broken by the operator manually. The action of main feedwater isolation was triggered by the high-high water level signal of 3# steam generator. The main feedwater pump shut down automatically, leading to automatic start-up of two motor-driven auxiliary feedwater pumps 1ASG001/002PO. The operator responded quickly to stabilize the level of steam generator and then maintained the reactor in the hot shutdown state.

INES level: 0

Cause analysis

The limit feedback was lacked during closing of discharge valve 1GCT121VV, resulting in unlocking of control signal sent to the steam generator water level control system from the turbine bypass system and interference to steam generator water level control system, thus the high-high water level signal of 3# steam generator appeared and triggered isolation of main feedwater.

The closing limit for bypass discharge pneumatic valves (both 1GCT121VV and 1GCT117VV) of the Turbine Bypass system was set to be very close to the full closing position of the valve. When the bypass discharge valve are actuated, the valve might be closed without triggering of the full closing limit switch due to vibration of pipeline or jamming of valve.

Corrective actions:

1) To recalibrate the valves closing limit switch for 1GCT121VV.

2) To modify the valve closing feedback logic for the bypass discharge valve, add an analog signal <5% of valve position in addition to valve limit switch.

3) To change the installation method of the radar guided-wave level meter of Moisture Separator Reheater system.

4) To make overhaul inspection for 1# and 2# bearing bushes of the turbine;

5) To inspect the outer clearance of 1# and 2# gland seals of the turbine;

6) To develop a plan for parameters necessary to be monitored during start-up of the unit.

7) Continue to communicate with the manufacturer to further identify the root cause of rub and impact in the turbine;

8) To make clear description in the operation procedure for the unit shutdown upon high vibration.

(3) Pressurizer low level triggered a shutdown protection signal in Unit 3 of Tianwan NPP

Event description:

On 31 October 2017, Unit 3 of Tianwan NPP was heating up after maintenance. Reactor operators were adjusting level of the pressurizer with the target level was 5.1m. The level of the pressurizer was below 4m, triggering the shutdown protection signal. Field inspection showed rupture of the bypass of the pressurizer spray pipeline of Unit 3, and the rupture was located on t the BOSS head pre-fabricated nozzle body of the pipe, about 14mm from the front of the field weld seam of the pipe.

INES level: 0

Cause analysis:

The rupture of pipeline were caused by the machined scratch on the surface of the small size pipe (BOSS) connection head, the thinner thickness of partial pipe and alternating load generated by pipeline vibration.

Corrective actions:

1) To conduct failure analysis on the rupture and base material of the bypass of pressurizer spray lines.

2) To verify the manufacturing quality of the base material and welding quality of the small size pipe (BOSS) connection head of the bypass of pressurizer spray lines.

3) To verify the pre-service inspection results of the bypass of pressurizer spray lines.

4) To conduct tensil-strength test for stainless steel bars of the same batch of the ruptured BOSS head.

5) To develop the repair process to fix the ruptured part;

6) According to the requirement, to implement post-welding inspection and pre-service inspection on newly welded joint.

7) To add vibration proof support to the bypass of pressurizer spray lines of Units 3 and 4.

8) To monitor vibration and stress of t the bypass of pressurizer spray lines of Units 3 and 4;

9) To remove the scratch on the BOSS head of Units 3 and 4.

(4) Air leakage at the connection of pressure gauge resulted in high radioactivity alarm during discharge through stack of nuclear auxiliary building ventilation system in Unit 4 of Fuqing NPP

Event description:

On July 17, 2018, Unit 4 of Fuqing NPP was under power operation mode at 96%. The maintainer worked with the operator to adjust nitrogen pressure reducing valve, then, the operator opened the downstream valves which were connected with the volume control tank to verify adjustment function and purge the tank. However, level 1 and level 2 alarms of iodine radioactivity monitoring channel were triggered. Responding as per the alarm procedure, the operator took emergency isolation measures immediately. After that, the effluent and RP measurement for aerosol and iodine returned to normal value, and the radioactivity monitoring returned to normal.

INES level: 0

Cause analysis:

The installation of instrument 4RAZ012LP was not up to standard. The copper gasket with wrong size and surface scratch was used, and the PTFE gasket was incorrectly added, which caused untight sealing of the connection of 4RAZ012LP instrument lead to leakage.

The control for the troubleshooting process was ineffective. The leaking equipment that has been identified was put into operation without taking any substantive action.

Corrective actions:

1) To repair the leaking connection and readjust the instrument 4RAZ024VZ;

2) To check and repair 4RCV215VY;

3) To check instruments in Units 1-4 of Fuqing NPP of any possible air leakage of radioactive gas through;

4) To update procedures related to I&C air leakage inspection and handling;

5) To prepare procedure for operating nitrogen pressure reducing valve, adding leakage inspection procedure;

6) To improve the risk analysis in purging procedure;

7) To streamline the purging operation during maintenance, and add procedures to look at the radiation monitoring alarm.

(5) Unit 4 of Yangjiang NPP fall back to shutdown status due to failure of 6.6kV AC emergency power system

Event description:

On July 13, 2017, Unit 4 of Yangjiang NPP was at RP mode and in full power operation. While monthly low power test of diesel engine of 6.6kV AC emergency power system was implemented, the diesel generator malfunction stopped. The on-site inspection found that A1 and B1 cylinders of the diesel generator in malfunction. After confirmation that they cannot be repaired and restored within the period specified in Operating Technical Specification (14 days), the unit falls back to shutdown status on July 16, 2017 as per Operating Technical Specification.

INES level: 0

Cause analysis:

Based on test parameter curve review and inspection and analysis of diesel engine of the damaged parts, the fault of diesel engine was caused by scrap metal particles left in manufacturing stage. The particles entered crankshaft bearing and connecting rod big end bearing along with machine operation and lubrication oil flow, caused abnormal damage of connecting rod big end bearing and thus loss of connecting operation stability, and further resulted in damage of A1/B1 cylinder power unit assemblies.

Corrective actions:

1) To replace faulty parts of diesel engine, perform diesel engine test after maintenance, including single system test, no-load excitation test, and improve the risk analysis related to diesel engine operation.

2) To work out emergency repair plan for diesel engine malfunction, start procurement of strategic (cyclic) spare parts for diesel engine, require the manufacturer to strengthen the quality control of the cleanliness in the manufacture and assembly of major parts and components, and to carry out rectification and improvement. Add quality control points for cleaning inspection of crankshaft and other important parts in construction company's inspection documents.

B.5 Response to VDNS

The IAEA issued the VDNS at the diplomatic conference on *Convention on Nuclear Safety* held in February 2015, which indicated the concerns and efforts of all Contracting Parties to the international society on improve nuclear safety. The VDNS

encompasses the following principles to guide all Contracting Parties in the implementation of the objectives of the *Convention on Nuclear Safety* to prevent accidents with radiological consequences and mitigate consequences should they occur:

(1) New NPPs are to be designed, sited, and constructed, consistent with the objective of preventing accidents in the commissioning and operation and, should an accident occur, mitigating possible releases of radionuclides causing long-term off site contamination and avoiding early radioactive releases or radioactive releases large enough to require long-term protective measures and actions.

(2) Comprehensive and systematic safety assessments are to be carried out periodically and regularly for existing installations throughout their lifetime in order to identify safety improvements that are oriented to meet the above objective. Reasonably practicable or achievable safety improvements are to be implemented in a timely manner.

(3) National requirements and regulations for addressing this objective throughout the lifetime of nuclear power plants are to take into account the relevant IAEA Safety Standards and, as appropriate, other good practices as identified *inter alia* in the Review Meetings of the *Convention on Nuclear Safety*.

China has continually taken a series of actions to implement the objectives in the VDNS.

B.5.1 Continually Revising and Improving Nuclear Safety Regulations

China has established its own nuclear safety regulation system with reference to the IAEA safety standards, and by extensively studying the nuclear safety laws and regulations of other countries with nuclear power.

The MEE (NNSA) closely tracked and participated in the review and revision of IAEA safety standards. After the Fukushima nuclear accident, the MEE (NNSA) organized systematic and in-depth study on the 77 topics issued by IAEA, and performed analysis on lessons learned from the Fukushima nuclear accident and the current nuclear safety regulations in China. Meanwhile, preparation and revision of the nuclear safety regulation system of China was proposed in conjunction with the nuclear safety practice in China after the Fukushima nuclear accident, and the preparation and revision work was carried out by phases.

China responded to the VDNS in time. In October 2016, with reference to the latest safety standards issued by the IAEA and by summarizing the improvement experience in China after the Fukushima nuclear accident, the MEE (NNSA) revised and issued *Code on the Safety of Nuclear Power Plant Design*, which was added with concepts and

requirements such as practical elimination, design extension condition, integrity of design throughout the full lifetime of NPP, portable equipment, requirements on accident involving multiple units in NPP, and resistance to malicious crash of commercial aircraft. Adjustment was made to the requirements on defence in depth, categories of plant states, internal and external hazard, structure integrity of containment, on-site emergency facilities, escape routes, control room, remote control panel, spent fuel pool, design to cope with loss of off-site power supply, effluent discharge and radioactive wastes treatment.

B.5.2 Design Features and Safety Improvements of New Units

The new units constructed in China include four AP1000 units, two EPR units, and four HPR1000 units which are independently developed by China.

(1) AP1000

For AP1000 units, the prevention and mitigation of severe accidents has been taken into account at design stage. The passive safety technology has been applied in AP1000, including the adoption of passive core cooling system, passive containment cooling system, passive main control room habitability system, high capacity rapid depressurization valves for reactor coolant system, passive reactor cavity water injection system and containment isolation system. Passive hydrogen recombiners, hydrogen igniters and hydrogen analyzers are installed in the containment to effectively control the hydrogen concentration in the containment. Level gauges are provided in the spent pool which has taken into account the diversity and reliability.

After the Fukushima nuclear accident, more improvements were made for AP1000 units as required by the safety inspection of the MEE (NNSA), including the addition of water proof gates, portable electric power supplies and portable pumps, and matching connection ports, setup of backup emergency commanding center, and updating SAMG. These improvements have further enhanced the capacity of AP1000 units to prevent and mitigate severe accidents.

The design of the steel containment structure of AP1000 units has taken the design basis accidents into account, which include the rupture of reactor coolant pipe or main steam or feedwater line, with ultimate load analysis performed. To prevent reactor vessel failure, the accident management strategy to flood the reactor cavity and submerge reactor vessel has been adopted. Outside the steel containment is the concrete shielding building, able to effectively resist missiles produced by typhoon and tornado. For Sanmen NPP and Haiyang NPP, additional portable power supplies and portable pumps are stored at places that cannot be flooded under the condition of beyond design

basis flood water level.

(2) EPR

For EPR units, the corresponding design extension conditions has been taken into account in design, including the beyond design basis accident series that will not lead to core melting (DEC-A) and the beyond design basis accident series that can possibly lead to core melting (DEC-B). Systems including the containment spray system, emergency core cooling system, and emergency feed water system and core catcher are provided. The passive hydrogen recombiners arranged in a distributed pattern can avoid hydrogen gathering and burning explosion in the containment.

When severe accident occurred, the primary pressure can be reduced by the primary rapid depressurizing system to prevent jetting of molten material at high pressure, the containment pressure and temperature can be controlled by the containment spray system and the hydrogen concentration can be controlled by the combustible gas control systems. The core catcher is equipped to prevent core molten material-concrete interaction so as to maintain the integrity of containment. The leaking materials in the annular space between the inner and outer containment and the leaking materials in containment not vented to the stack are collected by the filtration and ventilation system provided in the annulus between the two containment layers, to reduce discharge.

After the Fukushima nuclear accident, more improvements were made for EPR units as required by the safety inspection of the MEE (NNSA), including the addition of portable power supplies, portable pumps, matching interfaces, improvement of the level and temperature monitoring system for spent fuel pool, addition of portable makeup water for spent fuel pool, addition of portable makeup water for containment heat removal system, and addition of passive steam venting channel in the fuel building.

(3) HPR1000

Units adopting the HPR1000 technology are provided with both active and passive safety provisions. In the design, the critical event sequences that may lead to severe accidents are identified on the basis of combining deterministic assessment, probabilistic assessment and engineering judgment and the corresponding accident prevention and mitigation features are provided, so it has complete features to prevent and mitigate severe accidents. The secondary side passive residual heat removal system, passive containment heat removal system and combined active and passive cavity water injection cooling system are provided, and provisions such as containment passive hydrogen recombination, containment filtration and vent, and pressure vessel high level vent are adopted to prevent and mitigate severe accidents. The reactor coolant system

rapid depressurizing valve with redundant design, the two-layer containment against malicious crash of commercial aircraft, the sealing function for reactor coolant pump shutdown and diversified drive system with anti-seismic class I are provided.

After the Fukushima nuclear accident, more improvements were made for HPR1000 units as required by the safety inspection of the MEE (NNSA), including the addition of portable power supplies, portable pumps, and matching interfaces, improvement of the level and temperature monitoring system for spent fuel pool, addition of portable makeup water for spent fuel pool, and addition of portable makeup water for containment heat removal system.

B.5.3 Implementing Practically Feasible Improvement for Operating Nuclear Power Units through PSR and Post-Fukushima Improvements

For operating NPPs in China, PSR is performed every ten years after operation according to requirements of nuclear safety regulations. Systematic safety re-assessment is performed for NPPs according to the current safety standards and practice and with reference to operation experience and relevant safety research information. The scope of review covers all important aspects of nuclear safety, including 14 safety factors in 5 subject areas: plant design, actual conditions of SSCs, equipment qualification, ageing, deterministic safety analysis, probability safety analysis, hazard analysis, safety performance, use of experience from other plants and research findings, organization and administration, procedures, human factors, emergency planning and radiological impact on environment. During PSR, according to result of review, NPPs identify reasonably practical corrective actions/safety improvements and their implementation plans, and assess the effect of corrective actions/safety improvements on all safety factors.

By December 31, 2018, as required by regulations, the second PSR had been completed for Qinshan NPP and Daya Bay NPP, and the first PSR had been completed for Units 1 and 2 of Qinshan Phase II NPP, Units 1 and 2 of LingAo NPP, Third Qinshan NPP and Units 1 and 2 of Tianwan NPP. The third PSR have already been launched for Qinshan NPP in 2018, and the PSR for Units 3 and 4 of Qinshan Phase II NPP will be initiated in 2019, the first PSR for Units 3 and 4 of LingAo NPP, the third PSR for Daya Bay NPP and the second PSR for Units 3 and 4 of LingAo NPP will also be launched in 2019. Weaknesses for NPPs have been identified through PSR, and improvement plans and implemented relevant actions have been developed.

After the Fukushima nuclear accident, the MEE (NNSA) issued the *General Technical Requirements on the Improvement of Nuclear Power Plants after the*

Fukushima Nuclear Accident in order to standardize the common improvement actions of NPPs in China. The time limits of completion for all NPPs have been set according to the effect and urgency of the improvement projects on safety. The improvement projects in operating NPPs are classified as short-term, medium and long-term actions. Short-term and medium improvement actions such as adding portable equipment and water proof blocking were respectively completed in 2011 and 2013, and for detailed improvement, please refer to the Seventh National Report. The long-term improvement actions include:

(1) Improving emergency plans;

(2) Enhancing the nuclear accidents emergency response capacity;

(3) Updating NPP information release procedures and strengthening nuclear power science popularization.

At present, all long-term improvement actions are implemented on schedule. All NPP operating organizations will carry out further associated researches according to the implementation result, continually optimize and implement the improvement plans.

B.6 Main Safety Issues Identified Since the Previous National Report

(1) Deviation of chemical composition of cast and forging parts in main equipment of some NPPs

Based on the relevant issues notified by France, the MEE (NNSA) set up a special working group, and put forth clear requirements and deadlines on the operating organizations of all NPPs for the checking and treatment. By clarifying the application of casting and forging parts supplied by CF France and JCFC Japan, the work was focused on the 16 steam generator water chamber heads supplied by CF and JCFC in 8 nuclear power units, and the 4 reactor pressure vessel upper and lower heads supplied by CF for the EPR units in Taishan NPP.

The nuclear power plants concerned carried out various testing and analysis for total 16 steam generator water chamber heads and 4 reactor pressure vessel heads, the result showed partial carbon segregation in carbon concentration testing, and no anomaly in NDT and metallographic structure. For the 4 reactor pressure vessel heads of Taishan NPP, the carbon segregation testing result was better than that in units of the same type in France. The operating organizations carried out mechanical performance tests on witness parts and actual products of the same category, and fracture mechanics assessment, the analysis results meet the requirements of the RCC-M.

At the end of 2018, special working group was organized for in-depth review. The review initially accepted the analysis and assessment conclusions from the operating

organizations, however, considering the carbon concentration exceeding limit that reducing the design margin, management requirements have been put forth on operation limit measures and supplementary in-service inspection for relevant NPPs.

(2) Crack in steam-driven pump stage II impeller of auxiliary feedwater system in some NPPs

In the maintenance of NPPs, defective cracks were found in stage II impeller of steam-driven pump in auxiliary feedwater system supplied by the Clyde UK. The subsequent inspection indicated that it was a common issue. A notice for troubleshooting was issued by the MEE (NNSA) when design deficiency was identified as the common issue after being analyzed. All NPP operating organizations carried out analysis and assessment or replaced impeller as required.

(3) Screen blockage of water intake in some NPPs

A number of events occurred with marine creatures or foreign matter blocking the water intake system and affecting the safety of water intake for NPPs in China. The MEE (NNSA) organized relevant investigation and analysis, and concluded that there were various causes for such events, including the design problems in the water intake and filtration system, and unexpected problems resulted from the ocean eutrophication in recent years. The NPP operating organizations also took improvements, including: analyzing the effect of marine creatures or foreign matter on seawater system, especially the essential service water system; cooperation with relevant entities to study the rule of generation and movement of marine creatures or foreign matter; establishing early warning and prevention mechanism; improving the possible design problems in the water intake and filtration systems to enhance the resistance to marine creatures or foreign matter; performing periodical cleaning and desilting for water intake structures, systems and equipments; completing the response plans and enhancing drills to cope with marine creatures or foreign matter blocking the water intake system.

B.7 Lessons Learned from the Emergency Drills and Exercises Since the Previous National Report

Since the previous national report, China responded to the IAEA Initiatives and participated in Convention Exercises, and organized on-site and off-site joint exercises, on-site comprehensive exercises and various single drills according to laws and regulations. In June 2017, the Chinese government organized over 10 national level nuclear emergency teams to participate in the ConvEx-3 (2017) exercise organized by the IAEA, obtaining high confirmation by the IAEA. The national nuclear emergency teams organized and conducted over 100 comprehensive exercises and single drills.

Through these emergency exercises and drills, the effectiveness of the emergency plans and procedures has been verified, the response capability of the emergency teams has been exercised, and the technical skill of operating personnel has been improved. The problem such as insufficient public communication ability in the network age has also been identified, that is how the public can obtain true and effective information promptly in the network age.

NPP operating organizations in China arranged and conducted various types of exercises according to plan, and for the first time conducted emergency exercise for severe accidents in dual units. These exercises have verified the effectiveness of emergency plans, tested the nuclear emergency preparedness and response ability of NPPs, and also the following experience and lessons have been summarized:

(1) External detrimental conditions such as severe weather should also been taken into account in emergency exercises. By setting various detrimental external environmental conditions such as night, non-working day and heavy rain, it can not only fully verify the reliability of personnel and equipment, but also help NPPs to obtain conservative data about response time.

(2) The behavior of some staff needed to be improved, for example, respective non-emergency personnel failed to correctly take personal protection measures, and some doors and windows were not closed before escape; there was still deficiency in the use of three-way communication tool for some emergency personnel,.

(3) The emergency preparation was insufficient for some contractors, including failure to timely revise the emergency plan, lack of detailed training contents in the emergency training records, and training and qualification record were not found for few emergency duty personnel.

(4) A small amount of emergency facilities and equipment were not ready, for example, interference between the microphone and loudspeaker in the emergency headquarters resulted in inability to turn on the loudspeaker.

The NPP operating organizations in China developed and implemented relevant corrective actions for the problems identified in the exercises, and kept on tracking the improvement effects.

B.8 Good Practices and Challenges Since the Previous National Report

B.8.1 Good Practices

The good practices achieved by China on nuclear safety since the previous national report includes:

(1) Further improving management system and documents of nuclear and

radiation safety regulation

A thorough regulatory principle, concept and approach have been formulated with the 30 years practices of nuclear and radiation safety regulation in China. The MEE(NNSA) improved the management system and relevant documents such as the *Integrated Management System Manual for Nuclear and Radiation Safety Regulation*, based on sorting, analysis and summary of Chinese practical experience and in combination with the IAEA safety standards, aiming at optimizing the regulatory concept, method and process, so as to provide comprehensive support and guidance for further regulatory activities and modernize the regulation and internal management.

(2) Further refining experience feedback system

The MEE (NNSA) has established a fairly completive experience feedback system, and strengthened the active feedback mechanism. The operation events inland and overseas, internal and external events and important anomalies in domestic NPPs are screened periodically, and for significant information, regular announcements, regulatory suggestions or management requirements are prepared and notified to NPPs promptly. The MEE (NNSA) OE platform has been established for sharing operation and construction events of NPPs inland and overseas and regulatory information, performing activities related to OE for the regulatory body, technical support organizations, NPP operating organizations, nuclear power associated engineering companies, design institutes and colleges and universities. The status of NPPs operational safety is summarized on a quarterly basis, and NPP operating experience feedback is compiled to quarterly publication. Experience feedback meeting for quarterly routine has been established to discuss recent hot topics or important issues and to develop joint actions by regulatory body and NPP operating organizations. The MEE (NNSA) has established independent event investigation mechanism, in which 3-5 typical events were selected for independent investigation every year, and the investigation results were compared with those of the operating organizations for improving the accuracy of the root cause of events and the rationality of corrective actions. The MEE (NNSA) is organizing revision of event reporting rules to improve the quality of event reports.

(3) Paying high attention to the projects with the first-of-a-kind reactor during review and inspection

To perform the review and inspection for first-of-a-kind reactor of AP1000, EPR and HPR1000 in the worldwide properly, the MEE (NNSA) paid high attention to the assessment and selection of the first-plant-only tests for new units. The first-plant-only tests and inspection items were determined through assessment. In addition, high

attention has been paid to the design characteristics of new units, especially the special features of the passive systems.

The MEE (NNSA) performed the review of commissioning procedures for the first unit of AP1000 and EPR and prepared targeted commissioning inspection procedures, and conducted field inspection and witness for the selected pilot plant tests, comprehensively tracked the major deviations and anomalies in the tests, and reviewed the test results.

The MEE (NNSA) promptly summarized the experience in inspection for AP1000, EPR and HPR1000, and updated the relative NPP inspection programs and procedures.

(4) Setting up special working groups in nuclear power industry to carry out research on common issues

During the construction, commissioning and operation of NPPs, special working groups were set up for difficult and common issues, including NPP commissioning and startup, maintenance, equipment reliability, nuclear power risk management, nuclear grade pumps, turbine, emergency diesel generator, valves, in-service inspection and NDT, and cyber security, etc. The working groups organized the whole industrial chain including universities, research and design institutions, production and manufacturing entities, operation and maintenance organizations to solve difficult and hot issues, and achieved substantial results in exchange of technical and management experience, research of common issues, preparation of standards and specifications, organizing training activities, carrying out consultancy and evaluation and fostering specialists. Every year, the special working groups of the industry carry out research on over 20 subjects, over 40 exchange activities in the industry, and seminar lectures and training participated by over 2000 person-times. The special working groups in the nuclear power industry have become an important platform for research, experience exchange and talents training in all areas of nuclear power.

(5) Sharing Chinese experience under the MDEP framework

Under the MDEP framework, China carried out international exchanges, conducted in-depth discussions on important technical issues, to reach consensus, issue common position papers and maintain regulatory consistency.

China had shared the supervision experience and commissioning result for the first EPR reactor at Taishan NPP, and provided the Chinese solution for the regulation and handling of major anomalies for the EPR units. In March 2017, in the multilateral international cooperation framework of the EPR working group of MDEP, 6 experts from foreign regulatory bodies were invited to China to participate as observers in the joint witness of the first EPR reactor test for measuring the RPV reactor internals

flow-induced vibration at Unit 1 of Taishan NPP. At the summary meeting of joint witness, the regulator from foreign regulatory bodies highly appraised the regulation capability, regulation effectiveness and openness attitude and the contribution made to international cooperation of the MEE (NNSA), saying that this joint witness work was a paradigm of international cooperation.

Under the AP1000 working group mechanism of MDEP, the regulatory bodies of both China and the US dispatched regulators to each other to observe and exchange inspection experience for 27 items in the commissioning tests for the first AP1000 reactor. Through the AP1000 working group meeting, China shared the review of the commissioning testing plans and design change for the first AP1000 reactor, especially the commissioning inspection experience for passive safety systems, and participated in the preparation of the common position document of the MDEP AP1000 working group, including: the common position on issues related to the Fukushima nuclear accident and on containment condensate back flow, the technical reports on rupture valves and on reactor coolant pump, and the AP1000 commissioning inspection report.

In the meantime, China advocated setting up the HPR1000 working group under the MDEP mechanism to make exchange on inspection practices and experience in the design and construction phases of HPR1000 units with relevant countries. To deeply discuss special issues of key concerns by each country, technical subgroups for internal and external hazards and for severe accidents were set up under the HPR1000 working group, and a number of special topic symposiums were called. On the basis of the above work, China has also conducted technical exchanges and cooperation with nuclear safety regulatory body of the UK, Pakistan, South Africa, Romania, Argentina and other countries, with bilateral exchange and discussion on relevant issues of HPR1000 as concerned by various countries.

(6) Innovating forms of knowledge and experience propagation

During the construction of the first EPR reactor, operating organization of Taishan NPP developed the sharing, experiential and heuristic learning model, built dedicated learning studios, developed learning courses, and started the first international online live broadcast sharing of EPR family among China, France, UK and Finland in 2018. Furthermore, the approach of rapid learning was proposed to convert the experience in the EPR construction into knowledge of employees, that is, the knowledge points were continually broken down to narrow the focusing, and knowledge points of the same category were combined (summarizing and generalizing identical or similar points), to form a new and integrated knowledge point, and popularization was made via E-mail, network and WeChat official account, to facilitate rapid learning and accumulation of

knowledge by plant personnel.

(7) Making full use of previous other NPP operating experience to guarantee the operation of the first AP1000 reactor

During the commissioning and operation preparation for Unit 1 of Sanmen NPP, CNNP organized a technical support team with rich experience on commissioning and operating to provide all-round technical support for the AP1000 unit at the stage of initial fuel loading, initial criticality, initial grid connection, commercial operation and safe operation of the first cycle. This specialized team consisted of 56 specialty groups, with a total of 325 experts. The support team continually mastered new technologies and fostered professional teams in the course of solving problems, and finally formed specialized technical platforms with core capability. The safety operation of first AP1000 unit is ensured by communication of advanced experience, management system and culture concept via technology transfer.

(8) Application of network and information technology on NPP safety management

In some NPPs of China, such as Daya Bay NPP, work management center has been established. The network and information technology such as wireless network and mobile terminals are used as measures for NPP safety management, which can perform remote monitoring, remote expert support and malfunction diagnosis for field activities. The tracking of the maintenance work process are carried out for effectively manage and control risks in field work. The real-time mobile monitoring for individual exposure in nuclear island buildings is conducted to realize online auxiliary monitoring for radiation exposure, allowing the plotting of 2D or 3D real-time exposure map of nuclear island building in conjunction with locating information to optimize radiation exposure. QR code labels are used in the field to achieve good effect in the prevention of operation on the wrong unit, automatic counting of work hours and electronization of field work procedures.

B.8.2 Challenges

The challenges faced by China on nuclear safety at present mainly include:

(1) Ambiguities and divergences on the understanding of new concepts proposed by IAEA in the relevant safety standards

In recent years, IAEA has proposed some new concepts in the new versions of safety standards, such as large radioactive release to be practically eliminated and independence of levels of defence in depth etc. However, the details of such new concepts in the standards have not been interpreted by IAEA yet. China needs to further

study on the relevant concepts and specified in preparing relevant nuclear safety guidelines.

(2) Commissioning of DCS systems in NPPs

The DCS, as the core system for monitoring, control and management of the whole plant operation, is one of the critical systems to ensure the safety and has been widely used. However, the difference between the DCS and analog control and protection systems in the commissioning will still need to be studied and paid more attention.

(3) Issues on NPP equipment quality

In recent years, quality issues on NPP equipment from some foreign suppliers occurred one after another, such as the forging parts from CF France and the steam-driven pumps supplied from Clyde UK, indicating that the equipment quality management on the transnational supply chains should be further strengthened. The lower bid price is taken more focuses during the construction and procurement bidding of equipment because of pressure on competition of nuclear power. This results in that the cost is reduced in the procurement of original material, installation and manufacturing, which may cause potential quality issues of NPP equipment.

(4) Regulation for the operation of new safety systems and facilities

With the operation of the third generation nuclear power units, how to regulate the new safety systems and facilities is the new challenge, such as the performance monitoring of passive safety systems.

(5) Enhancement of human resources and capacity building for regulation

China commits itself to develop nuclear power with safety and high efficiency, the capacity of nuclear power keeps increasing and so as the manpower for nuclear safety supervision. However, there is still the shortage for regulation personnel, and also the experienced personnel are diluted. The capacity of new hire personnel shall be improved as early as possible.

6. Existing Nuclear Power Plants

Each Contracting Party shall take the appropriate steps to ensure that the safety of nuclear installations existing at the time the Convention enters into force for that Contracting Party is reviewed as soon as possible. When necessary in the context of this Convention, the Contracting Party shall ensure that all reasonably practicable improvements are made as a matter of urgency to upgrade the safety of the nuclear installation. If such upgrading cannot be achieved, plans should be implemented to shut down the nuclear installation as soon as practically possible. The timing of the shut-down may take into account the whole energy context and possible alternatives as well as the social, environmental and economic impact.

6.1 List of Existing NPPs

Up to December 31, 2018, there were 45 units in operation and 11 units under construction in China. From 2016 to 2018, no construction permit was issued, permissions for the initial fuel loading were issued for 14 units, and 15 units were put into operation. The list of existing NPPs is shown in Appendix 1. The distribution of NPPs in China is shown in Fig. 1.

6.2 General Situation of Existing NPPs

The existing units in operation in China include 43 light water reactors and 2 heavy water reactors. Units under construction include 10 light water reactors and 1 high temperature gas-cooled reactor.

Qinshan NPP has 20 operating years above with initially connected to grid in December 1991, and the validity period of operation license is up to July 30, 2021. At present, Qinshan NPP is requesting to extend the validity period of license up to July 30, 2041.

Daya Bay NPP, Qinshan Phase II NPP, LingAo NPP, Hongyanhe NPP, Ningde NPP, Yangjiang NPP, Fangjiashan NPP, Changjiang NPP, units 1~4 of Fuqing NPP, Units 1 and 2 of Fangchenggang NPP and Units 5 and 6 of Tianwan NPP are generation plus PWR units, which were designed on the basis of reference NPPs with successful experience and good performance, and improvements were made based on the operating experience from similar units at home and overseas, further improving the safety standard of the NPPs. After the Fukushima nuclear accident, significant improvements were made in all these units according to the requirements proposed by the MEE (NNSA), including the installation of portable power supplies and portable pumps,

installation or modification of hydrogen elimination facilities, and installation of monitoring and makeup water for spent fuel pool, thus further improving the capability of the units to prevent and mitigate severe accidents. All these safety improvements are described in detail in the Seventh National Report.

Units 1 and 2 of Third Qinshan NPP are CANDU-6 PHWR units, using natural uranium as fuel. For these two units, a number of design improvements were made on the basis of mature CANDU-6 design in foreign countries. In addition, after the Fukushima nuclear accident, improvements were made such as installation of portable power supplies and portable pumps as required by the MEE (NNSA).

Units 1-4 of Tianwan NPP are based on VVER technology, with Units 1 and 2 as reference units. For the new units, design modifications were made in conjunction with the operating experience and requirements in latest regulations, including the modification of nuclear island low radioactivity wastewater collection and monitoring and discharge system, modification of containment pit screen, installation of gas detection system for damaged fuel assemblies, and portable equipment used to make up water to the primary circuit, steam generators and spent fuel pool. In addition, 6kV and 400V portable power supplies are added. These improvements have further upgraded the safety level of units.

Units 1 and 2 of Sanmen NPP and Units 1 and 2 of Haiyang NPP are all AP1000 units, which provided complete prevention and mitigation measures against severe accident with passive design concept. Now these 4 AP1000 units have been put into operation, and Unit 1 of Sanmen NPP is the first AP1000 unit put into operation in the world.

Units 1 and 2 of Taishan NPP are based on EPR technology, and in their design, the corresponding design extension conditions were taken into account, and complete prevention and mitigation measures against severe accident are provided. Unit 1 of Taishan NPP is the first EPR unit put into operation in the world.

Units 5 and 6 of Fuqing NPP and Units 3 and 4 of Fangchenggang NPP are based on the HPR1000 technology independently developed by China. For HPR1000, the combined active and passive safety measures are provided, including the passive residual heat removal on the secondary side, passive heat removal of containment and passive reactor cavity water injection for cooling. The complete prevention and mitigation measures against severe accident are provided. Double-layer containment is provided with ventilation and filtration system arranged in the annulus, to reduce the release of radioactive substance to the environment under accident conditions. The inner and outer layer of containment has relatively independent functions, and the outer layer

can resist crash by large commercial aircraft. At present, construction is in steady progress for all 4 units according to schedule.

For the design features and safety improvements of AP1000, EPR and HPR1000 units, please refer to B.5.2.

The demonstration project of Shidao Bay NPP is a pebble-bed modular high temperature gas-cooled reactor, with the features of heat resistance of fuel elements, large heat capacity of the reactor core, passive residual heat removal and high negative reactivity coefficient. The core melting accident similar to that of PWR reactor isn't possibly occurred. The construction of the demonstration project of Shidao Bay NPP is in steady progress according to schedule.

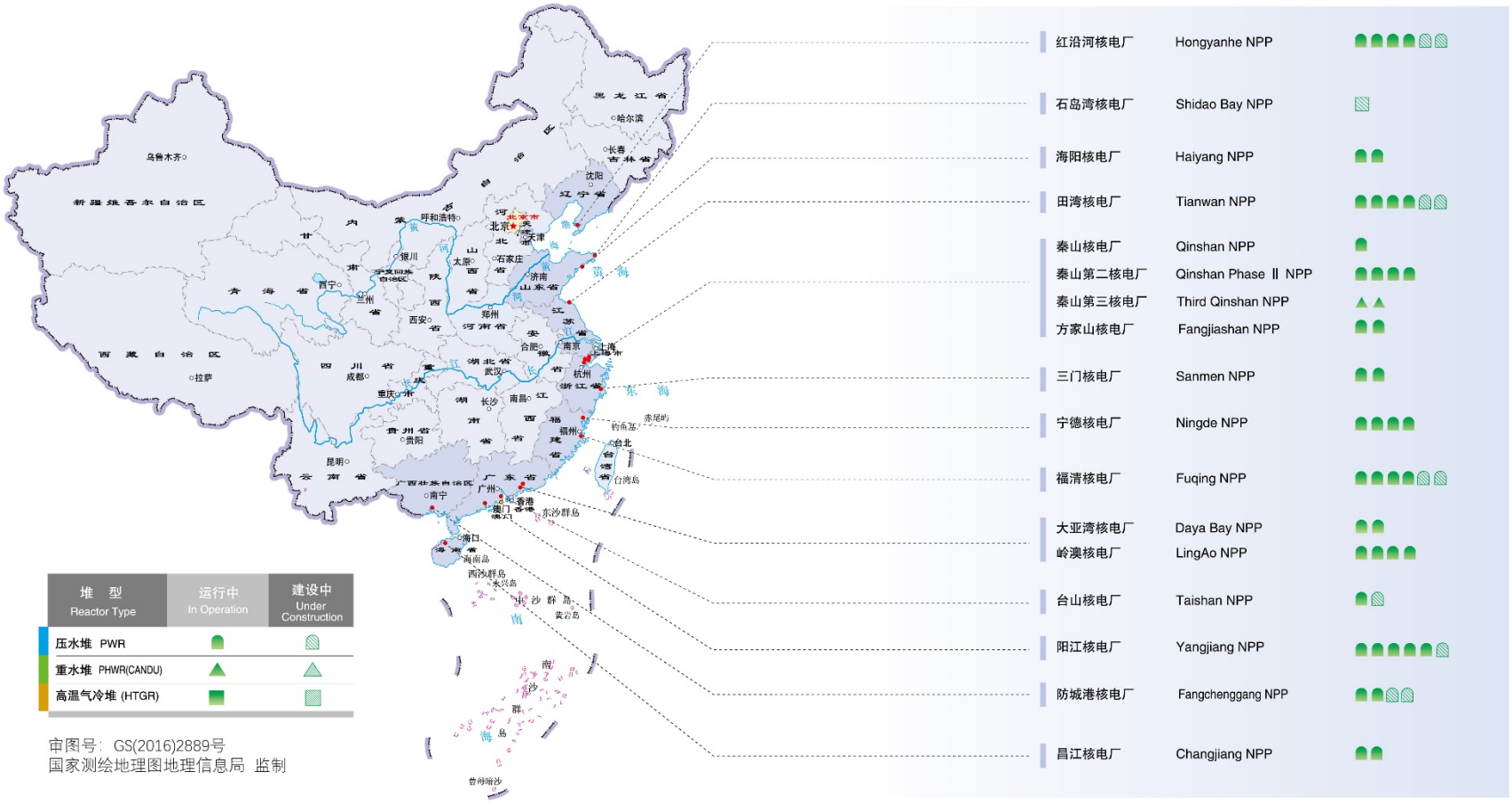


Fig. 1 Distribution of NPPs in China (up to December 31, 2018)

6.3 Overall Safety of NPPs in China

All operating nuclear power units in China have maintained safe and steady operation. The integrity of fuel element cladding in all units meets the requirements of technical specifications for NPPs, the leak rate of reactor coolant system and containment is also far below the limits set in technical specifications. All safety barriers of units are effective.

The MEE (NNSA) has developed the NPP operation safety performance indicator system, which covers four areas: reactor safety, radiation safety, emergency preparedness and safeguards. Seven safety cornerstones are established for the four areas, with a total of 23 performance indicators. From 2016 to 2018, the performance indicators of all operating NPPs in China were generally in good level. Most of the indicators were acceptable, with only 10 deviations, mainly occurring in safety cornerstones such as barrier integrity, site emergency and physical protection. The causes are as follows:

(1) The leakage rate of primary coolant increases due to the expansion of water before the pressurizer steam cavity is built up;

(2) Leakage existed in containment isolating valve;

(3) There was inconsistency between the actually implemented and planned emergency exercises resulted from the change of organizations;

(4) The compensation time of intrusion detection system exceeded the standard value because of lacking of spare parts for fiber optical fence detectors of the security monitoring system;

(5) In plant access management, there are cases of contractor swiping cards for others and using cards of others.

From 2016 to 2018, there were 152 operating events submitted from the NPP operating organizations to the MEE (NNSA). The statistics of operating events are presented in Appendix 2.

The performance indicators of operating nuclear power units in China are generally at fairly good level in the world, with some units at the advanced international level. In 2016, there were respectively 19 and 7 units with at least 10 of the 14 WANO individual indicators reaching the WANO first quartile and median values (a total of 30 units were covered in indicator calculation). In 2017, there were respectively 25 and 7 units with at least 10 of the 14 WANO individual indicators reaching the WANO first quartile and median values (a total of 36 units were covered in indicator calculation). In 2018, there were respectively 27 and 5 units with at least 10 of the 14 WANO individual indicators

reaching the WANO first quartile and median values (a total of 38 units were covered in indicator calculation).

The WANO performance indicators of all operating NPPs in China from 2016 to 2018 are listed in Appendix 3, these data presented a good overall trend in the WANO indicators within these three years.

7. Legislation and Regulatory Framework

1. Each Contracting Party shall establish and maintain a legislative and regulatory framework to govern the safety of nuclear installations.

2. The legislative and regulatory framework shall provide for:

(i) The establishment of applicable national safety requirements and regulations;

(ii) A system of licensing with regard to nuclear installations and the prohibition of the operation of a nuclear installation without a license;

(iii) A system of regulatory inspection and assessment of nuclear installations to ascertain compliance with applicable regulations and the terms of license;

(iv) The enforcement of applicable regulations and of the terms of licenses, including suspension, modification or revocation.

7.1 Legislation and Regulatory Framework

Since 1982, China has established the nuclear safety regulation system of China step by step with reference to the nuclear safety standards of IAEA and by extensively studying the laws and regulations on nuclear safety in other nuclear power countries. With continuous increase of operating NPPs, and cumulated practical experience on siting, design, construction, commissioning, safety operation and other aspects of NPPs, China has kept on improving its system of nuclear safety laws and codes on the basis of closely tracking and participating in the formulation and revision of IAEA safety standards and in conjunction with the nuclear safety practices in China.

7.1.1 General Description of Nuclear Safety Laws and Administrative Regulations, Department Rules and Guides

The nuclear safety laws and regulations system of China consists of laws, administrative regulations, department rules, guiding documents and reference documents.

(1) Laws

The laws applicable to nuclear safety field in China are enacted by the National People's Congress and its Standing Committee, issued through the Presidential Decree and have legal effects higher than administrative regulations and department rules.

(2) Administrative regulations of the State Council

Administrative regulations of the State Council, which have legal binding effects, are promulgated by the State Council according to the Constitution and laws. The

existing administrative regulations applicable to nuclear safety field are regulations to stipulate the scope of nuclear safety management, administrative organization and its rights, principles and procedures of regulation and other important issues.

(3) Department rules

The nuclear safety provisions and the detailed rules of Administrative Regulations of the State Council are department rules; they are prepared and promulgated by related departments of the State Council within the extent of their authority according to the laws and the administrative regulations of the State Council. Nuclear Safety Provisions are department rules enacting nuclear safety objectives and basic safety requirements, and the detailed implementation rules of Administrative Regulations of the State Council stipulate specific implementing measures to be adopted by relevant departments of the State Council according to these regulations on nuclear safety management.

(4) Guiding documents

Nuclear safety guides, which are prepared and promulgated by related departments of the State Council, are guiding documents that explain or supplement nuclear safety provisions and recommend relevant methods or procedures to implement safety provisions.

(5) Reference documents

Nuclear safety technical documents are reference documents in the technical fields of nuclear safety, and are promulgated by related departments of the State Council or their entrusted institutions.

The hierarchy of nuclear safety laws, codes and guides is listed in Figure 2.

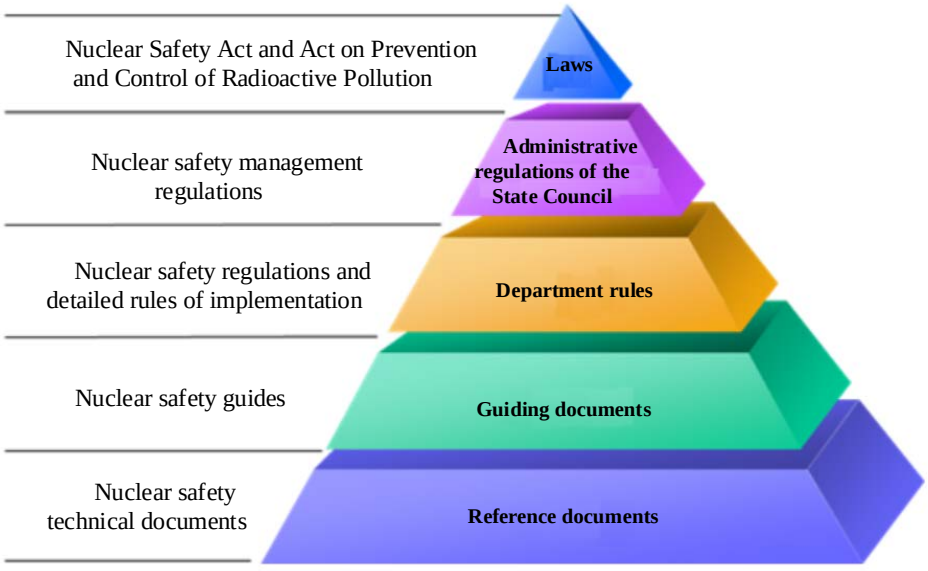


Fig. 2 Hierarchy of nuclear safety laws, regulations and guides of China

7.1.2 Issued nuclear safety laws, Administrative Regulations, Department Rules and Guides

The Chinese government always attaches high importance to nuclear safety. Since October 1986 when the State Council promulgated the *Regulations on the Safety Regulation for Civilian Nuclear Installations of the People's Republic of China*, China has already enacted a series of laws, regulations and guides, which cover NPPs, reactors other than those in NPPs, installations for nuclear fuel production, processing, storage and reprocessing, and facilities for radioactive waste processing and disposal, etc. All these constitute an available law system that shall be obeyed by civilian nuclear installations in siting, design, construction, operation and decommissioning.

In addition, the MEE (NNSA) and the related departments have promulgated in succession a series of nuclear safety provisions and detailed rules for implementing administrative regulations of the State Council, which are divided into different series according to the technical fields covered by them, related to siting, design, operation and quality assurance of NPPs.

After the Fukushima nuclear accident, the public paid daily increasing attention to the safety of nuclear power, and raised increasing requirements on nuclear safety. China carried out legislation of *Nuclear Safety Act* and *Atomic Energy Act*. The *Nuclear Safety Act* was issued on September 1, 2017, and came into force on January 1, 2018. As the top-level law in the nuclear safety area and an important part of the law system of China, the *Nuclear Safety Act* is the result of China fully learning the advanced experience in the world and summarizing the good practices in nuclear safety regulation for more than 30 years in all aspects. For the *Atomic Energy Act*, public opinion soliciting was completed on October 19, 2018.

Refer to Appendix 4 for the list of current laws, regulations, rules and guides of China on nuclear safety in NPPs.

7.1.3 Newly Issued Laws, Administrative Regulations, Department Rules and Guides on Nuclear Safety since the Previous National Report

The MEE (NNSA) carries out preparation and revision of regulations according to its preparation and revision working mechanism for nuclear and radiation safety regulations and the regulation preparation and revision plan, and four regulation and standard review meetings are held yearly, to review relevant regulations and standards.

From 2016 to 2018, China promulgated a series of new laws, regulations and guides on nuclear safety, and carried out the following work:

On September 1, 2017, China issued the *Nuclear Safety Act*, which consists of 94

articles in 8 chapters. This law has specified the policy, principle, responsibility and scientific, technological and institutional guarantee to ensure nuclear safety, specified the qualification, responsibilities and obligations of nuclear facility operating organizations, specified licensing system for nuclear facility and nuclear material and the radioactive wastes management system, defined the system of nuclear emergency coordination committee, the emergency plan system and nuclear accident information releasing system, defined the nuclear safety information disclosure and public participation system; defined the main bodies and scope for nuclear safety information disclosure and public participation; clearly stipulated on the specific approaches of nuclear safety inspection, penalty clauses for behavior violating the law and the compensation for damages caused by nuclear accidents.

In March 2016, the MEE (NNSA) revised and issued the *Measures for Regulation of Transportation Safety of Radioactive Materials*. In October 2016, the MEE (NNSA) revised and issued the *Code on the Safety of Nuclear Power Plant Design* with reference to the latest IAEA safety in response to VDNS. In October 2016, the MEE (NNSA) issued the *Radioactive Wastes Minimization of Nuclear Installations*. In December 2017, the MEE (NNSA) revised and issued the *Classification of Radioactive Wastes*. In December 2017, the MEE (NNSA) issued the *Development and Application of Computer Software for Safety Analysis in Nuclear Power Plants*. In February 2018, the MEE (NNSA) revised and issued the *Physical Protection for Nuclear Installations*.

7.1.4 Participation relating to International Conventions of Nuclear Safety

China signed the *Convention on Early Notification of a Nuclear Accident* and *Convention on Assistance in the Case of Nuclear Accident or Radiation Emergency* on September 26, 1986, and these two conventions were effective to China as of October 11, 1987; the *Convention on the Physical Protection of Nuclear Materials* was effective to China as of February 9, 1989. On October 28, 2008 China approved the Amendment to the *Convention on the Physical Protection of Nuclear Materials*; China signed the *Convention on Nuclear Safety* on September 20, 1994, and approved to join it as of March 1, 1996; China approved to join the *Joint Convention on the Safety of Spent Fuel Management and on the Safety of Radioactive Waste Management* on April 29, 2006. China implemented the international obligations in these conventions, and has played a full and constructive role within the mechanism of these conventions.

7.2 Nuclear Safety Licensing System

7.2.1 Administrative Licensing on Nuclear Safety

China has established the safety licensing system for nuclear installations. For activities such as siting, construction, operation and decommissioning of nuclear installations, NPP operating organizations shall apply for licensing from the nuclear safety regulation and administration department of the State Council. The nuclear safety license is legal document that the nuclear safety regulation and administration department approving the applicant to perform specific activities (such as siting, construction, operation and decommissioning of NPPs) related to nuclear safety.

The *Nuclear Safety Act* came into force January 1, 2018 made adjustment to the initial fuel loading approval and operation licensing system. It stipulates that NPP operating organizations shall apply for operation to the MEE (NNSA) before the initial fuel loading. The fuel can be initially loaded only after obtaining operation license for NPP operating organizations.

The *Nuclear Safety Act* stipulates that:

In the siting phase of NPP, the nuclear facility operating organization shall make scientific assessment of geology, seismology, meteorology, hydrology, environment, population distribution and other factors, submit a nuclear plant siting safety analysis report to the MEE (NNSA) on the premise of conforming to the requirements of nuclear safety technical appraisal, and obtain the *Review Opinion on Nuclear Power Plant Siting* after the review finds it conform to the nuclear safety requirements.

Before the construction of NPP, the NPP operating organization shall apply for construction to the MEE(NNSA), and submit the *Application for Nuclear Power Plant Construction, Preliminary Safety Analysis Report, Environmental Impact Assessment Report,* quality assurance document and other documents required by laws and administrative regulations. The NPP operating organization can start concrete casting for foundation of structures important to safety only after obtaining the construction permit, and shall perform related construction activities according to the conditions and scope specified on the permit.

Before the initial fuel loading of NPP, the NPP operating organization shall apply for operation to the MEE (NNSA), and submit the *Application for Nuclear Power Plant Operation, Final Safety Analysis Report*, quality assurance document, emergency plan and other documents required by laws and administrative regulations. The NPP operating organization can initially load nuclear fuel only after obtaining the operation license. The operating organization shall perform initial fuel loading, commissioning and operation activities within the conditions and scope specified on the permit.

Before the decommissioning of NPP, the NPP operating organization shall apply for decommissioning to the MEE (NNSA), and submit the *Application for Nuclear*

Power Plant Operation, *Safety Analysis Report*, *Environmental Impact Assessment Report*, quality assurance document and other documents required by laws and administrative regulations. The *Instrument of Ratification for Decommissioning of Nuclear Power Plant* will be issued after the review.

Organizations providing nuclear safety equipment design, manufacturing, installation and NDT services for NPPs shall apply license to the nuclear safety regulation and administration department of the State Council. Overseas organizations providing nuclear safety equipment design, manufacturing, installation and NDT services for domestic NPPs shall apply registration to the nuclear safety regulation and administration department of the State Council. NPP operators and nuclear safety equipment welding personnel, NDT personnel and other personnel for special techniques shall obtain the corresponding licenses or qualifications as required by the state.

7.2.2 Nuclear Safety Review

Nuclear safety review constitutes the technical basis of the administrative license for NPPs. The license is issued by the MEE(NNSA) only after the safety review.

It is specified in the *Nuclear Safety Act* that, the nuclear safety regulation and administration department of the State Council shall entrust a technical support organization having no interest relationship with the applicant when organizing safety and technical review. The entrusted technical support organization shall be responsible for the authenticity and accuracy of its technical assessment result. The MEE (NNSA) is the organizer and responsible entity for safety review of nuclear facilities in China, and organizes technical support organizations to perform overall review and assessment for the application documents submitted by the nuclear facility operating organization. The Nuclear and Radiation Safety Center is the permanent technical support organization of the MEE (NNSA).

The MEE (NNSA) confirms upon review and assessment that:

(1) Information submitted by the operating organization is accurate, and is in conformity with regulatory requirement;

(2) The existing information proves that the activities proposed by the operating organization meet the safety requirement;

(3) Technical solutions, especially new solutions, can meet the safety requirement as proved or examined by inspection / test or the combination of both.

The MEE (NNSA) has prepared the standard safety review program for NPPs to formalize the nuclear safety review activities. The review covers nuclear safety related

activities for the full lifetime of NPP, such as siting, construction, operation and decommissioning.

Nuclear safety regulations require that NPP operating organizations must review the NPP status periodically or in case of any major change, and submit specified documents and information for review by the MEE (NNSA), to ensure that the basis for issuance of license remains valid.

7.2.3 Types of Nuclear Safety Licenses for NPP

The nuclear safety license is legal document that the regulatory body approving the applicant to perform specific activities (such as siting, construction, operation and decommissioning of NPPs) related to nuclear safety. The current nuclear safety licenses for NPP in China include:

(1) Review Comments on NPP Siting;

(2) Construction permit of NPPs;

(3) Operation license of NPP;

(4) License for NPP operators;

(5) Instrument of Ratification for Decommissioning of NPPs and other documents subjected to approval.

7.2.4 Issuance of Nuclear Safety Licenses for NPP

The processes of application and issuance of nuclear safety licenses are as shown in Fig. 3.

The applicant shall submit the application, safety analysis report and other related documents required by laws and administrative regulations. The related nuclear activities can be performed only after obtaining permit following the approval by the MEE (NNSA).

The MEE (NNSA) organizes the technical review after receiving the application documents. For any question proposed in the review, the applicant shall timely reply, interpret and make corresponding supplement or modification to the documents. The MEE (NNSA) will organize nuclear safety field inspection for construction projects. When necessary, the MEE (NNSA) shall solicit opinions from relevant departments of the State Council and the provincial government where the nuclear facility is located. The MEE (NNSA) will organize a state nuclear safety committee to make deliberation and provide the consultancy comments.

For those complying with conditions, the MEE (NNSA) will make the decision of approval and notify the applicant in written form; for those not complying with conditions, the MEE (NNSA) will make the decision of not approval and notify the

applicant in written form.

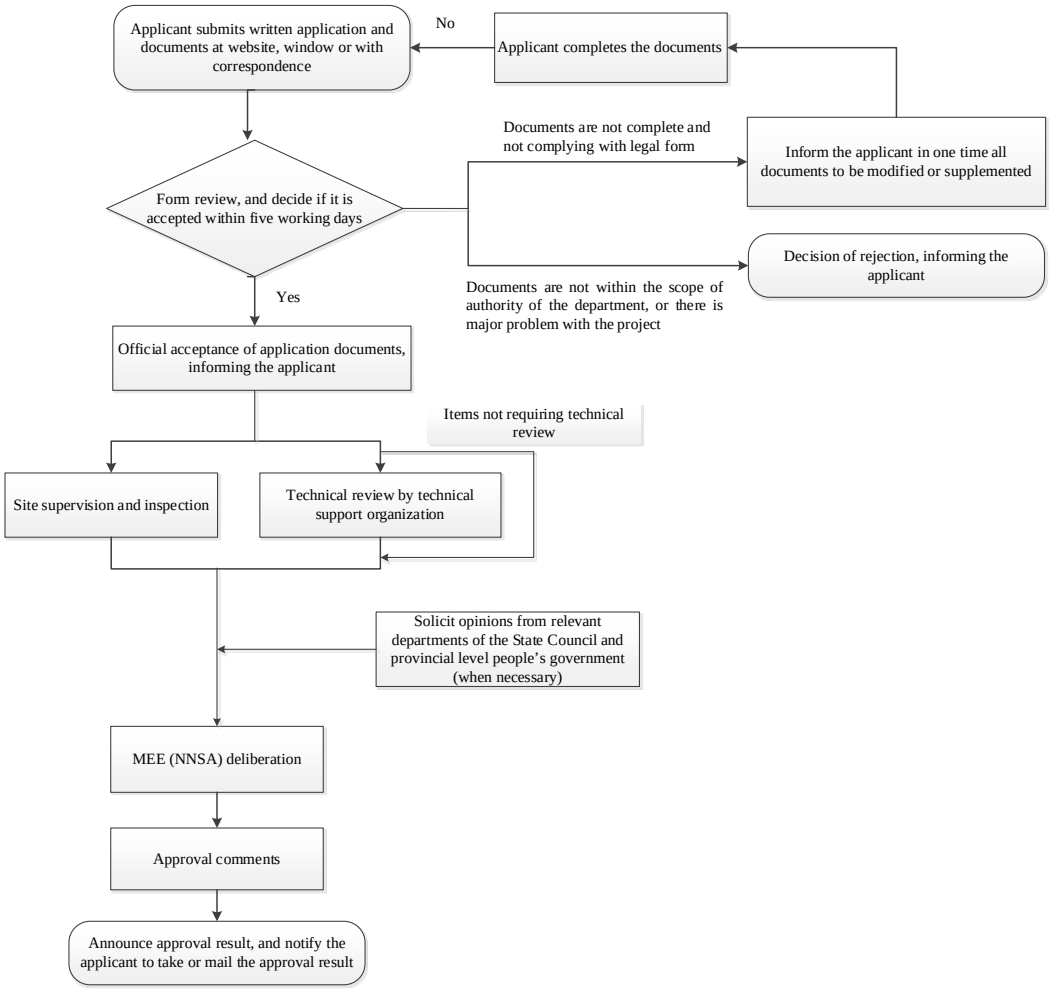


Fig. 3 Process of application and issuance of licenses

Applicants for NPP "Operator License" and "Senior Operator License" shall be examined by the competent department of NPPs or their entrusted entities, and the MEE (NNSA) will supervise and verify the process and issue the corresponding licenses.

7.2.5 Newly Issued Nuclear Safety Licenses during this Implementation Period

The MEE (NNSA) has newly issued the following nuclear safety licenses to NPPs from 2016 to 2018.

(1) Siting review comments

1) On March 1, 2016, the *Siting Review Comments for Units 3 and 4 of Sanmen NPP was issued*.
2) On October 10, 2016, the *Siting Review Comments for Phase I Project of Zhangzhou NPP was issued*.
3) On September 18, 2018, the *Siting Review Comments for Units 5 and 6 of Ningde NPP was issued*.

4) On December 19, 2018, the *Siting Review Comments for Phase I Project of Taipingling NPP of CGN in Guangdong was issued*.

(2) The instrument of ratification for the initial fuel loading[1]

1) On January 15, 2016, the *Instrument of Ratification for the Initial Fuel Loading for Unit 4 of Hongyanhe NPP in Liaoning* was issued.
2) On April 1, 2016, the *Instrument of Ratification for the Initial Fuel Loading for Unit 3 of Fuqing NPP in Fujian* was issued.
3) On May 6, 2016, the *Instrument of Ratification for the Initial Fuel Loading for Unit 2 of Changjiang NPP in Hainan* was issued.
4) On May 17, 2016, the *Instrument of Ratification for the Initial Fuel Loading for Unit 2 of Fangchenggang NPP in Guangxi* was issued.
5) On November 15, 2016, the *Instrument of Ratification for the Initial Fuel Loading for Unit 4 of Yangjiang NPP* was issued.
6) On June 9, 2017, the *Instrument of Ratification for the Initial Fuel Loading for Unit 4 of Fuqing NPP in Fujian* was issued.
7) On August 18, 2017, the *Instrument of Ratification for the Initial Fuel Loading for Unit 3 of Tianwan NPP* was issued.
8) On April 9, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 1 of Taishan NPP* was issued.
9) On April 17, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 5 of Yangjiang NPP* was issued.
10) On April 25, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 1 of Sanmen NPP* was issued.
11) On June 21, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 1 of Haiyang NPP* was issued.
12) On July 4, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 2 of Sanmen NPP* was issued.
13) On August 07, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 2 of Haiyang NPP* was issued.
14) On August 23, 2018, the *Instrument of Ratification for the Initial Fuel Loading for Unit 4 of Tianwan NPP* was issued.

(3) Operation license

1) On October 21, 2016, operation license for Units 1 and 2 of Ningde NPP in Fujian was issued.

[1] For the units that the initial fuel had not been loaded, the Instrument of Ratification for the initial fuel loading would be issued if the application was submitted before January 1, 2018 according to the previous nuclear safety regulations.

2) On February 2, 2018, operation license for Units 1 and 2 of Hongyanhe NPP in Liaoning was issued.
3) On February 14, 2018, operation license for Units 1 and 2 of Yangjiang NPP was issued.
4) On February 21, 2018, operation license for Units 1 and 2 of Fangjiashan NPP was issued.
5) On November 2, 2018, operation license for Units 1 and 2 of Fangchenggang NPP in Guangxi was issued.
6) On November 13, 2018, operation license for Units 1 and 2 of Fuqing NPP in Fujian was issued.

7.3 Nuclear Safety Supervision System

China has established the nuclear safety supervision system. The nuclear safety regulation and administration department of the State Council and other related departments shall perform supervision and inspection on organizations engaged in nuclear safety related activities for their compliance with nuclear safety laws, administrative regulations, rules and standards.

The MEE (NNSA) has developed the "Work Guideline for Nuclear and Radiation Safety Inspection" to formalize the organization and management of nuclear safety inspection. Nuclear safety inspection program and procedures have been worked out to maintain standardization and consistency for inspection. The MEE (NNSA) organizes regional offices to formulate annual inspection plans to define the forms, contents and frequency of inspections.

The specific tasks of nuclear safety inspection include:

(1) Review whether the documents related to safety submitted reflect the actual situation;

(2) Inspect whether the construction is carried out in accordance with the approved design;

(3) Inspect whether the management is performed in accordance with the approved quality assurance program;

(4) Supervise whether the construction and operation of NPPs and related nuclear activities comply with the nuclear safety regulations and the conditions specified in the licenses;

(5) Investigate whether the operating organization and personnel have an adequate capability for safe operation and implementation of emergency response plan;

(6) Other inspections to be carried out as authorized.

The MEE (NNSA) and regional offices mainly adopt the following approaches to carry out nuclear and radiation safety inspection for nuclear installations and activities:

(1) Daily inspection

It includes site visit, special topic investigation, review of anomaly or nonconformity management, activity witness and periodical report review. For important nuclear facilities, daily inspection is mainly performed by field inspect group (inspector).

(2) Routine inspection

It refers to the inspection activities carried out according to the specified procedures for the inspection items determined in the inspection program or annual inspection plan. The inspectors perform inspection according to the inspection procedures, including documents review, field personnel interview and field visit, and independent measurement and testing will be made when necessary.

(3) Non-routine inspection

It is the monitoring performed as required in work, as response to unexpected, non-scheduled or abnormal circumstance or events. Non-routine inspection can be either notified or not notified in advance depending on the specific conditions of the inspection items.

The nuclear safety inspectors have the right to access the sites of equipment manufacturing, construction and operation of NPPs to investigate and collect information related to nuclear safety when performing the inspect activity.

7.4 Enforcement

By nuclear safety laws enforcement, the nuclear safety regulation and administration department of the State Council or other related departments enforce administrative penalties on behavior of license holders violating the nuclear safety laws and regulations or license conditions. The nuclear safety regulation and administration department of the State Council or other related departments have the appropriate power to ensure compliance of laws and regulations and license requirements, including requiring the nuclear facility operating organization to modify or correct procedures related to nuclear activities, and to make necessary modification to the structures, systems and components of nuclear facility in order to ensure the implementation of conditions of the license. When authorized by law, the nuclear safety regulation and administration department of the State Council or other related departments may promulgate and issue detailed rules for implementation for law enforcement actions, and stipulate the rights and obligations of nuclear facility operating organizations.

According to the laws and regulations such as the *Nuclear Safety Act*, administrative penalties related to the nuclear and radiation safety include: ordering to correct; warning; fine; confiscating the illegal gains; ordering to stop construction or stop production for rectification; and suspending or revoking the license. The nuclear safety regulation and administration department of the State Council or other related departments make the penalties on illegal behavior in the nuclear safety area.

From 2016 to 2018, in accordance with legal stipulations, the MEE (NNSA) revoked license of two operators, gave warning to four operators, and enforced administrative penalties of ordering to make correction, or fine or revoking license for thirteen organizations involved in activities on nuclear safety equipment.

8. Regulation Organizations

1. Each Contracting Party shall establish or designate a regulatory body entrusted with the implementation of the legislative and regulatory framework referred to in Article 7, and provided with adequate authority, competence and financial and human resources to fulfill its assigned responsibilities.

2. Each Contracting Party shall take the appropriate steps to ensure an effective separation between the functions of the regulatory body and those of any other body or organization concerned with the promotion or utilization of nuclear energy.

8.1 Regulation System

The MEE (NNSA) is the regulatory body for nuclear safety and radiation of the Chinese Government, implementing unified and independent regulation on the nuclear safety of NPPs. The licensing system is one of main measures of the MEE (NNSA) in regulation, and inspection is also implemented for NPPs, nuclear materials and nuclear activities.

The CAEA is the competent authority for nuclear industry of the Chinese Government, which is responsible for research and drawing up policies, regulations planning, programme and industry standards related to China's peaceful uses of nuclear energy. It is also responsible for exchanges and cooperation with other foreign governments and international organizations in the nuclear field, and takes a leading role in the national nuclear emergency response and management.

The NEA is the competent authority for energy of the Chinese government, which is responsible for drawing up and implementing the development plan, conditions for access and technical standards of nuclear power; putting forward the review comments on major nuclear power projects; and organizing coordination and guidance for scientific research of nuclear power.

The National Health Commission, together with relevant departments, formulate laws and regulations on prevention of radioactive occupational diseases, organize the formulation and issuance of standards related to radioactive occupational diseases and conduct emergency medical rescue in nuclear and radiation accidents.

8.2 MEE (NNSA)

8.2.1 Organization Structure

The nuclear and radiation safety regulation system of China consists of the

headquarters of the MEE (NNSA) (including Department of Nuclear Facilities Safety Regulation, Department of Nuclear Power Safety Regulation and Department of Radiation Source Safety Regulation), regional offices and technical support organizations. Local ecology and environment departments at provincial level and municipality level and in autonomous region shall also perform part of the nuclear and radiation safety regulation functions. The MEE (NNSA) is headquartered in Beijing and has established six regional offices, namely Eastern Regional Office of Nuclear and Radiation Safety Inspection (located in Shanghai), Southern Regional Office of Nuclear and Radiation Safety Inspection (located in Shenzhen), South-Western Regional Office of Nuclear and Radiation Safety Inspection (located in Chengdu), Northern Regional Office of Nuclear and Radiation Safety Inspection (located in Beijing), North-Western Regional Office of Nuclear and Radiation Safety Inspection (located in Lanzhou) and North-Eastern Regional Office of Nuclear and Radiation Safety Inspection (located in Dalian), which are responsible for the routine inspection of nuclear and radiation safety in corresponding regions. The MEE (NNSA) has established the Nuclear and Radiation Safety Center in Beijing as its technical support organization, and has designated Zhejiang Environmental Radiation Monitoring Station as the radiation monitoring technical center of the MEE to provide technical support in terms of nationwide environmental radiation monitoring.

In order to better perform its regulatory function, the MEE (NNSA) also established the State Nuclear Safety Committee made up of experienced experts to provide independent advice to the MEE (NNSA) on major issues related to nuclear safety and environment, and provide support for the decision-making of major issues. Technical support organizations such as the Equipment Reliability Center of China Academy of Machinery Science and Technology, Beijing Nuclear Safety Review Center, and Suzhou Nuclear Safety Center are under the management by contract, which provide technical support on nuclear safety inspection and the licensing review of nuclear facilities and nuclear safety equipment. In addition, contracts of technical support are signed with universities and scientific research institutes as necessary. The MEE (NNSA) has established the Radiation Safety Regulation and Standard Review Committee to provide technical review for nuclear safety regulations and standards.

Other departments and bureaus of the MEE (NNSA) related to nuclear and radiation safety regulation include the General Office, Department of General Affairs, Department of Science & Technology and Finance, Department of Laws, Regulations and Standards, Department of Institutional Administration and Human Resources Management, Department of Communications and Education, Department of

International Cooperation, etc., which are responsible for works related to daily management, financial management, legislation, human resources, nuclear safety technical standards, international cooperation and publicity and education.

The organization structure of MEE (NNSA) is shown in Fig.4.

8.2.2 Main Duties and Responsibilities

The main duties and responsibilities of the MEE (NNSA) in the area of nuclear and radiation safety include:

(1) Responsible for regulation of nuclear and radiation safety, drawing up and organizing and implementing policies, plans, laws, administrative regulations and departmental rules, systems, standards and specifications related to nuclear and radiation safety, electromagnetic radiation, radiation environmental protection as well as nuclear and radiation accident emergency;

(2) Responsible for unified regulation of nuclear facility safety, radiation safety and radiation environment protection;

(3) Responsible for regulation of design, manufacture, installation and non-destructive testing activities for nuclear safety equipment and the safety inspection of imported nuclear safety equipment;

(4) Responsible for control of nuclear materials and regulation of physical protection;

(5) Responsible for regulation of radiation safety and radiation environment protection of nuclear technology application projects and uranium (thorium) mines and associated with radioactivity mines, and taking charge of radiation protection;

(6) Responsible for regulation of the safety and radiation environment protection of treatment and disposal of radioactive waste, and for supervisory inspection of radioactive pollution prevention and control;

(7) Responsible for regulation of safety transport for radioactive materials;

(8) Responsible for nuclear and radiation emergency response, investigation and treatment, of the MEE (NNSA) and participating in prevention and handling of nuclear and radiation terrorism event;

(9) Responsible for qualification management of reactor operators, special process personnel of nuclear equipment, etc;

(10) Organizing and developing radiation environment monitoring and regulatory monitoring of nuclear equipment and key radiation sources;

(11) Responsible for the implementation of *Convention on Nuclear Safety* and *Joint Convention on the Safety of Spent Fuel Management and on the Safety of*

Radiation Waste Management.

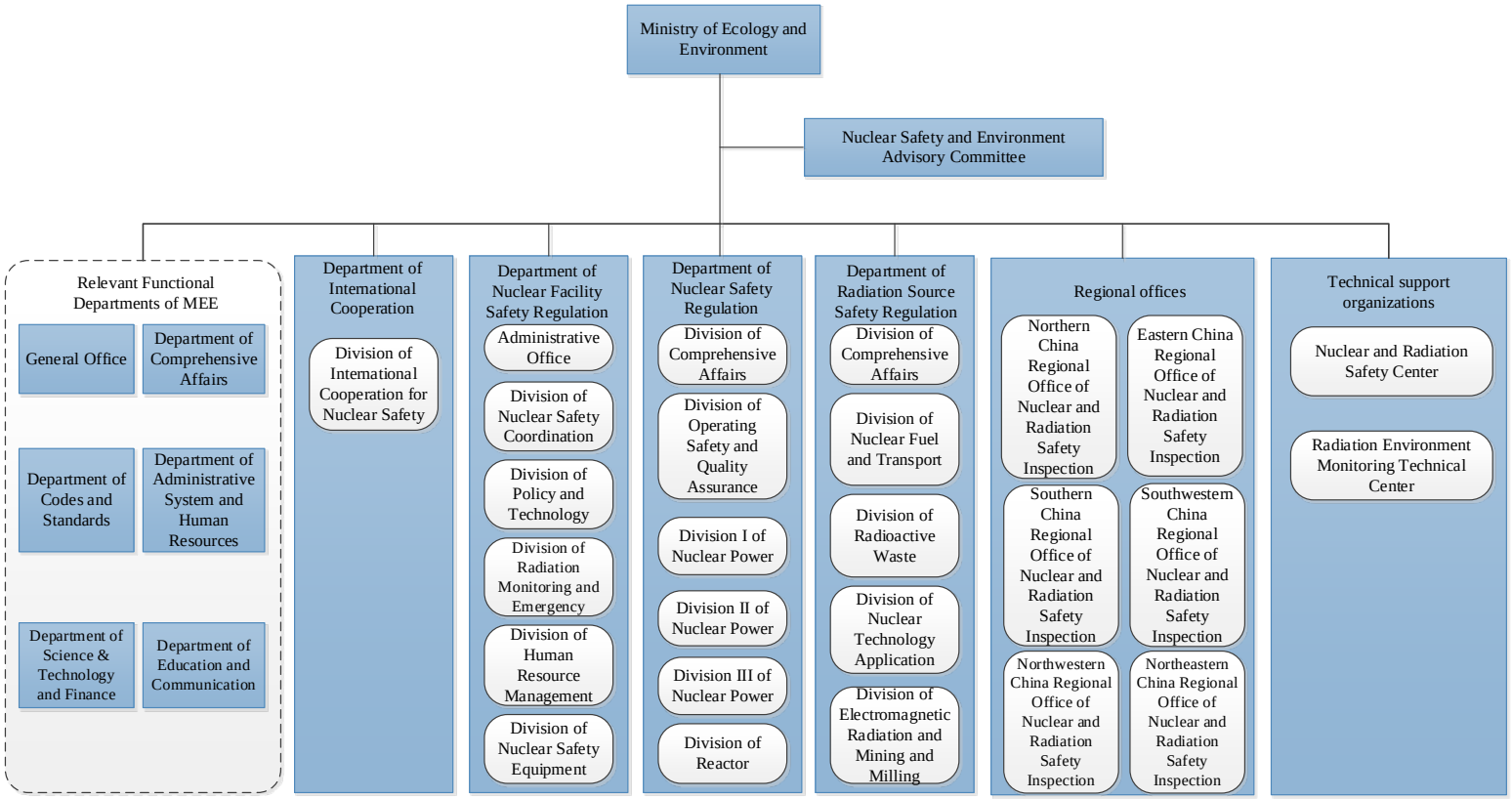


Fig.4 Organizational chart of the MEE(NNSA)

8.2.3 Human and Financial Resources

In order to meet the needs of nuclear power development, the Chinese government has continuously increased the manpower and financial resources in nuclear safety regulation to ensure the implementation of the functions in nuclear safety regulation. By 2018, there were 1106 established employees in the MEE (NNSA) for nuclear and radiation safety regulation, including 85 employees in the headquarters, 331 employees in the six regional offices, 600 employees in the Nuclear and Radiation Safety Center and 90 employees in the Environmental Radiation Monitoring Technical Center.

The MEE (NNSA) attaches high importance to the training of nuclear and radiation safety inspectors. It has established the multi-level nuclear and radiation safety training system covering the junior, intermedium and senior training, and conducted regular and irregular training activities. It continually improved the training system and mechanism for nuclear and radiation safety regulatory personnel, and made full use of training resources to implement trainings and discussion in various forms, such as special subject training (including simulator, non-destructive testing, civil nuclear safety equipment regulation, quality assurance, and commissioning and inspection of NPP), off-the-job training, dispatching young people to participate in the on-site inspection of regional offices, technical meetings and overseas exchanges.

In order to ensure the quality of nuclear safety inspection, the main requirements for nuclear safety regulatory personnel are specified in the *Rules for the Implementation of Regulations on the Safety Regulation for Civil Nuclear Installations of the People's Republic of China*, including education background, working experience, capability, basic professional quality, etc. The MEE (NNSA) performs personnel selection, training and examination in accordance with laws and regulations and work needs. At present, 382 people in the MEE (NNSA) have the valid Certificate of Nuclear Safety Inspector.

From 2016 to 2018, the financial budget of the MEE (NNSA) was stable, with a budget for daily regulation and management about 400 million yuan and 110 million yuan for nuclear safety regulatory capacity building in 2018. In addition, the MEE (NNSA) has expanded its resources through cooperation with IAEA, EU, and other national regulatory bodies and international organizations. The MEE (NNSA) standardizes and instructs the financial resources management of all departments and organizations of nuclear and radiation safety regulation according to *Work Guideline for Financial Resource Management*.

8.2.4 Integrated Management System

Based on over thirty years of practices in nuclear and radiation safety regulation,

China has gradually developed a set of relatively complete regulatory concepts, mechanisms and approaches, which constitute the nuclear and radiation safety management system of China. The nuclear and radiation management system includes the policies, goals and plans, organization structures and their management responsibilities, safety culture, resources and management required by the implementation of regulatory duties, main works and requirements of critical processes of the management system, and measures required for realizing the system continuous improvement of the MEE (NNSA). It can provide explicit working standard and basis, help the establishment of the systematic perspective and global view and improve the strategic thinking and working efficiency for staffs of the regulatory system. Furthermore, it is conducive to further improving the work quality of the regulatory activities, enhancing the authority and effectiveness of nuclear and radiation safety regulation to continuously improving the nuclear and radiation safety regulatory system of China.

The nuclear and radiation safety management system covers all relevant elements of the nuclear and radiation safety regulatory functions and management systems undertaken by the MEE(NNSA); covers all relevant departments/organizations of the nuclear and radiation safety regulatory system and their employees, including the central authorities, regional offices and technical support organizations. It is required to take the elements of safety, health, environment, security, quality, staff and organization, society, economy and health into comprehensive consideration to ensure that the principle of “ensuring safety” is always implemented and nuclear safety is always prioritized during the regulatory activities.

The nuclear and radiation safety management system of China mainly consists of four parts: management responsibility, support and guarantee, process implementation, assessment and improvement. Figure 5 shows the detailed structure of the management system, covering all nuclear and radiation safety regulatory activities of the MEE (NNSA).

(1) Management responsibility

The MEE (NNSA) has published the policy, mission, vision, core value, safety goal, regulatory principle and management commitment for nuclear and radiation safety; developed organization strategies and plans; defined the organization structure and responsibility; advocated safety culture, and paid attention to the needs of stakeholders. It is responsible for planning, establishing and implementing the management system, providing the required resources, and performing periodic assessment for self-improvement.

(2) Support and guarantee

Sufficient resources and the effective management help to enhance the capacity building for regulatory body and continually increase the satisfaction of stakeholders by providing powerful support and guarantee for the implementation, maintenance and improvement of the management system. These resources include infrastructure and working environment, human resources, information knowledge, international cooperation, financial resources, scientific and technological research and development, suppliers and external technical support organizations. All categories of resources are properly planned to ensure the resources necessary for the implementation of regulation activities and improvement of the regulatory capability.

(3) Process implementation

Process management involves planning, implementation, control and coordination for general management process and core regulatory process to ensure the quality and efficiency of each activity. Procedures are formulated for interrelated activities in processes to define the transition, interface and completion of activities among different departments to avoid redundancy, conflict or omission.

(4) Assessment and improvement

Based on the self-assessment of the management at all levels, internal and external independent assessment and operating experience, the monitoring and assessment system and self-improving mechanism for management system are established to build a learning organization, and promptly identify issues and deficiencies existing in the management system for the continuous improvement.

The documents of nuclear and radiation safety management system have been improved by the MEE (NNSA) based on the summarization of the practices and experience of China in nuclear and radiation safety regulation and by reference to the IAEA safety standards. The document is developed for systematically describing the nuclear and radiation safety management system, improving the management concepts and methods, optimizing the management process, promoting the participation of all relevant personnel and continuous improvement, and providing support for the modernization of regulatory system and capacity building.

“The process method” advocated by the IAEA safety standards have been fully referred to ensure that each work is a process that can be planned, implemented, evaluated and continuously improved during the documents preparation for nuclear and radiation safety management system. The management system consists of a set of interrelated process networks which has logic and link between each process. The management system shall be continuously improved by the results of identifying,

controlling and evaluating process and by making use of the obtained information and operating experience.

The document of nuclear and radiation safety management system is divided into three levels. Level I is the general introduction that describes the general structure, content and requirements of the management system. Level II is the work guideline and technical management program of various management elements and functional areas, which is further divided into three parts of “work guideline for general management”, “work guideline for business management” and “general technical management program”. Level III includes the working instructions/working procedures, supporting procedures of general technical management program, special technical management program and its supporting procedures, appropriate rules and regulations quoted or referred by documents of Level II.

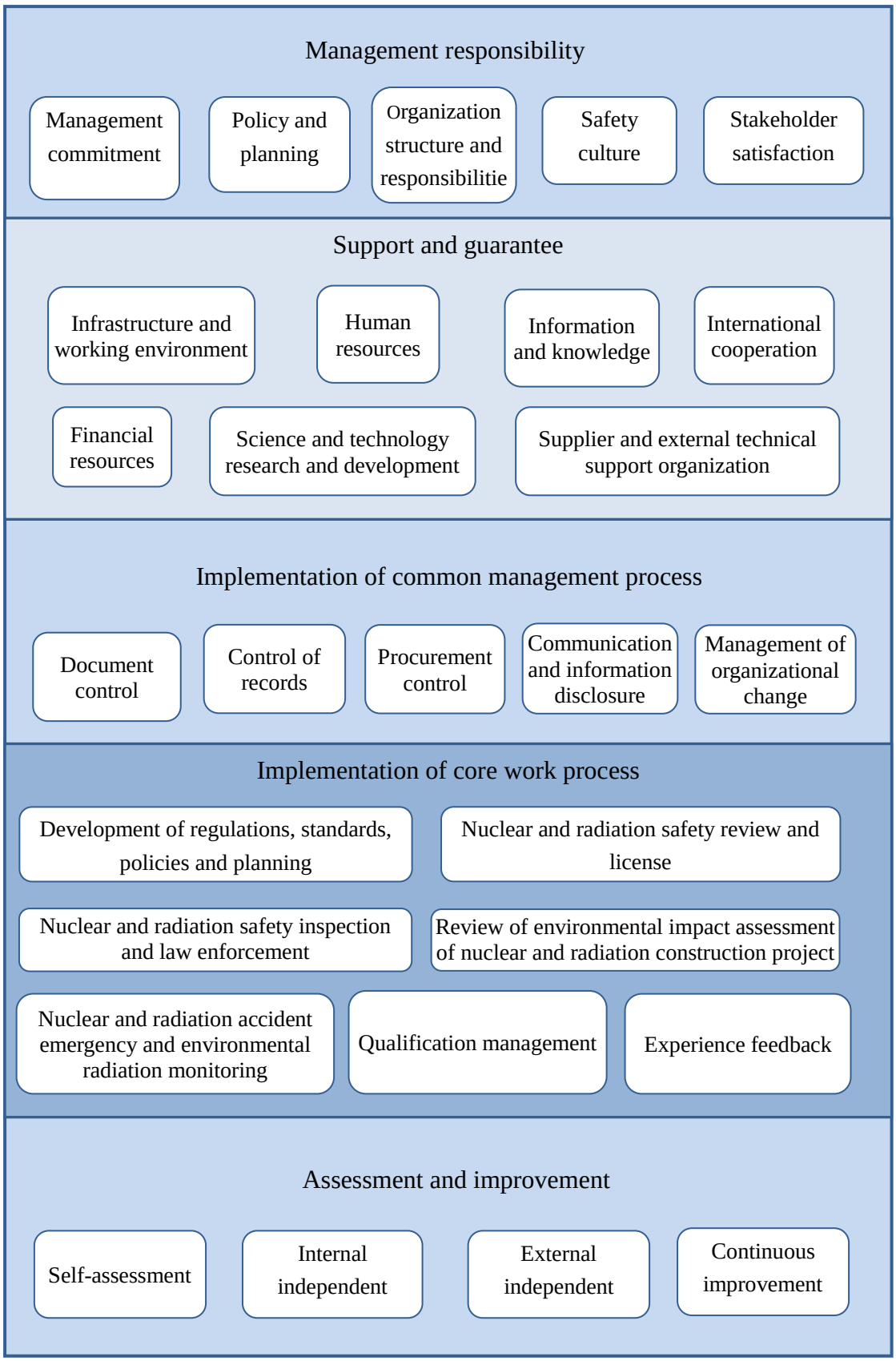


Figure. 5 Schematic diagram of Nuclear and Radiation Safety Management System structure

8.2.5 Public Communication

China attaches great importance to information disclosure and public communication to ensure the public right to know, participate and supervise. It is stipulated by the *Nuclear Safety Act* that the nuclear safety regulation and management departments of the State Council shall disclose the administrative license related to nuclear safety, as well as the safety inspection report, overall safety status, radiation environment quality, nuclear accident and other information on nuclear safety-related activities in accordance with the law. Nuclear safety information shall be disclosed to the public in a timely manner through government announcements, websites and other means convenient for the public to know about the information.

The MEE (NNSA) has continuously promoted the works related to the information disclosure and issued the *Nuclear and Radiation Safety Regulation Information Disclosure Plan* and *Work Guideline for Public Communication on Nuclear Power Projects*, and developed the *Working Plan of the MEE (NNSA) for Public Communication on Nuclear and Radiation Safety* and *Management Measures for Information Disclosure in Nuclear and Radiation Safety Regulation* in accordance with the *Nuclear Safety Act*, clearly specifying the scope, responsibilities, and the contents, timing, methods and channels of information disclosure.

The main contents of nuclear and radiation safety regulation information to be disclosed include the laws and regulations, standards, policies and planning of the state on nuclear and radiation safety, administrative licensing procedures and licensing documents for nuclear and radiation safety, safety inspection reports on relevant activities of nuclear and radiation safety, general safety status of nuclear facilities, environmental radiation monitoring results, nuclear and radiation emergency plans, significant nuclear and radiation events (accidents) and the investigation conclusions.

The MEE (NNSA) communicates with the public and media mainly by the following methods:

(1) Important information about regulatory activities is issued in time on the official website of the MEE (NNSA), such as license issuance, important review and inspection activities and their results, reports on construction and operation events of nuclear facilities, environmental radiation monitoring results, and relevant information on nuclear and radiation accident emergency. The “Administrator’s Mailbox” is also set up on the official website to directly communicate and interact between the MEE (NNSA) and the public, and to respond to public concerns in a timely manner.

(2) The MEE (NNSA) has disclosed WeChat Official Account, which introduces

the organizational structure and responsibilities of the MEE (NNSA), safety culture, policies and regulations, popular science knowledge, radiation environment, annual nuclear safety report, public announcement, etc. In addition, WeChat Official Account also regularly issues and recommends dynamic news in nuclear safety industry, popular science publicity and other related content.

(3) Publicizing the knowledge and information related to nuclear and radiation safety on websites, newspapers, periodicals, TV, publications and publicity materials.

(4) Soliciting public opinions by information disclosure, distributing investigation questionnaires, and holding symposium and hearings before publishing important regulatory documents or decisions, to accept public inquiry and supervision.

(5) Inviting media to participate in important experience exchange activities on nuclear safety regulation, answering and explaining questions of public concerns at media.

8.3 China Atomic Energy Authority

The CAEA is the competent authority for nuclear industry of the Chinese Government. Duties of the CAEA include:

(1) Responsible for research and drawing up policies and regulations related to China's peaceful uses of nuclear energy;

(2) Responsible for research and drawing up planning, programme, and industry standards related to China's peaceful uses of nuclear energy;

(3) Responsible for China's peaceful uses of nuclear energy except nuclear power related projects, assessment and approval, supervision, and coordination of the implementation of projects;

(4) Responsible for nuclear security and nuclear material control;

(5) Responsible for nuclear import and export review and management;

(6) Responsible for exchanges and cooperation with other countries and international organizations in the nuclear field, and participate in the affairs of the IAEA on behalf of the Chinese government;

(7) Undertake daily work of the National Nuclear Emergency Response Office, responsible for study and formulating national nuclear emergency plan and organize its implementation;

(8) Responsible for the decommissioning of nuclear facilities and radioactive waste management.

8.4 National Energy Administration

The NEA is the competent authority for energy in China, whose duties in the area

of nuclear include:

(1) Responsible for the nuclear power management and taking the lead in drawing up laws and regulations related to nuclear power;

(2) Drafting the development planning, access conditions and technical standards of nuclear power, and organizing the implementation;

(3) Putting forward the layout of nuclear power and review comments on major projects;

(4) Organizing the coordination and guidance of scientific research of nuclear power;

(5) Responsible for the international cooperation and exchange among governments in the area of nuclear power; and negotiating and contracting for agreements on peaceful use of nuclear energy with foreign countries.

8.5 National Health Commission

Main duties of the National Health Commission in the area of nuclear include the following:

(1) Formulating laws and regulations on the prevention of radioactive occupational diseases together with relevant departments, and organizing the formation and issuance of relevant standards;

(2) Organization and coordination of national nuclear emergency medical preparedness and rescue, and guiding local health departments in proper nuclear emergency medical preparedness and disposal.

9. Responsibilities of the Licensees

Each Contracting Party shall ensure that prime responsibility for the safety of a nuclear installation rests with the holder of the relevant licence and shall take the appropriate steps to ensure that each such license holder meets its responsibility.

China has made clear the safety responsibility of the personnel or organizations accountable for the nuclear installations or activities that may cause radiation impacts. The government issues nuclear safety licenses to the operating organizations or individuals responsible for operating the installations or carrying out such activities.

The *Nuclear Safety Act* stipulates that nuclear installations operating organizations are fully responsible for nuclear safety. In addition, the *Nuclear Safety Act* stipulates legal liability for the relevant illegal acts of licensees and proposes new requirements for licensees in terms of information disclosure and public participation.

It is stipulated by the *Regulations on the Safety Regulation for Civilian Nuclear Installations of the People's Republic of China* that the operating organization of the NPPs is directly responsible for the safety of the NPPs in operation. Its main duties are as follows:

(1) To comply with the relevant laws, administrative regulations and technical standards to ensure the safety of NPPs;

(2) To accept the safety regulation from the MEE (NNSA) and other departments, report the safety situation in a timely and faithful manner, and provide relevant information;

(3) To take overall responsibility for the safety of the NPPs operated, the safety of nuclear materials, and the safety of the employees, the public and the environment.

The responsibility for the safety of the NPPs which was fully undertaken by operating organizations will not be mitigated or transferred due to the activities and responsibilities of the design, manufacture, construction and regulatory personnel. Organizations performing design, manufacture and construction also undertake corresponding responsibilities within their respective scopes of work.

The operating organization shall make commitments to the nuclear safety of NPPs. All important activities related to the safety shall accord with the requirements in safety regulations. Nuclear safety is placed at the top priority, which shall not be compromised by production schedule and economic benefit.

The NPP operating organizations take various ways to carry out information disclosure and public communication. The major public concerns about the safety of nuclear power development are responsed through press conferences, new media and

other forms to guarantee the public right to know and supervise the safe operation of NPPs. Through the information platform, operating data are regularly and publicly released, and relevant information such as environment monitoring, radioactive wastes management and radiation protection of NPPs is released to the public. The public understanding of nuclear energy and nuclear safety has been strengthened through the disclosure of micro blog and WeChat account for science popularization, online intelligent Q&A program for online interaction with netizens and offline interactive Q&A between robots and netizens. In addition, facilities such as nuclear power science and technology museum and science popularization exhibition have been built in the NPPs, which are open to the public throughout the year. In addition, media and the public are also invited to visit NPPs to promote science popularization of nuclear power by means of live broadcast of new media, picture, text and video.

The NPPs operating organization has established on-site rescue team for the nuclear emergency. All neighboring NPPs have signed mutual support agreement, prepared mutual support action plans, established the nearby rapid mutual support mechanism and implemented the requirements and mechanisms of mutual emergency support for nuclear accidents. Such measures ensure that licensees of NPPs have appropriate resources for the effective on-site management of accidents and mitigation of their consequences.

10. Priority to Safety

Each Contracting Party shall take the appropriate steps to ensure that all organizations engaged in activities directly related to nuclear installations shall establish policies that give due priority to nuclear safety.

10.1 Safety Policies and Arrangements for Safety Management

In activities related to nuclear safety, China has always adhered to the policy of ensuring safety and the principle of "safety first, prevention first, clear responsibilities, strict management, defense in depth, independent regulation, and comprehensive safeguard", and taken effective measures to ensure nuclear safety. In all activities of siting, design, construction, manufacturing, operation and decommissioning of NPPs, safety shall be put an overriding priority.

The NPP operating organizations formulate and issue nuclear safety policy in accordance with the requirements of regulations. In all activities of NPPs, the nuclear safety management goal is to ensure safety of employees, the public and the environment. The goal of first and foremost safety and pursuit of excellence has been emphasized in all nuclear safety policies.

10.2 Safety Culture of NPPs

It is required by the *Nuclear Safety Act* that nuclear installations operating organizations shall cultivate and develop safety culture and integrate the safety culture into all aspects of production, operation and management.

In order to improve the level of safety culture and achieve good safety performance, all NPP operating organizations have set a long-term plan for safety culture development, including diversified communication and cultivation of safety culture, safety culture self-assessment and external independent assessment, to raise the personnel accountability and act consciously to ensure nuclear safety for the decision-making level, management level and execution level. In the past three years, NPP operating organizations have mainly carried out the following continuous improvements for the development of safety culture:

(1) Training and communication of safety culture

Nuclear power corporations and NPP operating organizations have organized the development of safety culture training and publicity materials, including lectures, videos, etc., and held a series of safety culture communication and exchange activities such as

meeting for exchanging experiences of safety culture development, safety culture knowledge contest, safety culture forum, etc. The safety culture was communicated by the middle and senior management of the NPP. The safety culture has been effectively passed on among the employees through the commitment and demonstration of the managers, the practice of the management and the participation of all staffs.

The effectiveness of safety culture development has been further developed through the activities such as “Year of Safety Procedure”, “Year of Safety Management Promotion”, “Year of Nuclear Safety Act Implementation” and “One Safety Message per Day”.

(2) Safety culture assessment

Nuclear power corporations have established safety culture assessment system, and regularly carried out safety culture assessment. In view of the general observations, negative observations and weaknesses identified in the assessment, corresponding corrective actions have been proposed and implemented step by step as planned. The NPP operating organizations conduct questionnaire surveys and interviews on safety culture to collect employees understanding and awareness towards safety culture, and continuously foster the atmosphere where all employees can discuss and pay attention to safety culture. The corporations also collect, classify and analyze the data of safety culture assessment in NPPs over the years, develop analysis reports based on the results of safety culture assessment, and identify the concerns in the development of safety culture as the opportunities for improvement.

(3) Improvement of safety management

The safety management systems, including technical management safety systems and safety oversight systems, have been established with the application of advanced safety management concepts and methods. Multilevel and in-depth safety management mechanism has been established for such links as the organization, rules, control, oversight, feedback, emergency and improvement of safety management, which integrates safety culture into all kinds of rules through advocating safety culture and improving the fundamental quality of employees. Safety committee of NPPs have been set up to guide, coordinate and comprehensively control the safety production, regularly discuss important issues of safety production and corrective measures and plans for safety issues, and review the implementation. In addition, the external expert committee has been set up and experts from the nuclear power industry have been invited to attend special meetings on major safety issues.

(4) Keeping on open mind

NPP operating organizations attach importance to international exchanges and

cooperation. On the one hand, they take initiative to apply for the IAEA and WANO peer review activities, and participate in the operating experience feedback system of international organizations including the IAEA and WANO to contribute the experience and information; on the other hand, they identify gaps for further improvements through international benchmarking and exchange.

(5) Formalizing human behaviors to prevent human errors

All NPP operating organizations attach great importance to foster good behavioral pattern of employees, integrate the requirements on using human error prevention tools in daily work, improve the human behavior by various means, and communicate management standards and expectations, including the continuous promotion of the development and application of human error prevention tools, strengthening the training on use of human error prevention tools and observation& coaching, and implementation of contests on human error prevention skills, etc.

(6) Improvement NPP operating experience system

The relatively complete operating experience system has been established for NPPs. The contractors' experiences have been integrated into the system to supervise and guide the contractor to carry out the experience feedback. All nuclear power corporations have issued corporation-level guidelines for experience feedback activities and procedures for standardized process of experience feedback, developed unified operating experience information platform, and established management mechanism for fleet experience feedback. In addition, they have actively joined the operating experience system established by the MEE (NNSA), NEA and CNEA to share information in the nuclear power industry.

(7) Maintaining good communication relationship

Effective interaction has been established between regulatory body and NPPs through the periodical coordination meetings, nonscheduled dialogs, symposiums, exchange visits and communication every year, to enhance the transparency and openness of regulation.

10.3 Peer Review and Self-assessment of NPPs

To ensure safe and reliable operation, NPPs in China continually learn the advanced management experience from foreign NPPs, and constantly improve their internal and external assessment system in combination with their own practices, as shown in Fig. 6.

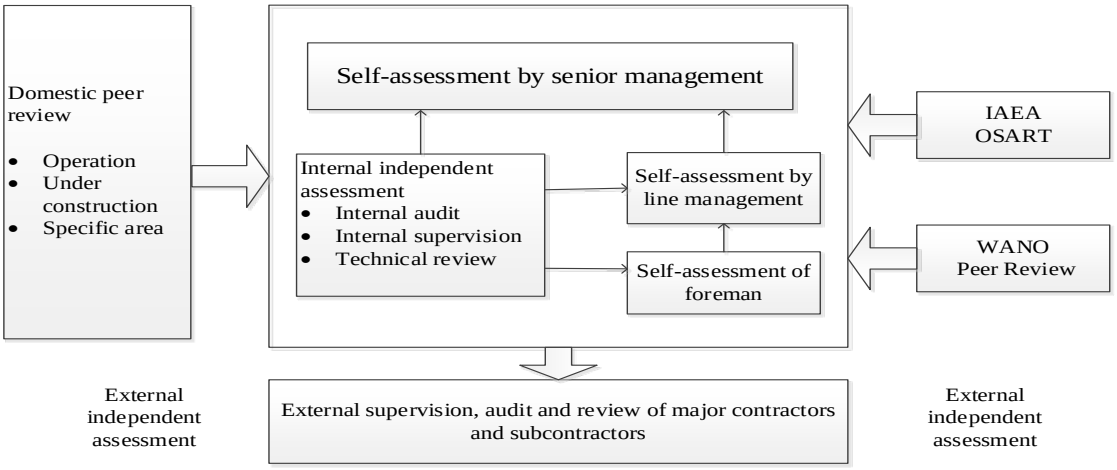


Fig. 6 Assessment system of NPPs

The internal assessment system of NPPs includes independent internal assessment and self-assessment at all levels of management. Independent assessment is conducted by authorized departments or organizations to check and verify each job done by plant employees or contractors through audit, supervision and technical review, etc. The results of independent assessment are important inputs for self-assessment.

Self-assessment at different management levels is performed in routine jobs so as to determine the effectiveness in establishing, promoting and achieving the objectives of nuclear safety, and identify and correct management weaknesses and obstacles to achieve nuclear safety objectives. Self-assessment of senior management focuses on meeting the strategic objectives of the organization, including safety objectives. The line management pays more attention to supervision and review of the working process, including the supervision of items, services and processes, review and verification of design documents, review of procedures and records, observation of independent assessment, and periodical walk-down of facilities.

Peer review is an important means to improve the safety performance of NPPs. The NPP operating organizations in China actively perform peer review and vigorously carry out specific area assessment activities on operation, maintenance, engineering and safety culture of NPPs. Through the relevant specific area assessment and comprehensive review activities, areas for improvements in the operation and project construction of NPPs are identified and then improvement actions are implemented after analyzing the causes to upgrade the management level.

In the past three years, China has continued to carry out domestic comprehensive assessment and specific assessment activities for the NPPs in operation. At the industry level, peer review activities have been carried out by CNEA. The five-year rolling peer

review plans (2017-2021) respectively for operating NPPs and nuclear power projects under construction have been developed and completed in 2016 to comprehensively arrange the review activities and optimize the resources allocation for review activities. From 2016 to 2018, CNEA organized and completed 24 peer reviews and 7 training sessions for reviewers, including comprehensive assessment and specific area assessment such as public communication, PSA, severe accident management, commissioning and start-up, outage contractor management and safety culture. The peer review for maintenance contractors of NPPs was performed first time of this kind in China. In 2016, CNEA issued *Performance Objectives and Criteria for Management of Nuclear Power Projects under Construction* to further improve the peer review of nuclear power projects under construction.

At the level of nuclear power corporations, review activities are carried out for specific areas according to the actual demand of NPPs, so that the operation assessment activities in China continually develop in depth, including the specific review in areas of foreign material exclusion, outage management, operating experience, training, fire protection, crew performance observation, etc.

In addition, NPP operating organizations in China conduct review activities held by WANO and IAEA. From 2016 to 2018, NPP operating organizations in China received 7 WANO peer review activities, 16 WANO PSUR activities and 6 follow-up activities, including 1 CPR follow-up. NPP operating organizations in China have conducted three IAEA missions, including the Pre-OSART mission conducted by Taishan NPP in January 2017, the Safety Aspects of Long-term Operation and Ageing management (SALTO) mission conducted by Qinshan NPP in June 2017, and the International Physical Protection Advisory Service (IPPAS) mission conducted by Qinshan nuclear power base in August 2017.

The results of all the above reviews have shown that the NPPs were in good safety condition with good safety management in general, and the quality of project construction was under control. In addition, review activities helped the NPPs or projects being reviewed identify the gaps and set the goal for improvement and standards to be achieved, and promoted the further improvement of safety and quality management. The list of domestic and international reviews performed by NPPs in China (from 2016 to 2018) is presented in Appendix 5. List of scheduled domestic and international reviews for NPPs in China (from 2019 to 2022) is presented in Appendix 6.

10.4 Regulatory Review and Control Activities

10.4.1 Safety review, inspection and licensing

According to the relevant nuclear safety laws and regulations, the MEE (NNSA) implements the safety review, inspection and licensing activities for nuclear facilities/activities under its regulation. The MEE (NNSA) will not issue the license until confirming that the NPP operating organization complies with the requirements of national nuclear safety laws, regulations and standards and that there are sufficient safety measures to protect the employees, the public and the environment from hazard of excessive radiation.

The MEE (NNSA), by inspecting licensees the performance of nuclear and radiation safety management requirements and conditions specified in licenses, urges the licensees to promptly correct defects and anomalies for complying with the nuclear safety management requirements and conditions specified in licenses. When necessary, compulsory actions shall be taken to protect the safety of nuclear facilities/activities. The MEE (NNSA) mainly adopts daily inspection, routine inspection and non-routine inspection, including:

(1) Formulate and effectively implement safety inspection programs and plans for various nuclear facilities/nuclear activities, organize the implementation of site inspection, and preparing corresponding inspection records and reports. Formulate and effectively implement the training program for nuclear safety and radiation safety inspectors.

(2) Timely report, assess and track problems identified in inspection until their effective resolution. Investigate and analyze important non-conformities and construction and operation events.

(3) For serious violation of nuclear safety laws and regulations, the administrative penalty procedure shall be launched, including issuing written warning/instructions, terminating or reducing specific activities, suspending or cancelling permit/license and fining.

(4) Inspect and verify the inspection processes and quality, and perform regular review on the independence and work quality of external technical support organizations and external experts.

(5) Establish nuclear and radiation safety regulation database and management system, promote the application of IT technology, carry out internal and external information exchange and operating experience feedback, and improve the effect of inspection.

10.4.2 Promotion of Safety Culture

In order to implement the national nuclear safety outlook and safety strategy, advocate the cultivation of safety culture and promote the overall improvement of the national nuclear safety level, China has promoted the construction of safety culture in the whole industry. In the *Safety Culture Policy Statement*, China has expounded the basic attitude towards safety culture, the principles and requirements for cultivating and practicing safety culture, and the eight characteristics of safety culture, including: the safety value and commitment of decision-making level; the attitude and exemplary role of the management level; personnel participation and accountability; cultivation of learning organization; establishment of comprehensive and effective management system; fostering suitable working environment; formation of questioning attitude, report and feedback mechanism of safety related problems; maintenance of harmonious public relations.

Since the release of the *Safety Culture Policy Statement*, China has taken various measures to continuously promote the safety culture.

(1) Incorporating the development of safety culture into laws

The *Nuclear Safety Act* incorporates the safety culture into the laws. It is stipulated in article 9 that the state shall develop nuclear safety policies and strengthen the development of safety culture. The nuclear safety regulation department, the nuclear industry department and the energy department of the State Council shall establish a mechanism for fostering safety culture. The nuclear facilities operating organizations and organizations which provided equipment, engineering and services shall vigorously foster and build safety culture and integrate it into each aspect of production, operation, research and management.

(2) Publishing *Traits of Safety Culture*

In February 2017, the MEE (NNSA) published the *Traits of Safety Culture*. The *Traits of Safety Culture* which is the detailed interpretation for the the *Safety Culture Policy Statement*, takes the eight characteristics of the safety culture as the framework and specifies the characteristics into 36 attributes and 154 practices and examples. The document is prepared by the reference of relevant documents of IAEA, WANO and other countries. In addition, 30 years of practices and experience in nuclear safety regulation was considered to highlight the good practices advocated by the MEE(NNSA).

The *Traits of Safety Culture* is the main reference for carrying out the safety culture assessment, which is applicable to all organizations involved in nuclear facilities/nuclear activities. It can be used to guide and carry out the activities related to safety culture for the MEE(NNSA) and relevant departments, various nuclear energy

and nuclear technology application organizations, engineering and service organizations and interested organizations.

(3) Continuously carrying out safety culture communication

In order to implement the requirements and propositions of the *Nuclear Safety Policy Statement*, the MEE (NNSA) has drawn up action plan to promote the safety culture communication, compiled relevant regulations and laws on nuclear safety, analyzed typical cases and other materials; and carried out a wide range of special activities for the safety culture communication. In addition, the MEE (NNSA) conducts multi-level communication and training on safety culture for the industry. In combination with the communication and education activities of the National Security Education Day, it promotes the safety culture communication into schools and communities, publicizes and promotes the safety culture to the public, strengthens the popularization of safety culture, and increases public attention to nuclear safety.

(4) Performing safety culture assessment

The MEE (NNSA) regards safety culture assessment as an important part of the development of safety culture and entrusts CNEA to organize a comprehensive assessment team to carry out safety culture assessment. From 2016 to 2018, CNEA conducted safety culture assessment in Huaneng Shidao Bay and Sanmen NPP.

(5) Promoting the organizations involved in civil nuclear safety equipment activities to develop safety culture

The MEE (NNSA) organizes the compilation of *Recommendations on the Development of Safety Culture in Organizations Engaged in Civil Nuclear Safety Equipment Activities*. Since 2016, the MEE (NNSA) has promoted the joint development of safety culture between nuclear power engineering companies and nuclear equipment licensees. In addition, it has carried out safety culture assessment in combination with daily inspection, collected good practices in human errors prevention, and further promoted the development of safety culture in the area of nuclear equipment.

10.4.3 Safety Culture of Regulatory body

Facing the rapid development of nuclear power in China and the increase of the public demands for safety, and in order to implement the rational, coordinated and balanced nuclear safety outlook, as well as perform the national nuclear safety responsibilities and international nuclear safety obligations, managers at all levels of the MEE (NNSA) attach more importance to the development of safety culture of the regulators and strive to foster good safety culture environment.

In recent years, the MEE (NNSA) has continuously strengthened the training, exchange and institutional building, and enhanced the safety culture development itself. The training of safety culture has been strengthened by preparation of training materials of safety culture and holding several trainings on safety culture for regulatory staff. The safety culture has been incorporated in the training system of nuclear and radiation safety regulation, and the cultivation and development of safety culture inside the regulatory body has been continually enhanced through annual regular trainings.The awareness of law, accountability and risk as well as the regulation transparency has been enhanced during the regulation to foster the environment of fearing, defending and cherishing nuclear safety. The exchange with nuclear industry has been increased to share experience and seek improvements. The study on the achievements of international safety culture has been strengthened by translating IAEA technical documents related to safety culture and compiling the good practices on safety culture of international organizations to train regulatory staff. The integrated management system for nuclear and radiation safety regulation has been updated, and the training of the management system has been carried out for all employees to further improve the regulatory capacity.

11. Financial and Human Resources

1. Each Contracting Party shall take the appropriate steps to ensure that adequate financial resources are available to support the safety of each nuclear installation throughout its life.

2. Each Contracting Party shall take the appropriate steps to ensure that sufficient numbers of qualified staff with appropriate education, training and retraining are available for all safety-related activities in or for each nuclear installation, throughout its life.

11.1 Financial Resources

The Chinese government allocates funds of certain amount for use in the development of nuclear power technologies and researches on nuclear safety technologies. The Chinese government gives full play to the guiding role of government, establishes effective fund guarantee mechanism, increases financial input to nuclear safety and radioactive contamination prevention and control, and promotes implementation of programmed projects; improves fund management and control mode in nuclear safety management; clearly defines fund source, method of capital contribution, approval process and application of funds for expenses to be jointly undertaken by the government and enterprises involving nuclear emergency, nuclear insurance, nuclear compensation, radioactive contamination prevention and treatment of civil nuclear facilities and construction of non-profit nuclear safety infrastructure; and strictly audits the fund flow to ensure appropriate fund raising and use for the designated purposes.

For status of the financial resources of the MEE (NNSA), please refer to Section 8.2.3. After the NPP has been put into operation, a defined percentage of the revenue from generating electricity is preserved for safety improvement, radioactive waste management and final decommissioning of the plant. Costs for the safety improvements have the priority in the annual plan and financial budget of NPPs.

It is stipulated by the *Nuclear Safety Act* that nuclear facilities operating organizations shall pay spent fuel treatment and disposal expenses in accordance with state regulations and include them in production costs. The operating organizations shall reserve the expenses for decommissioning and radioactive waste disposal and include them in the investment budget and production cost. The state shall establish the system of emergency reserve for nuclear accident to ensure the funds needed for nuclear

emergency preparedness and response.

It is clearly specified by the *Act of the People's Republic of China on Prevention and Control of Radioactive Pollution* that the NPP operating organization shall draw up plans for decommissioning. The administrative department for environmental protection under the State Council shall be responsible for regulatory monitoring of the NPPs. The expenses for the establishment, operation and maintenance of the regulatory monitoring shall be covered by the financial budget.

It is stipulated by the *Regulation on the Emergency Management Regulations for Nuclear Accident at Nuclear Power Plant* that the emergency preparedness fund for on-site nuclear accidents shall be undertaken by the NPPs and included in the investment budget and operation cost of construction projects. The emergency preparedness fund for off-site nuclear accidents shall be jointly undertaken by the NPPs and the local people's government. The amount of funds shall be approved by the department designated by the State Council in conjunction with the relevant departments. The fund to be undertaken by the NPPs shall be determined according to designed capacity of the NPPs before operation and the actual generated electricity after the operation, and be used for emergency preparedness of off-site nuclear accidents after comprehensive balance by the planning department of the State Council; the rest shall be raised by the local people's government. In addition, the *Rules on Management of Special Revenue for Emergency Preparedness of Nuclear Accident* was promulgated in 2007, which further specify the sources of special revenue for nuclear emergency preparedness, the paying standard, proportion, time and method of payment, the scope of payment application, budget and final accounting mechanism, and supervising mechanism.

China has established a liability insurance system for nuclear accidents and required that NPPs place insurance on the third-party liability. It is stipulated by the *Nuclear Safety Act* that nuclear facilities operating organizations shall make appropriate financial guarantee and arrangements to ensure timely and effective implementation of nuclear damage compensation by means of liability insurance and participation in mutual assistance mechanisms.

The Chinese government has specified a maximum compensation limit of 300 million yuan for nuclear accident damage caused by one-time nuclear accident for operating organizations. If the total amount of compensation payable for nuclear accident damage exceeds the stipulated maximum compensation amount, the state shall provide financial compensation up to 800 million yuan. Compensation for nuclear accidents caused by extraordinary nuclear accidents shall be decided by the State

Council after assessment if extra financial compensation is required.

11.2 Human Resources

For status of the human resources of the MEE (NNSA), please refer to Section 8.2.3.

The nuclear safety regulations and guides require that NPP operating organizations shall clearly define the staffing of personnel directly engaged in operation and supporting personnel, and specify their duties and responsibilities.

NPP operating organizations have set up human resources management departments in accordance with nuclear safety regulations, and are staffed with competent management personnel and sufficient number of qualified personnel. The safety analysis report of the NPPs has clearly specified the organizational structure and responsibilities of the NPP operating organizations. According to the requirements of nuclear safety guide *Periodic Safety Review of Nuclear Power Plants*, the adequateness of staffing of NPPs should be reviewed during PSR.

11.2.1 Human Resource Assurance Measures

The Chinese government is preparing talents education and cultivation plan to meet the increasing demand for human resources in the development of nuclear power. The state, enterprises, colleges and universities and academies of science vigorously strengthen talents cultivation and increase resources in talent reserve for nuclear energy technology, nuclear power design, equipment manufacture, construction of nuclear facilities, nuclear and radiation safety, nuclear fuel cycle, operation, maintenance and other functional areas.

(1) Improve the talent cultivating system: each nuclear power corporation has established a set of complete talent cultivating system, including the establishment of training centers, school-enterprise joint cultivation, establishment of nuclear universities for corporations, to foster and select professional personnel for nuclear power management, operation, construction, etc. An improved curriculum system and training facilities have been established; targeted training materials have been developed; and the combination of classroom training and on-the-job training have been implemented.

(2) Broadening the way to talent cultivation and recruitment: the demands for human resources are met by enrolling more students in colleges and universities, social recruitment, and engaging experts at home and abroad. Different patterns of cultivation have been built up according to the characteristics of professionals required by development of nuclear power in design, construction and operation.

(3) Attaching importance to the demand of high-end talents: before launching a

new project, all kinds of high-end core talents are selected and trained in advance. Key talents with international vision in management and technology are cultivated through expanding exchange and cooperation of training in nuclear power field. In addition, social resources are exploited to introduce high-end talents required by nuclear power industry.

(4) Actively building up the nuclear power experts support system: by setting up nuclear power experts committee at different levels and technical working groups in special fields, the talents related information is extensively collected at home and abroad or inside and outside the nuclear power industry for the establishment of talent information database and platform for nuclear power talents. Various talents and resources in nuclear power are used to provide expert consultancy and suggestions for major activities and decision-making of regulatory body, industry authorities and the operating organizations.

(5) Exchange of talents: job rotations are arranged between departments in NPPs, between different plants in nuclear power corporations and between senior managements of nuclear power corporations, to realize circulation of nuclear power talents for accumulation and exchange of management experience.

11.2.2 Training and Examination of Personnel in NPPs

The *Code on the Safety of Nuclear Power Plant Operation* puts forward specific requirements for personnel qualification and training of NPP operating organizations. The nuclear safety guide *Recruitment, Training and Qualification of Personnel for Nuclear Power Plants* provides detailed requirements for the competency and qualification, recruitment and selection, training and qualification of personnel for NPPs.

NPP operating organizations determine the qualification requirements of positions and prepare and implement the training/retraining programs and procedures for various personnel in accordance with the requirements of relevant regulations, guides and standards, and in consideration of specific duties and task analysis. Staff in NPPs are appropriately trained, qualified and authorized before undertaking the relevant work.

The validity periods of the personnel qualification and authorization have been specified for NPPs. In case the expiration of validity period, the renewal or extension of certificate shall be conducted through retraining and re-qualification to ensure that personnel meet the requirements of specific positions. NPP operating organizations manage the training, authorization and qualification of personnel of contractors with the same requirements, implement strict control and supervision in accordance with the

contractor management policy, include the training of contractors and suppliers into the training program of NPPs for enhanced training, to ensure effective performance of contractors.

In the past three years, NPP operating organizations made continual improvements on the training and examination of personnel, specifically as follows:

(1) Continuously enriching training forms

Based on the job training programme, NPP operating organizations develop and improve the corresponding curriculum system. In addition to the original traditional training methods such as classroom training, simulator training, on-the-job training, etc., new training methods including network training, mobile application schools, etc. are adopted for training.

(2) Continually improve the capability of simulator instructors

The NPP operating organizations have established the training, evaluation and qualification mechanisms; developed qualification courses and carried out competency training for simulator instructors to ensure that they have the necessary knowledge and skills to perform their duties and to ensure all-round improvement of capabilities of simulator instructors.

(3) Establishing a series of training facilities of various types

The nuclear power corporations have successively built a multi-level and a series of training facility systems, with full-scale simulators, principle simulators, multi-functional simulators, severe accident demonstration devices, and operator self-service training platforms. The training facilities cover a series of demands for new employees, operators, unit supervisor, shift supervisors and emergency commanders.

(4) Strengthening the training and examination of operators

The NPP operating organizations have established and promoted a multi-base joint development mechanism for the operator simulator retraining course, effectively integrating domestic and foreign significant operating experience, human factor traps, important unit transients, conservative decision-making methods, etc. into the retraining course. In accordance with the *Operator Fundamentals Weakness* (WANO SOER 2013-1), the NPPs have integrated the knowledge and requirements of the operator fundamentals into the design, analysis, development, implementation and evaluation of the operator training course, thus improving the basic knowledge and skills of operators.

(5) Further improving the application of Systematic Approach to Training

Based on systematic approach to training, NPP operating organizations have carried out job task analysis, training demand analysis, development of training objectives and training programme, development of training materials, training

implementation and training effect evaluation and other activities. The effectiveness of the training is verified based on the improvement of the trainees' ability and changes of human behaviors on site after training. NPP operating organizations continue to improve training materials based on operating experience feedback.

(6) Strengthening the training of occupational technicians

NPP operating organizations have set up the occupational skills testing and training facilities. For various types of nuclear power occupational skills, the question bank for occupational skills testing has been developed. Every year, the occupational skills testing is carried out regularly to promote the testing of intermediate labor, senior labor and technicians in nuclear-specific occupation, and strengthen the cultivation of high-level technical and skilled personnel. Workshops for skilled talents in the areas of machinery, I&C, electricity and other technologies have been set up to create a demonstration effect of skilled personnel cultivation. Skills competitions at different levels in the nuclear industry and nuclear power corporations are organized, such as pump maintenance, safety valve maintenance, crane operation and other skills competitions.

11.2.3 Examination and License Management for Operators

It is stipulated by the *Nuclear Safety Act* that operators of nuclear facilities shall obtain corresponding qualification certificates in accordance with state regulations. It is stipulated by the implementing rules *Application and Issuance of Safety License for Nuclear Power Plant* of the *Regulations on the Safety Regulation for Civil Nuclear Facilities* that the reactor control systems can only be allowed to operate by holders of Operator License or Senior Operator License of NPPs. In addition, clear requirements have been made for the issuance and management of operator licenses in the detailed rules of *Issuance and Management Procedures for Operator License of Nuclear Power Plants*. In September 2018, the MEE (NNSA) revised and completed the *Management Measures for Operators Licenses of Civil Nuclear Facilities (Draft for Comments)* and openly solicited comments from the public to further strengthen the license management of operators for NPPs.

The NEA has issued the standard *License Examination of Nuclear Power Plant Operators* to regulate the examination and license management of NPP operators. The *Health Standards and Medical Surveillance Specification for Nuclear Power Plant Operators* issued by the former National Health and Family Planning Commission defines clearly the health requirements for operators and the specific requirements for medical surveillance on operators.

The operators shall receive strict training, and pass the license examination and the qualification review organized by the Review Committee on Qualification for Operators of NPPs of the NEA. After the review and approval of the Authorization Committee on Qualification for Operators of NPPs of the MEE (NNSA), the Operator License or the Senior Operator License will be issued by the MEE (NNSA). The MEE (NNSA) shall supervise and inspect the training, examination and qualification management for operators.

According to the regulations of the MEE (NNSA), there shall be no less than 3 operators in each operating crew of the NPP, including at least one senior operator. The number of licensed operators of operating NPPs by the end of 2018 is showed in Appendix 7.

11.2.4 System of Registered Nuclear Safety Engineer

In 2002, the Chinese government issued the *Interim Rules on Practicing Qualification for Registered Nuclear Safety Engineers.* According to the regulation, the state has implemented the practicing qualification system for professional technical personnel on the positions critical to nuclear safety in the organizations providing technical services to the nuclear energy application and nuclear safety, which is included into the national occupational qualification system for professional technical personnel for centralized planning and management. In 2003, the Ministry of Personnel and the State Environmental Protection Administration jointly promulgated the *Implementing Measures for the Practicing Qualification Examination of Registered Nuclear Safety Engineer* and *Authentication Methods for the Practicing Qualification Examination of Registered Nuclear Safety Engineer.* In 2004, the *Interim Rules on Practicing Qualification Registration Management of Registered Nuclear Safety Engineers* was issued. In 2005, the *Interim Rules on Continuing Training of Registered Nuclear Safety Engineers* was issued. In addition, the *Serial Books for On-the-job Training of Registered Nuclear Safety Engineers* were compiled and published, involving laws and regulations relevant to nuclear safety, comprehensive knowledge of nuclear safety, professional practice of nuclear safety and case analysis of nuclear safety. In 2018, the MEE (NNSA) revised the *Examination Outline for Practicing Qualification of Registered Nuclear Safety Engineers* and revised the *Serial Books for On-the-job Training of Registered Nuclear Safety Engineers* according to the outline, mainly adding information related to the *Nuclear Safety Act* to examination contents and further clarifying and refining the examination requirements.

The examinations will be organized by the state every year. Those qualified in the

examination are issued with the *Certificate of Practicing Qualification for Registered Nuclear Safety Engineer* and can start practicing after registration. The validity period of registered nuclear safety engineer is two years. Continuing training system is adopted for registered nuclear safety engineers.

The scope of practicing for registered nuclear safety engineers includes: nuclear safety review, nuclear safety inspection, NPP operation, quality assurance, radiation protection, environmental radiation monitoring and other areas closely related to nuclear safety as specified by MEE (NNSA).

Since the examination of practicing qualification for the first batch of registered nuclear safety engineers in 2004, by the end of 2018, the number of nuclear safety engineers with practicing qualification had reached 4293. From 2016 to 2018, the MEE (NNSA) arranged 3 practicing qualification examinations for registered nuclear safety engineers in the whole country, where 563 people passed examinations, and 2552 people went through registration.

12. Human Factors

Each Contracting Party shall take the appropriate steps to ensure that the capabilities and limitations of human performance are taken into account throughout the life of a nuclear installation.

12.1 Regulatory Requirements on Prevention and Correction of Human Errors

China has made clear requirements for human factors management in nuclear safety regulations and guides such as *Code on the Safety of Nuclear Power Plant Design* and *Code on the Safety of Nuclear Power Plant Operation*. The main management requirements include:

(1) The concept of defence-in-depth in design, construction and operation management shall be implemented to ensure that all activities related to safety (including those aspects related to organization, design or personnel behavior) are defended by multi-level measures, and compensation or correction measures shall be taken in case of failure of one means of defence.

(2) Systematic consideration of human factors, including the human-machine interface, shall be included at an early stage in the design process and shall be continued throughout the entire design process. Verification and validation, including by the use of simulators, of features relating to human factors shall be included at appropriate stages.

(3) The design of workplaces and the working environment of the operating personnel shall be in accordance with ergonomic concepts. The design process shall give due consideration to plant layout and equipment layout, and to procedures, including procedures for maintenance and inspection, to facilitate interaction between the operating personnel and the plant, in all plant states.

(4) The human-machine interface shall be designed to provide the operators with comprehensive but easily manageable information, in accordance with the necessary decision times and action times. The design shall be such as to promote the success of operator actions with due regard for the time available for action, the conditions to be expected and the psychological demands being made on the operator.

(5) A suitable working environment shall be provided and maintained so that the staff can work safely and satisfactorily without having unnecessary physical and psychological pressure. The design shall be such as to ensure that, following an event affecting the plant, environmental conditions in the control room or the remote control panel and in locations on the access route to the remote control panel do not

compromise the protection and safety of the operating personnel.

(6) Important positions in NPP shall be staffed with sufficient number of qualified personnel, whose duties, responsibilities and contact channels shall be clearly specified. Sufficient and effective training and examination shall be provided for operators. Operators shall hold qualification licenses issued or recognized by the MEE (NNSA). Duties shall be performed in strict accordance with the procedures and operation procedures of NPP; and procedures shall be strictly reviewed and approved, periodically examined and updated in a timely manner.

(7) The operation status of NPP shall be periodically reviewed by independent internal and external assessment and self-assessment, to consolidate and strengthen the safety awareness and prevent overconfidence and self-complacence; the systematic assessment and operating experience on internal and external human factors shall be applied, and technical or management measures shall be timely taken to prevent or correct human errors for continuous improvement.

(8) Human factor is one of the factors of PSR of NPPs. The main purpose of the review is to evaluate the current human factors that may affect the safe operation of NPPs and to develop corrective actions.

12.2 Measures to be Taken by Licensees

According to requirements of nuclear safety regulations and guides of China, NPP operating organizations in China adopt the following measures to strengthen human factors management at the stage of operation during the design and operation:

(1) Clearly defining duties of organizations and positions for human factors management; constantly strengthening positions accountability system and surveillance system, and establishing and executing response and decision-making mechanism for unexpected events, and straightening out various interface relations and working process, to reduce human errors in coordinating and decision-making process. A human factors management program is developed to standardize human factors management in NPPs to prevent human errors and reduce events caused by human errors. It is required to continuously improve various management procedures, including procedures of code of conduct, work permit, human root cause analysis of human factor event, internal and external operating experience procedures, etc., to strengthen human factors management in NPPs.

(2) Improving training on human factors management and strengthening the construction of human factors laboratories. It is required to establish and improve the training curriculum system for human error prevention based on human factors

management theory and combined with singular training of human error prevention tools; further enhance the construction of the human error prevention training room. Through the development of a series of equipment for the human error prevention training room and the development of supporting training materials and instructor lesson guides, systematic training on human error prevention in NPPs is realized. In addition, human error traps in actual work are integrated into the curriculum design to enhance the ability of personnel to identify human errors in the field for improvement. The construction of human factors laboratories, personnel in different areas such as power plant operation, maintenance and technology can carry out real-life scenario training on human error prevention and integrate theoretical knowledge with skills and operation specifications, which effectively strengthens the effect of practical training on human error prevention.

(3) Continuous promotion and application of human error prevention tools. The human error prevention tool cards are developed. It is required to prepare detailed rules on each tool card to specify and emphasize implementation steps and precautions, and explain common failure symptoms in the process of using the tool card. In addition, a new concept of human error prevention is introduced to develop tools for managers to prevent human errors. The development and promotion of these human error prevention tools have played a good role in standardizing personnel behavior and reducing human error.

(4) Continuously carrying out management observation on personnel behaviors, and establishing a "personnel behavior analysis system". Through the collection, collation and analysis of personnel behavior related data, the organization defects are deeply investigated, and corresponding corrective actions are taken in combination with the analysis results, so as to realize closed-loop management and improve the personnel performance management of NPPs.

(5) Exploring researches on new technologies to prevent human errors and increasing physical protection measures to reduce human errors. It is necessary to use wireless cameras and other information equipment to realize the remote and real-time monitoring of on-site observation, improve the efficiency of personnel behavior observation, and achieve the traceability of on-site behavior; establish the working point identification and application platform system; develop a mobile application system and use the QR code signs on site to prevent entering wrong cells. It is required to add physical protection measures at the site, for example, painting and printing the accesses of the buildings, adjacent equipment and other areas with theme colors, and installing projection spotlights on the ground. Shields are provided for all high-risk and

easy-to-touch switches on site; add physical protection measures, such as acoustics system, to prevent entering wrong cells.

(6) Carrying out all kinds of seminars and workshops as to human error prevention topics. It is required to share the experience in applying new technologies to prevent human errors, discuss common issues of human errors and jointly discuss solutions; carry out human error prevention related competitions and other activities to cultivate and improve the staff awareness and skills of human error prevention.

(7) Making effective use of operating experience to continuously improve human factors management. It is required to make full use of the results of PSR and peer review. The NPP operating organizations develop and effectively implement corrective action plans for the human errors found in PSR and peer review, such as insufficient effect evaluation and utilization of management on-site observation, lack of clear requirements for training of short-term contractor on human error prevention, and insufficient analysis and utilization of low-level human errors; carry out operating experience on human factor events; carry out thematic analysis on human factor events; develop corrective actions for problems found, perform follow-up and implement such actions; develop the problem-oriented special courses for human error prevention, such as *Prevention of Entering Wrong Cells*, *Prevention of Accidental Touch* and *Foreign Material Exclusion*, etc.

(8) Carrying out research on human factor event analysis, including methods of human factor event statistical analysis (including correlation analysis, cluster analysis, mining of associated rules, etc.), research on human errors characteristics of digital NPPs, and methods of human induced event investigation and analysis, etc.

12.3 Regulatory Review and Control Activities

The regulatory body has clearly specified the technical and management requirements related to human factors in relevant nuclear safety codes and guides, and ensure that all requirements related to human factors are effectively implemented in the course of design, construction and operation through nuclear safety review and supervision. Main contents of review include: technical and administrative measures related to human factors in the license application documents; NPPs organization; staffing, training, examination and authorization of personnel at positions related to quality and safety; report, analysis and feedback on defects/events related to human factors in NPPs.

The regulatory activities of the MEE (NNSA) on human factor aspects include:

(1) Organizing workshops for human factors regularly to analyze and feedback

human factor events.

(2) Performing in-depth research in the aspects of individual, system, organization management, culture, etc., to explore and seek for approaches to reduce human errors.

(3) Requiring that the safety analysis report of the NPPs should include chapters on human factors engineering and control room in the safety analysis report of the NPP. The MEE (NNSA) review the details of human factors engineering and control room, and reviews the verification of human factors related characteristics.

(4) Reviewing human factor as one of the essential elements in the PSR.

(5) Paying attention to the analysis and evaluation of human factors when reviewing operational events. In addition, human factor events of NPPs at home and abroad shall be tracked and collected, and analysis and research on typical human factor events of NPPs shall be performed.

13. Quality Assurance

Each Contracting Party shall take the appropriate steps to ensure that quality assurance programmes are established and implemented with a view to providing confidence that specified requirements for all activities important to nuclear safety are satisfied throughout the life of a nuclear installation.

13.1 Overview

NPP operating organizations in China always adhere to the policy “Safety First and Quality First”. It is stipulated by the *Nuclear Safety Act* that nuclear facilities operating organizations and organizations providing equipments, engineering and services shall establish and implement a quality assurance system to effectively ensure the quality of equipment, engineering and services, to ensure that the performance of equipment meets the requirements of nuclear safety standards, and that engineering and services meet the nuclear safety related requirements. The NPP operating organizations shall establish and implement a quality assurance system for each phase of the NPP throughout the lifetime in accordance with the requirements of the nuclear safety regulation of *Code on the Safety of Nuclear Power Plant Quality Assurance.* The quality assurance system shall make provisions for the management of relevant work affecting the quality of items important to safety in the NPP, and provide appropriate control conditions for the completion of the above work.

The top management of the NPP is fully responsible for the establishment and effective implementation of the quality assurance system. All personnel and organizations engaged in quality-related work or activities affecting the items important to safety in NPPs shall comply with the requirements of the quality assurance system, and shall also undertake the responsibility and obligation to report the quality problems. All NPP operating organizations have set up independent functional departments for quality assurance, which shall be responsible for the establishment of quality assurance system and effectively execution, and to verify the effectiveness of the system implementation by inspection, surveillance and audit. These departments have the authority, including sufficient independence from cost and schedule, to handle quality problems until the quality problems are addressed and resolved effectively.

13.2 Basic Principles and Requirements of Quality Assurance

The basic principles and requirements of quality assurance are clearly defined in the *Code on the Safety of Nuclear Power Plant Quality Assurance*, mainly including:

(1) Establishing and effectively implementing the quality assurance system at each stage of the NPP throughout the lifetime; establishing the written procedures, instructions and drawings, and periodically reviewing and revising them (if necessary); performing management review regularly to evaluate the status and appropriateness of quality assurance system, and taking corrective actions if necessary.

(2) Establishing a documented organizational structure, clearly defining the functional responsibilities, authority and lines of internal and external communication; controlling and coordinating the working interfaces between organizations; controlling the selection, staffing, training and qualification of personnel to ensure that sufficient proficiency of work is achieved and maintained by working personnel.

(3) Controlling the development, review, approval, distribution and change of the documents of quality assurance system to preclude the use of outdated or inappropriate documents.

(4) Controlling design input, design analysis and calculation, design output, design interfaces and design changes, and performing design verification when necessary. Taking measures to ensure that approved design input and relevant requirements are correctly reflected in technical specifications, drawings, procedures or instructions.

(5) Controlling the development of procurement documents which cover the evaluation and selection of the suppliers as well as the control the procured items and services to ensure that the requirements of procurement documents are satisfied.

(6) Controlling the identifications of materials, parts and components; taking measures to control the handling, storage and shipping of items, and appropriately maintaining items important to safety so as to ensure that the quality is not compromised.

(7) Controlling the processes affecting the quality, especially the special processes, adopted in the manufacturing, installation and maintenance of NPPs. The purpose of the processes control is to ensure that these processes are completed by qualified personnel in the appropriate environment using qualified materials and equipments according to approved procedures.

(8) Establishing and effectively implementing inspection program and test program to verify that items and activities meet specified requirements, and to demonstrate that the structures, systems and components can satisfactorily execute the expected functions; controlling the selection, calibration, adjustment and usage of measuring and testing devices, and indicating the inspection, test and operating status.

(9) Identifying items that do not conform to the requirements when practical. The non-conforming items shall be reviewed and disposed according to the written

procedures. The responsibility to review and the authority to deal with non-conforming items shall be defined. Repaired and reworked items shall be re-inspected according to appropriate procedures.

(10) Identifying and correcting situations adverse to quality. For significant conditions adverse to quality, the quality assurance system shall provide that the cause of such conditions be determined and corrective action taken to prevent repetition.

(11) Establishing and executing quality assurance recording system. The system shall provide for identification, collection, indexing, filing, storing, maintenance and disposal of records. The system ensures that records are legible, complete and correct, and can provide sufficient and objective evidence of quality of items and/or activities.

(12) Establishing and implementing the internal and external quality assurance audit system to verify the effectiveness of the quality assurance system implementation. Corrective actions shall be taken for the deficiencies that identified in audit and follow-up actions shall be taken for tracking and verification.

In addition, ten safety guides on quality assurance have been formulated to put forward specific instructions for the above-mentioned basic requirements.

13.3 Establishment, Implementation, Assessment and Improvement on Quality Assurance System of NPPs

All NPP operating organizations in China attach great importance to the establishment of the quality assurance system. Massive manpower and material resources are invested every year to ensure the effective implementation of the system and the achievement of various objectives. The dedicated functional departments of quality assurance are set up in the NPPs and authorized with adequate authorities, to effectively control activities endangering safety and quality until the problems are effectively resolved. Based on the importance to safety, complexity and maturity of items and activities, quality assurance classification shall be carried out for activities that affect the quality of items important to safety.

13.3.1 Establishment of Quality Assurance System

The NPP quality assurance system is mainly divided into stages of design and construction, operation and decommissioning. The operating organization of the NPP has prepared the *Quality Assurance Program for Nuclear Power Plants* according to requirements of nuclear safety codes and relevant guides, and submitted it to the MEE (NNSA) for review and approval as one of application documents for corresponding licenses. Important contractors of the NPP, according to requirements of nuclear safety codes and relevant contracts, shall establish and implement separate *Quality Assurance*

Program applicable to the work undertaken by them. The *Quality Assurance Program* prepared by the contractors shall be submitted to the NPP operating organization for review and approval. For organizations engaged in the design, manufacture, erection and non-destructive test of civil nuclear safety equipment, their *Quality Assurance Program* shall be submitted to the MEE (NNSA) for review and approval as one of the application documents for licenses of nuclear safety equipment.

NPPs operating organizations shall also revise and update the *Quality Assurance Program* according to significant changes of organization and implementation of the quality assure system, and timely submit it to MEE (NNSA) for review to ensure the appropriateness and effectiveness of the quality assurance system.

13.3.2 Implementation of Quality Assurance System

Quality assurance is an important tool for effective management in the NPPs of China. The quality assurance system is implemented effectively through thorough analysis of the tasks to be performed, identification of the skills required, the selection and training of appropriate personnel, the use of appropriate equipment and procedures, the creation of a satisfactory working environment, defining the responsibilities of the individuals who are to perform the tasks, verifying the correct performance of activities, and preparing documentary evidence to demonstrate that the required quality has been achieved.

13.3.3 Assessment and Improvement of Quality Assurance System

Managers at all levels and the department performing the quality assurance functions of the NPPs are responsible for the supervision, evaluation and improvement of the quality assurance system. The departments performing the quality assurance functions are independent from other departments, which can directly report to the top management. The departments performing the quality assurance functions can identify problems and deficiencies of the quality assurance system by the planned internal and external quality assurance surveillance, audit, verification and evaluation, and timely take corrective actions for improvement. In addition, they will collect all kinds of quality assurance related information and analyze the quality trend, and report to the senior management on a regular basis.

In the past three years, the following work and improvements in the area of quality assurance have been completed:

(1) The NPP operating organizations have carried out effective internal and external quality assurance audit. According to written quality assurance audit procedures, and in consideration of characteristics and work practices of the

organizations, the departments performing the quality assurance functions have prepared annual plans for quality assurance audit, performed internal audit on each department, evaluated the effectiveness of system implementation within the scope of liability of each department, with the audit scope covering all departments and activities of quality and safety in the NPPs, and conducted external quality assurance audit to verify the implementation of contracts and the quality assurance systems on contractors.

(2) Quality control group activities have been widely carried out in all NPPs, providing solutions to on-site problems, including that of power plant equipments and systems. Achievements of quality control groups have been exchanged at various announcing conferences on quality control group achievements organized by the power industry, power group corporations, provincial and municipal quality associations or quality management organizations. The achievements of the two quality control groups of Changjiang NPP won the gold medal at the 42nd International Convention on Quality Control Circles; and the achievements of the two quality control groups of Changjiang NPP and the two quality control groups of Tianwan NPP won the gold medals at the 43rd International Convention on Quality Control Circles.

(3) Nuclear power corporations have integrated and optimized the resources of qualified suppliers for their respective NPPs, established the list of qualified suppliers, and performed unified, classified and graded management on suppliers of equipments, materials, spare parts, services, etc., and evaluated the qualification of qualified suppliers periodically within the corporation. All NPP operating organizations supervise, assess and share the information of the suppliers quality assurance system, working process and actual performance, perform on-site supervision for the manufacturing of the key nuclear safety equipments, and strengthen supervision, validation and acceptance in key processes.

(4) The operating organizations of NPPs in operation carried out quality assurance audit, focusing on activities for units, with emphasis on maintenances and modifications, and strengthen the awareness of adhering to the rules and regulations and strictly implementing the procedures; For each refueling outage, a quality assurance organization of refueling is established, and the refueling quality assurance surveillance plan is developed to perform quality assurance supervision during refueling outage preparation and implementation.

(5) To enhance the effectiveness of the quality assurance system, the operating organizations of NPPs under construction have prepared human resources demand plans, allocated the staff required for the positions, and conducted overall quality assurance trainings, according to the project management mode and construction progress, to meet

the needs on project construction. The non-conforming items have been divided into different types for NPPs. The disposition of different types is reviewed and approved by personnel with different responsibilities and authorities respectively.

13.4 Regulatory Review and Control Activities

The MEE (NNSA) and regional offices perform a series of supervision and inspections on important activities relating to safety and quality in each NPP by strictly following the requirements of the nuclear safety codes and relevant policy documents, conscientiously fulfilling the function on nuclear safety supervision. The following regulatory review and control activities have mainly been carried out over the three years:

(1) Reviewing and approving the quality assurance program or revision of NPPs.

(2) Carrying out the nuclear safety regulatory inspection to ensure the effectiveness of the quality assurance systems in operating organizations, with emphasis on the implementation of quality assurance system on design and construction and quality assurance system on operation of NPPs, the implementation effects of quality assurance system on operation of NPP under the combined licensing mode, the implementation and effectiveness of in-service inspection program, regular test program and maintenance program and procedures. For significant safety and quality related activities, hold-points have been selected from relevant quality plans for supervision and witness; technical review and verification have been organized for the results.

(3) For significant quality issues in NPPs, such as deviation of chemical composition of castings and forgings supplied by CF and JCFC, cracks in secondary impellers of steam-driven pumps of auxiliary feedwater system, etc., the analysis and experience feedback have been timely carried out.

14. Safety Assessment and Verification

Each Contracting Party shall take the appropriate steps to ensure that:

(i) Comprehensive and systematic safety assessments are carried out before the construction and commissioning of a nuclear installation and throughout its life. Such assessments shall be well documented and filed, subsequently updated in the light of operating experience and significant new safety information, and reviewed under the authority of the regulatory body;

(ii) Verification by analysis, surveillance, testing and inspection is carried out to ensure that the physical state and the operation of a nuclear installation continue to be in accordance with its design, applicable national safety requirements, and operational limits and conditions.

14.1 Regulatory Requirements on Safety Assessment and Verification for NPPs

It is required by the *Code on the Safety of Nuclear Power Plant Design* that comprehensive deterministic safety assessment and probabilistic safety assessment shall be carried out throughout design process for NPPs to ensure that all safety requirements on the design of the NPPs are met throughout all stages of the lifetime of the NPPs, and to confirm that the design, as delivered, meets requirements for manufacture and for construction, as built, as operated and as modified. Safety assessments shall be commenced at an early point in the design process, with iteration between design activities and confirmatory analytical activities, and shall increase in scope and level of detail as the design programme progresses. Safety assessment shall be documented in a formed that facilitates independent evaluation.

It is required by the *Code on the Safety of Nuclear Power Plant Operation* that the operating organization shall establish procedures to ensure proper design, review, control and implementation of all permanent and temporary modifications. Modifications to structure, system and component important to safety which affect the bases on which the operating license was issued, to the operational limits and conditions, and to procedures and other documents originally approved by the regulatory body shall be submitted to the regulatory body for prior approval. Systematic safety reassessments of the plant in accordance with the regulatory requirements shall be performed by the operating organization throughout its operational lifetime, with account taken of the operating experience and significant new safety information from all relevant resources.

The nuclear safety guide *Safety Assessment and Verification for Nuclear Power*

Plant provides specific recommendations for carrying out a safety assessment during the design process, as well as to the operating organization in carrying out independent verification of the safety assessment. The methods and recommendations in *Safety Assessment and Verification for Nuclear Power Plant* also applies to regulatory body for the conduct of the regulatory review and assessment. Other nuclear safety guides provide guidances and recommendations for safety assessment and verification of important systems of NPPs, such as reactor design, safety system design, protection system design, emergency power system design, etc., and also for safety assessment and verification during the operation of NPPs.

14.2 Practices on Safety Assessment and Verification for NPPs

According to the requirements in relevant nuclear safety regulations of China, safety analysis of the design for NPPs shall be conducted in which methods of both deterministic and probabilistic analysis shall be applied.

14.2.1 Deterministic Safety Analysis

The current nuclear safety regulations and the safety assessment and analysis of NPPs in China are mainly established on the basis of deterministic assessment. The deterministic safety analysis methods are a set of methods based on the concept of defence in depth, and aimed at ensuring three fundamental safety functions of reactivity control, residual heat removal and radioactivity containment, adopting conservative assumptions and analysis methods for the determined design basis, which meet the specific acceptance criteria. The deterministic safety analysis shall provide: (1) establishment and confirmation of the design basis for all items important to safety; (2) characterization of the postulated initiating events that are appropriate for the site and the design of the NPPs; (3) analysis and evaluation of event sequences that result from postulated initiating events, to confirm the qualification requirements; (4) comparison of the results of the analysis with acceptance criteria, design limits, dose limits and acceptable limits for purposes of radiation protection; (5) demonstration that the management of anticipated operational occurrences and design basis accidents is possible by safety actions for the automatic actuation of safety systems in combination with prescribed actions by the operator; (6) Demonstration that the management of design extension conditions is possible by the automatic actuation of safety systems and the use of safety features in combination with expected actions by the operator.

The safety analysis report of NPPs forms an important part of the basis on for licensing and an important part of the basis for the safe operation of NPPs. There is PSAR and FSAR for NPPs. The contents of PSAR include the analysis and

demonstration of the reliability and safety of the NPP design, and the safety measures to protect the personnel, the public and the environment against excessive radiation hazard. It covers the characteristics of the site, the design of structures, systems and components, reactor, reactor coolant and associated system, engineered safety features, radioactive waste management, safety analysis for events, human factor engineering and PSA, etc. PSAR is one of the application documents prepared by the designers organized by operating organizations and submitted to the regulatory body for applying construction permit. FSAR is one of the application documents prepared by the designers organized by operating organizations and submitted to the regulatory body for applying operating license. FSAR and PSAR are the same in writing format, only with different in the depth of the contents.

Safety analysis reports of NPPs include accurate and sufficiently precise information of NPPs and the operating conditions, typically include information on, for example, safety requirements, the design basis, site and plant characteristics, operational limits and conditions, and safety analyses in such a way that the regulatory body will be able to evaluate independently whether the safety level of the NPPs meets the requirements of regulations.

14.2.2 Probabilistic Safety Analysis

It is required by the *Code on the Safety of Nuclear Power Plant Design* that the design shall take due account of the PSA for all modes of operation and for all plant states, including shutdown, with particular reference to establishing that a balanced design has been achieved such that no particular feature or postulated initiating event makes a disproportionately large or significantly uncertain contribution to the overall risks, and that, to the extent practicable, the levels of defence in depth are independent; providing assurance that the cliff edge effects will be prevented; and comparing the results of the analysis with the acceptance criteria for risk where these have been specified. The *Safety Assessment and Verification for Nuclear Power Plant* further specify the methods, scope and objectives of PSA.

MEE (NNSA) encourages and supports the implementation of PSA. In 2010, *Policy of Technology: Application of PSA Technology in the Area of Nuclear Safety* was issued, and in 2017, *Policy of Technology for Improving Maintenance Effectiveness of Nuclear Power Plants* was issued.

NPP operating organizations in China have carried out PSA at the design stage to assess the overall safety level of NPPs and identify weaknesses in design by referring to relevant laws and regulations and PSA industry technical standards, and fully

considering the latest development in PSA technology. At the design stage, the PSA related activities have completed or are carrying out at present includes Level I PSA of internal event under operation and low power operation and shutdown conditions, Level I PSA of internal fire, Level I PSA of internal flooding , Level I PSA of external hazard, Level II PSA, spent fuel pool PSA, etc. At the preliminary design stage, due to the limitation of NPP information, simplified and conservative models and data are usually adopted for PSA to demonstrate that NPPs meet safety objectives. At the final design stage, the NPP operating organizations will update the PSA model and data according to the improvement of relevant design information. At the operation phase, NPPs will appropriately review the conformity of PSA model with the actual conditions of NPPs on the basis of considering modification in design and operation phases, new technical information, current analysis methods and new operation data.

At the operation phase, PSA is widely used in daily operation activities, for real-time risk monitoring of NPPs using PSA tools such as risk monitors, mitigation system performance indicators, and significance determination process for items important to safety. It is required to track and calculate the performance of the safety system, collect and analyze operating experience and update PSA model input data, perform demonstration of important documents of NPP (such as technical specifications, supervision requirements, in-service inspection program, etc.) for optimization, etc., so as to improve the safety management level of NPP and realize the optimization of resource utilization.

At present, the pilot projects of the risk-informed application are being implemented for operating NPPs. Several NPP operating organizations have carried out the optimization of risk-informed technical specifications and risk-informed in-service inspections, and have completed the optimization of technical specifications for auxiliary feedwater system, safety injection system and residual heat removal system.

14.2.3 Periodic Tests

As required by nuclear safety regulations, surveillance program for items important to safety have been developed for NPPs on the basis of the operating experience of NPPs at home and abroad and the surveillance requirements for equipment supplied by nuclear equipment manufacturers. The contents cover the surveillance of plant parameters and system status, surveillance of chemistry and radiological chemistry sampling, calibrations of the instrumentation, inspection and functional testing for structures, systems and components important to safety, and assessment of monitoring and calibration results.

Periodical tests are the main measures for implementing surveillance program, which are used for determining whether or not the safety-related systems and components can continuously perform their functions as required by design. The procedures of periodical tests are implemented after the system and equipment are transferred to the operating organizations.

The NPP operating organizations perform surveillance, inspection and testing of safety-related systems in strict accordance with the items and frequencies specified in the operation technical specification, the test results are recorded and evaluated. The defects and abnormalities identified are repaired and corrected in time, and test was made again for verification until the functions and parameters meet the acceptance criteria. The recovery of the system is strictly complied with the time limit requirements of the operating technical specification. The implementation of periodic tests related to nuclear safety is independently reviewed and witnessed by nuclear safety engineers or quality assurance personnel. The implementation of periodic tests is under control. From 2016 to 2018, the annual cumulative one-time success rate of periodic tests in all NPPs exceeded 96%, reaching the target value required by management. For some NPPs, the value had been kept above 99%. The results of safety system surveillance, inspection and tests show that the functions of safety system which is operated stably and reliably meet the design technical requirements, supporting the safe operation of NPPs.

The NPP operating organizations in China constantly improve the technical means and managing processes of surveillance, develop and apply on-line performance monitoring system and platform to analyze all sorts of collected monitoring data, timely identify adverse trend and take corresponding measures for improvement in combination with the management system of specific areas such as operation, maintenance, chemistry and radiation protection.

14.2.4 In-Service Inspection

According to relevant regulations, guides and standards for in-service inspections and by drawing on good practices of foreign NPPs, In-service inspection programs and procedures have been formulated for NPPs to systematically carry out various pre-service and in-service inspections, covering that of nuclear island, conventional island and BOP. Inspection methods mainly include visual inspection, radiographic examination, eddy current inspection, ultrasonic inspection, liquid penetration inspection, etc. For systems and equipment important to safety, such as reactor pressure vessel, containment components, pressure pipes, heat transfer branches and steam

generator heat transfer tubes, and for systems of conventional island and BOP, such as components of steam turbine generators, various pressure vessels, pressure pipes, condenser titanium tubes and other metal parts, corresponding in-service inspection plans have been prepared and implemented during operation and outage as planned.

From 2016 to 2018, a total of 72 in-service inspections were carried out in NPPs during refueling outage. Defects identified during refueling outage and daily in-service inspection have been recorded, evaluated and corrected and verified. The defects identified in inspections include damage of bolt and screw threads of main components such as reactor pressure vessel, steam generator, pressurizers and reactor coolant pump, wearing of thimbles of in-core instrumentation system, defects in steam generator heat transfer tubes, dent and weld seam flaws in condenser titanium tubes, weld seam flaws in supporting rod of HP casing, weld seam of small diameter tubes in the conventional island exceeding the standard, evidently thinning of steam pipelines of hydrotest pump turbine generator, carbon deposit and fatigue cracks in turbine generator bearing bush/thrust shoe and so on.

Pre-service and in-service inspections are all performed by qualified inspection personnel using qualified inspection equipment in accordance with approved inspection procedures; and quality assurance program and quality control procedures are strictly implemented in inspections, to ensure the effectiveness of inspection results. Through the in-service inspections, weaknesses in NPPs have been identified and corrected, ensuring the integrity of safety barriers and safety operation of NPPs. The results of pre-service and in-service inspection are subject to the review by MEE (NNSA).

In order to reduce the possibility of lack or omission of inspection or misjudgment during in-service inspection, China has established the in-service inspection capability verification system based on the latest industry standards and practices.

In-service inspection capability verification has been actively carried out for NPPs according to the requirements of relevant laws and regulations. For example, Tianwan NPP has carried out the verification of the VVER unit in-service inspection capability for Units 3 and Unit 4. The verification plan and technical specification of VVER unit in-service inspection capability have been developed. The verification of important equipment and projects has been conducted, such as automatic ultrasonic inspection for reactor pressure vessel, eddy current inspection for steam generator, and automatic ultrasonic inspection for main pipe. The non-destructive testing technology for VVER unit has been effectively verified to improve the reliability of inspection technology and the inspection capability of technical personnel.

In addition, the development of new technologies and processes for in-service

inspection has been actively tracked by NPPs. According to OLE requirements, the ultrasonic inspection technology for the reactor internals baffle bolts and the inspection technology for the defects of the lower head of the reactor pressure vessel was developed by Qinshan NPP, which were successfully applied during the 18th refueling outage. With the combination of gamma radiation source and digital radiographic imaging system, a comprehensive scheme including the initial scanning with high frequency guided wave and detailed fixed-point scanning with phased-array was designed by Daya Bay Nuclear Power Base to realize rapid detection of pipeline corrosion in fire fighting water production system. It has been applied to the inspection of fire fighting water pipelines of 6 units in Daya Bay Nuclear Power Base and has been extended to other NPPs of CGNPC. Automatic ultrasonic diffraction inspection technology for reactor pressure vessels and automatic ultrasonic phased-array inspection technology for steam generator welds has been developed by Tianwan NPP, and the high definition television technology, replacement of ray by ultrasonic wave and eddy current inspection for bolt are being researched to improve the effectiveness and reliability of in-service inspection. During the pre-service inspection of Unit 1 and Unit 2 in Sanmen NPP, the newly developed automatic inspection equipment was used for the welding seams of the reactor pressure vessel head, steam generator and reactor coolant pump casing; and phased-array ultrasonic inspection technology was used for parts of pipeline welding seams.

14.2.5 Ageing management

The ageing management of NPPs is mainly performed with reference to *Ageing management for Nuclear Power Plant* and *Periodic Safety Review for Nuclear Power Plant* as well as good practices in the nuclear power industry at home and abroad. Various ageing management related activities have been systematically carried out at all stages of the construction, commissioning and operation of NPPs. It is required to establish and improve the ageing management system for NPPs, including the ageing management organization and document system; perform monitoring, testing, sampling and inspection to assess the expected ageing mechanisms in the design of NPP and identify the unexpected possible conditions or performance degradation during operation.

The ageing management program have been established for all NPPs in operation clearly defining the responsibilities, organizational structure, methods and processes of ageing management, which are continuously improved based on the results of PSR and operating experiences of domestic and foreign NPPs.

Screening and classification of ageing sensitive equipment have been carried out for NPPs. For example, for Tianwan NPP, the screening of objects of ageing management has been completed for Unit 1 and Unit 2, specific ageing management programs for reactor pressure vessels, steam generators, containment and power cables have been developed, the monitoring and testing of the status of important equipment have been clearly stipulated, the screening and grading of ageing sensitive equipment have been conducted, relevant testing have been conducted and management strategies have been formulated.

Samples have been retained for ageing management of structures, equipment and other parts of the facilities. For example, cable samples have been retained to assess the cable ageing condition in Qinshan Nuclear Power Base. The inspection on ageing condition of structures has been completed for Qinshan NPP. For Tianwan NPP additional cable samples have been retained during the operation of Unit 1 and Unit 2, the cable sample have been retained during commissioning phase of Units 3 and 4; and environmental monitoring device were installed to record temperature and irradiation data. The thermal ageing test is under way to obtain ageing life curve of cable. Preliminary preparation for sample of containment has been carried out. Fuqing NPP has retained cable samples for Unit 1-4. For Unit 1 of Sanmen NPP, the placement of cable samples was completed during the construction phase. It is planned to conduct monitoring tests on cable sample every 5 years. The 1E cables samples have been retained for ageing management of Unit 1 of Haiyang NPP. The cable samples have been retained for for Unit 1 of Taishan NPP. For Fang chenggang NPP, cable samples for Unit 1 and the material samples of the main equipment of Unit 1 and Unit 2 have been retained, including steam generator, pressurizer, reactor pressure vessel, reactor internals, main pipe, etc. The ageing test of concrete for marine structures of Units 1 and 2 and buildings in the nuclear island was completed; and the concrete samples of structures of units 3 and 4 were retained as planned.

The operating organizations have carried out ageing management of civil structures and pipelines. For example, Daya Bay Nuclear Power Base has carried out ageing management of civil structures, including ageing detection and evaluation of nuclear safety structures, detection and repair of outer wall concrete of containment, corrosion prevention and bolt replacement of gantry steel structure, and modification of BONNA pipe lining of essential service water system. For Tianwan NPP, ageing management program for containment has been established, ageing analysis of other structures were carried out, pipelines defects caused by ageing are identified through non-destructive testing, and the content of harmful ions is controlled through water chemistry to prevent

the stress corrosion. The ageing test of containment has been performed for Unit 1 of Taishan NPP, through which the “zero point data” required for the ageing test of structures prior to service is obtained for comparison with subsequent tests to analyze the ageing trend.

The operating organizations have actively verified the effectiveness of the neutron absorption in spent fuel pools, designed and manufactured verification equipment, and formulated verification methods. For example, in 2016, Qinshan Nuclear Power Base lifted and cut the neutron absorber assembly with the worst storage conditions in the spent fuel pool, sampled boron carbide-polyethylene neutron shielding materials, and conducted post-irradiation inspection to carry out corresponding verification and supervision. The samples of the neutron absorbing materials in the spent fuel pool have been retained for Fuqing NPP to ensure that the neutron absorbing capacity of the materials in the spent fuel pool can be maintained throughout the life time of the NPP. The neutron absorb monitoring program and verification method for the spent fuel pool have been developed for Unit 1 and Unit 2 of Fangchenggang NPP.

In the PSR of NPPs, ageing management has been reviewed as a factor to confirm that ageing has been effectively managed, all required safety functions have been delivered, corrective plans have been development and implemented for identified weaknesses for effectively ageing management. For example, the operating organization of Daya Bay NPP and LingAo NPP has carried out thermal fatigue monitoring and research, the preparation of ageing management programs for ventilation systems and storage batteries, and the effectiveness evaluation of ageing management programs based on PSR results. The operating organization of Tianwan NPP has set up dedicated positions, upgraded the ageing management documents and added ageing management measures according to PSR results.

In addition, the operating organization of Qinshan NPP has implemented the review of ageing management and time-limited ageing analysis in coordination with OLE. The operating organization of Qinshan NPP has completed OLE safety assessment and submitted OLE application to the MEE (NNSA). It is planned to complete all review and dialogue in 2019; FSAR upgrade and submission, implementation of modifications for OLE in 2020. The OLE application is planned to get the approval in 2021. The operating organization of Daya Bay NPP launched OLE demonstration and evaluation in 2016. It has completed the formulation of OLE demonstration and evaluation guidelines; basically completed the definition and screening of OLE safety assessment scope and the consistency review of ageing management; completed the preparation of part of the ageing management program, the

effectiveness review of ageing management program and part of the time-limited ageing analysis; completed the retrospective assessment of the environmental impact assessment, and started the revision and supplement of FSAR and technical specifications and the preparation of modification plan as well as the predictive evaluation of environmental impact.

The NPP operating organizations have developed ageing management database. CNNP has developed unified ageing management database, including modules of basic equipment information, ageing management, obsolete management, action management, serving as ageing management platform for all NPPs belong to CNNP.

14.2.6 Periodical Safety Review

Based on the requirements of regulations, the periodical safety review of NPPs in accordance with the regulatory requirements shall be performed by the operating organization (every ten years in general) throughout the operational lifetime, with account taken of the changes of regulations, operating experience and significant new safety information from all relevant resources. The PSR is required to assess the cumulative effects of NPPs in ageing, modification, operating experience, technical developments and siting aspects. The PSR includes an assessment of plant design and operation against applicable current safety standards and operating practices, with the objective of ensuring a high level of safety throughout the lifetime of the NPPs. 14 safety factors which were divided into five subject areas of plant status, safety analysis, performance and feedback of experience, management and environment to be reviewed in the PSR, are plant design, actual condition of structures, systems and components, equipment qualification, ageing, deterministic safety analysis, probabilistic safety analysis, hazard analysis, safety performance, use of experiences from other plants and research findings, organization and administration, procedures, human factors, emergency planning, and radiologic impact on the environment. Each safety factor is reviewed using current methods and the findings are assessed against current safety standards and practices. Reasonable and practicable corrective actions and/or safety improvements are determined and an implementation plan is agreed, with account taken of the interactions and overlaps between safety factors. Overall assessment of any weakness that cannot be reasonably and practicably corrected is made. The risks associated with the unresolved shortcomings should be assessed and an appropriate justification for continued operation should be provided. The results of relevant studies, safety reviews and PSA should be utilized in the PSR to minimize any duplication of effort.

Based on the results of the second PSR review, twelve major modifications have been carried out for Qinshan NPP, including the modification of pipe hangers and the modification of Class 1E cable tray, ten of which have been completed and the remaining two are planned to be completed by 2020. Based on the results of the first PSR review, four major modifications have been carried out for Unit 1 and Unit 2 of Qinshan Phase II NPP, two of which have been completed and the remaining two are planned to be completed by the end of 2019. Based on the results of the first PSR review, a major modification has been carried out for Unit 1 and Unit 2 of Qinshan Phase III NPP. In 2018, the third PSR has been launched for Qinshan NPP.

The Daya Bay Nuclear Power Base conducted PSR reviews from 2015 to 2018. In 2018, the preparation of PSR review programs for Daya Bay NPP and LingAo NPP was completed.

The first PSR review for Unit 1 and Unit 2 of Tianwan NPP was completed in 2015. Based on the results, three major modifications are implemented, two of which have been completed and the remaining one is ongoing.

14.2.7 Re-evaluation of Hazard Assumptions

The overall evaluation was performed for the hazard assumption that may affect the safety of NPPs caused by external environment in the license review and approval during the siting, construction and operation stages of the NPPs, and the corresponding re-evaluation has been carried out in the PSR once every ten years. The review results show that all NPPs have the capacity against the impact of the hazard assumption.

After the Fukushima nuclear accident, the NPP operating organizations have carried out safety margin assessment on significant external events in accordance with nuclear safety management requirements, and improved and revised analysis and assessment reports on safety margin.

PSA analysis has been performed for each unit in Qinshan Nuclear Power Base, including Level I PSA under power/low power conditions, Level II PSA under power condition, internal flooding PSA, seismic PSA, internal fire PSA, and spent fuel pool PSA. In the development of internal fire PSA for Fangjiashan NPP, the fire risk level was evaluated; the consequences of cable failure caused by fire were analyzed; each analysis area was modeled, and the risk assessment was carried out for equipment and ignition sources mainly causing the fire risk. In development of the seismic PSA for Fangjiashan NPP, the analysis of structures response was completed, the structural model of floor response spectrum was established, and the risk under earthquake was evaluated.

The Daya Bay Nuclear Power Base has established the Level I and Level II seismic PSA models for six units to identify the weaknesses and use in the assessment of the main feedwater isolation modification and the treatment scheme for the reactor coolant pump damper oil leakage.

The external flooding margin evaluation and anti-seismic margin evaluation have been carried out for Tianwan NPP. For Units 3 and 4, the Level I seismic PSA models under power, low power and shutdown conditions are being developed.

The seismic margin analysis and seismic PSA analysis and development have been completed for Unit 1-4 of Fuqing NPP. For the hazard that may lead to external flooding, such as high water level and tsunami, screening and analysis has been conducted.

The seismic margin assessment based on PSA at the design stage has been carried out for Sanmen NPP and Haiyang NPP. The results show that the core damage and large release of radioactive substances caused by the earthquake can well be mitigated for Sanmen NPP and Haiyang NPP. Currently, external seismic PSA models are being developed for both NPPs.

The design standards for the flood drainage ditch and rain drainage system in case of external floods beyond design basis have been re-evaluated for Taishan NPP. The tsunami numerical simulation calculation and impact assessment research have been carried out. The seismic margin assessment has been carried out, and the results show that the actual seismic design of main structures and equipment has a great margin.

The seismic margin has been assessed for Fangchenggang NPP to identify the weaknesses in seismic design, and the corrective actions have been developed. The analysis of external hazard PSA is completed. The reactor core damage frequency caused by major external hazards is assessed through qualitative/quantitative screening and detailed analysis of external hazards and the quantitative calculation and envelope analysis, which indicates the distribution of reactor core damage frequency in different external hazards and the priority with importance.

14.3 Regulatory Review and Control Activities

The MEE (NNSA), by way of formulating regulations, guides, policies and standards of nuclear safety, put forward requirements for safety assessment and verification of NPPs, and by way of nuclear safety review, nuclear safety inspection, PSR, etc., perform inspection on related activities within the lifetime of NPPs, to confirm that the NPPs and their activities are in conformity with the safety objectives, principles and criteria.

In addition, the MEE (NNSA) inspects the implementation of the safety

management requirements and the conditions stipulated in licenses, urges the correction of items not in line with the nuclear safety management requirements and the conditions stipulated in licenses through nuclear safety regulatory inspection. Throughout the lifetime of the NPPs, in consideration of regulation changes, operating experience and significant new safety information from relevant resources, the MEE (NNSA) requires the operating organization to perform systematic safety re-assessment by way of PSR. The review strategy and safety factors to be assessed are subjected to review by the MEE (NNSA), for the purpose of determining the extent of validity of the existing safety analysis report.

From 2016 to 2018, main regulatory review and control activities conducted by the MEE (NNSA) are as follows:

(1) Reviewing the safety analysis report, in-service inspection program and PSR report of the NPPs.

(2) Establishing the equipment reliability database. In order to regulate the acquisition and processing of equipment reliability data, the MEE (NNSA) has revised and improved the *Guide on Acquisition of Equipment Reliability Data in Nuclear Power Plant*, and acquired the equipment reliability data of all NPPs by using the existing equipment reliability database of operating NPPs. The *Report on Equipment Reliability Data of Nuclear Power Plants in China* is issued every year based on the analysis and collation of relevant data. Equipment reliability data related activities plays a significant role on the guidance for NPPs and related research and design organizations to carry out PSA, implement maintenance rules and perform reliability-centered maintenance and other related activities.

(3) Promoting the application of risk-informed regulation. The MEE (NNSA) requires operating organizations to carry out pilot work of PSA application based on their own actual conditions, laying a good foundation and accumulating experience for the subsequent establishment of risk-informed regulatory models and regulatory standards. Each operating NPP has selected its own pilot project for PSA application according to the actual situation, such as the application of in-service inspection optimization in Daya Bay NPP and Tianwan NPP. The MEE (NNSA) is carrying out relevant activities according to the pilot application plans, and gradually promoting the pilot application of risk-informed regulation in operating NPPs. The MEE (NNSA) has approved the optimization report of the technical specifications for the operation of auxiliary feedwater system/auxiliary power supply system/containment spray system for Units 3 and 4 of LingAo NPP, the revision of the supervision requirements for periodic tests of safety-related systems and equipment of Daya Bay NPP and LingAo NPP, and

the demonstration report for online maintenance of heat exchangers of equipment cooling water system and essential service water systems of Daya Bay NPP and LingAo NPP. It has approved the optimization project of risk-informed in-service inspection and the online maintenance project for safety system for Units 1 and 2 of Tianwan NPP.

(4) Implementing dedicated inspections for important activities such as installation of important equipment and commissioning of safety systems by intensifying the process surveillance for construction and commissioning; and made more efforts to the investigation and handling of construction events and major nonconformity, to promptly identify and properly address problems in the construction of NPPs.

(5) Continuously promoting OLE activities in NPPs. The MEE (NNSA) has issued OLE technology policy, which specifies OLE's route, scope, object, method, review basis, etc. The review of OLE application of Qinshan NPP is underway. In addition, it has learned from the relevant regulatory experience of the NRC, and organized a Sino-U.S. NPP exchange seminar to exchange relevant work.

(6) Promoting the inspection and safety review of HPR1000. The MEE (NNSA) has organized the development of the nuclear safety inspection program at the construction phase of HPR1000 Project. It has summarized the review experience of HPR1000, promoted the compilation of technical insights on issues important to safety and determined the list of issues important to safety and technical insights.

15. Radiation Protection

Each Contracting Party shall take the appropriate steps to ensure that in all operational states the radiation exposure to the workers and the public caused by a nuclear installation shall be kept as low as reasonably achievable and that no individual shall be exposed to radiation doses which exceed prescribed national dose limits.

15.1 Basic Requirements of Radiation Protection

The Chinese government has promulgated a series of laws, regulations and national standards to ensure the implementation and achievement of radiation protection objectives.

(1) It is stipulated by *Nuclear Safety Act*:

The nuclear facility operating organizations shall strictly control radiation exposure and ensure that relevant personnels are not exposed to radiation exceeding the dose limit specified by the state and that radiation exposure maintains at a level as low as reasonably achievable.

The nuclear facility operating organizations shall monitor the type and concentration of radionuclides in the environment surrounding the nuclear facility as well as the total amount of radionuclides in the effluents of the nuclear facility and regularly report the monitoring results to the environmental protection department of the State Council and the environmental protection department of the people's government of the province, autonomous region or municipality directly under the Central Government where it is located.

(2) Provisions of the *Act of the People's Republic of China on Prevention and Control of Radioactive Pollution*:

The NPP operating organizations shall be responsible for prevention and control of radioactive pollution in their own organization, accept the supervision by the administrative department of environmental protection and other relevant departments, and bear the liabilities in accordance with the law for the radioactive pollution it has caused.

The NPP operating organizations shall monitor the type and concentration of radionuclides in the environment surrounding the NPP as well as the total quantities of radionuclides in the effluents of the NPP and regularly report the monitoring results to the environmental protection department of the State Council and the environmental

protection department of the people's government of the province, autonomous region or municipality directly under the Central Government where it is located.

The NPP operating organizations shall minimize the quantities of radioactive waste generated. The discharge any of radioactive waste in gaseous and liquid into the environment shall meet the national standards on the prevention and control of radioactive pollution; the results on the measurement of discharge shall be reported to the administrative department for environmental protection on a regular basis.

(3) Provisions of the *Law of the People's Republic of China on the Prevention and Control of Occupational Diseases:*

The employers shall equip protective equipment and alarm devices, and ensure that their personnel exposed to radiation carry a personal dosimeter.

The employers shall designate persons responsible for the daily monitoring of occupational disease hazard factors and ensure the normal condition of the monitoring system.

(4) On October 8, 2002, the national standard *Basic Safety Standard on the Ionization Radiation Protection and Radioactive Source* was revised and promulgated, which specifies the individual dose limit as follows:

Occupational exposure

- The annual average effective dose for five consecutive years determined by the regulatory body (no retroactive average shall be made) is 20mSv;
- The effective dose limit in any single year is 50mSv;
- The annual equivalent dose limit to the eye lens is 150mSv;
- The annual equivalent dose limit to the extremities (hands and feet) or the skin is 500mSv;
- In special circumstances, the period of annual average dose of 20mSv can be extended to ten consecutive years, and in this period, the annual average effective dose for any worker shall not exceed 20mSv, and shall not exceed 50mSv in any single year; and the circumstances shall be reviewed when the dose accumulated by any worker since the start of the extended period reaches 100mSv; the temporary change of the dose limit shall not exceed 50mSv in any single year, and the period of the temporary change shall not exceed five years.

Public exposure

- The annual effective dose limit is 1mSv;
- In special circumstances, the effective dose limit in a single year is up to 5mSv if the annual average dose does not exceed 1mSv over five consecutive years;
- The annual equivalent dose limit to eye lens is 15mSv;

- The annual equivalent dose limit to skin is 50mSv.

Emergency exposure

When undertaking intervention under these circumstances, all reasonable efforts shall be made to keep doses to workers below twice the maximum single year dose limit, except for life saving actions, in which every effort shall be made to keep doses below ten times the maximum single year dose limit in order to avoid deterministic effects on health. In addition, workers undertaking actions in which their doses may approach or exceed ten times the maximum single year dose limit shall do so only when the benefits to others clearly outweigh their own risk.

- Saving life or preventing serious injury;
- Averting large collective dose;
- Preventing the development of catastrophic conditions.

(5) In 2016, the MEE (NNSA) revised the *Code on the Safety of Nuclear Power Plant Design*, which specified the fundamental safety objective that to protect individuals, society and the environment against radiological hazards by establishing and maintaining effective defences in NPPs. To achieve the fundamental safety objectives, it is required to ensure that for all operational states of NPPs and for any associated activities, doses from exposure to radiation within the installation or exposure due to any planned radioactive release from the installation are kept below the dose limits and as low as reasonably achievable. In addition, it is required to take measures for mitigating the radiological consequences of any accidents, if they were to occur. Radiation protection in design is also required to keep all sources of radiation under strict technical and administrative control.

(6) On September 1, 2011, the MEE (NNSA) revised the *Regulation on the Environmental Radiation Protection of Nuclear Power Plant*, clearly specifies the effective dose caused to any individual (adults) of the public by the release of radioactive materials of a NPP to the environment, and the annual discharge limits of the airborne and liquid radioactive effluents.

The effective dose of radioactive materials released by all nuclear power reactors on any site to any individual of the public shall be less than the dose constraint value of 0.25mSv every year. Operating organizations of NPPs shall set respective dose management target values for airborne and liquid radioactive effluents according to the dose constraint values approved by examination and regulatory department.

The NPP operating organizations shall control the annual total discharge of radioactive effluents for every reactor. The control values of the reactors with thermal power of 3,000 MW are shown in Table I and Table II. For the reactors with the thermal

power higher than or lower than 3,000 MW, the control values shall be adjusted appropriately according to their power.

Table I Control of Airborne Radioactive Effluents (Unit: Bq/a)

	Light water reactor	Heavy water reactor
Inert gas	6×10^{14}	
Iodine	2×10^{10}	
Particles (Half life8d)	5×10^{10}	
Carbon-14	7×10^{11}	1.6×10^{12}
Tritium	1.5×10^{13}	4.5×10^{14}

Table II Control of Liquid Radioactive Effluents (Unit: Bq/a)

	Light water reactor	Heavy water reactor
Tritium	7.5×10^{13}	3.5×10^{14}
Carbon-14	1.5×10^{11}	2×10^{11} (except tritium)
Other nuclides	5.0×10^{10}	

For a site with multiple reactors of the same type, the annual total discharge from all units shall be controlled within 4 times the values specified in Table I and Table II. For a site with multiple reactors of different types, the annual total discharge from all units shall be approved by the regulatory authorities.

(7) China requires that all NPPs shall develop and implement radiation protection programs and procedures before the initial fuel loading. The radiation protection programs shall meet the safety requirements of national nuclear safety regulatory authorities and comply with relevant national standards on radiation protection and safety of radiation sources. The radiation protection programs shall be updated at least every 5 years. The radiation protection programs or procedures shall set reasonable dose management target values respectively for different types of activities as to the dose constraint values.

(8) After the Fukushima nuclear accident, the MEE (NNSA) organized the preparation of the *General Technical Requirements for Improvement Actions in Nuclear Power Plants after Fukushima Nuclear Accident*, stipulating that the arrangement of monitoring facilities and monitoring points in NPPs shall be reasonable and representative to enable the facility functions specified in the emergency monitoring plan for environmental radiation under the accident conditions of NPPs. When extreme external events lead to the unavailability of environmental monitoring facilities, there shall be appropriate backup wide-range monitoring means or means to restore the

availability of monitoring facilities in a timely manner, ensuring that on-site monitoring data can be provided for environmental quality evaluation of NPPs and the surroundings.

15.2 Application of ALARA Principle in NPPs

15.2.1 Application of ALARA Principle in NPP Design

(1) General design considerations

1) Proper layout and shielding shall be adopted for the SSCs which contain radioactive materials.

2) In design, importance shall be attached to minimizing the number of activities and staying time of personnel in radiation areas, and reducing the possibility of personnel within the plant area being exposed to contamination.

3) The radioactive wastes shall be processed into proper forms to facilitate transportation, storage and disposal;

4) Measures shall be taken to reduce the quantity and concentration of radioactive materials dispersed in the plant and released to the environment.

(2) Design consideration for equipment

1) Reliable and durable equipment, components and materials shall be selected to reduce or eliminate the need for maintenance;

2) The coating materials for equipment and components shall be selected to facilitate flushing and decontamination.

3) Modularized designs shall be adopted for equipment and components to facilitate disassembly and replacement, or moving to a low radiation area for repair;

4) Redundant equipments and components shall be prepared to reduce the demands for immediate repair when radiation levels are too high and there is no feasible method to reduce the radiation level.

5) It shall be possible to perform operation, repair, maintenance, monitoring and inspection for equipment and components remotely.

(3) Design consideration for equipment layout

1) Improving the accessibility of equipment;

2) Providing shielding for radioactive equipment;

3) Providing proper and sufficient ventilation and lighting;

4) Providing proper monitoring system and equipments;

5) Controlling contamination: performing distinct isolation between contaminated area and non-contaminated area, and decontaminating the contaminated area;

6) The processing technology and detection of radioactive materials;

7) Arranging equipments, instrumentations and sampling points in low radiation area.

15.2.2 Application of ALARA Principle in NPP Operation

For three years, operating NPPs have implemented the ALARA by improving source terms control technology and volume-reduction processing of radioactive wastes; all possible and reasonable radiation protection measures have been adopted to ensure realization of the radiation protection objectives:

(1) Improving radiation protection management systems: while maintaining the effective operation of existing management systems, according to the previous experience and practice, NPPs have constantly adjusted and improved the organizational structure, radiation protection programs and relevant procedures. The ALARA Committee has been established to discuss and make decisions on major issues of radiation protection and radioactive waste management, coordinate various line departments, provide support and guidance for radiation protection and radioactive waste management, and provide necessary resources and financial support. Relevant working groups in the area of radiation protection have been established to urge the implementation of radiation protection related activities, track and evaluate radiation protection actions, promote the implementation of dose reduction measures and enhance the capability in supervision of radiation protection. The virtual reality scenario for radiation protection has been designed to improve the training effect on contamination prevention by applying new technologies.

(2) Dose target management: the operating organizations periodically monitor and assess the target of radiation protection management and constantly optimize dose limits through management improvement and technical transformation. The system of radiation protection performance indicator has been established to continuously track and analyze the trend, identify the management weakness and develop effective management measures.

(3) Source terms control and reduction of radiation levels: on-line chemical decontamination of nuclear auxiliary systems has been implemented in some NPPs during refueling outage. The analysis results show the effectiveness of the decontamination.

(4) Technical modification and pollution prevention measures: the capacity of radioactive materials monitoring has been improved by installing on-line tritium monitoring system, regional shielding, overall shielding of equipment and shielding of equipment parts and adding vehicles and radiation monitoring channels for personnel,

etc. The surveillance of the prerequisite on the activities related to radioactive systems and equipments has been established, and the radiation protection witness points of open activities have been added.

Radiation exposure of employees mainly occurs in the refueling outage of the NPPs, therefore the operating organizations have paid full attention to the radiation protection activities during the refueling outage. Measures have been effectively applied and strengthened during the refueling outage, such as following up major items by designated person, enhancing contamination control on site, preparing and implementing ALARA plan; training and examination of maintenance contractors; and strengthening the boundary management of radiation protection, control of item transfer, contamination control, field shielding, area isolation and exercises. After the refueling outage, the operating organizations have systematically assessed the radiation protection for the refueling outage, compared it with the preset radiation protection management target value, analyzed the causes for deviations and developed relevant corrective actions. By strictly performing these measures, the operating organizations have guaranteed the integrity of the boundary of radiation control zone during the refueling outage, effectively controlled radioactive materials in the course of transfer and reduced exposure dose of workers.

15.3 Exposure Control for Workers in NPPs

In addition to considering the radiation control for personnel in view of design and source terms control, the operating organization also regulates human behaviors in the control area by formulating a complete radiation protection programs and relevant management procedures. In addition, the dose target value has been monitored and assessed regularly; and the annual dose indicator and refueling outage dose indicator are set for assessment. Special supervision has been carried out for major activities related to radiation; observation of radiation protection behaviors of on-site personnel and optimization and improvement of activities with high radiation risks have been carried out. New protection measures have been continuously researched to ensure effective control of personnel occupational exposure in the NPPs.

The monitoring results of the occupational exposure show that the annual average effective dose for on site personnel in the operating NPPs is far below the dose equivalent limit set by the national standards. For details, please refer to Appendix VIII.

15.4 Environmental Radioactivity Monitoring

On the basis of the critical nuclides, critical exposure pathway and critical groups determined in the environmental impact report (EIR), operating organizations have

prepared environmental monitoring programs to monitor the radioactivity level in the environment, to ensure that the requirements of national laws and regulations are met, the discharges of effluents are kept within discharge limits, and protect the public against radiation caused by operation of NPPs. The NPP operating organizations provide environmental radiation monitoring data to perform the following assessment and analyses:

- The effectiveness of controlling the release of radioactive materials to the environment;
- The public exposure by the radioactive effluents from NPPs;
- The long-term variation tendency of environmental radioactivity level;
- The diffusion and migration of radioactive nuclides in the environment;
- The effectiveness of assessment models adopted and the accuracy of environment parameters.

(1) Background survey before operation

Two years of radioactive background investigation shall be completed before the NPP is put into operation. Before operation, the operating organizations have monitored and recorded the level of the environmental background, to ensure the scope and frequency of environmental monitoring are representative and meet the requirement of related regulations.

(2) Environmental radiation monitoring

The NPP operating organizations make full use of the data obtained from the pre-operation survey to optimize the environmental monitoring while meeting the requirements of environmental impact assessment. The keys to environmental monitoring are the critical pathways and critical nuclides that have significant contributions to the dose of critical groups. Based on the national environmental protection regulations and environmental radiation monitoring standards, the NPP operating organizations have effectively monitored and evaluated the environment by preparing the environmental monitoring programs. For three years, the measurement and analysis results of radiation level, living organisms, air, soil and sea mediums in the environment surrounding the NPPs indicate that the operating NPPs in China have not caused bad influence to the environment and the environmental radiation level is within the scope of natural background.

(3) Monitoring of radioactive effluents

After the NPP is put into operation, airborne and liquid radioactive effluents shall be monitored. The measuring contents include the total discharge amount, the discharge concentration and the main nuclides to be analyzed. The monitoring results indicate that

the radioactive effluents from each NPP during operation are below the control values specified by national standards.

(4) Meteorological observation

To get information about atmospheric diffusion, the NPP operating organizations have developed meteorological monitoring programs; representative locations are selected to perform continuous monitoring of the wind direction, wind speed and air temperature at different elevations above ground, as well as conventional meteorological elements of wind, temperature, pressure, humidity, and precipitation at ground meteorological stations. In addition, NPPs have established communication channels with the meteorological department of provinces where they are located, to obtain necessary meteorological data of larger range.

(5) Emergency environmental monitoring in case of accident

The NPP operating organizations established emergency environmental monitoring plans before initial fuel loading, which specify the derived intervention level to enable assessment of monitoring results and determine as early as possible whether it is necessary to take relevant actions.

Under the emergency situation of nuclear accident, the operating organization of NPPs and local nuclear and environmental radiation monitoring departments mainly undertake the environment radiation monitoring responsibilities; the off-site Emergency Response Committee (the local provincial government usually) will take the lead in sufficiently coordinating the resources and activities of all related parties and unify the action. In the early phase of a nuclear accident, it mainly relies on the on-site emergency monitoring resources and strength of the NPP operating organization, which is responsible for environmental radiation monitoring in the vicinity of the site. In the later stage of the accident, environmental radiation monitoring mainly relies on off-site monitoring resources and strength due to the recovery activities in a large area. In the intermediate state of the accident, environmental monitoring is jointly carried out by both on-site and off-site parties.

(6) Monitoring of radiation effects on the public

The NPP operating organizations evaluate the radiation effect on the general public during the operation by using the data obtained from the monitoring of the accumulative radiation dose at the boundary of plant area and the analysis of samples of the environment media such as the atmosphere dust, the land-living organisms, the soil and water.

The environmental radiation monitoring stations of provinces where NPPs are located have performed the monitoring of the environment around NPPs. The results

indicate that the maximum individual annual effective dose exposed to the general public is far below the dose constraint values specified in national standards.

15.5 Regulatory Review and Control Activities

The regulation activities of the MEE (NNSA) on radiation protection of NPPs include:

(1) Formulating, issuing and revising laws, regulations, guides and standards on radiation protection, radioactive waste management and discharge of radioactive effluents;

(2) Assessing whether a NPP conforms to the related regulations and standards by reviewing the design, construction and operation of the radioactive waste management and discharge of radioactive effluents, as well as the personnel qualifications and records;

(3) Requiring remedial and corrective measures for the items discordant with the requirements of the related codes and standards;

(4) Reviewing and approving the Environmental Impact Report submitted by the operating organization;

(5) Reviewing the environmental monitoring report submitted by NPP operating organizations, and organizing the environmental radiation monitoring station of the province where the NPP is located to perform supervision environmental monitoring.

(6) Supervising and inspecting the radiation protection, radioactive waste management and the discharge of radioactive effluents;

The ecology and environment department of the province where the NPP is located adopts the off-site regulatory monitoring system near the NPP to monitor and evaluate the surrounding environment of the NPP. The obtained measurement data are compared with those of NPPs; and the results are analyzed and compared with those of NPPs at home and abroad. The MEE (NNSA) and the environmental protection competent department in province where the NPP is located are responsible for reviewing the monitoring reports respectively submitted by NPP operating organizations and local environmental radiation monitoring stations, to ensure the accuracy and authenticity of measurement results.

For three years, the MEE (NNSA) has mainly carried out the following activities:

(1) Promoting the revision and management of regulations and standards. In order to implement the latest requirements of the *Nuclear Safety Act*, the amendment of *Regulations for Environmental Radiation Protection of Nuclear Power Plant* have been launched.

(2) Further strengthening the construction of environmental radiation monitoring network. The national environmental radiation monitoring network, nuclear and radiation emergency response technical center and environmental radiation monitoring technical center have been built to monitor routine environmental radiation in all key cities in China. The number of environmental radiation quality monitoring stations in China has been increased to 1488, including 161 automatic environmental radiation air monitoring stations, 328 land radiation level monitoring stations, 521 water body activity level national control and monitoring sections, 34 marine organism monitoring stations, 359 soil radioactivity level monitoring stations, and 85 electromagnetic radiation monitoring stations; nuclear safety pre-warning and environmental radiation regulatory monitoring has been carried out at 46 important nuclear and radiation facilities and surrounding uranium mining and milling venues, which basically realize all-round monitoring and pre-warning for radiation environment quality and important nuclear facilities in the whole country. The national radiation monitoring data acquisition center and 31 provincial data acquisition centers have been completed; the daily operation management system and periodical reporting mechanism based on the principle of “daily monitoring and monthly walk-down” have been established and implemented. The monitoring result of the air absorbed dose rate of the automatic station has been regularly issued on the website of the MEE.

16. Emergency Preparedness

1. Each Contracting Party shall take the appropriate steps to ensure that there are on-site and off-site emergency plans that are routinely tested for nuclear installations and cover the activities to be carried out in the event of an emergency.

For any new nuclear installation, such plans shall be prepared and tested before it commences operation above a low power level agreed by the regulatory body.

2. Each Contracting Party shall take the appropriate steps to ensure that, insofar as they are likely to be affected by a radiological emergency, its own population and the competent authorities of the States in the vicinity of the nuclear installation are provided with appropriate information for emergency planning and response.

3. Contracting Parties which do not have a nuclear installation on their territory, insofar as they are likely to be affected in the event of a radiological emergency at a nuclear installation in the vicinity, shall take the appropriate steps for the preparation and testing of emergency plans for their territory that cover the activities to be carried out in the event of such an emergency.

16.1 Basic Requirements for Emergency Preparedness

China has issued the *Act of the People's Republic of China on Prevention and Control of Radioactive Pollution*, *Emergency Response Act of the People's Republic of China*, *Nuclear Safety Act*, *Emergency Management Regulations for Nuclear Accident at Nuclear Power Plant* and a series of regulations and guides, which clearly specify the national emergency management system. The *National Nuclear Emergency Plan* issued by the State Council further defines the work principles and main bodies of responsibilities for nuclear emergency, enhances the commanding mechanism, details rules for emergency preparedness and guarantee measures, regulates the information reporting and release procedures, and also specifies the work after accidents in nuclear facility.

For the nuclear emergency management in China, the guidelines of constant vigilance, versatile compatibility, unified direction, active co-ordination, public safeguard and environmental protection are implemented; and the basic principles of unified leadership, graded responsibilities, linear and local leadership combined, rapid

response and scientific treatment are adhered to.

After the Fukushima nuclear accident, the MEE (NNSA) issued the *General Technical Requirements for Improvement of Nuclear Power Plant after Fukushima Nuclear Accident*, proposing the technical requirements on emergency water makeup and associated equipments, on the habitability of emergency control center and its functions, and requirements at technical level such as emergency improvement after a number of units at the same site enter the emergency status at the same time. In addition, it supplements requirements for the emergency plan assessment, emergency power supply, emergency control center and emergency preparedness and actions.

China has followed up and studied the international development of nuclear safety codes and standards, and has promptly revised relevant nuclear safety codes and standards of China. The National Nuclear Emergency Coordination Committee started the revision of the *Emergency Management Regulations for Nuclear Accident at Nuclear Power Plant* in 2014. The MEE (NNSA) started the revision of guidelines including the *Emergency Preparedness and Response for Operating Organization of Nuclear Power Plant* in 2017. The MEE (NNSA) issued the technical document of nuclear and radiation safety laws and regulations of *Design Criteria for Emergency Facilities in Nuclear Power Plants* in 2017, clarifying the design criteria for major emergency facilities in NPPs including the emergency control center and technical support center in China, reflecting the latest requirements for earthquake resistance, flood control and habitability of major emergency facilities and equipments, such as emergency control center, after the Fukushima nuclear accident.

16.2 Emergency Organizational System and Responsibilities

A three-level emergency organization system has been formed for nuclear emergency in China, which consists of national nuclear emergency organizations, nuclear emergency organizations in the provinces where the NPP is located and nuclear emergency organizations of NPP operating organizations, as shown in Fig. 7.

In the three-level nuclear emergency organization system, the National Nuclear Emergency Coordination Committee is responsible to organize and coordinate nuclear emergency preparation and responding actions at the national level, and the National Nuclear Emergency Response Office work as its standing office. If necessary, the State Council shall establish the national nuclear accident emergency command post to uniformly lead, organize and coordinate responding activities to a nation-wide nuclear emergency. An expert committee has been assigned by the National Nuclear Emergency Coordination Committee to provide advices and recommendations for major

decision-making and important planning of national nuclear emergency management and nuclear accident response. The National Nuclear Emergency Coordination Committee also set up a liaison group to handle matters assigned by the Committee.

Provincial nuclear emergency coordination committees are made responsible, within their respective jurisdictional areas, for emergency management for nuclear accidents occurring within boundaries and to uniformly command the off-site emergency response actions. The provincial nuclear emergency coordination committee should set up an expert group for decision-making consultancy, and a provincial nuclear emergency response office to cope with the routine business of the provincial committee. If necessary, the provincial government may take the lead to organize and coordinate directly the off-site nuclear emergency responses and support the on-site responding actions.

The emergency command of NPP operating organization is responsible to organize on-site nuclear preparation and response; to implement unified command of emergency responses on-site; to cooperate and support the departments assigned by the provincial government, dealing with the off-site nuclear emergency preparation and response; and to propose the recommendations for entering the general emergency status and taking off-site emergency protective measures in a timely manner. The nuclear power corporation, to whom the nuclear facility is subordinate, is responsible to lead and coordinate the facilities nuclear emergency preparation and deploys its emergency resources or strengths to support the operators responding actions.

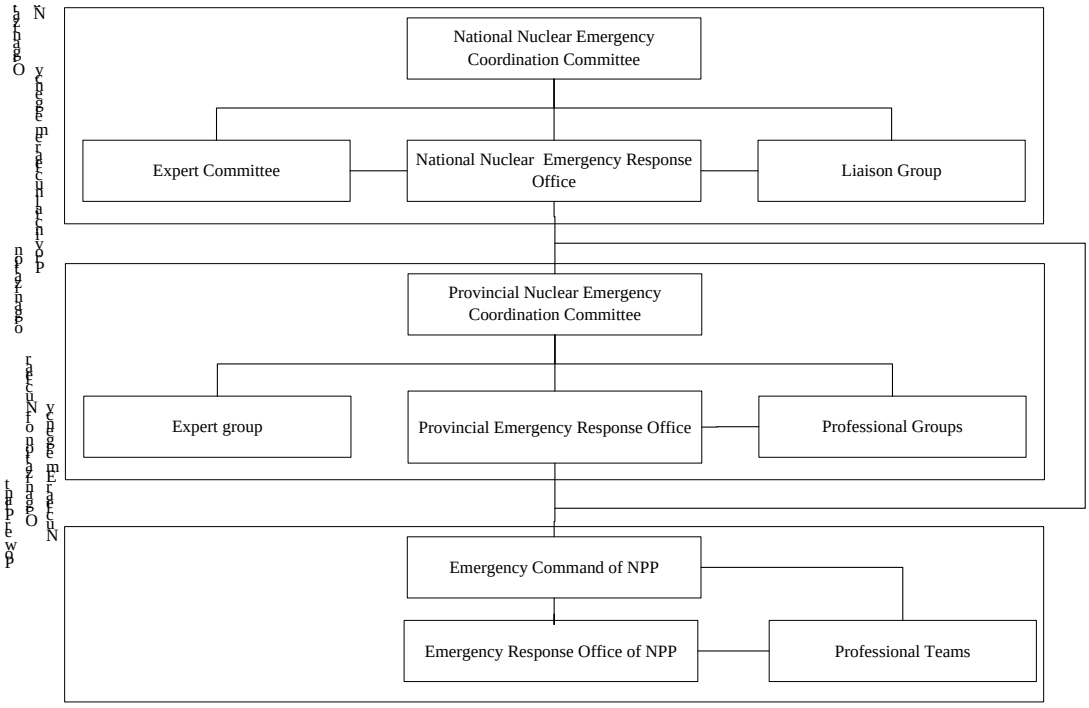


Fig. 7 Organization Chart of National Nuclear Emergency System

The main responsibilities and duties of nuclear emergency organizations at all levels are as follows:

(1) The National Nuclear Emergency Coordination Committee is responsible to organize and coordinate nuclear emergency preparation and responding actions at the national level.

- Carrying out the national policies on emergency to nuclear accidents, and working out national nuclear emergency policies.
- Organizing and coordinating the accident emergency of relevant departments of the State Council, competent departments of the nuclear industry, local government, NPPs and other nuclear facilities;
- Reviewing the national nuclear emergency planning and annual work plan.
- Organizing the preparation and implementation of the national nuclear emergency plan; reviewing and approving off-site emergency plans.
- Approving the commencement and termination of off-site emergency status in case of emergency response when appropriate.
- Deciding, organizing and commanding emergency support and response actions on a unified basis, and reporting to and requesting instructions from the State Council at any time.
- Proposing suggestions to the State Council on implementing special emergency actions when appropriate.
- Fulfilling relevant international conventions and bilateral or multilateral cooperation agreements on nuclear emergency; reviewing and approving bulletin and international notification on nuclear accident, and proposing schemes for requesting international aids.
- Handling other affairs assigned by the State Council.
- When necessary, the State Council will lead, organize, and coordinate national nuclear emergency management.

(2) The National Nuclear Emergency Response Office is an administrative organization for national nuclear emergency. It is a subordinate department of CAEA. Its main duties include:

- Carrying out nuclear emergency policies specified by the State Council and the National Nuclear Emergency Coordination Committee.
- Taking charge of daily works of the National Nuclear Emergency Coordination Committee;
- Implementing national nuclear emergency plan; knowing, coordinating and urging emergency preparedness activities of member organizations of the National

Nuclear Emergency Coordination Committee; inspecting, guiding, and coordinating emergency preparedness of local governments, NPPs and their superior organizations;

- Receiving, verifying, handling, transmitting, notifying, and reporting nuclear emergency information of the state; managing the national nuclear emergency response center; implementing relevant international conventions, bilateral or multilateral cooperation agreements, and requesting international aids as a national nuclear emergency liaison point with foreign countries.
- Preparing national nuclear emergency planning and annual work plan; working out scientific research plan and scheme of technical support system for emergency;
- Organizing the review of the off-site emergency plans, off-site integrated drill plans, and on-site and off-site joint drill plan; proposing the review comments.
- Organizing activities of liaison group and experts group.
- Organizing trainings and exercises on nuclear emergency.
- Collecting information, putting forward reports and proposals, timely communicating and executing decisions and orders from the State Council and the National Nuclear Emergency Coordination Committee, inspecting and reporting the implementation results in case of emergency response.
- Handling related affairs decided by the National Nuclear Emergency Coordination Committee after the termination of emergency status.

(3) The Province Nuclear Emergency Coordination Committee at which the NPP is located is responsible for nuclear emergency management in its administrative area. Its main duties include:

- Implementing the national nuclear emergency policies and principles.
- Preparing off-site nuclear emergency plans and making preparation for nuclear emergency.
- Commanding the off-site nuclear emergency response actions in its administrative area on a unified basis.
- Organizing supports to on-site nuclear emergency response actions.
- Timely notifying nuclear accident situations to the neighboring provinces, autonomous regions and municipalities directly under the Central Government or special administrative regions.
- When necessary, the provincial government shall lead, organize and coordinate nuclear emergency response management within its administrative area.

(4) The main duties of nuclear emergency organizations of NPPs include:

- Implementing the national nuclear emergency policies and principles.

- Preparing on-site nuclear emergency response plans and performing nuclear emergency preparedness.
- Determining the grade of nuclear emergency status and commanding the nuclear emergency response actions for their own organizations on a unified basis.
- Timely reporting the accident situations to the state, provincial nuclear emergency organizations and the designated departments, and proposing suggestions for entering off-site emergency status and implementing emergency protection.
- Assisting and coordinating the provincial nuclear emergency response committee to conduct the nuclear emergency.

(5) Duties of member organizations of the National Nuclear Emergency Coordination Committee and relevant departments include:

The MEE (NNSA), the National Health Commission, Ministry of Emergency Management and other member organizations of the National Nuclear Emergency Coordination Committee as well as other related departments will perform corresponding nuclear emergency preparedness and response within their respective scopes of responsibilities.

16.3 Emergency Classification and Reporting

16.3.1 Emergency Classification

In China, NPP emergency situations are classified into the following four scales:

(1) Emergency Standby: in case of some specific conditions or external events that may endanger the safety of NPPs. Relevant personnels of the NPPs will be on standby, and some off-site emergency organizations may be notified. It corresponds to Level IV response.

(2) Plant Emergency: the radiation consequences are confined within a partial area of the NPP; personnels in the plant will take emergency response action according to the on-site emergency plan and relevant off-site emergency organizations will be notified. It corresponds to Level III response.

(3) On-Site Emergency: the radiation consequences are limited to the site. On-site personnels will take emergency response actions; off-site emergency organizations will be notified, and some off-site emergency organizations may also take emergency response actions. It corresponds to Level II response.

(4) Off-site Emergency (overall emergency): the radiation consequences have gone beyond the site boundary. Both on-site and off-site nuclear emergency plans will be implemented. It corresponds to Level I response.

The first three levels of response are mainly organized and implemented for emergency needs within the site. When release of large amount of radioactive materials has occurred or will possibly occur, and the consequences of accident will go beyond the site boundary and may possibly seriously endanger the public health and environmental safety, it will enter off-site emergency, and level I response will be launched.

16.3.2 Emergency Report

When the NPP enters emergency standby, the nuclear emergency organization of the NPP shall promptly report to its superior competent departments and the MEE (NNSA), and also report to the Province Nuclear Emergency Coordination Committee where the NPP is located. If radioactive materials may release or have released, it shall promptly decide to enter the plant emergency or on-site emergency status, and quickly report to superior competent departments, the MEE (NNSA) and the Provincial Nuclear Emergency Coordination Committee.

In case radioactive material may spread or have spread beyond the site boundary, suggestions on entering the off-site emergency status and taking corresponding emergency prevention measures shall be promptly proposed to the Provincial Nuclear Emergency Coordination Committee. Upon receiving emergency report from nuclear emergency organization of NPP, the Provincial Nuclear Emergency Coordination Committee shall promptly take corresponding emergency countermeasures and protective measures against the nuclear accident, and report promptly to National Nuclear Emergency Response Office.

Under the off-site emergency condition, relevant departments such as the National Nuclear Emergency Response Office and the MEE (NNSA) shall promptly send personnels to the site and direct the nuclear emergency response actions.

16.4 Emergency Plan

For nuclear accidents (events) that may occur in NPPs, nuclear emergency organizations at all levels have prepared their own nuclear emergency plans.

The on-site emergency plan is prepared by operating organizations of NPPs; the off-site emergency plan is prepared by local government, the national nuclear emergency plan is prepared by the National Nuclear Emergency Coordination Committee. The contents of the three levels of emergency response plans are mutually linked and coordinated. The emergency plans at each level are supported by the implementation procedures as the supplement and refinement. In addition, the main member organizations of the National Nuclear Emergency Coordination Committee and

the superior group corporations of the NPP operating organizations shall formulate their own emergency plans. Emergency plans at all levels shall be prepared, approved and revised on a regular basis as required.

The contents of emergency plans at various levels shall cover the emergency response organizations and their responsibilities, the detailed plans of emergency preparedness and response, facilities and equipments, coordination and supports from the organizations concerned, and other technical aspects.

The NPP operating organization is responsible for developing the on-site nuclear emergency plan and delivering it to the MEE (NNSA) for review and reporting it to the nuclear industry competent department under the State Council, the energy competent department and the departments designated by the people's governments of provinces, autonomous regions and municipalities directly under the Central Government for archiving. The departments designated by the people's governments of provinces, autonomous regions and municipalities directly under the central government shall be responsible for the preparation of off-site nuclear emergency plans within their respective administrative areas, and shall organize their implementation after being submitted to the National Nuclear Emergency Coordination Committee for approval. The national nuclear emergency plan shall be examined and approved by the State Council.

After the Fukushima nuclear accident, China has put forward requirements for the capability of coping with multiple unit severe accidents. All NPPs, based on their actual conditions, have implemented improvement actions and updated their on-site nuclear emergency plans and procedures, and established the emergency management system to cope with possible multiple unit accidents. The emergency response plan for multiple units entering emergency status simultaneously has been prepared, to ensure that rapid and effective emergency response actions can be taken under the emergency status of multiple unit accidents. The emergency decision-making and response, emergency status approval and termination processes have been formalized; the internal and external support and emergency capability coordination processes are clearly developed to optimize the allocation of emergency resources.

16.5 Trainings, Drills and Exercises

China attaches great importance to nuclear emergency exercises and drills. It has issued rules and regulations including *Nuclear Safety Act*, *Emergency Management Regulations for Nuclear Accident at Nuclear Power Plant*, *Management Measures for Emergency Plans against Sudden Events*, *Guideline on Emergency Exercise for Sudden*

Events, *Management Regulation for Nuclear Emergency Exercises*; organized emergency exercises according to codes and regulations; made clear requirements on nuclear emergency exercises at all levels and of various types, and specified the policy and principles, organization structure, contents and forms, classification and frequency, guarantee and preparation, and implementation procedures for nuclear emergency exercises. To meet the demand in nuclear energy development, China launches national joint nuclear emergency exercises periodically; relevant provinces (municipalities and autonomous regions) organize a joint on-site and off-site nuclear emergency exercise at its own level every 2 to 4 years; NPP operating organizations organize a comprehensive exercise every 2 years, and a number of special exercises every year; and a site with 3 or more operating units shall increase the frequency of exercises as appropriate.

In order to enhance the professionalism of the personnel engaged in nuclear emergency and provide enough manpower for nuclear emergency preparedness and response, the national and local emergency organizations at all levels carry out training activities such as providing training courses, technical trainings and examinations of emergency knowledge to strengthen training of personnel engaged in nuclear emergency. A three-level nuclear emergency training system has been established. The national nuclear emergency management institution is in charge of training of nuclear emergency management personnel nationwide. The nuclear emergency management institutions of provinces (municipalities and autonomous regions) provide trainings for nuclear emergency personnels in their own administrative areas; and nuclear facility operating organizations conduct professional and technical training for their own nuclear emergency workers.

Emergency training in the NPP includes basic emergency training and special emergency training, which are applied to personnels of NPP (including contractors), personnel engaged in emergency preparedness and response. The training involves many aspects related to emergency preparedness and response of NPPs. At present, management personnel for nuclear emergency and relevant professional and technical personnel have taken part in different levels of professional training. All emergency response personnel, including emergency commanders of NPPs, are trained and examined systematically before the initial fuel loading. A training and examination corresponding to their expected emergency response activities will be performed at least once a year during the service life of the NPP.

In recent years, NPP operating organizations have carried out single drills, comprehensive drills and joint drills for many times in accordance with the requirement of nuclear safety regulations. To inspect and verify the results of the development of

nuclear emergency support forces, all nuclear power corporations provide material guarantee, technical support, rescue assisting and other support forces in the comprehensive nuclear emergency exercises conducted in their NPPs, to cooperate in joint exercises, well inspecting and upgrading the emergency support capability. The operating NPPs regularly hold different types of emergency exercises in accordance with the requirements of nuclear safety regulations to test, improve and strengthen emergency preparedness and emergency response capabilities. In addition, in order to verify the effectiveness of nuclear emergency preparedness for newly-built nuclear power units, nuclear emergency comprehensive exercises have been held for the newly-put-into-operation units before the initial fuel loading, after which the comprehensive nuclear emergency exercises and individual exercises are held regularly. The state regularly organizes joint exercises at the national level participated by three-level nuclear emergency organizations.

In June 2017, the Chinese government organized more than 10 national nuclear emergency forces to participate in ConvEx-3 (2017) exercise organized by the IAEA, which was highly recognized by the IAEA.

In December 2018, Qinshan Nuclear Power Base held the first on-site comprehensive emergency exercise in the country with the scenario of multi-unit severe accidents. During the exercise, it was simulated that there was a large break in the reactor coolant system together with the black-out of Fangjiashan Unit 1, which entered the conditions of major accident management. In addition, the grounding failure of Unit 1 BUA bus and BUF bus of Qinshan Phase 3 NPP plus moderator leakage, break of core inlet header of the primary heat transfer system and failure of two emergency core cooling pumps caused the severe accident management condition. CNNC dispatched Sanmen and Tianwan Emergency Support Detachments to support Qinshan. Through emergency exercises, the effectiveness and practicability of the emergency plan for NPPs have been tested.

For details of NPPs comprehensive exercises and joint exercises, please refer to Appendix 9.

16.6 Information Publicity and Public Communication on Emergency

The Nuclear Safety Act clearly stipulates that the nuclear industry competent department under the State Council or the departments designated by the people's governments of provinces, autonomous regions and municipalities directly under the Central Government shall be responsible for releasing emergency information on nuclear accidents. The nuclear safety supervision and management department under

the State Council shall disclose the corresponding nuclear accident information according to law.

China attaches great importance to information disclosure and public communication in nuclear emergency, and has issued relevant regulations and management measures. It has made clear regulations on the release of nuclear accident information and the disclosure of nuclear emergency information from the top level. The National Nuclear Accidents Emergency Response Office has established information communicating network to enhance communication with relevant ministries and commissions, local governments, the NPPs and the public. China performs centralized, unified and standardized managements on the information of nuclear accidents and emergency response, with corresponding regulations and requirements for information channel, information classification, information disclosure, etc., so as to ensure the information of nuclear and radiation accidents is released and reported to the public timely, uniformly, transparently and accurately.

With the rapid development of China's nuclear power industry, the awareness of the public on caring and participating in nuclear power safety and nuclear emergency related work has been continuously increased. The nuclear emergency organizations at all levels have set up special nuclear emergency publicity teams to publicize the state nuclear energy policy and nuclear emergency policy to the whole society in due course, so as to increase public communication and ensure that the general public has the right to know about nuclear power safety, nuclear emergency preparedness and response.

At the national level, the white paper *China's Nuclear Emergency Preparedness* was released, which publicized the overall situation of nuclear emergency to the world and conveyed the message of rational, coordinated and balanced development of nuclear energy industry. It has organized and invited the media to nuclear power activities to promote public confidence in the development of nuclear energy by showing the media the advanced nuclear technology, safety culture and nuclear emergency management concepts. It has guided nuclear-related organizations at all levels to perform various forms of public communication activities on nuclear energy and nuclear emergency, which establishes a broad social foundation for nuclear emergency.

Local governments are responsible for the popularized education of the public around the NPPs on the basic knowledge of nuclear safety and radiation protection, and propagating knowledge on emergency protection, such as alarm, shielding, evacuation and taking preventive anti-radiation medicine in case of an emergency, and giving directions on how to take these actions.

The operating organization of the NPP takes various measures such as broadcasting,

TV, newspaper, network and inviting local public to visit plant and to take part in or to watch emergency exercises, to eliminate the public nuclear panic, and to courage them to effectively participate in emergency response activities in case of an emergency.

16.7 Capability Building of Emergency Response and Rescue Arrangement

16.7.1 Capability Building of National Emergency Response

China adheres to the ideas of being compatible, resource integration and professional support, constructs and maintains national nuclear emergency response capability adaptive to the safe and efficient development of the nuclear energy undertakings, and has formed a national nuclear emergency capability system that can effectively cope with serious nuclear accidents.

In accordance with the principle of being compatible and by making full use of existing conditions, China has set up 8 national nuclear emergency professional and technical support centers, including radiation monitoring, radiation protection, aviation monitoring, medical rescue, marine radiation monitoring, meteorological monitoring and forecasting, auxiliary decision-making and response actions, and three national nuclear emergency training bases, which basically form a complete, functional and effective nuclear emergency technical support and training system. Based on the principles of regional deployment, module setting and professional support, more than 30 national professional rescue teams have been set up for various professional rescue tasks for nuclear emergency response. A national nuclear emergency rescue team of more than 300 people is under construction. It is mainly responsible for emergency rescue and emergency disposal of major nuclear accidents under complex conditions and participating in international nuclear emergency rescue.

In the past three years, China has issued and released such normative documents as the *Method of Management for National Nuclear Emergency Special Technical Support Centers and Rescue Teams,* the *Guidelines for the Preparation of Nuclear Emergency Rescue Plans,* and the *General Plan for National Nuclear Emergency Rescue Forces to Participate in Nuclear Emergency Rescue,* which have clearly defined the responsibilities, mission and requirements, rescue forms, rescue targets, command relationships and related support topics of national nuclear emergency rescue forces, and has well coordinated them with plans at all levels. 8 construction norms, including the *Construction Norm for National Site Technical Support Team for Nuclear Emergency Rescue and Radiation Monitoring,* have been developed and issued to guide and strengthen the building of nuclear emergency forces. In addition, the *13th Five-Year National Nuclear Emergency Work Plan* has been developed and implemented to

promote the continuous improvement of relevant capability building.

In recent years, the MEE (NNSA) has strengthened the construction of a nuclear emergency response capability system. Radiation environmental emergency monitoring and dispatching platforms have been built at the MEE headquarters and six regional offices respectively. In addition, provincial nuclear and radiation emergency monitoring and dispatching platforms have been built at 31 provincial environmental radiation monitoring agencies across the country, basically forming a nuclear emergency monitoring and dispatching network system at the national-provincial-on-site levels, which strengthens the capability in acquiring information related to nuclear emergency disposal and the efficiency of organizing and coordinating nuclear emergency response forces. The nuclear emergency platform has realized the full process of accident response, integrated the multi-media and multi-scale nuclear accident consequence evaluation model to provide technical support for emergency decision-making, and expanded the mobile client application (APP) for providing services such as reception and delivery of emergency response information, query of national radiation monitoring data, real-time query of important operating parameters of nuclear power units, on-line learning of emergency plans and regulations and standards, etc.

16.7.2 Capability Building in Emergency Response of Nuclear Power Corporation and NPPs

In order to learn from the Fukushima nuclear accident, fully draw lessons from the good emergency practices of the international community in dealing with severe nuclear accidents, further improve nuclear emergency response mechanism, and comprehensively enhance the capability in nuclear emergency support at the level of nuclear power corporations, the MEE (NNSA) issued the *General Requirements on Construction of Nuclear Emergency On-site Rapid Rescue Teams for Nuclear Power Plants of Nuclear Power Corporation* and jointly issued the *Technical Requirements on Construction of Nuclear Emergency On-site Rapid Rescue Teams for Nuclear Power Plants of Nuclear Power Corporation* with the NEA to strengthen the policy guidance for the construction of nuclear emergency support forces of nuclear power corporation. Major nuclear power corporations have established nuclear emergency support teams and bases, jointly signed the *Agreement of On-site Support for Nuclear Emergency of Nuclear Power Corporation and Nuclear Power Plant*, and established a cross-group mutual support mechanism. Neighboring NPPs have also successively signed mutual support agreements, prepared mutual support action plans, and established a nearby rapid mutual support mechanism. The goal of joint building of capabilities of

nationwide NPPs in nuclear emergency and sharing of emergency resource has been basically achieved; and the emergency support system has been further improved.

In order to improve the emergency preparedness and response capability of NPPs, the MEE (NNSA) has taken corresponding management actions. After the Fukushima nuclear accident in particular, it issued the *General Technical Requirements for Improvement Actions in NPPs after Fukushima Nuclear Accident*, which puts forward specific requirements for measures to improve the capability of NPPs in dealing with severe accidents. The improvement actions implemented by NPP operating organizations include: additional portable power supplies and portable pumps for all plants, with at least two sets of portable pumps and portable power supplies for a multiple units site; the anti-seismic evaluation, upgrading and modification and reinforcement of nuclear emergency control center; strengthening the radiation emergency monitoring forces of NPPs, including the additional emergency mobile monitoring vehicles and on-board equipment, cast-type emergency continuous real-time monitoring equipment, and the dense distribution of environmental radiation automatic monitoring stations; the research and development of the nuclear accident related marine area monitoring and consequence assessment system; the development and application of scenario library for NPP nuclear emergency exercises for the purpose of improving the validity, feasibility and effectiveness of emergency exercises.

The NPP has established a reinforcement mechanism for multi-unit accidents, and has prepared and issued procedures such as *Emergency Response Plan for Multi-unit Nuclear Accidents* and *Emergency Response Support* to recruit backup emergency personnel to the site as soon as possible to participate in emergency response when the condition of unit accident exceeds the handling capacity of emergency organization currently on duty and the support for emergency and personnel for shifts are required. The pre-duty system has been established, where the personnel of the next shift will be the backup emergency personnel for this week. In addition, in order to meet the needs of response to multi-unit accident, sufficient facilities and equipment are provided according to the equipment and resource allocation standard of the existing technical support group for the response by technical support group reinforcements in case of emergency of multi-unit accident.

16.7.3 Capability Building in Environmental Radiation Monitoring

In order to enhance the capability in environmental radiation monitoring, the MEE (NNSA) has established the national environmental radiation monitoring network system to perform daily monitoring and emergency monitoring of the environmental

radiation nationwide; established the nuclear and radiation facility surrounding environment regulatory monitoring network system particularly under national regulation, which forms the dual-monitoring mechanism covering the on-site, off-site and surrounding environment of national important nuclear facilities; established the decision-making, support, commanding and dispatching system for nuclear and radiation accident emergency, covering all NPPs, allowing the real-time access to critical safety parameters, video liaison and consequence assessment; established a national and provincial nuclear and radiation emergency monitoring and dispatching platform systems, making it possible to obtain real-time data from national environmental radiation automatic monitoring network, to allocate and dispatch radiation emergency monitoring forces through overall planning, and to make video liaison; and established two national environmental radiation monitoring laboratories.

In the past three years, the MEE (NNSA) has upgraded and reconstructed the regulatory monitoring system of nuclear facilities, strengthened the capacity building of border automatic stations and aerial surveys, and carried out regulatory monitoring of 46 nuclear facilities particularly under national regulation, which basically realize all-round monitoring and early warning of the radiation environment quality and important nuclear facilities nationwide. It has completed the preparation and release of relevant construction standards for state-controlled automatic stations. In addition to 161 automatic stations that have been built and operating, 206 new automatic stations have been started and 27 automatic stations have been built and upgraded.

16.8 Regulatory Review and Control Activities

The MEE (NNSA) has completed the review and re-examination of emergency plans for nuclear accidents in civil nuclear facilities, including the AP1000 units of Sanmen NPP and Haiyang NPP and the EPR units of Taishan NPP. The review of emergency plan for high temperature gas-cooled reactor units of Shidao Bay NPP and HPR1000 of Fuqing NPP are ongoing. During the review of on-site emergency plans of NPPs, it has attached great importance to the case of simultaneous accident conditions at two units; studied and analyzed the emergency response capability of NPPs, with emphasis on the system, manpower, material resources and technical measures of emergency organizations in the NPPs, so as to ensure that NPPs can effectively take emergency response actions when two units enter emergency conditions simultaneously.

The MEE (NNSA) has carried out inspection of emergency preparedness of NPPs during its supervision and inspection activities. In addition, the emergency exercises in the NPPs have also been supervised. In the past three years, the MEE (NNSA) has

carried out a total of 48 inspections over emergency preparedness and supervision over emergency exercise for NPPs operating organizations.

16.9 International Arrangements for Nuclear Emergency

China is a member country of the IAEA, and is always endeavoring to establish the international nuclear safety emergency system together with all countries, promote all countries to share the achievements of peaceful utilization of nuclear energy, and unswervingly support and promote international cooperation and exchange in the area of nuclear emergency. China carries out multi-level and all-round cooperation in the area of nuclear emergency with IAEA and other international organizations, and keeps expanding the cooperation and exchange with relevant nuclear countries in the area of nuclear emergency. In addition, China has maintained an honest and open attitude to carry out cooperation and exchange with surrounding countries on nuclear emergency.

As a contracting party to the *International Convention on Early Notification of a Nuclear Accident* and the *Convention on Assistance in the Case of Nuclear Accident or Radiation Emergency*, China supports IAEA in playing a leading role in the area of international nuclear emergency, responds to IAEA initiatives and participates in relevant activities. In the past three years, Chinese delegation participated in the 8th and 9th Meeting of the Representatives of Competent Authorities identified under the Early Notification Convention and the Assistance Convention. In September 2018, China and IAEA signed the *Practical Arrangement on CAEA and IAEA Cooperation in Education, Training, Knowledge Network Construction and Management and Human Resources Development in the Field of Emergency Preparedness and Response*, and established the IAEA Nuclear and Radiation Emergency Preparedness and Response Capability Building Center (CBC-EPR); China has participated in various activities under the framework of the IAEA Response and Assistance Network (RANET). China has taken part in convention exercises organized by IAEA for many times. It has recommended hundreds of experts and scholars in China's nuclear emergency area to participate in the work carried out by IAEA.

China has carried out bilateral cooperation and exchanges in the area of nuclear emergency response. China has signed bilateral agreements on nuclear energy cooperation with 30 countries, including Brazil, Argentina, Britain, the United States, South Korea, Russia and France, to carry out cooperation and exchange on items including nuclear emergency.

Nuclear safety regulatory agencies of China, Japan and South Korea have established a mechanism of China, Japan and South Korea nuclear safety regulatory

senior officials meeting since 2008. In the China, Japan and South Korea nuclear safety cooperation initiative signed in 2011, the establishment of an information exchange framework (IEF) and the strengthening of emergency response capability were listed as action items. In November 2013, the Sixth China-Japan-Korea Senior Officials Meeting on Nuclear Safety Regulation approved the implementation plan for nuclear safety cooperation actions of the three countries, which strengthened the exchange and sharing of information on NPPs regulation. In addition, any party can participate as observer in the emergency exercise activities conducted by the other two parties, indicating that the three countries are endeavored with a pragmatic attitude to carry out priority actions in the two areas of information exchange and emergency response capability development and cooperation. In November 2014, China participated in the first China-Japan-Korea joint exercise for nuclear emergency preparedness as an observer. By 2018, China had participated in five joint exercises. Among them, the third joint exercise of the three parties was held at Daya Bay Nuclear Power Base in 2016, where the representatives of the nuclear safety regulatory authorities of Japan and South Korea were invited as observers. In November 2016, the first working group for emergency preparedness and response (WGEPR) meeting was held in Beijing. Since then, China has participated in another two WGEPR meetings. The above actions have promoted exchanges on nuclear emergency among China, Japan and South Korea.

China has expanded multilateral cooperation, adhered to the principle of win-win cooperation and carried out cooperation and exchanges with other countries in the area of nuclear emergency. The Fifth Asia-Europe Seminar on Nuclear Energy Safety has been held under the framework of ASEM. The Seminar on Nuclear Counterterrorism and Nuclear Emergency for Major Public Activities has been held under the framework of the Global Initiative on Nuclear Counterterrorism. Representatives from more than 20 countries and international organizations, including the United States, Russia, Britain, France and IAEA, had in-depth exchanges on nuclear counterterrorism and nuclear emergency related issues by focusing on major public activities.

17. Siting

Each Contracting Party shall take appropriate steps to ensure that appropriate procedures are established and implemented:

(i) for evaluating all relevant site-related factors likely to affect the safety of a nuclear installation for its projected lifetime;

(ii) for evaluating the possible safety impact of a proposed nuclear installation on individuals, society and the environment;

(iii) for re-evaluating as necessary all relevant factors referred to in sub-paragraphs (i) and (ii) so as to ensure the continued safety acceptability of the nuclear installation;

(iv) for consulting Contracting Parties in the vicinity of a proposed nuclear installation, insofar as they are likely to be affected by that installation and, upon request providing the necessary information to such Contracting Parties, in order to enable them to evaluate and make their own assessment of the possible safety impact on their own territory of the nuclear installation.

17.1 Evaluation on Site-related Factors

Currently, there are 13 NPPs in operation and under construction in mainland China, all of which are located in coastal areas. The siting of most of these NPPs started from the end of the last century; and the siting procedures and external event evaluation were all performed as required by the nuclear safety codes, and reviewed and confirmed by the MEE (NNSA).

17.1.1 Regulations and Requirements on NPP Siting

China has always adhered to the relevant IAEA standards in developing regulations and guidelines for siting evaluation. Currently, the main laws, regulations and guidelines applicable to the siting of NPPs include: *Nuclear Safety Act, Act of the People's Republic of China on Prevention and Control of Radioactive Pollution, Regulations on the Safety Regulation for Civil Nuclear Facilities, Application and Issuance of Safety License for Nuclear Power Plant, Code on the Safety of Nuclear Power Plant Siting, Earthquakes in Relation to Nuclear Power Plant Siting, Site Selection and Evaluation for Nuclear Power Plant with Respect to Population Distribution, External Human-induced Events in Relation to Nuclear Power Plant Siting, Evaluation of Extreme Meteorological Events for Nuclear Power Plant Siting, Determination of Design Basis Floods for Riverfront Nuclear Power Plants, Determination of Design*

Basis Floods for Coastal Nuclear Power Plants, Safety Aspects of the Foundation of Nuclear Power Plants, Atmospheric Dispersion in Relation to Nuclear Power Plant Siting, Hydrological Dispersion of Radioactive Material in Relation to Nuclear Power Plant Siting, The Relationship Between the Siting and Hydrographical Geology of Nuclear Power Plants, Site Survey of Nuclear Power Plants, and Design Basis of Tropical Cyclone for Nuclear Power Plants, etc.

17.1.2 Assessment Criteria and Assessment of Site-related Factors

The siting for NPPs in China is based on the requirements in the *Code on the Safety of Nuclear Power Plant Siting*. The following factors shall be taken into account in the assessment of a site for its suitability as a NPP.

(1) The effects of external events occurring in the region of the particular site (the external events could be of natural origin or human induced).

(2) The characteristics of the site and its environment that could influence the transfer of released radioactive material to persons.

(3) The population density and population distribution and other characteristics of the external zone in relation to the possibility of implementing emergency measures and the need to evaluate the risks to individuals and to the population.

The general assessment criteria include:

(1) Site characteristics that could affect the safety of the NPPs shall be investigated and assessed.

(2) The safety of proposed sites for NPPs shall be evaluated with regard to the frequency and severity of external natural and human induced events, and various phenomena that could affect the safety.

(3) The foreseeable evolution of natural and human-induced factors in the region that could have a bearing on safety shall be evaluated for a time period that encompasses the projected lifetime of the NPPs. These factors, and in particular population growth and population distribution, shall be monitored over the lifetime of the NPPs. If necessary, appropriate measures shall be taken to ensure that the overall risk remains acceptably low.

(4) Overall consideration shall be taken for the proposed site and NPP, to determine design basis external events.

(5) Design basis for external events used in the design of NPPs shall be determined.

(6) After an overall assessment of the site, if it is proved that the proposed measures cannot provide sufficient protection against the damage resulted from external

events within design basis, it shall be concluded that it is not suitable to build the proposed NPP on such site.

(7) In the determination of the design basis for external events, consideration shall be given to the combination of these with ambient conditions (e.g. hydrological, geological and meteorological conditions). Reactor operational states shall also be considered.

(8) Design basis related to the site shall be evaluated and written into the application documents for review by the regulatory body.

(9) The results of investigation and researches shall be documented in detail, for independent review by regulatory body.

(10) In the analysis to determine the suitability of the site, consideration shall be given to matters such as fresh fuel, spent fuel as well as the storage and transport of radioactive waste.

(11) The potential for interactions between radioactive and non-radioactive effluents shall be considered.

(12) For each proposed site, the potential radiological impacts in operational states and in accident conditions on people in the region, including impacts that could warrant emergency response actions, shall be evaluated with due consideration of relevant factors, including population distribution, dietary habits, uses of land and water, and the radiological impacts of any other releases of radioactive material in the region.

(13) The total nuclear capacity to be installed on the site shall be determined as far as possible at the first stages of the siting process.

(14) The quality assurance program shall be implemented for all activities that may influence the safety and for the determination of site design basis parameters.

In the selection of NPP sites, as required by relevant regulations, the MEE (NNSA) requires that non-residence area and planning restricted area shall be arranged around NPPs, to take into full account the possible effect of external human induced event sources in the development of site area, and to control the future human activities in the area during the lifetime of the NPP.

During the siting of NPPs, the feasibility of off-site emergency plans is demonstrated in consideration of the current environmental characteristics and expected development around the site.

17.1.3 Criteria for Effect of External Events

17.1.3.1 Criteria for Design Basis for External Natural Events

(1) Proposed sites shall be adequately investigated and studied with respect to all

site characteristics that could affect safety in relation to design basis natural events.

(2) The internal and external natural events in the region of a proposed site shall be classified according to their significance for the safe operation of NPP, to identify important natural phenomena that need to be considered in the design basis.

(3) Historical recorded information and records of the occurrences and severity of important natural phenomena shall be collected for the region and shall be carefully analysed for reliability, accuracy and completeness.

(4) Appropriate methods shall be adopted to determine the design basis against these major natural phenomena. The methods shall be justified in terms of being up to date and compatible with the characteristics of the region.

(5) The size of the region to which for determining design basis natural events shall be large enough to include all the features and areas that could contribute to the determination of the design basis natural events and to the characteristics of the events.

(6) In the assessment of design basis, site specific data shall be used, unless such data are unobtainable. In this case, data from other regions that are sufficiently relevant to the region of interest may be used.

17.1.3.2 Criteria for Design Basis for External Human-induced Events

(1) Proposed sites shall be adequately investigated and studied with respect to all site characteristics that could affect safety in relation to design basis external human induced events.

(2) All facilities and human activities that could affect the NPP safety in the area where the NPP site is located shall be identified, and classified according to their significance for the safe operation of NPP, to determine the important human induced events to be used for design basis.

(3) Information concerning the frequency and severity of those important human induced events shall be collected and analysed for reliability, accuracy and completeness.

(4) Appropriate methods shall be adopted to define design basis external human induced events. The methods should be up to date and compatible with the characteristics of the region.

17.1.4 Regulatory Review and Control Activities

The siting process of NPPs is divided into three stages, namely site survey, site assessment and pre-operation according to nuclear safety regulations and guides. All nuclear power corporations carry out NPP site selection and assessment in accordance with the requirements of nuclear safety regulations and guides, and prepare site safety

analysis reports and environmental impact assessment reports at siting stage.

The MEE (NNSA) will issue the *Review Comments on Siting* after the review and confirmation of the safety analysis report on siting and the environmental impact assessment report in the siting phase.

From 2016 to 2018, the MEE (NNSA) issued *Review Comments on Siting* to Units 3 and 4 of Sanmen NPP, Zhangzhou NPP Phase I Project, Units 5 and 6 of Ningde NPP, and Taipingling NPP Phase I Project respectively. Among them, for the determination of the design basis earthquake motion of Zhangzhou NPP and the emergent evacuation route on the island site of Units 5 and 6 of Ningde NPP, the MEE (NNSA) has organized experts from relevant industries in China to carry out reviews and consultations for many times to ensure the suitability of the site.

17.2 Impact of NPP on Individuals, Society and Environment

In the siting phase, the NPP license applicants in China perform assessment on the possible impact of the proposed NPP on individuals, society and environment and the feasibility of implementing emergency plans according to the requirements of the *Code on the Safety of Nuclear Power Plant Siting*.

17.2.1 Criteria for Determining Potential Impact of NPP on Residents and Environment

17.2.1.1 Criteria for Determining Potential Impact of NPP on the Region

(1) In the evaluation of a site for its radiological impact on the region for operational states and accident states which could lead to emergency measures, appropriate estimates shall be made of expected or potential releases of radioactive material, taking into account the design of the NPP and its safety features.

(2) During the evaluation of the direct or indirect pathways by which radioactive materials released from the NPP could reach and affect people, characteristics of the region and site shall be taken into account, with special attention given to the role of the biosphere in the accumulation and transport of radionuclides.

(3) The relationship between the site and the NPP design shall be examined comprehensively to ensure that the radiological risks to the public and the environment arising from the radioactive material release determined by the source terms is acceptably low.

(4) The design of the NPP shall be able to compensate for any unacceptable impacts of the NPP on the region, otherwise the site shall be deemed unsuitable.

17.2.1.2 Criteria for Considering Population Factor and Emergency Plan

(1) The population distribution characteristics and utilization of land and water in

the site area at present and in the predictable future shall be assessed; and any specific characteristics that could impact the potential consequences of radioactive release to individuals and groups shall be taken into consideration.

(2) With respect to characteristics and distribution of the population, the site and plant combination shall be such that:

During operational states the radiological exposure of the population remains as low as is reasonably achievable and in any case is in accord with national requirements;

The radiological risk to the population from accident states, including those which may lead to the implementation of emergency measures, is acceptably low and in accord with national requirements.

If, after thorough evaluation, it is shown that appropriate measures cannot be envisaged to meet the above requirements, the site shall be deemed unsuitable for the location of a NPP of the type proposed.

(3) The external zone(planning restricted area) for a proposed site shall be established with a view to the potential for radiological consequences to people and to the capability of implementing emergency plans, having due regard for any external event which may hinder implementation. During siting assessment, it shall be determined that no basic problems exist for establishing an emergency plan for the external zone, and the feasibility of the emergency plan shall be demonstrated.

17.2.1.3 Non-residence area and planning restricted area

It is stipulated by the *Nuclear Safety Act* that the people's governments of provinces, autonomous regions and municipalities directly under the Central Government shall delimit the planning restricted area around NPPs and other important nuclear facilities, which shall be implemented with the consent of the nuclear safety supervision and management department under the State Council. It is stipulated by the *Act of the People's Republic of China on Prevention and Control of Radioactive Pollution* that planning restricted area in the peripheral area of the NPP and other important nuclear facilities shall be delimited.

It is required by the *Regulations on Environmental Radiation Protection of Nuclear Power Plant* that non-residence area and planning restricted area shall be delimited around the NPPs. For the determination of the boundary of non-residence area and planning restricted area the radioactive consequences of the postulated accident as to the siting shall be considered. The non-residence area is not required to be circular and may be determined according to the specific conditions of the site, such as topography, geomorphology, meteorology and traffic; while the distance between the boundary of the non-residence area and the reactor shall not be less than 500 m; the radius of the

planning restricted area shall not be less than 5km. There shall be no villages and towns with more than 10,000 people within the planning restricted area; and there shall be no towns with more than 100,000 people within a radius of 10km from the site. For multi-reactor sites, the characteristics of each reactor shall be comprehensively considered to determine the boundaries of non-residence area and planning restricted area. In case of a postulated accident as to siting, in consideration of conservative atmospheric dispersion conditions, the effective dose received by any individual on the boundary of non-residence area within any 2h through external exposure to smoke cloud immersion and internal exposure of inhalation after the occurrence of accident shall not exceed 0.25Sv; and the effective dose received by any individual on the boundary of the planning restricted area during the entire duration of the accident (may be 30 days) via the above-mentioned two exposure paths shall not exceed 0.25Sv. During the entire duration of the accident, the collective effective dose received by the public groups within a radius of 80km from the site via the above-mentioned two exposure paths shall be less than 2x104 man·Sv.

All NPPs in China have delimited the non-residence area and planning restricted area.

17.2.2 Implementation of Criteria for Potential Impact of NPP on Residents and Environment

During NPP siting in China, the risks imposed by the potential releases of radioactive materials to the surrounding environment and the residents have been adequately considered, and the possible pathways leading to the risks have been studied and controlled.

Factors such as the dispersion of radioactive materials in the atmosphere, in the surface water and the ground water, the population distribution, the utilization of land and water, etc. have been extensively investigated, periodically monitored, studied and analyzed by using models, so as to effectively control the radiation risks caused by the potential radioactive releases to the surrounding environment and residents.

According to the requirements of the nuclear safety guides related to NPP siting, NPP operating organizations perform monitoring and assessment for factors affecting the site safety of NPP, such as seismological, meteorological, hydrological and geological phenomena, to ensure the safety of NPPs.

China has meteorological, hydrological and seismological observation stations and networks with fairly complete coverage. NPP operating organizations can timely know the changes of relevant natural phenomena in the areas where the NPPs are located with

the observation data of nearby stations. The *General Technical Requirements for Improvement Actions in Nuclear Power Plants after Fukushima Nuclear Accident* has further emphasized that NPPs shall establish cooperation relationship with meteorological and oceanological departments when coping with external natural hazards, to ensure obtaining forecast information in a rapid, accurate and timely manner; it has also further emphasized that NPPs shall strengthen communication with seismological departments, to obtain the latest data promptly and to evaluate the suitability of the seismic capacity of NPPs.

17.3 Re-evaluation of Site-related Factors

17.3.1 Re-evaluation of Site-related Factors after Siting

The relevant nuclear safety codes of China have set clear requirements on factors to be assessed in the siting for NPPs. The MEE (NNSA) will perform overall re-examination in the review and approval of licensing documents in different phases after siting (application for construction permit and operating license); and the re-assessment will be carried out in the PSR.

Safety Analysis Report mainly analyzes and demonstrates the reliability and safety of the NPP design, and the safety measures provided by the NPP to protect the site personnel, the public and the environment from excessive radiation hazard. The NPP shall submit PSAR and FSAR to the MEE (NNSA) respectively when applying for construction permit and operation license. The second chapter of the safety analysis report describes the geological, seismological, hydrological and meteorological data of the NPP site and its nearby areas, as well as the current population distribution, land use and various activities on the site and the management methods. It also evaluates and demonstrates the impact of site characteristics on the design and operation criteria of the NPP, and verifies the suitability of the site characteristics from the perspective of safety.

Code on the Safety of Nuclear Power Plant Operation clearly specifies that PSR shall be performed for systematic re-assessment of NPPs, and the NPPs in China perform a Periodic Safety Review every ten years. Hazard analysis is an important element to be considered in the ten-year safety review, by which the changes of site characteristics resulted from changes of external environment are analyzed on the basis of the assessment of historical operation records of the operating NPP, to determine the capability of the NPP in defending internal and external hazards.

17.3.2 Re-evaluation after Significant Accident or Extreme Event

According to the requirements of national nuclear safety regulations, if a significant accident or extreme event occurs, the NPP shall perform a corresponding re-evaluation. After the Fukushima nuclear accident, the MEE (NNSA) organized and carried out safety assessment of NPPs, including comprehensive safety inspection and safety margin assessment of external events.

The main purpose of the comprehensive safety inspection is to further review and evaluate the compliance of NPPs with the current nuclear safety regulations and standards, find out possible safety weaknesses, and propose improvement measures to further improve the safety level of NPPs. It is concluded from the inspection that international practical experience has been extensively learned for the NPP siting and has been earnestly implemented in the NPP siting. Therefore, the evaluation of extreme external events during the siting is appropriate and meets the relevant requirements of the current nuclear safety regulations. In addition, based on the conclusions of the inspection, the MEE (NNSA) has proposed the post-Fukushima improvement technical requirements for nuclear safety for all civil nuclear facilities; all NPPs have completed them as required.

The purpose of safety margin assessment for external events is to evaluate the safety margin of NPPs under given extreme external events (including earthquakes, floods and subsequent black-out) through quantitative analysis.

In order to further improve the capability of NPPs in dealing with extreme external events, for units under construction in China, the main improvement items are implemented before the first fuel loading, such as anti-flooding improvement for nuclear safety related buildings and equipment, addition of facilities such as portable power supply and portable pumps, enhancement of the seismic monitoring and anti-seismic response capability of NPPs, in-depth assessment of earthquake and tsunami risks, etc. In addition, the MEE (NNSA) has organized relevant professional and technical personnel to prepare the *Assessment Report on the Impact of Strong Typhoon on Coastal Nuclear Power Plant in China* (2017), etc., to analyze and evaluate the frequency of typhoons in the southeast coastal area and the impact of landing typhoons in the southeast coastal area on the design basis flood and design basis wind speed of relevant NPPs in the latest ten years, and has re-evaluated the design basis in response of sudden extreme natural events for several times. It is shown by the evaluation results that the extreme natural events of the existing NPPs can be enveloped by the design basis of the relevant NPPs. The relevant design basis of NPPs in China is reasonable and has a certain safety margin.

According to the requirements of the MEE (NNSA), the complete emergency

plans shall be prepared for NPPs under construction and in operation. In the case of any extreme natural event affecting a given site, the NPP operating organization and associated design institute shall make timely recheck and evaluation of the relevant design basis.

17.4 Consultation with Other Contracting Parties Likely to Be Affected by NPPs

NPPs in China are mainly distributed in the coastal areas in the east and south. For plant sites close to the border, China consulted with relevant countries to solve the nuclear emergency related issues involved in potential nuclear accidents via bilateral or multilateral cooperation in accordance with relevant requirements of the signed *Convention on Early Notification of a Nuclear Accident*, *Convention on Assistance in the Case of a Nuclear Accident or Radiological Emergency* and *Convention on Nuclear Safety*.

China has established bilateral or regional nuclear safety cooperation mechanisms with Japan, South Korea, Vietnam, Pakistan and other countries. According to the cooperation framework, China has carried out cooperation in the areas of regulatory capability, regional emergency response mechanism environmental radiation monitoring methods and so on.

China has set up environmental radiation automatic monitoring stations in key cities and at the border, and distributed monitoring stations around NPPs for the real-time and continuous monitoring of atmospheric environment, marine environment, ground water environment, soil, living organism and electromagnetic radiation. The radiation environment quality and radiation monitoring data of operating NPPs are released to the public periodically, to enable neighboring countries to promptly get information and raise the regional nuclear safety emergency response capability.

18. Design and Construction

Each Contracting Party shall take the appropriate steps to ensure that:

(i) The design and construction of a nuclear installation provides for several reliable levels and methods of protection (defence in depth) against the release of radioactive materials, with a view to preventing the occurrence of accidents and to mitigating their radiological consequences should they occur;

(ii) The technologies incorporated in the design and construction of a nuclear installation are proven by experience or qualified by testing or analysis;

(iii) The design of a nuclear installation allows for reliable, stable and easily manageable operation, with specific consideration of human factors and the human-machine interface.

18.1 Implementation of Defence in Depth

18.1.1 Regulations and Requirements for the Design and Construction of NPPs

18.1.1.1 Regulations and Requirements for the Design of NPPs

It is stipulated by the *Nuclear Safety Act* that the design of NPPs shall conform to nuclear safety standards, adopt scientific and reasonable parameters and technical requirements of structures, systems and components, provide diversified protection and multiple barriers, ensure reliable, stable and easy operation of NPPs, and meet nuclear safety requirements.

In according to Chinese laws and regulations as well as the practices and experience in nuclear safety regulation, the MEE (NNSA), with reference to the corresponding IAEA safety standards and other relevant national standards, have prepared the *Code on the Safety of Nuclear Power Plant Design* and a series of guides related to NPP design to guide relevant nuclear safety activities of siting, design and the associated review and inspection activities for new NPPs.

The MEE (NNSA) requires the applicant for the NPP Construction Permit to illustrate that the standards and specifications to be used comply with the requirements of the nuclear safety regulations and guides of China. The standards and specifications adopted shall be agreed by the MEE (NNSA).

The safety of a NPP relies on the guarantee of three fundamental safety functions (control of reactivity, residual heat removal, and the confinement of radioactivity). The concept of defence in depth is conducive to maintaining the reliability of these three fundamental safety functions and protecting of the public and the environment against

radioactive hazard.

The *Code on the Safety of Nuclear Power Plant Design* has clearly specified the safety objectives, concept of defence in depth, safety management requirements for design, main technical requirements and general design requirements for the design of NPPs.

(1) The safety objectives and concept of defence in depth set in the *Code on the Safety of Nuclear Power Plant Design*

The fundamental nuclear safety objective is defined as establishing and maintaining an effective defence in NPPs against radiological hazards to protect the individuals, society and the environment in the *Code on the Safety of Nuclear Power Plant Design*. The fundamental safety objective applies to all stages in the lifetime of a NPP, including planning, siting, design, manufacture, construction, commissioning and operation, as well as decommissioning. This includes the associated transport of radioactive material and the management of spent nuclear fuel and radioactive waste. In order to achieve the fundamental safety objective, the following measures shall be taken:

1) Controlling the radiation exposure of people and the release of radioactive material to the environment during operational states;

2) Restricting the likelihood of events that might lead to a loss of control over a nuclear reactor core, radioactive source, spent nuclear fuel, radioactive waste or any other source of radiation at a NPP;

3) Mitigating the consequences of such events, were they to occur.

It is clearly specified in the *Code on the Safety of Nuclear Power Plant Design* that the primary means of preventing accidents in a NPP and mitigating the consequences of accidents is the application of the concept of defence in depth. This concept is applied to all safety related activities, whether organizational, behavioral or design related, and whether in full power, low power or shutdown states. This is to ensure that all safety related activities are subject to independent layers of provisions so that if a failure were to occur, it would be detected and compensated for or corrected by appropriate measures. Application of the concept of defence in depth throughout design and operation provides protection against anticipated operational occurrences and accidents, including those resulting from equipment failure or human induced events within the plant, and against consequences of events that originate outside the plant.

(2) The design safety management requirements defined in the *Code on the Safety of Nuclear Power Plant Design*.

1) Responsibilities of design safety management

The operating organization shall ensure that the design submitted to the nuclear safety regulatory department of the State Council meets all applicable safety requirements. All organizations, including the design organization, engaged in activities important to the safety of the design of a NPP shall be responsible for ensuring that safety related items are given the highest priority.

2) Quality assurance

The quality assurance program, which describes the general arrangement for management, implementation and assessment of NPPs design, shall be established and implemented. This programme shall include provision for ensuring the quality of the design of each structure, system and component, as well as of the overall design of the NPP, including means for identifying and correcting design deficiencies, for checking the adequacy of the design and for controlling design changes.

3) Maintaining the safety and integrity of design of the NPP throughout the lifetime of the plant

The operating organization has overall responsibility for nuclear safety. It shall establish a formal system for ensuring the continuing safety of the plant design throughout the lifetime of the NPP.

(3) The main technical requirements defined in the *Code on the Safety of Nuclear Power Plant Design.*

1) Fundamental safety functions

Fulfillment of the following fundamental safety functions for a NPP shall be ensured for all plant states:

- Control of Reactivity;
- Removal of the heat of the reactor and from the fuel stored in spent fuel storage facility;
- Confinement of radioactive material, shielding against radiation and control of planned radioactive releases, as well as limitation of accidental radioactive releases.

A comprehensive and systematic approach shall be taken to identify those items important to safety that are necessary to fulfill the fundamental safety functions and to identify the inherent features that are contributing to fulfilling, or that are affecting, the fundamental safety functions for all plant states.

2) Radiation protection

The design of a NPP shall be such as to ensure that radiation doses to workers at the plant and to members of the public do not exceed the dose limits, that they are kept as low as reasonably achievable in operational states for the entire lifetime of the plant,

and that they remain below acceptable limits and as low as reasonably achievable in accident conditions. The design shall be such as to ensure that plant states that could lead to high radiation doses or to a large radioactive release have been “practically eliminated”, and that there would be no, or only minor, potential radiological consequences for plant states with a significant likelihood of occurrence. Acceptable limits for purposes of radiation protection associated with the relevant categories of plant states shall be established, consistent with the regulatory requirements.

3) Design management

The design for a NPP shall ensure that the plant and items important to safety have the appropriate characteristics to ensure that safety functions can be performed with the necessary reliability, that the plant can be operated safely within the operational limits and conditions for the full duration of its design life and can be safely decommissioned, and that impacts on the environment are minimized.

4) Application of defence in depth

The design of a NPP shall incorporate defence in depth. The levels of defence in depth shall be independent as far as is practicable so as to avoid that failure of one level of defence degrades the effectiveness of other levels of defence. The defence in depth concept shall be applied to provide several levels of defence that are aimed at preventing consequences of accidents that could lead to harmful effects on people and the environment, and ensuring that appropriate measures are taken for the protection of people and the environment and for the mitigation of consequences in the event that prevention fails. Safety features for design extension conditions (such as features for mitigating the consequences of accidents involving the melting of fuel) shall as far as is practicable be independent of safety systems.

5) Physical protection

Physical protection measures, including physical protection systems and related management measures, shall be provided to prevent, detect and cope with potential theft, malicious sabotage, unauthorized contact, illegal transfer and other malicious acts in relation to nuclear materials and NPP-related facilities, as well as to prevent the occurrence of nuclear materials being obtained and NPPs being damaged by terrorists.

6) Proven engineering practice

Codes and standards that are used as design rules for items important to safety shall be identified and evaluated to determine their applicability, adequacy and sufficiency, and shall be supplemented or modified as necessary to ensure that the quality of the design is commensurate with the associated safety function. Items important to safety for a NPP shall preferably be of a design that has previously been

proven in equivalent applications, and if not, shall be items of high quality and of a technology that has been qualified and tested.

7) Safety assessment

Comprehensive deterministic safety assessments and probabilistic safety assessments shall be carried out throughout the design process for a NPP to ensure that all safety requirements on the design of the plant are met throughout all stages of the lifetime of the plant, and to confirm that the design, as delivered, meets requirements for manufacture and for construction, and as built, as operated and as modified.

8) Provision for construction

Items important to safety in the NPP shall be so designed that they can be manufactured, constructed, assembled and installed according to the established processes to ensure the achievement of design specifications and the required level of safety.

9) Management and decommissioning of radioactive waste

Special consideration shall be given at the design stage of a NPP to the incorporation of features to facilitate radioactive waste management and the future decommissioning and dismantling of the plant.

(4) The general design requirements defined in the *Code on the Safety of Nuclear Power Plant Design.*

1) General design basis

The design basis for items important to safety shall specify the necessary capabilities, reliability and functionality for the relevant operating states, accident conditions and conditions arising from internal and external hazards, to meet the specific acceptance criteria over the lifetime of the NPP. The design shall be such that for design basis accident conditions, the key plant parameters do not exceed the specified design limits. It is necessary to perform the analysis of the design extension conditions of NPPs. The main technical objective of considering the design extension conditions is to provide assurance that the design of the plant is such as to prevent accident conditions that are not considered design basis accident conditions, or to mitigate their consequences.

2) Independence of safety systems

Interference between safety systems or between redundant elements of a system shall be prevented by means such as physical separation, electrical isolation, functional independence and independence of communication (data transfer), as appropriate. Safety system equipment (including cables and raceways) shall be readily identifiable in the plant for each redundant element of a safety system.

3) Safety classification

All items important to safety shall be identified and classified on the basis of their safety significance .The method for classifying the safety significance of items important to safety shall be primarily based on the deterministic methods, where appropriate, by probabilistic methods. The design shall be such as to ensure that any interference between items important to safety will be prevented, and in particular that any failure of items important to safety in a system in a lower safety class will not propagate to a system in a higher safety class. Equipment that performs multiple functions shall be classified in a safety class that is consistent with the most important function performed by the equipment.

4) Reliability of items important to safety

The reliability of items important to safety shall be commensurate with its safety significance. The design of items important to safety shall be such as to ensure that the equipment can be qualified, procured, installed, commissioned, operated and maintained to be capable of withstanding all conditions specified in the design basis for the items, with sufficient reliability and effectiveness.

5) Design for safe operation over the lifetime of the plant

Items important to safety for a NPP shall be designed to be calibrated, tested, maintained, repaired or replaced, inspected and monitored as required to ensure their capability of performing functions and to maintain the integrity of functions in all conditions specified in the design basis.

A qualification programme for items important to safety shall be implemented to verify that items important to safety at a NPP are capable of performing their intended functions when necessary, and in the prevailing environmental conditions throughout their design life.

The design life of items important to safety in NPPs shall be determined. Appropriate margins shall be provided in the design to take due account of relevant mechanisms of ageing, neutron embrittlement and wear out and of the potential for age related degradation, to ensure the capability of items important to safety to perform their necessary safety functions throughout their design life.

6) Human factors

Systematic consideration of human factors, (including the human-machine interface) shall be included throughout the entire design process. The human-machine interface shall be designed to provide the operators with comprehensive but easily manageable information, in accordance with the necessary decision and action times. At the appropriate stage, the relevant characteristics of human factors shall be verified and

confirmed (including the use of simulators) to confirm the actions that the operator really needs to take and to confirm that these actions can be implemented correctly. The design of the workplace and working environment of operating personnel shall conform to the concept of ergonomics.

7) Other design considerations

There are also design requirements for the safety system and safety features for design extension conditions in the multiple unit NPP, escape routes and means of communication, control of access to the plant, prevention of unauthorized access or interference with items important to safety and the prevention of harmful interactions of systems important to safety, etc.

8) Safety analysis

Safety analysis of NPPs design shall be carried out in which methods of both deterministic analysis and probabilistic analysis shall be applied to enable the challenges to safety in the various categories of plant states to be evaluated and assessed. Safety analysis shall demonstrate that the NPP as designed is capable of complying with authorized limits on discharges with regard to radioactive releases and with the dose limits in all operational states, and is capable of meeting acceptable limits for accident conditions. The safety analysis shall provide assurance that defence in depth has been implemented in the design of the plant. The safety analysis shall provide assurance that uncertainties have been given adequate consideration in the design of the plant and in particular that adequate margins are available to avoid cliff edge effects and early radioactive releases or large radioactive releases.

In addition, the *Code on the Safety of Nuclear Power Plant Design* has also made detailed stipulations on the design requirements for important systems in a NPP, such as reactor core, reactor coolant system, containment system, I&C system, emergency control, emergent power supply and radiation protection.

18.1.1.2 Codes and Requirements for Construction of NPPs

The basic requirements for NPP construction are mainly embodied in the *Code on the Safety of Nuclear Power Plant Quality Assurance* and its guides. For specific features of the construction activities, the nuclear safety guide *Quality Assurance in the Construction of Nuclear Power Plants* specifies the following requirements:

(1) General requirements include:

- Preparing and documenting plans for on-site construction (including the verification).
- Stipulating and fulfilling the required activities according to the written procedures, work instructions, specifications and drawings suitable to such activities.

- Implement housekeeping, to preserve the requisite quality of the item to be constructed or installed.
- Controlling the receiving, storage and handling of materials and equipment, and strictly avoiding abuse, misuse, damage, degradation or loss of identification.
- Specifying and implementing the cleaning requirements for fluid systems and relevant components and the management requirements of the cleanness.
- Performing the painting or coating activities for quality/safety-related items and surfaces according to the approved procedures.
- Managing the measuring and testing equipment, and controlling their selection, identification, calibration and application.
- Working personnel shall receive training as necessary in order to develop proficiency in the tasks they performed.

(2) Installation, inspection and testing of items

The construction of a NPP mainly involves three categories of activities: the installation, inspection and testing of soil, ground foundation, concrete and steel structure; the installation, inspection and testing of mechanical equipment and systems; and the installation, inspection and testing of monitoring instruments and electrical equipment.

All important links of the above activities must be strictly controlled, including:

- The verification of the prerequisites before construction and installation;
- The management and control during construction and installation;
- The inspection and testing of the completed structures and installed equipment and systems after the completion of construction and installation activities.

(3) Analysis and assessment of inspection and testing results

The inspection and testing results shall be collected, collated, analyzed and assessed, to judge whether the structures, equipment and systems have met the required operational level, so as to determine the subsequent actions.

18.1.2 Application of Defence in Depth

18.1.2.1 Five Levels of Defence in Depth

Defence in depth is implemented primarily through the combination of a number of consecutive and independent levels of protection to prevent harmful effects being caused to people or to the environment. If one level of protection or barrier were to fail, the subsequent level or barrier would be available. The independent effectiveness of the different levels of defence is a necessary element of defence in depth.

(1) The purpose of the first level defence is to prevent deviation from normal

operation and prevent failure of items important to safety. This leads to requirements that the plant be soundly and conservatively sited, designed, constructed, maintained and operated in accordance with appropriate quality levels and proven engineering practices. Design options that reduce the potential for internal hazards contribute to the prevention of accidents at this level of defence.

(2) The purpose of the second level defence is to detect and control deviations from normal operational states in order to prevent anticipated operational occurrences from escalating to accident conditions. This is in recognition of the fact that postulated initiating events are likely to occur over the operating lifetime of a NPP, despite the care taken to prevent them. This second level of defence necessitates the provision of specific systems and features in the design, the confirmation of their effectiveness through safety analysis, and the establishment of operating procedures to prevent such initiating events, or otherwise to minimize their consequences, and to return the plant to a safe state.

(3) For the third level of defence, it is assumed that, although very unlikely, the escalation of certain anticipated operational occurrences or postulated initiating events might not be controlled at a preceding level and that an accident could develop. This leads to the requirement that inherent and/or engineered safety features, safety systems and procedures be capable of preventing damage to the reactor core or preventing radioactive releases requiring off-site protective actions and returning the plant to a safe state.

(4) The purpose of the fourth level defence is to mitigate the consequences of accident caused by the failure of the third level defence in depth. The fourth level of defence is achieved by preventing the progress of the accident and mitigating the consequences of a severe accident. The safety objective in the case of a severe accident is that only protective actions that are limited in terms of lengths of time and areas of application would be necessary and that off-site contamination would be avoided or minimized. Event sequences that would lead to an early radioactive release or a large radioactive release are required to be practically eliminated.

(5) The purpose of the fifth and final level of defence is to mitigate the radiological consequences of radioactive releases that could potentially result from accidents. This requires the provision of adequately equipped emergency response facilities and emergency plans and emergency procedures for on-site and off-site emergency response.

18.1.2.2 Physical Barriers of the Defence in Depth

NPPs in China are designed with a series of physical barriers, as well as a

combination of active, passive and inherent safety features that contribute to the effectiveness of the physical barriers in confining radioactive material at specified locations. The number of barriers that will be necessary will depend up on the initial source term in terms of the amount and isotopic composition of radionuclides, the effectiveness of individual barriers, possible internal and external hazards, and the potential consequences of various failures.

18.1.2.3 Consideration in NPP Design

According to the requirements of national nuclear safety regulations, the design shall embody defence in depth. The levels of defence in depth shall be independent as far as is practicable so as to avoid that failure of one level of defence degrades the effectiveness of other levels of defence. The defence in depth concept shall be applied to provide several levels of defence that are aimed at preventing consequences of accidents that could lead to harmful effects on people and the environment, and ensuring that appropriate measures are taken for the protection of people and the environment and for the mitigation of consequences in the event that prevention fails. The design shall take due account of the fact that the existence of multiple levels of defence is not a basis for continued operation in the absence of one level of defence. All levels of defence in depth shall be kept available at all times and any relaxations shall be justified for specific modes of operation.

Therefore, in the design:

(1) Shall provide for multiple physical barriers to the release of radioactive material to the environment;

(2) Shall be conservative, and the construction shall be of high quality, so as to provide assurance that failures and deviations from normal operation are minimized, that accidents are prevented as far as is practicable and that a small deviation in a plant parameter does not lead to a cliff edge effect;

(3) Shall provide for the control of plant behavior by means of inherent and engineered features, such that failures and deviations from normal operation requiring actuation of safety systems are minimized or excluded by design, to the extent possible;

(4) Shall provide for supplementing the control of the plant by means of automatic actuation of safety systems, such that failures and deviations from normal operation that exceed the capability of control systems can be controlled with a high level of confidence, and the need for operator actions in the early phase of these failures or deviations from normal operation is minimized;

(5) Shall provide for systems, structures and components and procedures to control the course of and, as far as practicable, to limit the consequences of failures and

deviations from normal operation that exceed the capability of safety systems;

(6) Shall provide multiple means for ensuring that each of the fundamental safety functions is performed, thereby ensuring the effectiveness of the barriers and mitigating the consequences of any failure or deviation from normal operation.

To ensure that the concept of defence in depth is maintained, the design shall prevent, as far as is practicable:

(1) Challenges to the integrity of physical barriers;

(2) Failure of one or more barriers;

(3) Failure of one barrier as a consequence of the failure of another barrier;

(4) The possibility of harmful consequences of errors in operation and maintenance.

The design shall be such as to ensure, as far as is practicable, that the first, or at most the second, level of defence is capable of preventing an escalation to accident conditions for all failures or deviations from normal operation that are likely to occur over the operating lifetime of the NPP. Safety features for design extension conditions (especially features for mitigating the consequences of accidents involving the melting of fuel) shall as far as is practicable be independent of safety systems.

18.1.3 Application of Design Principles

NPPs in China are provided with engineered safety features and supplementary safety facilities, such as safety injection system, containment spray system, containment hydrogen concentration control and air monitoring system, steam generator auxiliary feed water system, containment isolation system, primary circuit rapid pressure release, reactor cavity water injection or core catcher, portable power supply and portable water injection facilities, etc. Engineered safety features are used to limit the consequences of a postulated accident that might lead to damage of radioactive product shielding, or design basis. Additional safety facilities are used to deal with design extension conditions; such design extension conditions or sequences are obtained through deterministic, probabilistic safety analysis, engineering judgment and other methods to strengthen the prevention and mitigation of severe accidents.

In the design of NPPs in China, consideration is taken for design principles such as inherent safety and fail-safe features, automatic operation, physical and functional isolation, redundancy and diversity, which are embodied mainly in the design of the following types of reactors, also with slight difference for different types of reactors:

For units with VVER technology, in order to improve the safety of the units, a series of important measures have been taken in design. In the passive safety design,

passive safety injection tank water makeup is provided to deal with large break loss of coolant accident; the passive containment hydrogen combiner is used to limit the hydrogen concentration. The core melt catcher is to prevent melting through of containment ground foundation. The units are designed as fully automatic operation by DCS. Four trains safety systems with completely independent and physical separation, double-layer containment for reactor building, high pressure emergency concentrated boron injection system for diversified reactivity control are equipped to ensure redundancy, independence and physical separation. Safety equipment and systems such as four emergency diesel generator sets, two emergency diesel generator sets for station blackout and the seventh channel power supply are provided to deal with common cause failures.

For AP1000 units, diversified instrumentation and control systems are designed to avoid common mode failure. Redundant automatic pressure relief system is designed to avoid the occurrence of high-pressure melt injection and steam explosion. A passive hydrogen combiner, a hydrogen igniter and a hydrogen analyzer are provided in the containment, so that the hydrogen concentration in the containment can be effectively controlled; the spent fuel pool is equipped with three 1E class level meters and one non-1E class level meter, which considers the diversity and reliability. AP1000 has passive safety features. It adopts passive core cooling system, passive containment cooling system, passive main control room habitability system, passive reactor cavity flooding system, etc., which can maintain long-term accident mitigation without operator action and AC power supply. The design of the AP1000 passive safety system meets the single failure criteria. In order to further enhance the capability of accident management over long time (after 72 hours) following an accident, additional portable pumps and portable power supplies are provided in Sanmen NPP and Haiyang NPP, and the corresponding accident procedures have also been further completed.

Taishan NPP was designed as the single reactor layout, and physical separation and protection are provided for the four trains of engineered safety features of each reactor. The safety system design to mitigate design basis condition accidents has taken into account redundancy or diversity; and the redundant equipments are physically or geographically isolated to mitigate the risk of loss of safety functions. In order to prevent and mitigate severe accidents, special systems including containment spray system, emergency core cooling system, emergency feed water system and core melt capture system are provided. Passive hydrogen combiners are distributed to avoid hydrogen accumulation and explosion in containment. In case of severe accidents, the primary pressure is reduced through the primary circuit quick pressure relief system to

prevent high-pressure melt injection. The containment spray system controls the pressure and temperature of the containment, and the combustible gas control system controls the hydrogen concentration. The design of mechanical, electrical and I&C systems that perform F1 safety functions in Taishan NPP meets the single failure criteria. In addition, after the Fukushima nuclear accident, a series of improvement measures have also been taken in Taishan NPP to improve the capability to prevent and mitigate severe accidents. In order to prevent possible common cause failures, the I&C structure, the Anticipated Transient Without Scram (ATWS) mitigation system, the design of protection system, the emergency shutdown breaker devices and the control and display in Taishan NPP have all been featured with diversity.

HPR1000 was designed as single reactor layout, realizing optimization and physical separation in layout. The double-layer containment is used to increase the safety of NPP. The reactor building and fuel building are fully protected with APC shell and the safety building has realized full physical separation. The diesel generator set building is geographically separated to ensure that at least one train of diesel generator set is available. The item of ultimate heat sink, such as essential service water pump station and the corresponding pipe trenches are protected with redundant trains and geographic separation. Prevention and mitigation measures for severe accidents such as passive hydrogen recombination of containment, filtration and discharge of containment, high-level exhaust of pressure vessel, etc. have been adopted. Secondary passive residual heat removal system, passive containment heat extraction system and reactor cavity injection and cooling system combined active and passive function have been added. Redundant design of quick relief valve of reactor coolant system, and double-layer containment against impact of large commercial aircraft have been considered. Function of main pump sealing under shutdown and Class I anti-seismic diversified drive system have been provided. In case of failure of auxiliary feed water system, the reactor can maintain safe shutdown states for 72 hours by the residual heat removal with passive provisions and the reactor coolant. Under the severe accident condition of the NPP, the reactor cavity injection and cooling system can remove the heat released from core melt by cooling the external surface of reactor vessel. The passive containment heat removal system is designed for passive long-term heat removal from containment under accident conditions. Emergency boron injection system is provided for diversified means of shutdown. Components with sufficient redundancy and independence are arranged to provide necessary safety actions to eliminate the effect of single failure of active components.

The high-temperature gas-cooled reactor has good negative feedback

characteristics. In case of some transients or the occurrence of accident, the negative reactivity introduced by temperature rise will realize automatic reactor shutdown or reduce the core power to a very low level, therefore core melting accident as with a PWR reactor will not occur. Shidao Bay NPP is designed with engineered safety features such as ventilated low pressure-resisting containment, passive residual heat removal system, passive reactor pressure vessel support cooling system, main control room habitability system, secondary circuit isolation system and steam generator accident discharge system. In the design of engineered safety features, full consideration has been taken for physical separation, single failure criterion and redundancy.

18.1.4 Response to Design Extension Conditions

NPPs in China are able to defence various risks within the scope of design basis accidents, and have certain ability to control and mitigate core melting and other design extension conditions. NPPs are designed to take preventive or mitigation measures against extreme natural hazards. For accident combined with a number of extreme natural events, provisions at various levels including emergency operation procedures, SAMG and emergency response systems, have been developed and improved to effectively mitigate consequences of accidents. Also, in-depth analysis and assessment of the provisions have been performed to prevent and mitigate accidents via PSR, and the capability to control and mitigate beyond design basis severe accidents has been improved by implementing improvement of plant flood-prevention capability, improvement of emergency water makeup and associated equipment, adding portable power supplies, improvement of hydrogen monitoring and control, improvement of elevated water tank and high energy batteries, and building new emergency control and commanding center.

(1) Requirements for newly-built NPP

After the Fukushima nuclear accident, China has put forth relevant requirements for the prevention and mitigation of severe accidents. The main contents of these requirements have been reflected in the *Code on the Safety of Nuclear Power Plant Design*.

1) A set of design extension conditions shall be derived on the basis of engineering judgement, deterministic assessments and probabilistic assessments for the purpose of further improving the safety of the NPP by enhancing the plants capabilities to withstand, without unacceptable radiological consequences, accidents that are either more severe than design basis accidents or that involve additional failures. These design extension conditions shall be used to identify the additional accident scenarios to be

addressed in the design and to plan practicable provisions for the prevention of such accidents or mitigation of their consequences.

2) An analysis of design extension conditions for the plant shall be performed. The main technical objective of considering the design extension conditions is to provide assurance that the design of the plant is such as to prevent accident conditions not considered design basis accident conditions, or to mitigate their consequences, as far as is reasonably practicable. This might require additional safety features for design extension conditions, or extension of the capability of safety systems to maintain the integrity of the containment. These additional safety features for design extension conditions, or this extension of the capability of safety systems, shall be such as to ensure the capability for managing accident conditions in which there is a significant amount of radioactive material in the containment (including radioactive material resulting from severe degradation of the reactor core). The plant shall be designed so that it can be brought into a controlled state and the containment function can be maintained, with the result that early radioactive releases or significant radioactive releases would be practically eliminated. The effectiveness could be analysed on the basis of the best estimate approach.

3) The design extension conditions shall be used to define the design basis for safety features and for the design of all other items important to safety that are necessary for preventing such conditions from arising, or, if they do arise, for controlling them and mitigating their consequences.

4) The analysis undertaken shall include identification of the features that are designed for use in, or that are capable of preventing or mitigating, events considered in the design extension conditions. These features shall be independent, to the extent practicable, of those used in more frequent accidents; shall be capable of performing in the environmental conditions pertaining to these design extension conditions, including design extension conditions in severe accidents, where appropriate; and shall have a reliability commensurate with the function that they are required to fulfill.

5) The containment and its safety features shall be able to withstand extreme scenarios that include, among other things, melting of the reactor core. These scenarios shall be selected using engineering judgement and input from PSA.

6) The design shall be such as to ensure that plant states that could lead to early radioactive releases or large radioactive releases are practically eliminated

7) For the design extension conditions, protective measures that are of limited scope in terms of area and time shall be necessary for protection of the public, and sufficient time shall be made available to implement these measures.

After the Fukushima nuclear accident, the MEE (NNSA) required that the following items for units under construction shall be completed before the initial fuel loading, including the anti-flooding improvement for nuclear safety related buildings and equipment, addition of facilities such as portable power supply and portable pumps, improvement of the seismic monitoring and anti-seismic response capacity of plants, improvement of SAMG for NPPs, in-depth assessment of earthquake and tsunami risks, implementation of PSA for external events, improvement of emergency plan and increasing capability in emergency response to nuclear accidents, preparation and improvement of information release procedures of NPPs, improvement of the anti-hazard plans and management procedures. All units that realized the initial fuel loading after the Fukushima nuclear accident have completed the improvements on schedule with required quality.

(2) Countermeasures taken by operating NPPs

All operating NPPs, with reference to the above requirements and international experience, and in consideration of their own actual conditions, have gradually adopted some reasonable and feasible prevention and mitigation measures:

1) Actively investigating and studying up-to-date development of severe accident research by foreign organizations and NPPs.

2) Initiating or improving the research and development of SAMG, including the SAMG covering all conditions of NPPs, so as to specifically mitigate the consequences of severe accidents, ensure the integrity of the pressure boundary of primary circuit and containment, reduce the radioactive releases to the environment, and to finally recover the NPP to a controllable and steady states.

3) Including the portable equipment into SAMG.

4) Performing engineering evaluations and modifications for systems and features designed to mitigate severe accidents, to enhance the capability of mitigating severe accidents.

5) Developing emergency response plan for multiple units nuclear accidents.

(3) Promoting management of severe accidents

The SAMG applicable to its own plant with reference to the practices of similar plants in foreign countries and in full consideration of the actual situation of the plant has been prepared or improved for the operating NPPs. By incorporating the latest research on the beyond design basis external event mitigating strategies both domestic and abroad, SAMG for all plant states and extreme accidents of spent fuel pools has been developed for some NPPs. In addition, the conditions of extensive damage have been incorporated into accident management guidelines for some NPPs.

After the Fukushima nuclear accident, the MEE (NNSA) required NPPs to work out emergency response plans and the allocation and coordination plans of emergency personnel and materials after multiple units on the same site entered the emergency status simultaneously and to complete the SAMG. The NPP operating organizations actively carried out research on countermeasures against emergency status with multiple units entering emergency status simultaneously, implemented corresponding improvements and updated the NPP emergency plans. Qinshan Nuclear Power Base has unified the emergency plans for the nine units to cope with multi-unit accidents, and also taken into comprehensive account the effect of multi-unit accidents on the emergency action level. The unified dispatching procedure for emergency portable equipment and the emergency management system to respond to multi-unit accidents for Qinshan Nuclear Power Base has been established. Emergency response exercises for multiple units are performed to verify the emergency response capability of Qinshan Nuclear Power Base to multi-unit accidents. Daya Bay Nuclear Power Base organized relevant forces for investigation, analysis and research, and completed the emergency response plan for possible multiple unit nuclear accidents that might exist for the six units in Daya Bay NPP and LingAo NPP, and later it will work out detailed implementation procedures for multiple units. The multiple unit nuclear emergency response plans for Sanmen NPP, Taishan NPP and Haiyang NPP have been prepared, the system of emergency organization, emergency facilities, equipment and materials, technical measures, etc. have been analyzed, emergency response guidelines for two units entering the emergency state at the same time have been provided, the emergency preparedness for multiple unit emergency response have been stipulated, and emergency exercises plans for multiple unit accidents have been prepared.

18.1.5 The Measures to Maintain the Integrity of Physical Barrier

NPPs in China are equipped with multiple barriers (such as containment) to enclose any radioactive materials released from the core, to limit the emission of radioactive materials to the environment, and to protect the public and the environment. Based on the characteristics of its own technical route, NPP operating organizations have taken appropriate measures to maintain the integrity of containment to prevent long-term off-site contamination. It mainly includes the prevention of overpressure and negative pressure from threatening the integrity of the containment, prevention of hydrogen combustion and high-pressure melt injection from posing serious threats to the containment, and prevention of core molten materials from burning through the containment base, etc.

For generation II+ PWR nuclear power units, the following measures are mainly adopted to maintain the containment integrity:

(1) Adopting concrete containment with conservative design. The pressure within the containment conservatively calculated under design basis accident conditions (loss of coolant accident and main steam line break accident) is lower than the design pressure, and also with some margin. The containment limit loading capacity evaluation has been performed, proving that the containment can withstand a limit pressure over 1.5 times its design pressure.

(2) Setting or perfecting the passive containment combustible gas control system. The passive hydrogen recombiner has been added to control the hydrogen concentration inside containment under the design basis accident and beyond design basis accident conditions, to prevent explosion resulted from local accumulation of hydrogen.

(3) Setting the containment filtration and discharge system. It has been provided to depressurize the containment to ensure its integrity when the pressure inside the containment has increased to its opening set value following a severe accident of core meltdown.

(4) The safety-class containment spray system has been provided. When the pressure in the containment has reached the actuating setting value of the containment spray system, the containment spray will be actuated to control the containment pressure.

(5) Considering extension for the relief function of the pressurizer, the primary circuit pressure is reduced to avoid high pressure core melting threatening the containment integrity in case of a severe accident.

(6) Adding portable diesel generators, portable pumps, etc.

(7) Preparing SAMG. It is to further improve the management system for NPPs to deal with severe accidents under the power and shutdown conditions, protect the pressure boundary and the containment barrier under severe accident conditions, specifically mitigate the consequences of severe accidents, reduce the radiological releases, and finally recover the unit with accident to a controllable and stable state.

For units based on AP1000 technology, the following measures are mainly adopted to maintain the containment integrity:

(1) The structural design of steel reactor containment has considered the design basis accident conditions such as rupture of primary pipe, main steam pipe and main feed water pipe. Under the design basis accident conditions, especially in the condition of a big break in the reactor coolant system, the passive containment cooling system can cool the containment to prevent it from over-temperature and over-pressure, and thus

maintain the containment integrity.

(2) In order to prevent the failure of the pressure vessel, the accident management strategy of flooding the reactor cavity and the reactor pressure vessel with water in the refueling water storage tank in the containment is considered. During a postulated serious accident, water will be used to cool the external surface of pressure vessel and to prevent the core molten debris at the bottom head from melting through the vessel wall and entering the containment. Containing the core molten materials inside the reactor vessel can prevent severe accidents outside the reactor vessel, such as steam explosion outside the reactor and the chemical reaction of core materials with the concrete. The hydrogen igniters and hydrogen recombiners are provided to avoid hydrogen combustion and explosion inside the containment. Redundant large capacity rapid pressure relief valves are provided to reduce the primary pressure, to avoid reactor melting under high pressure, and also increase the possibility of successful reactor cavity water injection to protect the containment integrity.

(3) For natural hazards, the on-site meteorological conditions of Sanmen NPP and Haiyang NPP can be enveloped within the scope of natural hazard analysis considered for AP1000 standard design. The outer side of the steel containment is a concrete shielding building, which can effectively resist missile caused by typhoons, tornadoes and so on. The impact of negative pressure caused by extreme weather on the integrity of containment is also considered in the design. In addition, the containment is also equipped with a vacuum breaking subsystem to prevent the external pressure from exceeding the internal pressure, thus maintaining the integrity of the containment. For beyond design basis accidental conditions, additional portable power supplies and portable pumps that are stored to ensure containment cooling in Sanmen NPP and Haiyang NPP.

For units based on EPR technology, the following measures are mainly adopted to maintain the containment integrity:

(1) The containment of an EPR unit with the volume of about 80000cu.m can maintain the containment pressure below its design value without starting any system to reduce pressure within 12 hours after a severe accident.

(2) In severe accident conditions, the dedicated pressure relief system of primary circuit can release pressure of the reactor coolant system to prevent high pressure core melting and its injection.

(3) In severe accident conditions, the hydrogen inside the containment can be controlled by the combustible gas control system to prevent hydrogen explosion.

(4) In severe accident conditions,, the containment spray can control the pressure

inside the containment.

(5) The in-vessel retention system is designed to prevent molten core-concrete interaction under the severe accident conditions.

(6) The filtration and ventilation system collects the leaks in the annulus between the inner and outer layers of the containment and those in the safety building not discharged into the stack for filtration and discharge.

(7) After the Fukushima nuclear accident, the portable water makeup was provided for the containment heat removal system of the EPR nuclear power units to strengthen the control of containment pressure when the containment spray is not available and in case of station blackout so as to maintain the containment integrity.

For HPR1000, in the event of a severe accident, the integrity of the containment can be ensured through measures such as large volume containment, containment hydrogen recombiner system, containment spray system, reactor cavity water injection and cooling system and containment filtration and discharge system. In case of overpressure in the containment, the containment filtration and discharge system can actively relieve the pressure so that the pressure in the containment will not exceed the bearing limit; and the radioactivity of the discharged gas can be filtered by the filtering device. The passive containment hydrogen elimination system and containment hydrogen monitoring system shall be provided to remove hydrogen and monitor the containment, so as to limit the concentration of hydrogen in the containment below the safety limit and avoid the hydrogen combustion and explosion.

For high-temperature gas-cooled reactors, the containment used is a ventilating low-pressure containment, which is different in principle from the containment of traditional pressurized water reactor NPPs and boiling-water reactor NPPs. It is based on the fact that high-temperature gas-cooled reactors are equipped with highly reliable coated particulate fuel elements, which can realize the containment function of radioactive materials. Through reasonable core design, under normal operation and accident conditions, the normal operating temperature of the fuel element is lower than 1200, and the temperature after the accident is lower than 1620. After the accident of the loss of pressure in primary circuit, it will take quite a long time for fuel temperature to reach the maximum temperature because of the high thermal capacity of reactor, which provided a long tolerance time for accident management measures. The analysis shows that even if there is no negative pressure exhaust system in the nuclear island, the dose caused by the radioactivity released into the environment without filtration is lower than the acceptance dose of accident conditions, and will not cause serious off-site consequences.

18.1.6 NPP Design Improvement

On the basis of the results of deterministic safety assessment and PSA and the requirements of current nuclear safety codes, national and industrial technical standards of China, the operating organization for NPPs in operation have carried out highly effective technical modifications for their structures, systems and components with respect to the design and equipment problems identified during the commissioning and operation period by using proven and qualified technologies to increase safety and reliability since the commercial operation of the units.

Since the unit was put into operation, the operating organization of Qinshan NPP has carried out and completed dozens of major technical modification projects, such as the replacement of pressure vessel head and related component systems, the digital modification of the reactor protection and associated equipment and I&C systems, modification of emergency chilled water system and chilled water tank, modification of cement solidification system, addition of passive hydrogen recombiner and modification of containment sump strainer. The operating organization is carrying out OLE and will carry out major technical modification projects such as hanger modification and cable tray modification.

The operating organization of Daya Bay NPP has successively completed major technical modification projects such as improvement to prevent boron water from being diluted by mistake, improvement to prevent core from being exposed, modification of the blocked pipe section at the inlet of the residual heat removal system, improvement to install passive hydrogen combiner in the reactor building, improvement to install water level meter and make-up water in the recirculation sump of the safety injection system/containment spray system, improvement to replace the whole rubber lined pipe of the safety service water system, modification of adding continuous monitoring of the temperature and liquid level of the spent fuel pool, etc., and is working on major technical modification projects such as the modification of blocked pipes of the safety injection system, the modification of heat shielding and isolation of the reactor coolant pump, the improvement of the cooling capacity of the reactor refueling pool and spent fuel pool cooling and processing systems, and the modification as to the remodeling and expansion of the 220V AC uninterrupted power supply system.

During the design of Units 1 and 2 of Qinshan Phase 2 NPP, a series of design improvements were implemented according to the construction and operating experience of similar power plants. Further improvements were made to Units 3 and Unit 4 on the basis of fully absorbing the experience of Unit 1 and Unit 2, such as the use of new AFA3G type fuel assemblies, the adoption of containment sump filter of new

model, the addition of hydrogen removal measures under beyond design basis accident, the addition of prevention of inadvertent boron dilution, etc.

Dozens of significant design improvement projects on the basis of Daya Bay NPP and its own operating practice have been implemented in LingAo NPP, such as multi-point hydrogen measurement in containment, installation of water level gauge and make-up water modification to the recirculation sump of safety injection system/ containment spray system, etc., and is currently carrying out overall replacement and modification of rubber-lined carbon steel pipeline of safety service water system, modification of diesel generator of diesel generator system with new hydraulic test pump, expansion and modification of uninterrupted power supply system for reactor protection, etc.

A number of technical improvements with reference to Units 1 and Unit 2 have been carried out for Units 3 and Unit 4 of LingAo NPP. The safety design improvements as to safety include the adoption of advanced AFA 3G fuel assemblies, the extension of pressure relief function of pressurizer to prevent high pressure core melt condition, the modification of pressure relief and discharge system of the containment sand filter, etc. passive hydrogen recombiner in the reactor building have been installed Units 1 and 2, and the detailed design are being conducted for Units 3 and 4. For Unit3 and 4, modification projects such as modification of emergency cooling water inlet interface under severe accident conditions, modification of emergency diesel generator, displacement of outgoing voltage transformer and modification of mobile test load of the diesel generator, optimization and modification of non-safety-class digital I&C platform have been completed.

Since 2003 when its first unit was put into operation, design improvements have been carried out for Qinshan Phase Ⅲ NPP according to the operation experience of CANDU-6 HWR reactor and the actual conditions of China, such as the spent fuel dry storage project, the addition of backup repair system for re-circulating cooling water, the addition of two data acquisition systems for reactor shutdown system, and the addition of auxiliary feed water pumps, modification of sump strainer of emergency core cooling system, upgrading and modification of excitation system and protection device of the standby diesel generators.

Improvements have been conducted for Units 1 and Unit 2 of Tianwan NPP in the design based on VVER-1000 PWR and their own operating experience, such as modification of ventilation units of negative pressure system in safety building and overall modification of rotary screen, etc. Necessary design improvements have been carried out for Units 3 and 4 with reference to Units 1 and 2, includes the modification

of collection and monitoring system for nuclear island low radioactive wastewater, improvement of nuclear island fire protection system and improvement of coolant purification system, etc. In addition, the containment sump strainer has been improved and a gas detection system for damaged fuel assemblies has been added according to the new regulations, standards and technical requirements. Major technical modifications have been completed for Tianwan NPP such as the modification of excitation regulation system for emergency diesel generators of Units 1 and 2, the modification of process waste delivering interfaces of Units 1 and 2, the modification of intermediate dry storage facilities of spent fuel, the modification of secondary side header flange and side manhole flange of steam generator, the flange of pressurizer manhole and the flange of electric heater, and the upgrade and modification of velocity measurement system of ventilation stack of Units 1 and 2 on nuclear island.

From 2016 to 2018, a number of improvements have been implemented for Fangjiashan NPP, including eighteen important technical improvement projects such as changes of long fuel cycle, replacement of C-type sealing ring of reactor pressure vessel with domestic products, addition of ventilation system in main stream isolation valve room, replacement of bypass isolation valve for temperature measurement, addition of Class 1E DCS engineer station and supporting equipment, optimization and upgrading of Class 1E DCS software, etc. Important technical improvement projects such as CF3 improved domestic fuel assembly irradiation test, reliability improvement of fuel loading and unloading machine, change of safety valve model of nuclear sampling system, modification of differential pressure gauge of plate heat exchanger of safe service water system, modification of backwashing pipeline of train A/B shellfish trap of safe service water system, application for change of operation station of power station control computer system, change of newly added alarm signal of LOCA monitoring system, etc. are under way.

Among the NPPs under construction in China, a series of design improvements have been made to the independently designed generation II improved pressurized water reactor units based on the introduction, digestion and absorption of mature technologies in foreign countries and in conjunction with years of operating experience of similar units at home and abroad and the results of safety research and continuous improvement. A series of design improvement have been made to further improve the safety level.

The safe and stable operation of operating NPPs in China has been effectively ensured through continuous technical improvement. After the Fukushima nuclear accident, improvements have been actively carried out for the operating NPPs in accordance with the plan formulated by the MEE (NNSA) in consideration of the results

of the national comprehensive safety inspection and self-inspection. All short-term safety improvement projects and medium-term safety improvement project packages have been completed on schedule. The long-term safety improvement projects include improving emergency plans, raising the emergency response capacity of NPPs, completing the NPP information release procedures and strengthening popularization of nuclear knowledge. All operating organizations have actively carried out relevant researches, taken a series of effective measures and achieved significant progresses.

The NPPs in China under construction basically meet the requirements of national nuclear safety codes, and have implemented effective management in all links such as siting, design, manufacturing, construction, installation and commissioning. The quality assurance system and quality monitoring system are in normal operation, and the engineering construction meets the design requirements.

18.1.7 Regulatory Review and Control Activities

The MEE (NNSA) requires that defence-in-depth shall be incorporated in the design process of NPPs. The levels of defence in depth shall be independent as far as is practicable so as to avoid that failure of one level of defence degrades the effectiveness of other levels of defence. Safety features used for design extension conditions shall be as independent as possible from safety systems. PSAR and FSAR of NPPs shall be reviewed to confirm that the principle of defence-in-depth is implemented. In addition, important modifications as to safety of NPPs shall also be reviewed and approved by the MEE (NNSA).

18.2 Incorporation of proven technologies

18.2.1 Codes and Regulatory Requirements

It has been more than 30 years since China began to design and construct the first NPP, Qinshan NPP. During this period, the nuclear power technologies have been developing continuously both home and abroad. In the meanwhile, some new codes and standards have been issued. In order to promote the progress of nuclear power technologies and to continuously improve the safety and economic performance of nuclear power units, China has made continual design improvements for the nuclear power units both in operation and under construction according to the operating experience at home and overseas and relevant progress in technologies. Especially for new nuclear power units, new technologies are adopted to ensure the advanced level and safety of design based on the construction and operation experience of similar plants and reference units.

In China, the requirements for NPPs adopting proven technologies in design, construction and improvement include the following:

(1) The MEE (NNSA) requires NPP operating organizations to adopt the proven and qualified process technologies in the design. Documents (e.g., the safety analysis report) submitted to the MEE (NNSA) by operating organizations of NPPs shall describe that the technologies adopted have been validated and verified.

(2) It is necessary to identify and assess the codes and standards of the design criterion for items important to safety in NPPs to determine their applicability, appropriateness and adequacy, and make supplement or modification of such codes and standards as necessary to ensure that the design quality is commensurate with the required safety functions.

(3) Items important to safety in NPPs shall have been previously verified under equivalent conditions of application, otherwise the items shall be of high quality and their technologies have been verified or tested.

(4) When unverified design or facilities are introduced, or there is any deviation from existing engineering practice, the safety shall be proved suitable by appropriate backup research plan or performance test with particular acceptance criterion or by verification with operating experience obtained from other relevant applications. New designs, facilities or practices shall be fully tested before being put into operation and monitored during the application to verify that the expected results are achieved.

(5) For various types of improved reactors, the design of structures, systems and components that have been successfully applied in the operating NPPs shall be adopted as far as possible, and relevant operating experiences obtained in other NPPs shall be at least used for reference.

(6) The manufacture and construction methods shall be laid down carefully. The personnels shall be selected correctly and trained, and their qualification reviewed. The manufacture and construction of SSCs shall be performed by experienced domestic and foreign contractors or suppliers. Operating organizations shall review their contracting and supplying capability, engineering experience of manufacturing and construction, and documents and records that can demonstrate their qualification.

18.2.2 Measures to be taken by Licensees

According to the requirements of current nuclear safety codes, national and industrial technical standards of China, technical modifications for their structures, systems and components have been effectively carried out for operating NPPs with respect to the design and equipment problems identified during the commissioning and

operation period by using proven and qualified technologies to increase safety and reliability since the commercial operation of the units.

Among the NPPs under construction in China, a series of design improvements have been made to the independently designed pressurized water reactor units based on the introduction, digestion and absorption of mature technologies from foreign countries and in combination with years of operating experience of similar units at home and abroad as well as the results of safety research and continuous improvement. Compared with the same units in other countries, a higher safety level is achieved.

Dozens of major safety-related technical improvements have been implemented for Units 5 and 6 of Tianwan NPP based on the latest safety standards at home and abroad after the Fukushima nuclear accident with reference to Units 1 and 2 of Fuqing NPP that have been put into operation.

For generation II+ nuclear power projects, including Hongyanhe NPP, Ningde NPP, Yangjiang NPP, Fuqing NPP, Fangchenggang NPP, Units 5 and 6 of Tianwan NPP, the improvements and relevant operation experience of the reference power plant have all been adopted in the construction, mainly including the change of long fuel cycle, extension of pressure relief function of pressurizer, improvement of dual-unit common hydraulic test pump, improvement of blocked pipe of safety injection system, improvement of primary circuit leakage detection method and device, analysis and measures of non-condensable gas accumulation in safety-related systems, fire hazard analysis and improvement of fire protection design, validation of neutron poison effectiveness in fuel storage pool, modification of continuous monitoring of temperature and liquid level of fuel storage pool, improvement of sump recirculation system and strainer of containment, measures to improve the safety of water intake affected by marine organisms or foreign materials, adjustment and optimization of containment hydrogen monitoring in severe accidents, addition of emergency cooling water inlet interfaces under severe accident conditions, addition of standby emergency diesel generator sets and continuous improvement of minimization of radioactive waste, etc.

The HPR1000 technical scheme adopted by Units 5 and 6 of Fuqing NPP and Units 3 and 4 of Fangchenggang NPP is based on mature technologies that China has mastered, absorbs the experience feedback of Fukushima nuclear accident, and considers perfect measures for prevention and mitigation of severe accidents. A series of important technical improvements have been implemented. It mainly includes 177 sets of fuel assemblies in the reactor core, with innovative and improved reactor structure, double-layer containment which is adapted to improve accident prevention

capability, no high-pressure safety injection, and with medium-pressure safety injection to mitigate the consequences of steam generator tube rupture accident, and low-pressure safety injection and residual heat removal functions. A secondary passive residual heat removal system is provided to maintain core cooling when water supply is completely lost and AC power supply inside and outside the site is lost. In terms of mitigation of severe accidents, additional cooling systems are set up to realize diversified cold sources for containment and spent fuel pool cooling, and the post-severe accident monitoring system is improved to facilitate monitoring of unit status and effectiveness of mitigation measures after severe accidents. 3 emergency diesel generators and 2 station black out diesel generators are provided to improve the reliability of emergency power supply.

Major system and equipment problems identified during commissioning of Unit 1 of Taishan NPP were analyzed and solved by the designer. In addition, the design changes of Unit 1 during commissioning have been implemented to Unit 2 in an effective way.

18.2.3 New Technologies and Methods for Analysis, Inspection and Testing

The nuclear safety codes of China require when unproven design or feature is introduced, or there is any deviation from existing engineering practice in the design and construction of NPPs, safety shall be demonstrated to be adequate by appropriate supporting research programmes, or by examination of operational experience from other relevant applications. The development shall also be adequately tested before being brought into service and shall be monitored in service, to verify that the expected behavior is achieved.

Sanmen NPP and Haiyang NPP are adopted AP1000 nuclear power technology. For the passive design features and special engineered safety features of AP1000 units, China have carried out analysis for accidents such as rupture of reactor vessel direct injection pipeline, rupture of core makeup water tank pipeline, inadvertent actuation of automatic pressure release system, and rupture of passive residual heat removal system pipeline, in addition to traditional analysis on the design basis accident analysis. The main contents of severe accident analysis include: phenomenological analysis for severe accidents, analysis and justification of the effectiveness of severe accident mitigation measures, level I PSA success criterion analysis, level II PSA success criterion analysis and source term analysis. In addition, special analysis for AP1000 units has been made for the long-term station blackout exceeding 72 hours and the conditions of loss of normal heat sink by lessons learned from the Fukushima nuclear accident. The

improvement of long-term period water make up has been proposed based on the analysis results, and analysis and evaluation have been made for relevant improvements. China has also performed safety assessment independent verification and qualification tests for certain AP1000 units, the safety assessment independent verification is mainly focused on functions of passive safety systems, and the feasibility and effectiveness of design important to safety and new accident mitigation measures; the qualification tests are mainly to verify if the system and equipment functions meet the design requirements via testing of equipment design, equipment performance and system design. According to the relevant requirements in the conditions for construction permit, NPPs in China have carried out control rod driving mechanism anti-seismic test, which has verified the rationality of the seismic design of the control rod drive mechanism.

Taishan NPP is adopted the EPR technology, with similar units now under construction in France as reference, and also making reference to similar units under construction in Finland on certain systems and equipments, including the addition of passive steam exhaust channel in the fuel building. On the analysis of design basis accident, the EPR units have taken into account the initial states of each type of accident when it occurs. For many accidents, not only state A in the standard operational states is analyzed, the analysis has been made up to state B, C and D of refueling outage. For each design basis accident, not only the process to bring the plant to the controllable status has been analyzed, analysis is also made on how it can reach the safe shutdown status from the controllable status. In the analysis of design basis accident, consideration has been given not only to the single failure criterion, but also to the assumption that a train of safety system is in preventive maintenance. The N+3 principle has been applied in the design of safety systems. On analysis of beyond design basis accidents, analysis has been made for EPR on both DEC-A condition (the severe accident prevention condition) and DEC-B condition (the severe accident mitigation condition), the function demand analysis for a series of severe accident prevention and mitigation systems has been completed, the complete level I and level II PSA analyses have been carried out, and based on deterministic analysis, the severe accident management documents covering power operation and shutdown conditions have been prepared.

HPR1000 technology which is adopted in Unit 5 and 6 of Fuqing NPP and Units 3 and 4 of Fangchenggang NPP are designed according to the requirements in the *Code on the Safety of Nuclear Power Plant Design* and the relevant requirements of User Requirement Document (URD) and Europe User Requirement Document (EUR). Meanwhile, a lot of testing and verification work has been carried out for HPR1000.

HPR1000 is designed with reactor core with 177 fuel assemblies, safety systems with multiple redundancies, single reactor layout, double-layer containment, and combination of active and passive systems. The design principle of defence in depth has been fully carried out in a balanced way, and complete provisions to prevent and mitigate severe accidents have been made for HPR100. On the basis of the overall list of initial event conditions, accident analysis and PSA for all plant states have been carried out, including levels I and II PSA for internal and external events, also covering the spent fuel pool and other facilities containing large amount of radioactive materials. The accident analysis has indicated that HPR1000 meets the requirements of 15% thermal engineering margin and no intervention by operator for 30 minutes with the optimized core type and safety system design plans; the use of a series of provisions to prevent core damage and mitigate severe accidents can ensure fairly low core damage frequency and radioactive materials release.

For new NPPs, simulation design verification has been performed for DCS. The verification is backed up by the simulation platform software, which integrated technologies of simulation for NPP process systems, control strategy and human-machine interface. The simulation design verification was performed in the dynamic form for both classified verification and special verification.

18.2.4 Regulatory Review and Control Activities

The MEE (NNSA) has continuously improved the regulation procedures for safety review and test verification and has relatively perfect analysis software for safety review. It has promoted the safety review of NPPs under construction. By the end of 2018, FSAR reviews for Sanmen NPP, Haiyang NPP, Taishan NPP, Yangjiang NPP (Units 5 and 6) have been completed, and FSAR reviews for Fuqing NPP (Units 5 and 6) and Shidao Bay NPP (high-temperature gas-cooled reactor) are under way. The MEE (NNSA) has prepared the *Method of Management for Software Safety Analysis*, established a software assessment team and a software assessment related implementation method, and carried out the certification of safety analysis software for the first time. For AP1000, the MEE (NNSA) mainly carried out review calculation for accidents such as loss of coolant, reactor coolant pump shaft block, rupture of a single SG heat transfer tube, small break loss of coolant with breaking of both ends of direct injection pipeline and rupture of primary feed water pipe. The MEE (NNSA) mainly carried out verifying calculation on reactor physics, accident analysis, radiation protection, effluent discharge source term, probabilistic safety assessment and verification and stress analysis and structure anti-seismic analysis for HPR1000.

18.3 Design for Reliable, Stable and Manageable Operation

18.3.1 Codes and Regulatory Requirements

China requires that human factors, especially human-machine interfaces, shall be taken into systematic consideration for the design of NPPs. The *Code on the Safety of Nuclear Power Plant Design* sets forth the following design requirements on optimal operator performance:

(1) Systematic consideration of human factors, including the human-machine interface, shall be included at an early stage in the design process for a NPP and shall be continued throughout the entire design process.

(2) The minimum number of operating personnel shall be specified to meet the requirements that performing all the simultaneous operations necessary to bring the plant into a safe state.

(3) Operating personnel who have gained operating experience in similar plants shall, as far as is practicable, be actively involved in the design process, in order to ensure that consideration is given as early as possible in the process to the future operation and maintenance of equipment.

(4) The design shall support operating personnel in the fulfilment of their responsibilities and in the performance of their tasks, and shall limit the effects of operating errors on safety. The design process shall pay attention to plant layout and equipment layout, and to procedures, including procedures for maintenance and inspection, to facilitate interaction between the operating personnel and the plant.

(5) The human-machine interface shall be designed to provide the operators with comprehensive but easily manageable information, in accordance with the necessary decision times and action times. The information necessary for the operator to make a decision to act shall be simply and unambiguously presented.

(6) The operator shall be provided with the necessary information:

- To assess the general state of the plant in any condition;
- To operate the plant within the specified limits on parameters associated with plant systems and equipment (operational limits and conditions);
- To confirm that safety actions for the actuation of safety systems are automatically initiated when needed and that the relevant systems perform as intended;
- To determine both the need for and the time for manual initiation of the specified safety actions.

(7) The design shall be such as to promote the success of operator actions with due

regard for the time available for action, the conditions to be expected and the psychological demands being made on the operator.

(8) The need for intervention by the operator on a short time scale shall be kept to a minimum, and it shall be demonstrated that the operator has sufficient time to make a decision and sufficient time to act.

(9) The design shall be such as to ensure that, following an event affecting the plant, environmental conditions in the control room or the remote control panel and in locations on the access route to the remote control panel do not compromise the protection and safety of the operating personnel.

(10) The design of workplaces and the working environment of the operating personnel shall be in accordance with ergonomic concepts.

(11) Verification and validation, including by the use of simulators, of features relating to human factors shall be included at appropriate stages to confirm that necessary actions by the operator have been identified and can be correctly performed.

Furthermore, the standard *Design for Control Rooms of Nuclear Power Plants* has specified the main control room design principles, main control room function design methods and function design and staffing requirements for NPPs, and also specified the procedures to verify and validate the control room function designs. The standard *Design Criterion for Remote Control Panel of Nuclear Power Plants* has specified the design requirements for remote control panel of NPPs, including the functional selection, design and organization of human-machine interfaces, and also specified the procedure requirements for system inspection and approving remote control panel design.

The human-machine interfaces of NPPs are divided into four categories: main control room, remote shutdown station, technical support center and local control. Main control room can provide all process information, control and communication means under all conditions to monitor and control the state of the power plant. When the main control room is unavailable, the remote shutdown station can lead the power plant into and maintain a safe shutdown state. The technical support center provides information as well as information evaluation and communication means for the expert group under accident conditions. The local control function is independent of the main control room or operates independently. In the short term of accident conditions, various safety actions of the NPP are automatic, so that the operator's intervention is not required within a reasonable period of time when the anticipated operational occurrences or design basis accidents start. The operator can obtain sufficient information to monitor the effect of automatic actions.

18.3.2 Measures to be Taken by Licensees

Optimization design has been made for the operator working areas and equipment arrangement according to ergonomic principles in NPPs:

(1) Necessary measures are taken to ensure that the lighting, humidity and temperature of the working environment are satisfactory.

(2) Monitors and instruments are arranged on a unified basis, so that operators move the minimum distance when monitoring and controlling the plant.

(3) On the main control console, signals and operation push-buttons are grouped by functions; different function blocks are distinguished with different colors; and different signs shall be used to indicate the control of valves and pumps.

(4) Different sounds and videos are used, so that operators can distinguish alarms of different levels.

(5) Alarm signal sources are carefully selected and arranged in priority levels, to provide necessary function of alarm jumper, avoid too many and too complicated display and alarm system signals in the MCR.

(6) The operators have sufficient time to check and confirm automatic response, and implement specified procedures because of automatic safety actions. The need for intervention by the operator on a short time-scale is decreased as possible to reduce psychological burden of operator.

(7) It shall be quite easy to link the control and display function marks with the devices being monitored or controlled.

(8) Control devices and their function displays shall be designed where it is easy for operators to operate and observe.

The design organizations in China attach great importance to the control room since it is the area with most centralized human-machine interface and the direct working place of operating personnel. The following practices have been successfully employed according to the experience feedback and with reference of the design ideas of other advanced NPPs in the design of new NPPs and the modification of operating NPPs in recent years:

(1) Fully adopting the design ideas of DCS and advanced control room. Digital human-machine interfaces are used in the MCR, the logical relation of human-machine interface, operating display frame and procedures and the alarm systems were designed with the principle of ergonomics.

(2) In designing the backup panel, operating habits of operators were taken into full consideration for function zoning, function grouping and equipment standardization to reduce human errors.

(3) The human-machine interface equipments used in remote shutdown stations were the same with main control room to ensure that operators can learn the ropes quickly without accustoming to another interface when using the remote shutdown station.

(4) Carrying out the MCR environmental design by applying the basic research results of ergonomics, physiology and psychology.

(5) Carrying out the design verification and ensuring the independence of the verification team; formulating design rules relevant to human factors and integrating the design team.

(6) Fully considering the human factors and human-machine interface related issues in the design and modification of NPPs. In the design and modification of NPPs, voice prompts have been added for units and rooms, and unit labeling specification has been prepared, such as unit color distinction and direction indication. The “special work to prevent entering wrong interval” has been carried out.

18.3.3 Regulatory Review and Control Activities

The MEE (NNSA) has paid attention to the effect of human factors on safety, and issued in 1992 the nuclear safety regulatory technical document *Standard format and contents of nuclear power plant safety analysis reports Chapter 18: Ergonomics and control room*, which proposed the requirements on the organization and structure and design criteria, safety parameters and display functions for control room design. The ergonomic principle shall be considered and applied in the control room design and safety parameters display. The main control room design is described in PSAR and FSAR of NPPs according to the requirements in the above-mentioned nuclear safety regulatory technical document. In most NPPs of China, the advanced main control room with computer technology is designed and realized based on the operator structure and digital technology and many other technical means. The conventional analog monitoring measures are the backup to computerized monitoring means, and the application of both has taken into full consideration the ergonomic principle.

19. Operation

Each Contracting Party shall take the appropriate steps to ensure that:

(i) the initial authorization to operate a nuclear installation is based upon an appropriate safety analysis and a commissioning programme demonstrating that the installation, as constructed, is consistent with design and safety requirements;

(ii) operational limits and conditions derived from the safety analysis, tests and operational experience are defined and revised as necessary for identifying safe boundaries for operation;

(iii) operation, maintenance, inspection and testing of a nuclear installation are conducted in accordance with approved procedures;

(iv) procedures are established for responding to anticipated operational occurrences and to accidents;

(v) necessary engineering and technical support in all safety-related fields is available throughout the lifetime of a nuclear installation;

(vi) events significant to safety are reported in a timely manner by the holder of the relevant licence to the regulatory body;

(vii) programmes to collect and analyze operating experience are established, the results obtained and the conclusions drawn are acted upon and that existing mechanisms are used to share important experience with international bodies and with other operating organizations and regulatory body;

(viii) the generation of radioactive waste resulting from the operation of a nuclear installation is kept to the minimum practicable for the process concerned, both in activity and in volume, and any necessary treatment and storage of spent fuel and waste directly related to the operation and on the same site as that of the nuclear installation take into consideration conditioning and disposal.

19.1 Initial Authorization

19.1.1 Review and Approval of Operation Related Licenses of NPPs

It is stipulated by the *Nuclear Safety Act* that before the initial fuel loading of a NPP, the NPP operating organization shall apply to the MEE (NNSA) for operation and submit the following documents:

(1) NPP operation application;

(2) Final safety analysis report;

(3) Quality assurance related document;

(4) Emergency plan;

(5) Other materials required by laws and administrative regulations.

The final safety analysis report is one of the importance bases for the review and approval of NPP licenses, including accurate and precise information of NPPs and their operation status, such as safety principles, design basis, site and plant characteristics, operation limits and conditions, and accident analysis information, which enable regulatory body to independently assess the safety of plants. After obtaining the operating license, the NPP operating organization shall operate in accordance with the provisions of the license.

The NPP operating organization shall also submit the following documents to the MEE (NNSA) according to the NPP progress of construction, commissioning and operation preparation:

(1) Commissioning program;

(2) Maintenance program;

(3) In-service inspection program;

(4) Fuel loading and refueling program;

(5) Report on results of pre-service inspection;

(6) Commissioning report before initial fuel loading.

The NPP operating organization shall complete the commissioning test project according to the items, methods and steps determined by the approved commissioning program. The operating organization of the NPP shall submit the commissioning report to the MEE (NNSA) within two months after the completion of all commissioning items determined in the commissioning program.

The MEE (NNSA) will perform technical review on applications for NPP operating license by law. For application for NPP operating license meeting the nuclear safety requirements, the MEE (NNSA) shall make provisions for granting the license in accordance with the law and make an announcement within 20 working days from the date of completion of the technical review. For those not meeting the nuclear safety requirements, the applicants shall be notified in writing and the reasons shall be given. When reviewing and approving applications for operating license of NPPs, the MEE (NNSA) shall consult the relevant departments of the State Council and the people's governments of provinces, autonomous regions and municipalities directly under the Central Government where the NPPs are located.

19.1.2 Commissioning of NPP

Commissioning program is prepared and implemented by operating organizations

of NPPs in order to ensure that the commissioning activities are safely and effectively implemented according to the written procedures. The commissioning program shall get approval from the MEE (NNSA). During the preparation of the commissioning program, the upstream design documents shall be strictly followed and the operating experience of similar units shall be learned from. In addition, it is required to strengthen the management of interfaces between commissioning and design, manufacturing, installation and operation, and strengthen the coordination and communication among commissioning disciplines of engineering projects, so as to further improve the preparation quality of the commissioning program. The commissioning program shall be able to provide evidence that the constructed facilities meet the design requirements and safety requirements. The commissioning program includes all the tests mandatory to verify that the completed NPP meets the requirements of the safety analysis report and design requirements and can therefore operate according to the operational limits and conditions. In addition, operators shall be given the opportunity to become familiar with the operation of NPPs.

Commissioning activities of operating organizations shall strictly follow the provisions in the commissioning program. If the commissioning program needs to be modified due to changes in upstream design documents, it shall be submitted to the MEE (NNSA) for review and approval in advance.

The commissioning activities of the operating organization of NPP is divided into three phases, namely pre-loading, loading-criticality-low power, and power, so as to clarify the tests required to be finished in the expected period of each phase and define the “control points” at which the testing results shall be reviewed before continuing with the next phase. The implementation of commissioning activities shall be reviewed phase by phase. The next phase cannot be started by the operating organization until the evaluation and the examination of the obtained results in current commissioning phase are finished, and it is confirmed that all objectives have been achieved and all regulatory requirements of nuclear safety have been met.

The MEE (NNSA) sets control points during the commissioning of relevant units in the NPP. The purpose of control-point inspection is to identify whether all commissioning items are completed according to the requirements of the commissioning program before the control points and whether the release conditions of the control points are met. Only after the MEE (NNSA) approves the release of the control point can the NPP operating organization carry out the commissioning work of the next phase. For test items that do not meet the requirements of the commissioning program such as preconditions, process parameters and test results, the MEE (NNSA)

will urge the operating organization to carry out analysis, evaluation or re-implementation.

The MEE (NNSA) has organized the preparation of the commissioning supervision and inspection program. For the commissioning supervision and inspection of specific nuclear facilities, the general supervision and inspection program is expanded and refined based on the specific conditions of the facilities, and detailed implementation procedures are made for the commissioning supervision and inspection of systems and equipment important to safety. In January 2016, the MEE (NNSA) issued *Nuclear Safety Supervision and Inspection Program for Commissioning Phase of Units 1 and 2 of Sanmen Nuclear Power Plant*, *Nuclear Safety Supervision and Inspection Program for Commissioning Phase of Units 1 and 2 of Haiyang Nuclear Power Plant* and *Nuclear Safety Supervision and Inspection Program for Commissioning Phase of Units 1 and 2 of Taishan Nuclear Power Plant*, for the commissioning and supervision of new reactor types such as AP1000 and EPR, and to strengthen the supervision of the whole process of commissioning of new reactors.

As for the commissioning supervision and inspection of the first-of-a-kind AP1000 and EPR, there is no successful commissioning and operation experience available for reference in the world. The MEE (NNSA) has focused on many areas such as review, supervision and inspection of first-of-a-kind commissioning, including the integrity of commissioning test items, the appropriateness of acceptance criteria of commissioning test, the accuracy of division of commissioning test phases, and the effectiveness of commissioning test supervision and test result evaluation.

During the commissioning supervision and inspection of AP1000 and EPR units, the MEE (NNSA) has studied and determined the commissioning test items (85 for AP1000 units and 78 for EPR units) for which nuclear safety supervision is required for AP1000 and EPR first-of-a-kind units in the world. It has formulated the nuclear safety supervision plan for commissioning test items important to safety of AP1000 and EPR units, prepared the corresponding supervision and inspection procedures, completed the on-site supervision and inspection of the selected commissioning test items, and issued the supervision reports of these test items.

19.2 Operational Limits and Conditions

The *Code on the Safety of Nuclear Power Plant Operation* stipulates the operational limits and conditions of NPP. The *Operational Limits, Conditions and Procedures of Nuclear Power Plant* is a further supplement and explanation to the relevant provisions of *Code on the Safety of Nuclear Power Plant Operation* in order to

provide guidance for NPPs to establish and implement the operational limits and conditions.

It is required by the *Code on the Safety of Nuclear Power Plant Operation* that in order to ensure that the operation of NPPs meets the design requirements, operating organizations shall develop operational limits and conditions including technical and management aspects. Operational limits and conditions shall reflect the final design and shall be evaluated and approved by the MEE (NNSA) before the NPP starts operation.

Operational limits and conditions are important bases for operating organizations to operate NPPs. Operational limits and conditions include requirements for various operational states (including shutdown). The operational limits and conditions also include the actions taken by the operators and the restrictions observed. Operational limits and conditions shall determine operating requirements to ensure that the safety system, including engineered safety features, can perform necessary functions under all operational states and design basis accidents. Operating personnel directly responsible for operation in NPPs shall be familiar with the operational limits and conditions and comply with such limits and conditions.

The technical aspects of operational limits and conditions include the limits and operating requirements that shall be observed when all structures, systems and components important to safety perform the predetermined functions determined in the safety analysis report. As for operating personnel, the operational limits and conditions include the principled requirement of necessary supervision, correction or supplementary actions of the equipment that shall maintain the operational limits and conditions to perform its functions. Some operational limits and conditions may include a combination of automatic functions and manual maneuver.

The operational limits and conditions of NPPs include:

- safety limits;
- setting values of safety systems;
- normal operational limits and conditions;
- supervision requirements;
- actions to be taken in case of deviation from operational limits and conditions.

The operational limits and conditions are based on the analyses of specific NPP and its environment, and are in accordance with the provisions in the final design. Some necessary modifications are made according to the test results during the commissioning; the justifications and the necessities adopted each of the operational limit and condition shall be demonstrated in the written form.

NPP operating organizations shall carry out periodical review over the operational limits and conditions throughout the operating life of the NPP, in order to ensure that they remain applicable to the intended purpose, and shall make periodical modifications of the operational limits and conditions based on accumulated experience and technological developments. Even if no modification is made to the NPP, such periodic review shall also be carried out. NPP operating organizations shall prepare the working procedures for the modification of operational limits and conditions and modify the operational limits and conditions according to the procedures. Any modification to operation limits and conditions in NPPs of China is subjected to review and approval by the MEE (NNSA).

19.3 Procedures for Operation, Maintenance, Inspection and Testing of NPP

It is stipulated by the *Code on the Safety of Nuclear Power Plant Operation* and relevant guidelines that operating organization of NPP shall prepare operating procedures which are completely applicable for normal operation, anticipated operational occurrences and accidents. The operating organization of NPP shall prepare and implement programs of maintenance, testing, supervision and inspection for structure system and component important to safety, which shall be completed before fuel loading. Such programs shall consider the operational limits and conditions as well as any other applicable regulatory requirements as to nuclear safety. Re-evaluation is required in the light of operating experience.

Before the operation of NPPs in China, its operating organizations will work out detailed written operating procedures in cooperation with design organizations and suppliers. The preparation, review and modification of the operation procedures shall conform to the approved operational limits and conditions, and appropriate safety margin shall be reserved. The operating procedure prepared shall include actions to be taken by the NPP under normal operation, anticipated operational occurrence and accident. The operating procedures will instruct the operators to operate in the correct order. In case the operating personnel find that the state or conditions of the NPP system or equipment do not conform to the operating procedure, the operating organization shall clearly stipulate the responsibilities and communication channels of the relevant personnel. The operating procedures shall be regularly reviewed; any changes shall be notified to the operating personnel and other users of such documents. The changes shall be made according to the procedures specified in writing.

The operating organization of NPP in China have prepared and implemented programs of maintenance, testing, supervision and inspection for structure system and

component important to safety before fuel loading.

Prior to the maintenance, testing, verification and inspection of the structure system and component, the procedures for the maintenance, test verification and inspection of the structure system and component important to safety are prepared by the operating organization of NPP in cooperation with the NPP and the equipment suppliers, which clearly specify the standards and periods of the maintenance, testing, verification and inspection of structure system and component important to safety. After maintenance, the authorized personnel will inspect the structures, systems and components before resuming normal operation, and carry out corresponding verification tests when necessary.

For the in-service inspection (ISI) of NPP, some measures have already been taken at the design stage, and reviews have also been performed for the design of systems, components and relevant arrangement so that the inspecting personnel can access the components to be inspected and perform the required inspections and tests and to make the personnel exposure be as low as reasonably practical and achievable. The ISI program in which the systems and components requiring in-service inspection and the inspection frequency are determined according to the safety importance and the rate of the equipment degradation, etc. has been prepared by the operating organization before the operation of NPP. In addition, the integrity of the pressure-retaining components has to be verified through the in-service inspections.

NPP operating organizations assess all inspection results, to determine if they comply with the qualification standards; and all parts not suitable for further service as determined in the assessment are repaired or replaced.

In past three years, more efforts have been made by the nuclear safety regulatory body and operating NPPs in China on following aspects while ensuring the safe operation, maintenance and periodic tests:

(1) Promoting the effectiveness of NPP maintenance. The *Technical Policy for Improving Effectiveness in NPP Maintenance (for Trial Implementation)* has been issued. Pilot work has been carried out to improve the maintenance effectiveness of NPPs. A working group on maintenance rules for NPPs has been established to standardize the implementation of relevant domestic maintenance rules and provide a technical basis for the subsequent preparation and revision of nuclear safety codes, standards and guidelines.

(2) The reliability-centered maintenance optimization (RCM) method has been used continually to analyze plant systems; preventive maintenance programs and procedures are optimized for the purpose of ensuring system reliability. Based on the

experience of nuclear power industry at home and abroad, comprehensive equipment classification have been conducted and applied in some NPPs. The results of equipment classification can be applied to plant business processes including equipment reliability management, configuration management, spare parts management and plan management.

(3) PSA has been extensively applied to optimize resource allocation and improve nuclear safety. PSA pilot research project have been implemented in some NPPs, namely “Optimization of Risk-oriented Technical Specification and Optimization of Risk-oriented In-service Inspection” and completed optimization of in-service inspection programs for individual systems. In addition, PSA technology has also been popularized and applied in the areas of operation risk assessment, engineering modification, maintenance risk assessment, change of supervision period, etc.

19.4 Procedures for Responding to Operational Occurrences and Accidents

The *Code on the Safety of Nuclear Power Plant Operation* and related guidelines stipulate that NPP operating organizations shall develop event-oriented or symptom-oriented procedures for anticipated operational occurrence and design basis accidents, and shall also prepare management guidelines for serious accidents (beyond design basis accident).

The NPP operating organizations in China have worked out relevant procedures for anticipated operational occurrence and accidents, tried to verify accident procedures on full scale simulator and/or on site, and performed training to operators.

At present, the procedures for accident adopted by NPPs in China are divided into two categories, namely, event-oriented procedures and symptom-oriented procedures.

- Event-oriented procedures stipulate that the actions of operators are based on confirmed events. For event-oriented procedures, decisions and measures responding to accidents shall be made according to the state of the NPP related to the event pre-determined in the design and safety analysis report.
- For symptom-oriented procedures, the method of pre-determining and prioritizing the major critical safety functions is adopted. In the symptom-oriented procedure, the response measures for the event shall be determined according to the symptoms and states of the NPP (such as safety parameter values and critical safety functions), which is conducive to the processing of accidents involving multiple faults.

According to the upgrade and modification of NPP system, research results of PSR and PSA, experience in utilization of accident procedures and research on accident evolvement and active follow-up of the international development, NPP operating

organizations have assessed and modified the procedures for accidents. After Fukushima accident, in order to further improve the nuclear safety level of NPPs, MEE (NNSA) took the lead in organizing safety inspection and proposing rectification requirements for the management of serious accidents, requiring all NPPs to develop or optimize SAMG, analyze the availability and accessibility of important equipment and monitoring instruments in serious accidents in consideration of various accident conditions and common cause failure at a multi-reactor site. A series of work has been carried out by the plants as required; and a completed response and management system for serious accidents of NPPs will be formed. Detailed situation of fulfillment of this job is described in section 18.1.4 of this report.

Furthermore, according to the actual conditions, NPP operating organizations have developed the emergency response plans and response procedures in case a number of units enter the emergency status at the same time, considering the risk of simultaneous accident with both units due to common cause event and external common cause event (such as typhoon, earthquake and flood) and the corresponding emergency action level and emergency response process.

19.5 Engineering and Technical Support

After years of development and practice, China has basically established its own engineering and technical support system for NPPs.

Chinese government and each nuclear power company appropriately adjust and recombine the existing design and research organizations of nuclear power engineering to establish the NPP-oriented technical support system, including the areas of operational research, safety analysis, radiation protection, in-service inspection, plant modification, special tests, equipment maintenance and safety review. The technical support organizations include China Nuclear Power Engineering Co., Ltd., Nuclear Power Institute of China, China General Nuclear Power Design Co., Ltd., Shanghai Nuclear Engineering Research and Design Institute, China Academy of Machinery Science and Technology, Research Institute of Nuclear Power Operation, Suzhou Nuclear Power Institute, China Institute for Radiation Protection, and China Institute of Atomic Energy, etc.

The necessary technical support department has been set up for the NPPs to ensure operation safety and provide technical support for operation safety.

Through cooperation and information exchange channels with international organizations such as IAEA and WANO, NPP operating organizations can get technical supports from the international peers when necessary.

Under accident conditions, all round technical support for NPPs can be obtained. In the overall layout for nuclear emergency at three capability levels, the national nuclear emergency related professional technical support center and national nuclear emergency related professional rescue detachments will provide the technical support at a higher level; and in the meanwhile, the provincial and local governments will also provide corresponding coordination and support. NPP operating organizations can also obtain technical support from their respective groups. In addition, according to the cooperation agreement between domestic nuclear corporations on mutual rescue in case of nuclear accidents, the nuclear power corporations will provide technical support for each other when necessary.

Under their respective accident management systems, all NPP operating organizations actively cooperate with various organizations and sign support agreements with them, for example, with hospitals, fire brigades, police and anti-terrorism forces, and meteorological, geological and hydrological organizations, which will provide professional technical support when necessary.

19.6 Event Reporting System of Operating NPPs

The state has established a nuclear safety reporting system. The NPP operating organization shall report the operating events to the MEE (NNSA) in accordance with relevant laws and regulations.

The event reporting criteria at the operation phase include the followings:

(1) Any event violating the *Technical Specifications* of the NPP;

(2) Any event that seriously damages the performance of safety barriers or important equipment of the NPP, or causes one of the following conditions:

1) An unanalyzed working condition that would obviously endanger safety;

2) A working condition beyond the design basis of the NPP;

3) A working condition not being taken into account in the operation procedures or emergency procedures of the NPP.

(3) Any natural event or other external event that would pose actual threat to the safety of the NPP or clearly hinder the on-duty personnel in their performance of ensuring the safety operation of the NPP.

(4) Any event that would result automatic or manual activation of the engineered safety features and the reactor protection system (excluding the preplanned tests of this kind);

(5) Any event that may prevent a structure or system from realizing the following safety functions;

1) The shutdown and the maintenance of the safe shutdown conditions;

2) Residual heat removal;

3) The control of release of radioactive substance;

4) The mitigation of the accident consequences.

(6) Any common cause event that would lead to simultaneous failure of a number of independent systems, trains or channels with the following functions:

1) The shutdown and the maintenance of the safe shutdown conditions;

2) Residual heat removal;

3) The control of release of radioactive substance;

4) The mitigation of the accident consequences.

(7) Any event that would result in uncontrolled radioactive release;

(8) Any internal event that would pose actual threat to the safety of the NPP or clearly hinder on-duty personnel in their performance of ensuring the safety operation of the NPP.

(9) Other events.

The ways of event reporting include:

1) Oral notification, which shall be sent out within 24 hours after the occurrence of the event;

2) Written notification, which shall be submitted within three days in a specified format after the occurrence of the event;

3) Event report, which shall be submitted in the specified format within 30 days after the occurrence of the event;

4) Accident report of NPP under emergency status, seeing section 16.3 of this report.

The root cause analysis method recommended by the MEE (NNSA) has been adopted for NPPs in China to analyze the root cause of the operating events, prepare reasonable corrective actions, and complete the operating event report within the specified time and submit it to the MEE (NNSA). The MEE (NNSA) evaluates the submitted operating event report according to the operating event report and the mastered information about the operating event, and gives review opinions. If an operating event requires investigation, the MEE (NNSA) will organize an investigation team to investigate the operating event. NPP operating organizations modify the operating event report and implement corresponding corrective actions according to the review opinion on operating events given by the MEE (NNSA). The Regional Offices track and supervise the implementation of corrective actions.

In addition, according to *Measures for the Administration of Operating Experience*

Feedback in Operating Nuclear Power Plants (Trial), lists and abstracts of their internal events periodically for operating NPPs need to be submitted for screening, and the corresponding internal event reports as required by the MEE (NNSA) need to be submitted, for the convenience of the regulatory body to search other important events, and promptly find any detrimental trend and precursor event in the safety of NPPs.

From 2016 to 2018, the NPP operating organizations reported 152 operating events, including 3 events rated at INES level 1, the rest of which were rated at INES level 0.

19.7 Operational Experience Feedback

It is stipulated by the *Nuclear Safety Act* that the relevant departments of the State Council shall establish a nuclear safety operating experience feedback system and timely process nuclear safety report information to realize information sharing. Nuclear facility operating organizations shall establish a nuclear safety operating experience feedback system. It is required by the *Code on the Safety of Nuclear Power Plant Operation* that operating organizations shall systematically review the operating experience of NPPs. It is necessary to investigate and study abnormal events important to safety to determine their direct causes and root causes.

China has carried out operating experience feedback related activities among NPPs; at present, operating experience system has been established at all levels, including the operating experience feedback system of the MEE (NNSA), NEA, nuclear industry, nuclear power corporations and NPPs.

Operating event reports for NPPs are submitted to the corresponding operating experience feedback platforms on a regular basis according to the stipulations of MEE (NNSA), the management requirements of the affiliating nuclear power corporations and relevant stipulations of the nuclear industry, to ensure sufficient sharing of operating event information of all NPPs in the industry. In addition, as a supplement to the operating experience feedback platform, the MEE (NNSA), NEA, CNEA, nuclear power corporations and all NPPs prepare and release quarterly and annual operation reports, weekly reports of domestic and overseas events, annual operating experience report and report on critical performance indicators to further promote the exchange of operation experience.

The MEE (NNSA), the CNEA and the nuclear power corporations also hold regular feedback and exchange meetings on the experience of operating NPPs to discuss important operating events in order to find vulnerabilities or weaknesses in nuclear safety supervision or NPP safety management. In addition, in order to improve the professional capability of the staff engaged in operating experience feedback and to

enhance the awareness of ordinary staff on operating experience feedback, relevant organizations will organize relevant operating experience feedback training related activities on a regular basis.

While strengthening and improving the domestic feedback and exchange of operation experience, China has also strengthened feedback and exchange of international operation experience; maintained close ties with the national nuclear safety regulatory body and nuclear power operating organizations such as EU, the United States, Russia, Japan and Korea, as well as the international organizations including IAEA and WANO; carried out operating experience feedback and exchange activities on a regular or irregular basis; dispatched personnel to each other for study and training activities, and provided event reports according to the respective event report criteria and requirements, to share experience with each other. To further track and analyze international hot issues as to nuclear safety, the MEE (NNSA) also established special teams for international information follow-up, to enhance the analysis and research on international nuclear events, and provide important reference for the decision-making and implementation of nuclear safety supervision.

In the past three years, China mainly carried out the following activities in feedback of operation experience of NPPs:

(1) The MEE (NNSA) has organized a quarterly meeting on operating experience feedback of NPPs and annual national operating experience feedback and exchange meetings in consideration of the actual nuclear safety regulation and formed a regular exchange mechanism for operating experience feedback involving the nuclear safety regulation department, nuclear power enterprises, research and design institutes, industry associations and contractors. It has prepared operating experience related event notification to urge operating organizations to deal with common problems and typical events, and ensure the construction quality and operation safety of NPPs. In order to promote the event reporting of NPPs, standardize and deepen the event analysis and evaluation, promote the revision of the *Reporting System for Operating Organization of NPP*, it has organized the revision of the event reporting criteria at the construction and operation phases of NPPs, studied and drafted basic documents such as the *Method for Analysis of Root Causes of Nuclear Power Plant Operating Events*, and studied and prepared the *Work Guidelines for Operating Experience Feedback of Nuclear and Radiation Safety*.

The MEE (NNSA) comprehensively organizes and carries out works on feedback of NPPs related information, submits information such as operating event notification and reports to the operating experience feedback platform of the MEE (NNSA), and

organizes the preparation and publication of quarterly operating experience feedback on the operation of NPPs. In addition, it has organized the redevelopment of the NPP information feedback platform and integrated the operating experience feedback information of the NPP under construction.

Also, the MEE (NNSA) has continuously carried out topical research on experience feedback and special research and feedback for some operating events. In 2016, special investigations were carried out on common issues such as the unavailability of operator station of digital I&C information and control system in NPPs, the impact of marine organisms or foreign materials on water intake safety in NPPs, deformation of water storage containers under negative pressure in NPPs, water leakage-related operating events in NPPs or important abnormalities; special reports were developed and fed back to relevant organizations. In 2017, the MEE (NNSA) selected two typical NPP operating events to carry out independent review, and compared the results with those of NPP operating organizations to improve the ability of root cause analysis.

(2) The NEA has strengthened the development of operating experience feedback and performance indicator system. A nationwide operating experience feedback system has been established. It has carried out the summarization, collation, analysis and conclusion of all events at various stages of design, manufacturing, construction, commissioning, operation and decommissioning among various organizations in the industry; communicated and exchanged in the industry, and put forward suggestions, opinions and measures for improvement. The experience feedback platform of the NEA is being developed, including NPP operating events, internal events, management requirements of the National Energy Administration, etc.

(3) The CNEA has continuously promoted the exchange of experience and information sharing among NPPs in China. The annual experience feedback report is prepared every year. For repeated problems in the industry, special analysis is performed and reports such as *Summary and Analysis of Operation and Maintenance Events and Good Practices of Reactor Coolant Pumps in Nuclear Power Plants* and *Analysis of Water Intake Blockage Events and Good Practice Feedback in Nuclear Power Plants* are prepared. The exchange meetings and seminars on the experience feedback in the nuclear industry are convened; training courses on root cause analysis are held to promote methods and technologies for root cause analysis of equipment; rapid experience feedback is performed, etc.

(4) Each nuclear power corporation has established and improved a unified platform for operating experience feedback and carried out integrated management. The

status report standardization and sharing mechanism for operating experience feedback information are facilitated based on this unified platform to collect real-time event information of other NPPs both at home and abroad, learn experience and lessons and prepare them into operating experience feedback documents for release, and make adaptive analysis. All nuclear power corporations have collated and screened operating events within the corporations and can quickly respond to important common cause or common mode events, find out the root causes of events, help the management promptly improve safety management measures, and raise the operation safety level of all subordinated NPPs.

(5) NPP operating organizations have continuously improved and optimized the operating experience feedback related information processing scheme, procedure and external event feedback mechanism. At present, the nuclear power corporations have established an evaluation mechanism for the effectiveness of experience feedback while regularly organizing the evaluation of the effectiveness of experience feedback for each NPP. They have regularly carried out self-assessment and integrated it with internal performance evaluation and carried out quality rating of operating experience event reports. In addition, an event investigation and reproducing mechanism has been established. On the premise of ensuring the safety of the unit, through on-site simulation or observation, the process and phenomena of the event will be reproduced as truly as possible, so as to deeply analyze the root causes of the event and develop effective corrective actions to avoid recurrence of similar events. The comprehensive inspection and evaluation of contractors has been performed to evaluate the structure, management documents and operation of contractor's operating experience feedback system from the systematic perspective, so as to help contractors continuously improve the effect of operating experience feedback. They have participated in various peer reviews, seminars and training activities, to benchmark with and learn from international advanced management practices, improve event classification and graded management, optimize the allocation of resources, focus on important events, look for problems existing in experience feedback related work to continuously improve the level of operating experience feedback.

19.8 Management of Spent Fuel and Radioactive Wastes on the Site

19.8.1 On-site Spent Fuel Management

It is stipulated by the *Nuclear Safety Act* that organizations that produce, store, transport and post-process spent fuel shall take measures to ensure the safety of spent fuel and undertake nuclear safety responsibility for the spent fuel they hold. Nuclear

facility operating organizations shall pay for spent fuel treatment and disposal in accordance with state regulations and include such expense in production costs.

It is stipulated by the *Regulations on the Safety Regulation for Civilian Nuclear Installations of the People's Republic of China* that the nuclear facility operating organization is fully responsible for the safety of nuclear facilities under its responsibility, including the spent fuel management facility, and shall accept the supervision and management of the nuclear safety regulatory departments.

Codes and standards such as the *Code on the Safety of Nuclear Power Plant Design, Code on the Safety of Nuclear Power Plant Operation, Nuclear Safety Regulatory Requirements for Dry Storage System of Spent Fuel in Nuclear Power Plants* and *Design Criteria for Storage Facilities of Spent Fuel of PWR Nuclear Power Plants* have proposed clear safety requirements and suggestions, including ensuring that critical issues are properly solved; ensuring that the issues of residual heat removal is properly solved; ensuring that the production of radioactive waste is kept at the lowest practical level; the interdependence between different steps in spent fuel management is considered; ensuring effective protection of individuals, society and the environment; full consideration has been given to biological, chemical and other hazards that may be related to spent fuel management.

Each NPP operating organization in China has been provided with spent fuel storage facilities with a certain storage capacity to receive spent fuel generated from the operation of NPPs during a certain period and to ensure their safe storage.

Each NPP operating organization has prepared an operation plan for spent fuel storage facilities. Such plan includes commissioning, operation, maintenance, modification, inspection, testing, radiation protection, prevention of radioactive material release to the environment, unexpected accident and emergency preparedness, accident records, reporting and investigation, quality assurance and monitoring, personnel training, and nuclear material control and physical protection. In order to manage and control safety risks, each NPP operating organization has also prepared operation requirements for spent fuel storage facilities, including the minimum cooling capacity of the spent fuel cooling system and the minimum water level above the spent fuel pool, prohibition of storage of spent fuel in any place other than the designated location, minimum reserved storage capacity, sub-critical margin to be remained, and radiation monitoring requirements for spent fuel storage areas. Each NPP operating organization has strictly managed the spent fuel storage facilities in accordance with the program and procedures prepared and approved before the spent fuel storage facilities are put into operation.

19.8.2 On-site Radioactive Waste Management

China has established a sound system of laws, codes and standards, and has taken a series of measures to ensure the safety of radioactive waste management, so as to protect individuals, society and the environment from radiation and other hazards. It is required by Chinese laws and regulations to keep the production of radioactive waste at the lowest level that can be actually achieved.

It is stipulated by the *Nuclear Safety Act* that radioactive waste shall be disposed by categories. The operating organization of a nuclear facility shall treat the radioactive solid waste and the radioactive liquid waste that cannot be purified before being discharged, so as to convert them into stable and standardized solid waste, and send them to the radioactive waste disposal organization for disposal in a timely manner. The operating organization of a nuclear facility shall treat the radioactive gaseous waste produced by itself and discharge it only after it reaches the national standards for the prevention and control of radioactive contamination. The operating organizations of nuclear facilities shall withhold the expenses for decommissioning of the nuclear facility and radioactive waste disposal and include them in the investment budget and production cost for the decommissioning of nuclear facilities and radioactive waste disposal in particular.

It is stipulated by the *Act of the People's Republic of China on Prevention and Control of Radioactive Pollution* that nuclear facilities operating organizations shall reduce the amount of radioactive waste as far as possible through reasonable selection and utilization of raw materials and the adoption of advanced production processes and equipment. The *Regulations on the Management for Safety of Radioactive Waste* has defined the safe management of radioactive waste, which is required to adhere to the principle of reduction, harmlessness, proper disposal and permanent safety. It is clearly required by the *Rules on the Safety Regulation and Management for Radioactive Waste* that in all nuclear activities, the amount of waste generated shall be controlled so as to keep the activity and volume of radioactive waste at the minimum level actually achievable.

The MEE (NNSA) issued the nuclear safety guideline of *Minimization of Radioactive Wastes from Nuclear Installations* in 2016, which provided guidance for organizations involved in nuclear facilities design, construction, operation and decommissioning to realize the minimization of radioactive waste, that is, through measures such as source control, recycling and reuse of waste, clearance, optimization of waste treatment and strengthening management, and based on cost benefit analysis, the final production (volume and activity) of radioactive solid waste is ensured as low

as reasonably achievable.

All NPPs have been equipped with radioactive liquid waste and gas waste treatment facilities and solid waste storage facilities. The operational limits of radioactive waste management facilities have been specified according to the requirements of laws and regulations, including the operational limits of evaporation and concentration, the continuous treatment capacity of cement solidification process, the alarm limits and monitoring limits of radiation monitoring instruments (including effluent monitoring), etc. A program of radioactive waste management has been prepared to manage radioactive waste by category. The corresponding radioactive waste management procedures have been prepared according to the program. NPP operating organizations strictly follow the above programs and procedures to carry out related work.

NPP operating organizations usually divide their radioactive wastes into process wastes, technical wastes and other wastes according to their sources. According to the physical properties of waste, radioactive waste can be divided into radioactive airborne waste, radioactive liquid waste and radioactive solid waste.

NPP operating organizations continue to implement the minimization management of radioactive waste, control the production amount (activity and volume) of radioactive waste as low as reasonably achievable, implement the optimization of the overall control plan for all waste gas, waste liquid and solid waste and the optimization of the whole process from generation to treatment of waste, and strive to obtain the best technical, economic, environmental and social benefits and contribute to sustainable development. Each NPP operating organization has enhanced the awareness of all employees and contractors on waste minimization through training and publicity, optimized operation, restricted the use of items and carried out classified collection to control source items. The waste management has been optimized by recycling, clearance, volume reduction and strict effluent discharge control. In addition, it has also promoted continuous improvement through supervision and feedback. New waste treatment technologies have been introduced, such as pre-compression and super-compression and volume reduction technologies, and protective equipment made of degradable materials such as protective clothing and shoe covers. In the design of newly-built NPPs, the new waste treatment technologies and operation modes have been actively adopted, such as barrel drying, waste resin drying and hot pressing, high-integrity vessel, mobile waste liquid treatment devices and centralized waste treatment facilities, etc. In addition, radioactive waste gas, waste liquid and solid wastes are subject to informationizing management.

19.9 Cyber Security of NPPs

Recently, due to the continuous improvement of digitalization and networking of control systems in NPPs, attention is increasingly drawn to the threat suffered by the cyber security of NPPs.

In 2016, China officially issued the *Cyber security Act of the People's Republic of China*, which aimed to supervise cyber security, protect personal privacy and sensitive information, and safeguard the sovereignty/security of the national cyber space, as well as provide guidance for the management of cyber security of NPPs.

In September 2017, the MEE (NNSA) set up a NPP cyber security working group for the regulation of NPPs, started the preparation of *Technical Policy for Cyber Security Regulation in Nuclear Power Plants*, completed the preparation of the draft solicitation in January 2019, and is currently soliciting opinions from all parties.

Nuclear power corporations and NPP operating organizations have established an emergency safeguards system for cyber security, including the establishment of cyber security management organization, the preparation of cyber security related management procedures, and the establishment of cyber security laboratory or cyber security research and development center. NPP operating organizations have set up protection strategies through various means, and carried out cyber security research to continuously improve the level of cyber security in the meanwhile. Each NPP operating organization has also carried out self-assessment of cyber security, especially the assessment of anti-cyber attack of the digital control and protection system. Based on the assessment results, it has taken comprehensive measures to strengthen cyber security and prepared the emergency plan for cyber attacks.

Appendix 1: List of NPPs in China (By December 31, 2018)

NPP		Unit No.	Reactor Type	Rated power MW(e)	Date of Construction	Date of Initial Connection to Grid	Date of Commercial Operation
Qinshan NPP		CN-01	PWR	310	1985-03-20	1991-12-15	1994-04-01
Daya Bay NPP	Unit 1	CN-02	PWR	2×984	1987-08-07	1993-08-31	1994-02-01
	Unit 2	CN-03			1988-04-07	1994-02-07	1994-05-06
Qinshan Phase II NPP	Unit 1	CN-04	PWR	2×650	1996-06-02	2002-02-06	2002-04-15
	Unit 2	CN-05			1997-04-01	2004-03-11	2004-05-03
	Unit 3	CN-14		2×660	2006-04-28	2010-08-01	2010-10-05
	Unit 4	CN-15			2007-01-28	2011-11-25	2011-12-30
LingAo NPP	Unit 1	CN-06	PWR	2×990	1997-05-15	2002-02-26	2002-05-28
	Unit 2	CN-07			1997-11-28	2002-09-14	2003-01-08
	Unit 3	CN-12		2×1086	2005-12-15	2010-07-15	2010-09-15
	Unit 4	CN-13			2006-06-15	2011-05-03	2011-08-07
Third Qinshan NPP	Unit 1	CN-08	PHWR	2×728	1998-06-08	2002-11-19	2002-12-31
	Unit 2	CN-09			1998-09-25	2003-06-12	2003-07-24
Tianwan NPP	Unit 1	CN-10	PWR	2×1060	1999-10-20	2006-05-12	2007-05-17
	Unit 2	CN-11			2000-09-20	2007-05-14	2007-08-16
	Unit 3	CN-45		2×1126	2012-12-27	2017-12-30	2018-02-15
	Unit 4	CN-46			2013-09-27	2018-10-27	2018-12-22
	Unit 5	CN-53		2×1118	2015-12-27		
	Unit 6	CN-54			2016-09-07		
Hongyanhe NPP	Unit 1	CN-16	PWR	6×1118.79	2007-08-18	2013-02-17	2013-06-06

NPP		Unit No.	Reactor Type	Rated power MW(e)	Date of Construction	Date of Initial Connection to Grid	Date of Commercial Operation
	Unit 2	CN-17			2008-03-28	2013-11-23	2014-05-13
	Unit 3	CN-26			2009-03-07	2015-03-23	2015-08-16
	Unit 4	CN-27			2009-08-15	2016-04-01	2016-09-19
	Unit 5	CN-49			2015-03-29		
	Unit 6	CN-50			2015-07-24		
Ningde NPP	Unit 1	CN-18	PWR	4×1089	2008-02-18	2012-12-28	2013-04-15
	Unit 2	CN-19			2008-11-12	2014-01-04	2014-05-04
	Unit 3	CN-34			2010-01-08	2015-03-21	2015-06-10
	Unit 4	CN-35			2010-09-29	2016-03-29	2016-07-21
Fuqing NPP	Unit 1	CN-20	PWR	4×1089	2008-11-21	2014-08-20	2014-11-22
	Unit 2	CN-21			2009-06-17	2015-08-05	2015-10-16
	Unit 3	CN-42			2010-12-31	2016-09-07	2016-10-24
	Unit 4	CN-43			2012-11-17	2017-07-29	2017-09-17
	Unit 5	CN-51		2×1150	2015-05-07		
	Unit 6	CN-52			2015-12-22		
Yangjiang NPP	Unit 1	CN-22	PWR	6×1086	2008-12-16	2013-12-31	2014-03-25
	Unit 2	CN-23			2009-06-04	2015-03-10	2015-06-05
	Unit 3	CN-40			2010-11-15	2015-10-18	2016-01-01
	Unit 4	CN-41			2012-11-17	2017-01-08	2017-03-15
	Unit 5	CN-47			2013-09-18	2018-05-23	2018-07-12
	Unit 6	CN-48			2013-12-23		
Fangjiashan NPP	Unit 1	CN-24	PWR	2×1089	2008-12-26	2014-11-04	2014-12-15
	Unit 2	CN-25			2009-07-17	2015-01-12	2015-02-12
Sanmen NPP	Unit 1	CN-28	PWR	2×1250	2009-04-19	2018-06-30	2018-09-21
	Unit 2	CN-29			2009-12-15	2018-08-24	2018-11-05

NPP		Unit No.	Reactor Type	Rated power MW(e)	Date of Construction	Date of Initial Connection to Grid	Date of Commercial Operation
Haiyang NPP	Unit 1 Unit 2	CN-30 CN-31	PWR	2×1250	2009-09-24 2010-06-20	2018-08-17 2018-10-13	2018-10-22 2019-01-09
Taishan NPP	Unit 1 Unit 2	CN-32 CN-33	PWR	2×1750	2009-11-18 2010-04-15	2018-06-29	2018-12-13
Changjiang NPP	Unit 1 Unit 2	CN-36 CN-37	PWR	2×650	2010-04-25 2010-11-21	2015-11-07 2016-06-20	2015-12-25 2016-08-12
Fangchenggang NPP	Unit 1 Unit 2 Unit 3 Unit 4	CN-38 CN-39 CN-55 CN-56	PWR	2×1086 2×1180	2010-07-30 2010-12-23 2015-12-24 2016-12-23	2015-10-25 2016-07-15	2016-01-01 2016-10-01
Shidao Bay NPP	Demonstration Project	CN-44	HTGR	211	2012-12-09		

Appendix 2: Operational Events in NPPs of China (From 2016 to 2018)

NPP \ Year		2016	2017	2018
Qinshan NPP	Unit 1	0	1	0
Daya Bay NPP	Unit 1	1	0	0
	Unit 2	0	0	1
Qinshan Phase II NPP	Unit 1	1	1	0
	Unit 2	0	0	1
	Unit 3	0	1	0
	Unit 4	0	1	0
LingAo NPP	Unit 1	0	0	0
	Unit 2	4	0	0
	Unit 3	1	0	0
	Unit 4	0	0	0
Third Qinshan NPP	Unit 1	0	1	0
	Unit 2	0	0	0
Tianwan NPP	Unit 1	1	1	0
	Unit 2	1	0	1
	Unit 3	/	2	4
	Unit 4	/	/	0
Hongyanhe NPP	Unit 1	3	0	1
	Unit 2	2	1	0
	Unit 3	0	1	0
	Unit 4	1	0	0
Ningde NPP	Unit 1	7	4	0
	Unit 2	4	0	1
	Unit 3	5	1	1
	Unit 4	6	2	0
Fuqing NPP	Unit 1	1	2	0
	Unit 2	4	2	2
	Unit 3	0	2	1
	Unit 4	/	3	3
Yangjiang NPP	Unit 1	4	0	1
	Unit 2	0	3	0
	Unit 3	3	1	0

NPP \ Year		2016	2017	2018
	Unit 4	1	3	0
	Unit 5	/	/	1
Fangjiashan NPP	Unit 1	1	2	2
	Unit 2	2	0	0
Sanmen NPP	Unit 1	/	/	1
	Unit 2	/	/	2
Haiyang NPP	Unit 1	/	/	0
	Unit 2	/	/	2
Taishan NPP	Unit 1	/	/	12
Changjiang NPP	Unit 1	3	3	2
	Unit 2	7	0	1
Fangchenggang NPP	Unit 1	3	2	0
	Unit 2	6	0	0
Total		72	40	40

Appendix 3: WANO Performance Indicators of Operating Nuclear Power Units in China (from 2016 to 2018)

Table 1 WANO Performance Indicators of Operating Nuclear Power Units (2016)

Indicator (unit) / NPP		UCF (%)	UCLF (%)	FLR (%)	GRLF (%)	UA7	US7	SSPI			FRI (Bq/g)	CPI	CRE (man·Sv)	ISA	CISA
								High Pressure Safety Injection System	Auxiliary Feed-Water System	Emergency AC Supply System					
Qinshan NPP	CN01	91.38	0.01	0.02	0.00	0.00	0.00	0.0001	0.0021	0.0001	0.037	1.00	0.281	0.00	0.00
Daya Bay NPP	CN02	86.58	1.24	0.04	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.449	0.00	0.06
	CN03	87.42	0.30	0.30	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.583		
Qinshan Phase II NPP	CN04	90.45	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.283	0.00	0.00
	CN05	82.96	0.01	0.02	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.631		
	CN14	99.81	0.11	0.11	0.00	0.00	0.00	0.0000	0.0000	0.0005	0.037	1.00	0.020	0.00	0.00
	CN15	92.39	0.06	0.07	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.157		
LingAo NPP	CN06	99.81	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	25.462	1.00	0.080	0.00	0.17
	CN07	88.65	0.06	0.07	0.00	0.89	0.89	0.0000	0.0000		0.037	1.00	1.038		
	CN12	91.62	1.03	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.319	0.00	0.00
	CN13	87.84	2.24	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.709		
Third Qinshan NPP	CN08	94.91	5.08	5.08	0.00	0.00	0.00	0.0006	0.0000	0.0008	0.037	1.00	0.291	0.00	0.00
	CN09	79.28	0.02	0.02	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.718		
Tianwan NPP	CN10	81.87	2.84	0.71	0.74	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.579	0.00	0.15
	CN11	87.23	0.57	0.65	2.40	0.90	0.90	0.0000	0.0000		0.037	1.00	0.431		
Hongyanhe NPP	CN16	87.19	1.86	2.06	0.00	0.00	0.00	0.0000	0.0000	0.0001	0.037	1.00	0.286	0.07	0.00
	CN17	87.49	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.543		
	CN26	94.90	0.35	0.37	0.00	0.00	0.00	0.0002	0.0000	0.0000	0.037	1.00	0.055	0.00	0.00
	CN27	99.98	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.002		

Ningde NPP	CN18	98.13	1.85	1.85	0.00	0.00	0.00	0.0000	0.0001	0.0013	0.037	1.00	0.085	0.00	0.00
	CN19	86.38	1.23	0.31	0.00	0.00	0.00	0.0000	0.0000	0.0014	0.037	1.00	0.633		
	CN34	80.08	0.01	0.02	0.00	0.00	0.00	0.0000	0.0000	0.0004	0.037	1.00	0.729	0.00	0.00
	CN35	99.98	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.003	0.00	0.00
Fuqing NPP	CN20	99.31	0.20	0.20	0.00	0.00	0.00	0.0000	0.0000	0.0004	0.037	1.00	0.052	0.00	0.06
	CN21	81.55	0.24	0.29	0.00	0.98	0.98	0.0000	0.0000		3.229	1.00	0.857		
Yangjiang NPP	CN22	81.56	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0100	0.037	1.00	0.774	0.00	0.06
	CN23	77.68	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.900		
	CN40	91.24	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	89.664	1.03	0.449	0.00	0.06
Fangjiashan NPP	CN24	91.23	0.04	0.04	0.00	0.00	0.00	0.0001	0.0000	0.0006	0.037	1.00	0.235	0.00	0.00
	CN25	86.88	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.488		
Changjiang NPP	CN32	93.96	0.00	0.00	0.10	0.00	0.00	0.0000	0.0030	0.0000	/	1.03	0.009	0.00	0.00
Fangchenggang NPP	CN38	99.02	0.93	0.93	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.04	0.015	0.00	0.00

Table 2 WANO Performance Indicators of Operating Nuclear Power Units (2017)

NPP \ Indicator (unit)		UCF (%)	UCLF (%)	FLR (%)	GRLF (%)	UA7	US7	SSPI			FRI (Bq/g)	CPI	CRE (man·Sv)	ISA	CISA
								High-Pressure Safety Injection System	Auxiliary Feed-Water System	Emergency AC Supply System					
Qinshan NPP	CN01	99.97	0.00	0.00	0.02	0.00	0.00	0.0000	0.0001	0.0001	0.037	1.00	0.042	0.00	0.27
Daya Bay NPP	CN02	99.98	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.056	0.00	0.00
	CN03	88.74	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.656		
Qinshan Phase II NPP	CN04	99.52	0.36	0.36	0.00	0.80	0.80	0.0000	0.0000	0.0002	0.037	1.00	0.033	0.00	0.00
	CN05	88.83	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.399		
	CN14	88.42	0.33	0.38	0.00	0.00	0.00	0.0000	0.0001	0.0005	0.037	1.00	0.317	0.00	0.00
	CN15	90.62	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.190		
LingAo NPP	CN06	89.15	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.809	0.00	0.00
	CN07	96.32	0.46	0.48	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.108		
	CN12	86.99	0.02	0.03	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.729	0.00	0.00
	CN13	91.33	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.471		
Third Qinshan NPP	CN08	79.83	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0009	1.636	1.00	0.604	0.00	0.00
	CN09	99.96	0.03	0.03	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.098		
Tianwan NPP	CN10	92.05	0.37	0.40	0.00	0.86	0.86	0.0000	0.0000	0.0000	0.037	1.00	0.306	0.00	0.00
	CN11	99.90	0.03	0.03	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.018		
Hongyanhe NPP	CN16	88.92	0.38	0.43	0.00	0.00	0.00	0.0000	0.0000	0.0003	0.037	1.00	0.626	0.00	0.00
	CN17	98.08	1.53	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.029		
	CN26	83.07	0.10	0.12	0.00	1.13	1.13	0.0000	0.0000		71.410	1.00	0.548	0.00	0.00
	CN27	85.76	0.15	0.17	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.494		
Ningde NPP	CN18	83.66	4.78	4.24	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.875	0.00	0.00
	CN19	98.80	1.18	1.18	0.00	0.00	0.00	0.0017	0.0000		0.037	1.00	0.065		

	CN34	95.62	0.01	0.01	0.00	0.00	0.00	0.0016	0.0000		0.037	1.00	0.300	0.00	0.00
	CN35	84.38	0.10	0.11	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.724		
Fuqing NPP	CN20	89.46	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0006	0.037	1.00	0.469	0.00	0.00
	CN21	86.73	2.67	2.99	0.00	0.90	0.90	0.0000	0.0000		0.037	1.00	0.440		
	CN42	83.01	0.00	0.00	0.00	0.00	0.00	0.0002	0.0001	0.0000	N/A	1.00	0.695	0.00	0.00
Yangjiang NPP	CN22	99.61	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0059	0.037	1.00	0.031	0.00	0.00
	CN23	87.97	0.46	0.53	0.00	0.92	0.92	0.0000	0.0000		1787.100	1.00	0.623		
	CN40	86.49	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.298	0.00	0.00
	CN41	90.00	9.99	9.99	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.011		
Fangjiashan NPP	CN24	89.52	0.07	0.07	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.447	0.00	0.12
	CN25	85.72	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.721		
Changjiang NPP	CN36	79.25	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0012	0.037	1.06	0.530	0.00	0.00
	CN37	87.78	0.71	0.00	0.00	0.00	0.00	0.0011	0.0000		1.045	1.00	0.312		
Fangchenggang NPP	CN38	76.19	0.91	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0019	0.037	1.00	0.865	0.00	0.00
	CN39	80.70	1.33	1.62	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.512		

Table 3 WANO Performance Indicators of Operational Nuclear Power Units (2018)

NPP \ Indicator (unit)		UCF (%)	UCLF (%)	FLR (%)	GRLF (%)	UA7	US7	SSPI			FRI (Bq/g)	CPI	CRE (man·Sv)	ISA	CISA
								High-Pressure Safety Injection System	Auxiliary Feed-Water System	Emergency AC Supply System					
Qinshan NPP	CN01	56.46	0.00	0.00	0.00	0.00	0.00	0.0014	0.0008	0.0001	0.037	1.00	0.764	0.00	0.00
Daya Bay NPP	CN02	89.25	0.38	0.42	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.660	0.00	0.00
	CN03	99.72	0.27	0.27	0.00	0.80	0.80	0.0000	0.0000		0.037	1.00	0.092		
Qinshan Phase II NPP	CN04	87.00	0.14	0.16	0.00	0.00	0.00	0.0000	0.0000	0.0011	0.037	1.00	0.382	0.14	0.13
	CN05	97.66	2.29	2.29	0.00	0.82	0.82	0.0005	0.0000		0.037	1.00	0.036		
	CN14	87.93	1.41	1.57	0.00	0.00	0.00	0.0000	0.0000	0.0014	0.037	1.00	0.389	0.00	0.00
	CN15	88.16	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.342		
LingAo NPP	CN06	87.52	1.27	0.28	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	1.099	0.00	0.00
	CN07	92.09	0.00	0.01	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.524		
	CN12	89.32	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.590	0.00	0.00
	CN13	98.34	1.65	1.65	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.038		
Third Qinshan NPP	CN08	99.98	0.00	0.00	0.00	0.00	0.00	0.0012	0.0000	0.0008	0.037	1.00	0.075	0.00	0.12
	CN09	82.62	0.78	0.94	0.00	0.00	0.00	0.0013	0.0000		0.037	1.00	0.780		
Tianwan NPP	CN10	90.05	3.30	3.54	4.14	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.323	0.00	0.00
	CN11	92.50	0.06	0.07	0.86	0.00	0.00	0.0000	0.0000		0.037	1.00	0.258		
	CN45	99.04	0.96	0.96	1.67	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.011	0.00	0.00
Hongyanhe NPP	CN16	99.98	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0050	0.037	1.00	0.035	0.00	0.00
	CN17	89.80	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.587		
	CN26	91.17	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		3.049	1.00	0.392	0.00	0.00
	CN27	85.16	0.03	0.03	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.516	0.00	0.00

Ningde NPP	CN18	88.50	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.672	0.00	0.04
	CN19	89.13	0.37	0.41	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.460		
	CN34	92.54	0.20	0.22	0.04	0.85	1.70	0.0000	0.0000		0.037	1.00	0.283	0.00	0.04
	CN35	99.99	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.034		
Fuqing NPP	CN20	87.56	0.03	0.03	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.374	0.00	0.00
	CN21	85.47	2.74	3.11	0.00	0.00	0.00	0.0002	0.0000		0.037	1.12	0.523		
	CN42	86.27	2.04	0.48	0.00	0.00	0.00	0.0001	0.0000	0.0001	0.037	1.00	0.287	0.00	0.00
	CN43	78.22	2.27	1.04	0.00	1.01	1.01	0.0001	0.0005		0.037	1.00	0.584		
Yangjiang NPP	CN22	88.23	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.455	0.00	0.00
	CN23	99.98	0.00	0.00	0.00	0.00	0.00	0.0000	0.0000		175.750	1.00	0.048		
	CN40	91.61	0.20	0.22	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.288	0.00	0.00
	CN41	82.42	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.657		
Fangjiashan NPP	CN24	99.60	0.39	0.39	0.00	0.80	0.80	0.0000	0.0002	0.0000	0.037	1.00	0.037	0.00	0.00
	CN25	98.08	0.47	0.48	0.00	0.00	0.00	0.0000	0.0001		0.037	1.00	0.037		
Changjiang NPP	CN36	87.39	1.16	1.31	0.17	0.00	0.00	0.0002	0.0000	0.0026	0.037	1.01	0.220	0.00	0.11
	CN37	73.85	2.24	1.63	0.16	1.05	1.05	0.0017	0.0000		0.037	1.00	0.329		
Fangchenggang NPP	CN38	89.60	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000	0.0000	0.037	1.00	0.283	0.00	0.00
	CN39	99.98	0.01	0.01	0.00	0.00	0.00	0.0000	0.0000		0.037	1.00	0.015		

Appendix 4: Laws, Administrative Regulations Departmental Rules and Guides on Nuclear Safety for NPPs in China

I. National Laws

1. Nuclear Safety Act of the People's Republic of China

(Promulgated in the Twenty-Ninth Meeting of the Standing Committee, the Twelfth National People's Congress of the People's Republic of China, September 1, 2017; came into force on January 1, 2018)

2. Act of the People's Republic of China on Protection and Control of Radioactive Pollution

(Promulgated in the Third Meeting of the Standing Committee, the Tenth National People's Congress of the People's Republic of China, on June 28, 2003; came into force on October 1, 2003)

II. Decrees of the State Council

1. Regulations on the Safety Regulation for Civilian Nuclear Installations of the People's Republic of China (HAF001)

(Promulgated by the State Council on October 29, 1986)

2. Emergency Management Regulations for Nuclear Accident at Nuclear Power Plant (HAF002)

(Promulgated by the State Council on August 4, 1993)

3. Regulations on Nuclear Materials Control of the People's Republic of China

(Promulgated by the State Council on June 15, 1987)

4. Regulations on the Supervision and Management for Civilian Nuclear Safety Equipment

(Promulgated by the State Council, July 11, 2007)

5. Regulations on the Management for Transport Safety of Radioactive Materials

(Promulgated by the State Council, September 14, 2009)

6. Regulations on the Management for Safety of Radioactive Waste

(Promulgated by the State Council, December 20, 2011)

7. Regulations on the Safety and Protection of Radioisotopes and Radiation Devices

(Promulgated by the State Council, September 14, 2005)

8. Regulations of the People's Republic of China on Control of Nuclear Export

(Promulgated by the State Council on September 10, 1997; Revised by the State Council on November 9, 2006)

9. Regulations of the People's Republic of China on Control of Nuclear Dual-purpose Goods and Related Technologies Export

(Promulgated by the State Council on June 10, 1998; Revised by the State Council on January 26, 2007)

III. Department Rules

1. Rules for the Implementation of Regulations on the Safety Regulation for Civilian Nuclear Installations of the People's Republic of China, Part One

- Application and Issuance of Safety License for Nuclear Power Plant (HAF001/01)

(Issued by NNSA on December 31, 1993)

2. Rules for the Implementation of Regulations on the Safety Supervision and Management for Civil Nuclear Installations of the People's Republic of China, Part One

- Appendix 1: Issuance and Management Procedures for Operator License of Nuclear Power Plant (HAF001/01/01)

(Issued by NNSA on December 31, 1993)

3. Rules for the Implementation of Regulations on the Safety Supervision and Management for Civil Nuclear Installations of the People's Republic of China, Part Two

- Safety Regulation of Nuclear Installations (HAF001/02)

(Issued by NNSA on June 14, 1995)

4. Rules for the Implementation of Regulations on the Safety Supervision and Management for Civil Nuclear Installations of the People's Republic of China, Part Two

- Appendix 1: The Reporting System for Operating Organization of Nuclear Power Plant (HAF001/02/01)

(Issued by NNSA on June 14, 1995)

5. Rules for the Implementation of Emergency Management Regulations for Nuclear Accident at Nuclear Power Plant - Part One:

- Emergency Preparedness and Response for Nuclear Power Plant Operating Organization (HAF002/01)

(Issued by NNSA on May 12, 1998)

6. Code on the Safety of Nuclear Power Plant Quality Assurance (HAF003)

(Issued by NNSA as per Order No.1 on July 27, 1991)

7. Measures for Management on the Certificates of Nuclear and Radiation Safety Inspectors (HAF004)

(Issued by Ministry of Environmental Protection on December 30, 2013)

8. Code on the Safety of Nuclear Power Plant Siting (HAF101)

(Issued by NNSA as per Order No.1 on July 27, 1991)

9. Code on the Safety of Nuclear Power Plant Design (HAF102)

(Issued by NNSA on October 28, 2016)

10. Code on the Safety of Nuclear Power Plant Operation (HAF103)

(Issued by NNSA on April 18, 2004)

11. Code on the Safety of Nuclear Power Plant Operation, Appendix 1

- Management of Refueling, Modifications and Accidental Shut-down of Nuclear Power Plant (HAF103/01)

(Issued by NNSA on March 2, 1994)

12. Regulation on Safety of Radioactive Waste (HAF401)

(Issued by NNSA on November 5, 1997)

13. Measures for Administration on License of Storage and Disposal of Radioactive Solid Waste (HAF402)

(Issued by Ministry of Environmental Protection on December 30, 2013)

14. Rules for the Implementation of Regulations on Nuclear Materials Control of the People's Republic of China (HAF501/01)

(Issued by NNSA, the Ministry of Energy and Commission of Science, Technology and Industry for National Defence on September 25, 1990)

15. Rules on the Regulation and Management of the Design, Manufacture, Installation and Non-destructive Testing of Civilian Nuclear Safety Equipment (HAF601)

(Issued by National Environmental Protection Administration on December 28, 2007)

16. Rules for Qualification Management of Non-destructive Testing Personnel of Civilian Nuclear Safety Equipment (HAF602)

(Issued by National Environmental Protection Administration on December 28, 2007)

17. Rules for Qualification Management of Welders and Welding Operators of Civilian Nuclear Safety Equipment HAF603

(Issued by National Environmental Protection Administration on December 28, 2007)

18. Rules for the Regulation and Management of Imported Civilian Nuclear Safety Equipment (HAF604)

(Issued by National Environmental Protection Administration on December 28, 2007)

19. Measures for the Administration on License of Transportation Safety of Radioactive Materials (HAF701)

(Issued by Ministry of Environmental Protection on September 25, 2010; Issued by Ministry of Environmental Protection on December 20, 2017)

20. Measures for the Regulation and Management on Transportation Safety of Radioactive Materials (HAF702).

(Issued by Ministry of Environmental Protection on March 14, 2016)

21. Measures for the Management on Safety License of Radioisotopes and Radiation Devices (HAF801)

(Issued by Ministry of Environmental Protection on December 6, 2008)

22. Measures for the Management on Safety and Protection of Radioisotopes and Radiation Devices (HAF802)

(Issued by Ministry of Environmental Protection on April 18, 2011)

23. Measures for the Management of Electromagnetic Environmental Radiation Protection (HAF901)

(Issued by NNSA on March 25, 1997)

24. Classification of Radioactive Wastes

(Issued by Ministry of Environmental Protection on December 2017)

25. List of Classification Management for Environmental Impact Assessment of Construction Projects

(Issued by Ministry of Ecology and Environment on April 28, 2018)

26. Measures for Public Participation in Environmental Impact Assessment

(Issued by Ministry of Ecology and Environment on July 16, 2018)

IV. Nuclear Safety Guides

1. Emergency Preparedness and Response for Nuclear Power Plant Operating Organization (HAD002/01)

(Issued by NNSA on August 20, 2010)

2. Emergency Preparedness of Local Government for Nuclear Power Plant (HAD002/02)

(Issued by NNSA, the National Environmental Protection Administration and the Ministry of Health on May 24, 1990)

3. Principles and Levels for Public Protection Intervention during the Nuclear Accidental Radiation Emergency (HAD002/03)

(Issued by NNSA, the National Environmental Protection Administration on April 19, 1991)

4. Levels of Derived Intervention of Public Protection during the Nuclear Accident Radiation Emergency (HAD002/04)

(Issued by NNSA, the National Environmental Protection Administration on April 19, 1991)

5. Medical Emergency Preparedness and Response of Nuclear Accident

(HAD002/05)

(Issued by the Ministry of Health and NNSA on June 24, 1992)

6. Preparation of the Quality Assurance Program for Nuclear Power Plants (HAD003/01)

(Issued by NNSA on October 6, 1988)

7. Quality Assurance Organization in Nuclear Power Plants (HAD003/02)

(Issued by NNSA on April 13, 1989)

8. Quality Assurance in the Procurement of Items and Service for Nuclear Power Plants (HAD003/03)

(Issued by NNSA on October 30, 1986)

9. Quality Assurance Record of Nuclear Power Plants (HAD003/04)

(Issued by NNSA on October 30, 1986)

10. Quality Assurance Audit of Nuclear Power Plants (HAD003/05)

(Issued by NNSA on January 28, 1988)

11. Quality Assurance in the Design of Nuclear Power Plants (HAD003/06)

(Issued by NNSA on October 30, 1986)

12. Quality Assurance in the Construction of Nuclear Power Plants (HAD003/07)

(Issued by NNSA on April 17, 1987)

13. Quality Assurance in the Manufacturer of Items for Nuclear Power Plant (HAD003/08)

(Issued by NNSA on October 30, 1986)

14. Quality Assurance during Commissioning and Operation of Nuclear Power Plants (HAD003/09)

(Issued by NNSA on January 28, 1988)

15. Quality Assurance in the Procurement, Design and Manufacture of Nuclear Fuel Assemblies (HAD003/10)

(Issued by NNSA on April 13, 1989)

16. Earthquake Related Issues in Nuclear Power Plant Siting (HAD101/01)

(Issued by NNSA and National Seismological Bureau on April 6, 1994)

17. Atmospheric Dispersion Related Issues in Nuclear Power Plant Siting (HAD101/02)

(Issued by NNSA on November 20, 1987)

18. Population Distribution Related Issues in Nuclear Power Plant Siting and Evaluation (HAD101/03)

(Issued by NNSA on November 20, 1987)

19. External Human-induced Events in Nuclear Power Plant Siting (HAD101/04)

(Issued by NNSA on November 28, 1989)

20. Hydrological Dispersion of Radioactive Material in Nuclear Power Plant Siting (HAD101/05)

(Issued by NNSA on April 26, 1991)

21. Hydrogeological Aspects in Nuclear Power Plant Siting (HAD101/06)

(Issued by NNSA on April 26, 1991)

22. Site Survey for Nuclear Power Plants (HAD101/07)

(Issued by NNSA on November 28, 1989)

23. Determination of Design Basis Floods for Nuclear Power Plants on River Sites (HAD101/08)

(Issued by NNSA on July 12, 1989)

24. Determination of Design Basis Floods for Nuclear Power Plants on Coastal Sites (HAD101/09)

(Issued by NNSA on May 19, 1990)

25. Extreme Meteorological Phenomenon in Nuclear Power Plant Siting (HAD101/10)

(Issued by NNSA on April 26, 1991)

26. Design Basis Tropical Cyclone for Nuclear Power Plants (HAD101/11)

(Issued by NNSA on April 26, 1991)

27. Foundation Safety of Nuclear Power Plants (HAD101/12)

(Issued by NNSA on February 20, 1990)

28. General Design Safety Principles for Nuclear Power Plants (HAD102/01)

(Issued by NNSA on July 12, 1989)

29. Seismic Design and Qualification of Nuclear Power Plants (HAD102/02)

(Issued by NNSA on May 13, 1996)

30. Safety Functions and Component Classification for BWR, PWR, and Pressure Tube Reactor (HAD102/03)

(Issued by NNSA on October 30, 1986)

31. Protection against Internally Generated Missiles and Their Secondary Effects in Nuclear Power Plants (HAD102/04)

(Issued by NNSA on October 30, 1986)

32. External Human-induced Events Related to Nuclear Power Plant Design (HAD102/05)

(Issued by NNSA on November 28, 1989)

33. Design of the Reactor Containment System in Nuclear Power Plants (HAD102/06)

(Issued by NNSA on May 19, 1990)

34. Design for Reactor Core Safety in Nuclear Power Plants (HAD102/07)

(Issued by NNSA on July 12, 1989)

35. Reactor Coolant Systems and Associated Systems in Nuclear Power Plants (HAD102/08)

(Issued by NNSA on April 13, 1989)

36. Ultimate Heat Sink and Directly Associated Heat Transport Systems for Nuclear Power Plants (HAD102/09)

(Issued by NNSA on April 17, 1987)

37. Protection System and Related Facilities in Nuclear Power Plants (HAD102/10)

(Issued by NNSA on October 6, 1988)

38. Fire Protection in Nuclear Power Plants (HAD102/11)

(Revised in 1996)

39. Radiation Protection Design for Nuclear Power Plants (HAD102/12)

(Issued by NNSA on May 19, 1990)

40. Emergency Power Systems in Nuclear Power Plants (HAD102/13)

(Issued by NNSA on February 13, 1996)

41. Safety-related Instrumentation and Control Systems for Nuclear Power Plants (HAD102/14)

(Issued by NNSA on October 6, 1988)

42. Design of Fuel Handling and Storage Systems in Nuclear Power Plants (HAD102/15)

(Issued by NNSA on January 23, 2007)

43. Computer-Based Software Important to Safety in Nuclear Power Plants (HAD102/16)

(Issued by NNSA on December 8, 2004)

44. Safety Assessment and Verification for Nuclear Power Plants (HAD102/17)

(Issued by NNSA on June 5, 2006)

45. Development and Application of Computer Software for Safety Analysis in Nuclear Power Plants (for Trial Implementation) (HAD102/18)

(Issued by NNSA on December 20, 2017)

46. Operational limits and Conditions and Operating Procedures for Nuclear Power Plants (HAD103/01)

(Issued by NNSA on January 1, 2005)

47. Commissioning Procedures for Nuclear Power Plants (HAD103/02)

(Issued by NNSA on April 17, 1987)

48. Reactor Core and Fuel Management in Nuclear Power Plants (HAD103/03)

(Issued by NNSA on November 28, 1989)

49. Radiation Protection in Operation of Nuclear Power Plants (HAD103/04)

(Issued by NNSA on May 19, 1990)

50. Recruitment, Training and Qualification of Personnel for Nuclear Power Plants (HAD103/05)

(Issued by NNSA on May 24, 2013)

51. Organization and Safe Operation Management of Nuclear Power Plants Operating Organization (HAD103/06)

(Issued by NNSA on June 5, 2006)

52. In-service Inspection of Nuclear Power Plants (HAD103/07)

(Issued by NNSA on October 6, 1988)

53. Maintenance of Nuclear Power Plants (HAD103/08)

(Issued by NNSA on January 1, 1994)

54. Surveillance of Items Important to Safety in Nuclear Power Plants (HAD103/09)

(Issued by NNSA on January 1, 1994)

55. Fire Protection Safety in Operation of Nuclear Power Plants (HAD103/10)

(Issued by NNSA on December 8, 2004)

56. Periodic Safety Review of Nuclear Power Plants (HAD103/11)

(Issued by NNSA on June 5, 2006)

57. Ageing management of Nuclear Power Plants (HAD 103/12)

(Issued by NNSA on May 23, 2012)

58. Management of Radioactive Effluents and Wastes in Nuclear Power Plants (HAD401/01)

(Issued by NNSA on May 19, 1990)

59. Design of Radioactive Waste Management System for Nuclear Power Plants (HAD401/02)

(Issued by NNSA on January 16, 1997)

60. Design and Operation of Radioactive Waste Incinerators (HAD401/03)

(Issued by NNSA on February 15, 1997)

61. Decommissioning of Gamma Irradiation Installation (HAD 401/07)

(Issued by NNSA on May 24, 2013)

62. Minimization of Radioactive Wastes from Nuclear Installations (HAD 401/08)

(Issued by NNSA on October 21, 2016)

63. Physical Protection of Nuclear Facilities (HAD501/02)

(Issued by NNSA on February 11, 2018)

65. Intrusion Alarm System at the Periphery of Nuclear Facilities (HAD501/03)

(Issued by NNSA July 22, 2005)

65. Access Control of Nuclear Facilities (HAD501/04)

(Issued by NNSA on September 1, 2008)

66. Physical Protection of Nuclear Material Transportation (HAD501/05)

(Issued by NNSA on September 1, 2008)

67. Format and Content of Physical Protection of Nuclear Facilities, Nuclear Material Accounting and Safety Analysis Report (HAD501/06)

(Issued by NNSA on September 1, 2008)

68. Nuclear Material Accounting of Nuclear Power Plants (HAD501/07)

(Issued by NNSA on September 1, 2008)

69. Fabrication of mock-up pieces for Civil Nuclear Safety Machinery and Equipment (Provisional) (HAD601/01)

(Issued by NNSA on May 24, 2013)

70. Technical Conditions of Civil Nuclear Safety Equipment Installation License Applicant (Provisional) (HAD601/02)

(Issued by NNSA on May 24, 2013)

71. Format and Content of Safety Assessment (Analysis) Report for Design of Transport Containers of Radioactive Material(HAD701/01)

(Issued by NNSA on May 31, 2010)

72. Form and Contents of Nuclear and Radiation Safety Analysis Report for Transportation of Radioactive Materials(HAD701/02)

(Issued by NNSA on June 9, 2014)

Appendix 5: List of Domestic and Overseas Peer Review Received by NPPs in China (From 2016 to 2018)

No.	Time	Host Plant	Review Content	Review Organization
1.	2016.01.04-01.08	Fangjiashan NPP	201 outage special evaluation	CNNP
2.	2016.01.11-01.19	Changjiang NPP	Pre-startup peer review of Unit 2	WANO
3.	2016.03.07-03.18	Fuqing NPP	Pre-startup peer review of Unit 3	WANO
4.	2016.03.21-04.01	Fangchenggang NPP	Pre-startup peer review of Unit 2	WANO
5.	2016.04.11-04.29	Daya Bay NPP	10 major areas, such as operation, maintenance and technology, etc.	WANO CNEA
6.	2016.05.17-05.20	Sanmen NPP	Special review of fire fighting	CNNP
7.	2016.05.29-06.03	Sanmen NPP	CPO	CNNP
8.	2016.05.31-06.03	Sanmen NPP	Special review of radiation protection	CNNP
9.	2016.07.04-07.15	Taishan NPP	Pre-startup peer review of Unit 1	WANO
10.	2016.07.17-07.22	Tianwan NPP	Peer review of commissioning in the installation phase of Units 3 and 4	CNEA
11.	2016.07.04-07.22	Fangchenggang NPP	Review of Units 1 and 2 on eight areas, e.g. nuclear safety, operation, maintenance and technology, etc.	CGN
12.	2016.08.08-08.12	Hainan NPP	Special evaluation in training fields	CNNP
13.	2016.08.14-08.19	Fuqing NPP	Nuclear safety culture review	CNNP
14.	2016.08.22-08.26	Third Qinshan NPP	Peer review of severe accident management	CNEA
15.	2016.08.24-08.26	Fuqing NPP	Follow-up for special review of investment management	CNNP
16.	2016.09.05-09.09	Qinshan Nuclear Power Base	Peer review follow-up 7 units	CNEA

17.	2016.09.21-09.23	Sanmen NPP	Follow-up for special review of design management	CNNP
18.	2016.10.10-10.21	Yangjiang NPP	Pre-startup peer review of Unit 4	WANO
19.	2016.10.24-10.28	Qinshan NPP	PSA special review	CNEA
20.	2016.09.27-09.29	Tianwan NPP	Special review on documentation management	CNNP
21.	2016.10.31-11.11	Sanmen NPP	Pre-startup peer review of Unit 1	WANO
22.	2016.11.02-11.04	Tianwan NPP	Follow-up of special evaluation in the field of procurement management	CNNP
23.	2016.11.06-11.12	Haiyang NPP	Pre-startup peer review of Units 1 and 2	WANO
24.	2016.11.14-11.18	Fuqing NPP	CPO for Unit 4	CNNP
25.	2016.11.19-11.26	Tianwan NPP	Peer review follow-up for Units 1 and 2	WANO CNEA
26.	2016.11.21-12.02	Fangjiashan NPP	Peer review	WANO
27.	2016.12.05-12.16	Haiyang NPP	Pre-startup peer review of Unit 1	WANO
28.	2016.12.05-12.09	Jiangsu Nuclear Power Corporation	Follow-up of Corporate peer review (CPR)	WANO
29.	2017.01.09-01.26	Taishan NPP	Pre-OSART for Unit 1	IAEA
30.	2017.02.13-02.17	Changjiang NPP	outage special evaluation	CNNP
31.	2017.02.20-02.24	Changjiang NPP	Special assessment of operation	CNNP
32.	2017.03.06-03.10	Third Qinshan NPP	Peer review follow-up	WANO
33.	2017.03.19-03.24	Shidao Bay NPP	Special review on commissioning	CNEA
34.	2017.03.13-03.17	Fuqing NPP	Special review on foreign material control	CNNP
35.	2017.04.13-04.27	Qinshan NPP	Peer review	WANO
36.	2017.04.17-04.28	Fuqing NPP	Pre-startup peer review of Unit 4	WANO
37.	2017.05.08-05.12	Tianwan NPP	CPO for Unit 3	CNNP
38.	2017.05.29-06.14	Ningde NPP	Peer review	WANO
39.	2017.06.05-06.05	Sanmen NPP	Special review on work	CNNP

			management process	
40.	2017.06.06-06.16	Qinshan NPP	Peer Review of Safety Aspects of Long-term Operation and Ageing management (SALTO)	IAEA
41.	2017.06.12-06.23	Tianwan NPP	Pre-startup peer review of Unit 3	WANO
42.	2017.06.20-06.24	Haiyang NPP	Special review on public communication	CNEA
43.	2017.07.24-07.28	Sanmen NPP	Special review on foreign material control	CNNP
44.	2017.08.14-08.18	Fuqing NPP	Special review on operating experience	CNNP
45.	2017.08.21-08.25	Sanmen NPP	Special review on documentation management	CNNP
46.	2017.08.28-09.15	Hongyanhe NPP	Peer review	WANO
47.	2017.08.29-08.31	Qinshan Nuclear Power Base	International Physical Protection Advisory Service (IPPAS) special review	IAEA
48.	2017.09.03-09.08	Shidao Bay NPP	Follow-up for special review on commissioning	CNEA
49.	2017.09.11-09.15	Changjiang NPP	Nuclear safety culture review	CNNP
50.	2017.09.12-09.13	Qinshan Phase II NPP	PSA peer review follow-up	CNEA
51.	2017.09.25-09.29	Taishan NPP	Pre-startup peer review of Unit 1	WANO
52.	2017.10.15-10.24	Shidao Bay NPP	Special review on CPO	CNEA
53.	2017.10.16-10.20	Qinshan Phase II NPP	Peer review follow-up	WANO
54.	2017.11.06-11.08	Qinshan Nuclear Power Base	Review on culture excellence	CNNP
55.	2017.11.06-11.10	Qinshan Nuclear Power Base	Nuclear safety culture review	CNNP
56.	2017.11.27 -12.01	Fangchenggang NPP	Review of Units 1 and 2 on eight areas, e.g. nuclear safety, operation, maintenance and technology, etc.	CGN
57.	2017.12.04-12.08	Sanmen NPP	CPO for Unit 2	CNNP
58.	2017.12.10-12.15	Shidao Bay NPP	Nuclear safety culture review	CNEA

59.	2018.01.15-01.26	Yangjiang NPP	Pre-startup peer review of Unit 5	WANO
60.	2018.03.19-03.23	Fangjiashan NPP	Peer review follow-up	WANO
61.	2018.03.05-03.09	Changjiang NPP	CPO for Units 1 and 2	CNNP
62.	2018.03.18-03.29	Haiyang NPP	Pre-startup peer review of Units 1 and 2	WANO
63.	2018.03.26-03.30	Tianwan NPP	CPO for Unit 4	CNNP
64.	2018.04.12-04.26	Changjiang NPP	Peer review for Units 1 and 2	WANO
65.	2018.05.21-05.25	Changjiang NPP	Procurement management review	CNNP
66.	2018.05.28-06.05	Sanmen NPP	Pre-startup peer review of Unit 1	WANO
67.	2018.05.13-05.17	Daya Bay Nuclear Power Base	Peer review follow-up	WANO
68.	2018.05.27-06.14	Fangchenggang NPP	Peer review for Units 1 and 2	WANO
69.	2018.06.11-06.15	Fuqing NPP	CPO for Units 1- 4	CNNP
70.	2018.06.25-07.06	Tianwan NPP	Pre-startup peer review of Unit 4	WANO
71.	2018.06.19-06.25	Shidao Bay NPP	Follow-up of CPO	CNEA
72.	2018.07.02-07.13	Fuqing NPP	Peer review for Units 1 to 4	CNEA
73.	2018.06.26-06.29	Sanmen NPP	Special review on code of conduct of maintenance personnel	CNNP
74.	2018.06.26-06.29	Sanmen NPP	Special review on temporary changes	CNNP
75.	2018.07.23-08.03	Sanmen NPP	Pre-startup peer review of Unit 2	WANO
76.	2018.08.20-08.24	Fuqing NPP	Special review on spare parts	CNNP
77.	2018.08.20-08.24	Fuqing NPP	Nuclear safety culture review	CNNP
78.	2018.08.20-08.24	Fuqing NPP	Special review on culture excellence	CNNP
79.	2018.09.17-09.21	Qinshan Nuclear Power Base	Review on spare parts	CNNP
80.	2018.09.17-09.21	Tianwan NPP	Nuclear safety culture review	CNNP
81.	2018.10.14-10.19	Shidao Bay NPP	Pre-startup SOER Response review	CNEA

82.	2018.10.15-10.19	Tianwan NPP	Special review on spare parts	CNNP
83.	2018.12.09-12.14	Sanmen NPP	Nuclear safety culture review	CNEA

Appendix 6: List of Scheduled Domestic and Overseas Peer Review for NPPs in China (From 2019 to 2021)

No.	Time	Host Plant	Review Content	Review Organization
1	2019.01	Fangjiashan NPP	OSART	IAEA
2	2019.01	Taishan NPP	Pre-OSART follow-up	IAEA
3	2019.01	Fuqing NPP	Peer review for Units 1 to 4	WANO
4	2019.04	Taishan NPP	PSUR of Unit 2	WANO
5	2019.09	Qinshan NPP	Peer review follow-up	WANO
6	2019.09	Qinshan Phase II NPP	outage observation	WANO
7	2019.09	Third Qinshan NPP	CPO	WANO
8	2019.09	Qinshan NPP	CPO	WANO
9	2019.10	Qinshan NPP	outage observation	WANO
10	2019.10	Fangjiashan NPP	CPO	WANO
11	2019.10	Qinshan Phase II NPP	CPO	WANO
12	2019	Fuqing NPP	CPO for Unit 5	CNNP
13	2019	Fuqing NPP	Special review on fire fighting	CNNP
14	2019	Fuqing NPP	Pre-startup peer review of Unit 5	WANO
15	2019	Sanmen NPP	Special review on outage preparation	CNNP
16	2019	Sanmen NPP	Special review on personnel performance	CNNP
17	2019	Sanmen NPP	Special review on procurement management	CNNP
18	2019	Sanmen NPP	Special review on operator fundamentals	CNNP
19	2019	Sanmen NPP	Special review on operating procedures	CNNP
20	2019	Tianwan NPP	Peer review for Units 1 and 2	WANO

No.	Time	Host Plant	Review Content	Review Organization
21	2019	Tianwan NPP	Pre-startup peer review of Unit 5	WANO
22	2019	Tianwan NPP	Special review on SPV equipment management	CNNP
23	2019	Tianwan NPP	Special review of operation preparation for Units 5 and 6	CNNP
24	2019	Tianwan NPP	CPO for Unit 5	CNNP
25	2019	Haiyang NPP	Operation review for Units 1 and 2	CNEA
26	2019	Haiyang NPP	Construction review of Units 3 and 4	CNEA
27	2020	Haiyang NPP	Review for Units 1 and 2	WANO
28	2020.02	Qinshan Nuclear Power Base	Peer review of 9 units	WANO
29	2020.05	Fangjiashan NPP	OSART follow-up	IAEA
30	2020	Fuqing NPP	CPO for Unit 6	CNNP
31	2020	Sanmen NPP	Peer review	WANO
32	2020	Sanmen NPP	Special review on operation risks	CNNP
33	2020	Sanmen NPP	Special review on reactivity management	CNNP
34	2020	Sanmen NPP	Special review on operating experience	CNNP
35	2020	Jiangsu Nuclear Power Corporation	Corporate peer review (CPR)	WANO
36	2020	Tianwan NPP	Pre-startup peer review of Unit 6	WANO
37	2020	Tianwan NPP	CPO for Unit 6	CNNP
38	2021	Tianwan NPP	Peer review follow-up for Units 1 and 2	WANO CNEA
39	2021	Tianwan NPP	Peer review for Units 3 and 4	WANO CNEA
40	2021	Tianwan NPP	Nuclear safety culture review	CNNP
41	2021	Tianwan NPP	CPO for Units 3 and 4	CNNP
42	2021	Tianwan NPP	Special review on SPV equipment	CNNP

No.	Time	Host Plant	Review Content	Review Organization
			management	
43	2021	Fuqing NPP	Nuclear safety culture review	CNNP
44	2021	Fuqing NPP	Follow-up of peer review for Units 1 to 4	WANO
45	2021	Fuqing NPP	Pre-startup peer review of Unit 6	WANO
46	2021	Haiyang NPP	Peer review follow-up for Units 1 and 2	CNEA
47	2021	Haiyang NPP	Construction review follow-up for Units 3 and 4	CNEA
48	Before the initial fuel loading	Shidao Bay NPP	Pre-startup peer review	WANO

Appendix 7: Licensed Operators and Senior Operators of Operating NPPs in China (by December 31, 2018)

NPP / Item	Qinshan NPP	Daya Bay NPP	Qinshan Phase II NPP	LingAo NPP	Third Qinshan NPP	Tianwan NPP	Hongyanhe NPP	Ningde NPP	Fuqing NPP	Yangjiang NPP	Fangjiashan NPP	Changjiang NPP	Fangchenggang NPP
Reactor Operators	24	35	60	73	48	106	111	79	123	104	44	68	65
Senior Reactor Operators	39	79	129	159	64	117	147	115	96	160	54	48	73

Appendix 8: Occupational Exposure in NPPs of China (From 2016 to 2018)

NPP \ Item			Annual Average Effective Dose (mSv)	Annual Maximum Individual Dose (mSv)	Annual Collective Effective Dose (man·Sv)	Normalized Collective Effective Dose (man·mSv/GWh)
Qinshan NPP	2016		0.133	3.439	0.281	0.109
	2017		0.029	2.135	0.042	0.015
	2018		0.272	5.687	0.764	0.470
Daya Bay NPP	2016		0.303	8.277	1.032	0.068
	2017		0.242	6.756	0.712	0.043
	2018		0.260	5.114	0.753	0.046
Qinshan Phase II NPP	2016		0.307	7.171	1.092	0.052
	2017		0.251	7.639	0.941	0.044
	2018		0.295	9.730	1.149	0.055
LingAo NPP	2016	(Units 1 & 2)	0.348	6.071	1.117	0.070
		(Units 3 & 4)	0.305	6.834	1.028	0.063
	2017	(Units 1 & 2)	0.301	6.610	0.917	0.059
		(Units 3 & 4)	0.332	7.668	1.200	0.074
	2018	(Units 1 & 2)	0.517	10.323	1.623	0.109
		(Units 3 & 4)	0.223	5.247	0.628	0.037
Third Qinshan NPP	2016		0.474	7.167	1.009	0.093
	2017		0.303	6.033	0.702	0.064
	2018		0.456	6.801	0.855	0.076

NPP \ Item			Annual Average Effective Dose (mSv)	Annual Maximum Individual Dose (mSv)	Annual Collective Effective Dose (man·Sv)	Normalized Collective Effective Dose (man·mSv/GWh)
Tianwan NPP	2016		0.297	6.032	1.010	0.066
	2017		0.126	2.140	0.326	0.019
	2018		0.107	3.074	0.600	0.026
Hongyanhe NPP	2016	(Units 1 & 2)	0.159	3.386	0.829	0.068
		(Units 3 & 4)	0.023	1.089	0.071	0.010
	2017	(Units 1 & 2)	0.152	3.672	0.655	0.047
		(Units 3 & 4)	0.196	7.136	1.042	0.109
	2018		0.435	7.601	1.530	0.065
Ningde NPP	2016	(Units 1 & 2)	0.284	4.918	0.718	0.053
		(Units 3 & 4)	0.263	5.484	0.766	0.069
	2017		0.514	8.624	1.965	0.064
	2018		0.369	7.998	1.448	0.043
Fuqing NPP	2016		0.239	8.763	0.920	0.057
	2017		0.363	8.007	1.609	0.065
	2018		0.368	9.999	1.768	0.058
Yangjiang NPP	2016		0.443	13.078	2.124	0.092
	2017		0.256	7.889	0.974	0.030
	2018		0.265	8.112	1.473	0.042
Fangjiashan NPP	2016		0.234	6.595	0.723	0.045
	2017		0.352	8.503	1.168	0.073
	2018		0.034	0.984	0.074	0.004

NPP \ Item		Annual Average Effective Dose (mSv)	Annual Maximum Individual Dose (mSv)	Annual Collective Effective Dose (man·Sv)	Normalized Collective Effective Dose (man·mSv/GWh)
Sanmen NPP	2018	0.005	0.246	0.009	0.001
Haiyang NPP	2018	0.003	0.232	0.008	0.002
Taishan NPP	2018	0.007	0.288	0.014	0.004
Changjiang NPP	2016	0.011	0.945	0.018	0.031
	2017	0.320	5.890	0.842	0.113
	2018	0.221	4.335	0.549	0.071
Fangchenggang NPP	2016	0.011	0.432	0.022	0.002
	2017	0.431	8.034	1.377	0.109
	2018	0.135	3.588	0.298	0.019

Appendix 9: List of Drills of NPPs in China (From 2016 to 2018)

Year	Organization	Times of Joint Drills	Times of Comprehensive Drills	Times of Single Drills
2016	Qinshan Nuclear Power Base	0	1	104
	Daya Bay Nuclear Power Base	0	1	55
	Tianwan NPP	0	1	38
	Hongyanhe NPP	1	1	54
	Ningde NPP	0	1	49
	Fuqing NPP	0	1	58
	Yangjiang NPP	0	1	20
	Sanmen NPP	0	1	24
	Haiyang NPP	0	1	18
	Changjiang NPP	0	0	25
	Fangchenggang NPP	0	0	67
2017	Qinshan Nuclear Power Base	0	3	97
	Daya Bay Nuclear Power Base	0	1	68
	Tianwan NPP	1	1	39
	Hongyanhe NPP	0	1	73
	Ningde NPP	0	1	23
	Fuqing NPP	0	1	46
	Yangjiang NPP	0	1	21
	Sanmen NPP	1	0	38
	Haiyang NPP	1	0	20
	Taishan NPP	0	1	32
	Changjiang NPP	0	1	39
	Fangchenggang NPP	0	1	65
2018	Qinshan Nuclear Power Base	0	1	140
	Daya Bay Nuclear Power Base	0	1	67
	Tianwan NPP	0	1	43

	Hongyanhe NPP	0	1	86
	Ningde NPP	0	1	67
	Fuqing NPP	0	1	54
	Yangjiang NPP	1	1	25
	Sanmen NPP	0	1	47
	Haiyang NPP	0	1	17
	Taishan NPP	0	0	25
	Changjiang NPP	0	0	33
	Fangchenggang NPP	0	0	64

Note: According to the definitions in 9.2.2 of the *Emergency Preparedness and Emergency Response of Operating Organization of Nuclear Power Plants* (HAD 002/01-2010) issued by MEE (NNSA) on August 20, 2010, emergency drill includes single drill (exercise) and comprehensive drill of on-site emergency organization as well as joint drill of off-site emergency organization, and the exercise can be a part of the drill.

Annex: Contributors to the Eighth National Report of China

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Tang Bo, Yao Bin, Shi Lishan, Zhang Jing, Li Jingyun, Hou Wei, Wang Jinjun, Zhang Yue, Li Xiaomeng, Cui Hongyu, Han Yuhong, Yang Di, Cheng Jianxiu, Rao Shuang, Huang Xiaoheng, Chen Zhigang, Liu Zhiyong, Wang Changming, Cai Erling, Jiang Xiaohua, Zhu Delong, Zhen Xiaohui, Yi Fei, Zhu Yuanli, Gong Bing, Huang Fang, Sun Quanfu.

List of Compilers of the Eighth National Report of China (in the sequence of numerical strokes of surnames)

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